Theory of Statistics

Axiomatic Foundations, Proofs and Computational Applications

İbrahim Güney

October 26, 2026

Preface

This book aims to present the core ideas of mathematical statistics as a single whole, starting from measure-theoretic probability and continuing through point estimation, hypothesis testing, interval estimation and Bayesian inference. The final part unifies these topics within decision theory (minimaxity and admissibility) and extends them to three topics that arise often in practice: nonparametric methods, multiple testing and the false discovery rate, and the EM algorithm together with generalized linear models. Every chapter follows the same structure:

1. **Introduction and motivation:** the question the chapter sets out to answer.
2. **Definitions:** precise, axiomatic statements.
3. **Theorems and proofs:** the main results, with their assumptions stated explicitly.
4. **Worked examples:** what the theory says for the classical distributions.
5. **Computational applications:** Python code that checks the theoretical results by Monte Carlo simulation, together with the actual output of that code.
6. **Comments:** the limits of the results, common mistakes and links to other chapters.
7. **Chapter summary:** a box collecting the main results with their theorem numbers, and a note pointing to the corresponding chapters of standard textbooks.
8. **Exercises:** for consolidating the material.

Appendices. Appendix **??** at the end of the book is a reference table of the densities, means, variances and MGFs of common distributions, the relations between them and their conjugate priors. Appendix A gives solutions to all the exercises; the link at the end of each exercise leads to its solution, and the heading of each solution leads back to the exercise. The *Index* at the very end lists the pages on which the main concepts appear; the page on which a concept is defined is set in bold.

About the code. All code is written in Python 3 with `numpy`, `scipy` and `matplotlib`. Each code block is a separate file in the `code/` folder of the book (the file name is printed above the code box) and runs on its own; all of them can be run with `python3 code/run_all.py`. The simulations use numpy's recommended random number interface:

```
rng = np.random.default_rng(42)
```

Because the seed is fixed, the results are reproducible. The "Output" boxes directly below the code blocks were produced by actually running the code. Throughout the book the Gamma and exponential distributions are written with a *rate* parameter (see Notation), whereas `numpy` and `scipy` parametrise them by the *scale* $= 1/\text{rate}$. Expressions such as `scale=1/beta` in the code perform this conversion.

Notation

Symbol	**Meaning**
$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space
$\mathcal{B}(\mathbb{R})$	Borel σ-algebra on $\mathbb{R}$
F_X, f_X, p_X	Distribution function (CDF), density (pdf), probability mass function (pmf)
$\mathbb{E}$, Var, Cov	Expectation, variance, covariance
$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$	Independent and identically distributed sample
$\bar{X}_n$, S_n^2	Sample mean, unbiased sample variance (divided by $n-1$)
$X_{(1)} \le \cdots \le X_{(n)}$	Order statistics
$\xrightarrow{P}$, $\xrightarrow{d}$, $\xrightarrow{\text{a.s.}}$, $\xrightarrow{L^p}$	Convergence in probability, in distribution, almost surely, in L^p
$L(\theta; \mathbf{x})$, $\ell(\theta; \mathbf{x})$	Likelihood and log-likelihood function
$I_1(\theta)$, $I_n(\theta)$	Fisher information for one and for n observations
Φ, φ, z_α	Standard normal CDF, pdf and upper α quantile: $\Phi(z_\alpha) = 1-\alpha$
$\chi^2_{k,\alpha}$, $t_{k,\alpha}$	Upper α quantiles of the chi-square and t distributions
Gamma(α, β)	Shape α, *rate* β: $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, mean α/β
Exp(λ)	Rate λ: Gamma$(1, \lambda)$, mean $1/\lambda$; also $\chi^2_k = \text{Gamma}(k/2, \frac{1}{2})$
$\mathbb{I}_A$	Indicator function of the event A
$\perp\!\!\!\perp$	Independence

Contents

Preface **i**

I Foundations of Probability Theory **1**

1 Probability Spaces and Kolmogorov's Axioms **2**
1.1 Definitions . . . 2
1.2 Main Theorems and Proofs . . . 3
1.3 Worked Examples . . . 4
1.4 Computational Applications . . . 5
1.4.1 Verifying Continuity by Monte Carlo . . . 5
1.4.2 The Birthday Problem: Exact Value and Simulation . . . 6
1.4.3 Bayes' Formula: Simulating a Diagnostic Test . . . 6
1.5 Comments and Notes . . . 6
1.6 Chapter Summary . . . 7
1.7 Exercises . . . 7

2 Random Variables and Distribution Functions **8**
2.1 Definitions . . . 8
2.2 Main Theorems and Proofs . . . 9
2.3 Worked Examples . . . 11
2.4 Computational Applications . . . 11
2.4.1 Empirical CDF and the Kolmogorov–Smirnov Distance . . . 11
2.4.2 Inverse Transform and the Probability Integral Transform . . . 13
2.5 Comments and Notes . . . 14
2.6 Chapter Summary . . . 14
2.7 Exercises . . . 14

3 Expectation, Variance and Cauchy–Schwarz **15**
3.1 Definitions . . . 15
3.2 Main Theorems and Proofs . . . 16
3.3 Worked Examples . . . 17
3.4 Computational Applications . . . 18
3.4.1 Verifying the Cauchy–Schwarz Inequality . . . 18
3.4.2 Uncorrelatedness versus Independence; the Law of Total Variance . . . 19
3.4.3 Jensen's Inequality: $\mathbb{E}[1/X]$ and $\mathbb{E}[S]$ . . . 19
3.5 Comments and Notes . . . 20
3.6 Chapter Summary . . . 20
3.7 Exercises . . . 20

II Distributions and Limit Theorems 21

4 Joint Distributions and Transformations 22
4.1 Definitions . . . 22
4.2 Main Theorems and Proofs . . . 23
4.3 Worked Examples . . . 25
4.4 Computational Applications . . . 26
4.4.1 Box–Muller and Gamma–Beta Independence . . . 26
4.4.2 Multivariate Normal: Cholesky Generation, Conditional Distribution, Mahalanobis . . . 27
4.4.3 Order Statistics and a Vector That Is Not Jointly Normal . . . 27
4.5 Comments and Notes . . . 28
4.6 Chapter Summary . . . 29
4.7 Exercises . . . 29

5 Basic Inequalities and the Law of Large Numbers 30
5.1 Definitions: Convergence of Random Variables . . . 30
5.2 Probability Inequalities . . . 30
5.3 Laws of Large Numbers . . . 32
5.4 Relations Between the Modes of Convergence . . . 32
5.5 Worked Examples . . . 33
5.6 Computational Applications . . . 33
5.6.1 An Empirical Check of Chebyshev's Bound . . . 33
5.6.2 How Well Do the Sample-Size Rules Really Perform? . . . 34
5.6.3 The Law of Large Numbers and the Cauchy Counterexample . . . 34
5.7 Comments and Notes . . . 35
5.8 Chapter Summary . . . 35
5.9 Exercises . . . 36

6 Characteristic Functions and the Central Limit Theorem 37
6.1 Definitions . . . 37
6.2 Properties of the Characteristic Function . . . 37
6.3 The Central Limit Theorem . . . 40
6.4 Relations Between Modes of Convergence and Slutsky . . . 42
6.5 Worked Examples . . . 43
6.6 Computational Applications . . . 43
6.6.1 Convergence to the CLT: Skewness and Kurtosis . . . 43
6.6.2 Checking the Berry–Esseen Bound by Exact Computation . . . 44
6.6.3 Visualising the CLT and the Continuity Correction . . . 44
6.7 Comments and Notes . . . 45
6.8 Chapter Summary . . . 46
6.9 Exercises . . . 46

III Point Estimation Theory 47

7 Unbiasedness, Consistency and the MSE Decomposition 48
7.1 Definitions . . . 48
7.2 Main Theorems and Proofs . . . 49
7.3 Worked Examples . . . 49
7.4 Computational Applications . . . 50
7.4.1 Comparing the MSE of S^2 and the MLE . . . 50

7.4.2 Estimators for Uniform$(0,\theta)$ . . . 51
7.4.3 Inadmissibility: the James–Stein Estimator . . . 51
7.5 Comments and Notes . . . 52
7.6 Chapter Summary . . . 52
7.7 Exercises . . . 53

8 The Cramér–Rao Lower Bound and Fisher Information 54
8.1 Definitions . . . 54
8.2 Main Theorems and Proofs . . . 55
8.3 Worked Examples . . . 56
8.4 Computational Applications . . . 57
8.4.1 Poisson: $\bar{X}$ Attains the CRLB . . . 57
8.4.2 Moments of the Score: Checking Lemma 8.6 . . . 57
8.4.3 When the CRLB Is Not Attained and When It Is Violated . . . 58
8.5 Comments and Notes . . . 58
8.6 Chapter Summary . . . 59
8.7 Exercises . . . 59

9 Sufficiency and the Rao–Blackwell Theorem 60
9.1 Definitions . . . 60
9.2 Main Theorems and Proofs . . . 60
9.3 Worked Examples . . . 62
9.4 Computational Applications . . . 63
9.4.1 Bernoulli: $X_1 \to \bar{X}$ . . . 63
9.4.2 Poisson: Three Estimators of $\mathbb{P}(X=0)$ . . . 63
9.4.3 Uniform$(0,\theta)$: Rao–Blackwell by Simulation . . . 64
9.5 Comments and Notes . . . 64
9.6 Chapter Summary . . . 65
9.7 Exercises . . . 65

10 Completeness, Lehmann–Scheffé and the UMVUE 66
10.1 Definitions . . . 66
10.2 Main Theorems and Proofs . . . 66
10.3 Worked Examples . . . 67
10.4 Computational Applications . . . 69
10.4.1 The German Tank Problem . . . 69
10.4.2 Basu's Theorem: $\bar{X} \perp\!\!\!\perp S^2$ Only for the Normal . . . 69
10.4.3 Checking the Characterisation of the UMVUE . . . 70
10.5 Comments and Notes . . . 70
10.6 Chapter Summary . . . 70
10.7 Exercises . . . 71

11 Maximum Likelihood and the Invariance Principle 72
11.1 Definitions . . . 72
11.2 Main Theorems and Proofs . . . 72
11.3 Worked Examples . . . 73
11.4 Computational Applications . . . 74
11.4.1 Analytic and Numerical MLE in the Normal Model; Invariance . . . 74
11.4.2 Gamma MLE: Newton–Raphson . . . 75
11.4.3 Multimodality of the Cauchy Likelihood . . . 75
11.5 Comments and Notes . . . 76
11.6 Chapter Summary . . . 77

11.7 Exercises . . . 77

12 Asymptotic Theory of the MLE and the Delta Method 78
12.1 Definitions . . . 78
12.2 Consistency . . . 78
12.3 Asymptotic Normality . . . 79
12.4 The Delta Method . . . 80
12.5 Worked Examples . . . 81
12.6 Computational Applications . . . 82
12.6.1 The Delta Method for the Odds . . . 82
12.6.2 Asymptotic Normality of the MLE and Coverage of the Wald Interval 82
12.6.3 ARE, Variance Stabilisation and the Second-Order Delta Method . . . 83
12.7 Comments and Notes . . . 83
12.8 Chapter Summary . . . 84
12.9 Exercises . . . 84

IV Hypothesis Testing and Interval Estimation 85

13 The Neyman–Pearson Lemma and Most Powerful Tests 86
13.1 Definitions . . . 86
13.2 The Neyman–Pearson Lemma . . . 87
13.3 Worked Examples . . . 88
13.4 Computational Applications . . . 88
13.4.1 Power of the Test for a Normal Mean . . . 88
13.4.2 A Randomised Poisson Test . . . 89
13.4.3 Power Curves and the Distribution of p-Values . . . 89
13.5 Comments and Notes . . . 90
13.6 Chapter Summary . . . 91
13.7 Exercises . . . 91

14 The Karlin–Rubin Theorem and UMP Tests 92
14.1 Definitions . . . 92
14.2 Main Theorems and Proofs . . . 92
14.3 Worked Examples . . . 94
14.4 Computational Applications . . . 94
14.4.1 The Exponential Rate Test: Simulation and Exact Power . . . 94
14.4.2 The Two-Sided Problem: One-Sided Tests Cannot Be UMP . . . 95
14.4.3 Uniform: MLR and the Power of the UMP Test . . . 96
14.5 Comments and Notes . . . 96
14.6 Chapter Summary . . . 97
14.7 Exercises . . . 97

15 The Likelihood Ratio Test and Wilks' Theorem 98
15.1 Definitions . . . 98
15.2 Main Theorems and Proofs . . . 98
15.3 Worked Examples . . . 100
15.4 Computational Applications . . . 100
15.4.1 Verifying Wilks' Theorem (Poisson) . . . 100
15.4.2 The Holy Trinity: Exact Sizes in Small Samples . . . 101
15.4.3 Multinomial Goodness of Fit . . . 101
15.5 Comments and Notes . . . 102

15.6 Chapter Summary . . . 102
15.7 Exercises . . . 103

16 Confidence Intervals and the Pivotal Quantity Method 104
16.1 Definitions . . . 104
16.2 Normal Sampling Theory . . . 105
16.3 Methods for Constructing Confidence Intervals . . . 105
16.4 Worked Examples . . . 106
16.5 Computational Applications . . . 107
16.5.1 Coverage of the t Interval . . . 107
16.5.2 Binomial Proportion: Wald, Wilson, Clopper–Pearson, Agresti–Coull . 108
16.5.3 The Pivotal Interval for Uniform$(0, \theta)$ . . . 108
16.6 Comments and Notes . . . 109
16.7 Chapter Summary . . . 110
16.8 Exercises . . . 110

17 The Linear Model 111
17.1 Definitions . . . 111
17.2 Least Squares: Geometry and Moments . . . 112
17.3 Exact Inference in the Normal Linear Model . . . 113
17.4 Without Normality: Asymptotic Theory and Heteroscedasticity . . . 114
17.5 Worked Examples . . . 115
17.6 Computational Applications . . . 116
17.6.1 Verifying the Sampling Theory . . . 116
17.6.2 Gauss–Markov: LS versus the Endpoint Estimator; ANOVA . . . 117
17.6.3 Non-Normal Errors, Heteroscedasticity and Sandwich Standard Errors 117
17.7 Comments and Notes . . . 118
17.8 Chapter Summary . . . 119
17.9 Exercises . . . 119

18 The Bootstrap and Resampling Methods 120
18.1 Definitions . . . 120
18.2 Main Theorems and Proofs . . . 121
18.3 Worked Examples . . . 123
18.4 Computational Applications . . . 123
18.4.1 Bootstrap Standard Error of the Median . . . 123
18.4.2 Coverage of Bootstrap Confidence Intervals . . . 124
18.4.3 Failure: The Maximum of Uniform$(0, \theta)$ . . . 125
18.4.4 Jackknife and Permutation Test . . . 126
18.5 Comments and Notes . . . 127
18.6 Chapter Summary . . . 127
18.7 Exercises . . . 128

V Bayesian Inference 129

19 Bayesian Inference and Conjugate Priors 130
19.1 Definitions . . . 130
19.2 Main Theorems and Proofs . . . 131
19.3 Worked Examples . . . 132
19.4 Computational Applications . . . 133
19.4.1 Beta–Binomial: Posterior Summaries and the HPD Interval . . . 133

19.4.2 Prior, Likelihood and Posterior; Bernstein–von Mises 134
19.4.3 Frequentist Coverage of Bayesian Intervals 134
19.4.4 A Non-Conjugate Model: The Metropolis Algorithm 135
19.5 Comments and Notes 136
19.6 Chapter Summary 136
19.7 Exercises 137

VI Decision Theory and Advanced Topics 138

20 Statistical Decision Theory: Minimaxity and Admissibility 139
20.1 Definitions 139
20.2 Main Theorems and Proofs 140
20.3 Worked Examples 141
20.4 Computational Applications 142
20.4.1 Risk Curves for a Binomial Proportion 142
20.4.2 The Risk Set in a Two-Point Problem 143
20.5 Comments and Notes 145
20.6 Chapter Summary 145
20.7 Exercises 146

21 Nonparametric Methods 147
21.1 Definitions 147
21.2 Main Theorems and Proofs 148
21.3 Worked Examples 151
21.4 Computational Applications 151
21.4.1 Level and Power of Three Tests 151
21.4.2 Kolmogorov–Smirnov and the Lilliefors Problem 152
21.4.3 Kernel Density Estimation: Bandwidth and Rate of Convergence . . . 153
21.5 Comments and Notes 154
21.6 Chapter Summary 155
21.7 Exercises 155

22 Multiple Testing: Family-Wise Error Rate and False Discovery Rate 156
22.1 Definitions 156
22.2 Main Theorems and Proofs 157
22.3 Worked Examples 158
22.4 Computational Applications 159
22.4.1 FWER, FDR and Power 159
22.4.2 Dependence and a Visual Interpretation of BH 160
22.5 Comments and Notes 161
22.6 Chapter Summary 162
22.7 Exercises 162

23 The EM Algorithm and Generalised Linear Models 163
23.1 Definitions 163
23.2 Main Theorems and Proofs 164
23.3 Worked Examples 166
23.4 Computational Applications 167
23.4.1 EM: Linkage Data and a Normal Mixture 167
23.4.2 Logistic Regression: IRLS, Wald Intervals and Complete Separation . 168
23.5 Comments and Notes 169

23.6 Chapter Summary 170
23.7 Exercises 170

VII Appendices **172**

A Solutions to the Exercises **173**
A.1 Probability Spaces 173
A.2 Random Variables and Distribution Functions 174
A.3 Expectation and Variance 174
A.4 Joint Distributions and Transformations 175
A.5 Inequalities and the Law of Large Numbers 175
A.6 Characteristic Functions and the CLT 176
A.7 Unbiasedness, Consistency and MSE 176
A.8 Cramér–Rao and Fisher Information 177
A.9 Sufficiency and Rao–Blackwell 178
A.10 Completeness and UMVUE 178
A.11 Maximum Likelihood 179
A.12 MLE Asymptotics and the Delta Method 179
A.13 The Neyman–Pearson Lemma 180
A.14 Karlin–Rubin and UMP Tests 180
A.15 The Likelihood Ratio Test 181
A.16 Confidence Intervals 182
A.17 The Linear Model 182
A.18 The Bootstrap 183
A.19 Bayesian Inference 184
A.20 Decision Theory 184
A.21 Nonparametric Methods 185
A.22 Multiple Testing 186
A.23 EM and GLMs 186

Bibliography **188**

Part I

Foundations of Probability Theory

Chapter 1

Probability Spaces and Kolmogorov's Axioms

Probability theory is the mathematics of assigning numerical weights, in a consistent way, to the outcomes of experiments that involve uncertainty. The word "consistent" is crucial: unless we specify in advance which sets may receive a probability (the σ-algebra) and which rules the assignment must obey (Kolmogorov's axioms), arguments that look intuitively correct can lead to contradictions. In this chapter we lay the foundation on which the rest of the book is built.

1.1 Definitions

Definition 1.1 (Sample Space and Event)**.** The set of all possible outcomes of a random experiment is called the *sample space* and is denoted by Ω. Its elements $\omega \in \Omega$ are called *outcomes* (or elementary events), and the subsets $A \subseteq \Omega$ to which a probability can be assigned are called *events*.

Definition 1.2 (σ-Algebra)**.** A family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ of subsets of the sample space Ω is a σ-algebra if

1. $\Omega \in \mathcal{F}$,
2. $A \in \mathcal{F} \implies A^c = \Omega \setminus A \in \mathcal{F}$,
3. $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is called a *measurable space*. If the third condition is required only for finite unions, $\mathcal{F}$ is called an *algebra* (or field).

Remark 1.3. By De Morgan's law $\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c$, a σ-algebra is also closed under countable intersections, under differences ($A \setminus B = A \cap B^c$), and contains $\emptyset = \Omega^c$.

Definition 1.4 (Generated σ-Algebra)**.** Let $\mathcal{C} \subseteq \mathcal{P}(\Omega)$ be any family of sets. The intersection of all σ-algebras containing $\mathcal{C}$,

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{G} : \mathcal{G} \text{ is a } \sigma\text{-algebra and } \mathcal{C} \subseteq \mathcal{G}\},$$

is again a σ-algebra, and it is the *smallest* σ-algebra containing $\mathcal{C}$. (The intersection is over a non-empty family, since $\mathcal{P}(\Omega)$ always belongs to it.)

Definition 1.5 (Kolmogorov's Axioms of Probability [43])**.** A function $\mathbb{P} : \mathcal{F} \to [0, 1]$ defined on a measurable space $(\Omega, \mathcal{F})$ is a *probability measure* if it satisfies:

1. (Non-negativity) $\mathbb{P}(A) \geq 0$ for every $A \in \mathcal{F}$.
2. (Normalisation) $\mathbb{P}(\Omega) = 1$.
3. (Countable additivity) For every sequence $\{A_i\}_{i=1}^{\infty}$ of pairwise disjoint events, $\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a *probability space*.

Definition 1.6 (Conditional Probability)**.** For $\mathbb{P}(B) > 0$, the conditional probability of A given B is defined by

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For fixed B, the map $A \mapsto \mathbb{P}(A \mid B)$ is a new probability measure on $(\Omega, \mathcal{F})$.

Definition 1.7 (Independence)**.** Events $A, B \in \mathcal{F}$ are *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. A family $\{A_i\}_{i \in \mathcal{I}}$ is *mutually independent* if

$$\mathbb{P}\Big(\bigcap_{j \in J} A_j\Big) = \prod_{j \in J} \mathbb{P}(A_j)$$

for every finite $J \subseteq \mathcal{I}$. If the condition is required only for $|J| = 2$, the family is *pairwise independent*.

Definition 1.8 (Limit Superior and Limit Inferior of a Sequence of Events)**.** For a sequence of events $\{A_n\}$,

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \{A_n \text{ occurs infinitely often}\}, \qquad \liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k.$$

$\liminf A_n$ is the event "from some index on, all the A_n occur". Both belong to $\mathcal{F}$.

1.2 Main Theorems and Proofs

Proposition 1.9 (Basic Properties of a Probability Measure)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A, B \in \mathcal{F}$.*

1. $\mathbb{P}(\emptyset) = 0$.
2. *(Finite additivity) If $A_1, \dots, A_n$ are pairwise disjoint, then $\mathbb{P}(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i)$.*
3. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
4. *(Monotonicity) $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$ and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$.*
5. *(Inclusion–exclusion) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.*
6. *(Boole's inequality) For every $\{A_i\} \subseteq \mathcal{F}$, $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.*

Proof. (1) Applying Axiom 3 to the disjoint sequence $A_1 = \Omega$, $A_i = \emptyset$ $(i \geq 2)$ gives $1 = 1 + \sum_{i \geq 2} \mathbb{P}(\emptyset)$; a sum of non-negative terms equal to zero forces $\mathbb{P}(\emptyset) = 0$.

(2) Take $A_{n+1} = A_{n+2} = \cdots = \emptyset$ and use Axiom 3 together with (1).

(3) From the disjoint union $\Omega = A \cup A^c$ we get $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$.

(4) $B = A \cup (B \setminus A)$ is a disjoint union; hence $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.

(5) From the disjoint unions $A \cup B = A \cup (B \setminus A)$ and $B = (A \cap B) \cup (B \setminus A)$ we obtain $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

(6) The sets $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{i<n} A_i$ are pairwise disjoint, $B_n \subseteq A_n$ and $\bigcup B_n = \bigcup A_n$. By Axiom 3 and monotonicity, $\mathbb{P}(\bigcup A_n) = \sum \mathbb{P}(B_n) \leq \sum \mathbb{P}(A_n)$. $\square$

Theorem 1.10 (Continuity of Probability)**.** *If $A_1 \subseteq A_2 \subseteq \dots$ is an increasing sequence of events and $A = \bigcup_{n=1}^{\infty} A_n$, then*

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A).$$

Proof. Define disjoint rings: $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1} \in \mathcal{F}$ for $n \geq 2$. The sets are pairwise disjoint, $\bigcup_{i=1}^{n} C_i = A_n$ and $\bigcup_{i=1}^{\infty} C_i = A$. By Kolmogorov's Axiom 3 and finite additivity,

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} C_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(C_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{P}(C_i) = \lim_{n\to\infty} \mathbb{P}(A_n). \qquad \square$$

Corollary 1.11 (Continuity from Above)**.** *If $A_1 \supseteq A_2 \supseteq \dots$ is a decreasing sequence and $A = \bigcap_{n=1}^{\infty} A_n$, then $\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.*

Proof. The sequence A_n^c is increasing, and by De Morgan $\bigcup_n A_n^c = A^c$. By Theorem 1.10, $1 - \mathbb{P}(A_n) = \mathbb{P}(A_n^c) \to \mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. $\square$

Remark 1.12. Theorem 1.10 is in fact *equivalent* to Axiom 3: every finitely additive set function that is continuous from below is countably additive. Continuity is thus the "analytic face" of countable additivity. In Chapter 2, the right-continuity of the CDF will follow directly from Corollary 1.11.

Theorem 1.13 (Law of Total Probability and Bayes' Formula)**.** *Let $\{B_i\}_{i\geq 1}$ be a countable partition of Ω (pairwise disjoint with $\bigcup B_i = \Omega$) with $\mathbb{P}(B_i) > 0$. For every $A \in \mathcal{F}$,*

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i), \qquad \mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)} \quad (\mathbb{P}(A) > 0).$$

Proof. $A = \bigcup_i (A \cap B_i)$ is a disjoint union; by Axiom 3 and the definition of conditional probability, $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$. Bayes' formula follows by writing the numerator and denominator of $\mathbb{P}(B_j \mid A) = \mathbb{P}(A \cap B_j)/\mathbb{P}(A)$ in this way. $\square$

Theorem 1.14 (Borel–Cantelli Lemmas; see [22, Section 2.3])**.** *Let $\{A_n\}$ be a sequence of events.*

1. *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(\limsup A_n) = 0$.*
2. *If the $\{A_n\}$ are mutually independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(\limsup A_n) = 1$.*

Proof. (1) For every n, $\limsup A_k \subseteq \bigcup_{k\geq n} A_k$; by Boole's inequality $\mathbb{P}(\limsup A_k) \leq \sum_{k\geq n} \mathbb{P}(A_k)$. The right-hand side is the tail of a convergent series and tends to zero as $n \to \infty$.

(2) Since $(\limsup A_n)^c = \bigcup_n \bigcap_{k\geq n} A_k^c$, it suffices to show that $\mathbb{P}(\bigcap_{k\geq n} A_k^c) = 0$ for every n. By independence and the inequality $1 - x \leq e^{-x}$, for every $m > n$,

$$\mathbb{P}\Big(\bigcap_{k=n}^{m} A_k^c\Big) = \prod_{k=n}^{m} (1 - \mathbb{P}(A_k)) \leq \exp\Big(-\sum_{k=n}^{m} \mathbb{P}(A_k)\Big) \xrightarrow{m\to\infty} 0.$$

Corollary 1.11 then gives $\mathbb{P}(\bigcap_{k\geq n} A_k^c) = 0$. $\square$

1.3 Worked Examples

Example 1.15 (A Finite σ-Algebra)**.** For a roll of a die let $\Omega = \{1, \dots, 6\}$. If we are interested only in the information "odd or even?", then $\mathcal{F} = \{\emptyset, \{1,3,5\}, \{2,4,6\}, \Omega\}$ is a σ-algebra. This $\mathcal{F}$ does not contain the event $\{1\}$, so at this level of information no probability can be assigned to the event "a 1 was rolled". Interpreting a σ-algebra as a body of *information* is the basis of the conditional expectation in Chapter 3.

Example 1.16 (An Algebra That Is Not a σ-Algebra)**.** Let $\Omega = \mathbb{N}$ and $\mathcal{A} = \{A \subseteq \mathbb{N} : A \text{ is finite or } A^c \text{ is finite}\}$. $\mathcal{A}$ is closed under complements and finite unions, so it is an algebra. However, the set of even numbers, which is the union of the finite sets $A_i = \{2i\}$, is neither finite nor co-finite; hence $\mathcal{A}$ is not a σ-algebra.

Example 1.17 (The Birthday Problem)**.** Suppose the birthdays in a group of n people are independent and uniformly distributed over 365 days. By the complement rule, the probability that at least two people share a birthday is

$$p_n = 1 - \prod_{k=0}^{n-1} \frac{365 - k}{365}.$$

The approximation $1 - x \approx e^{-x}$ gives $p_n \approx 1 - \exp\{-n(n-1)/730\}$; for $n = 23$, $p_{23} \approx 0.507$. This counter-intuitive result is explained by the existence of $\binom{23}{2} = 253$ distinct pairs of people.

Example 1.18 (A Medical Diagnostic Test — Bayes' Formula)**.** Let the prevalence of a disease be $\mathbb{P}(D) = 0.01$, the sensitivity of the test $\mathbb{P}(+ \mid D) = 0.99$ and its specificity $\mathbb{P}(- \mid D^c) = 0.95$. By Theorem 1.13,

$$\mathbb{P}(D \mid +) = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0099}{0.0594} \approx 0.167.$$

A person with a positive result is actually ill with probability only 16.7%. A low prior probability (prevalence) lowers the positive predictive value of even a highly accurate test. This is the simplest instance of the discussion of "the role of prior information" in Chapter 19.

Example 1.19 (Events That Are Pairwise but Not Mutually Independent)**.** Toss two fair coins: $A = \{\text{first toss H}\}$, $B = \{\text{second toss H}\}$, $C = \{\text{both tosses equal}\}$. Since $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 1/2$ and every pairwise intersection has probability $1/4$, the events are pairwise independent. However, $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\{HH\}) = 1/4 \neq 1/8 = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$. Mutual independence is therefore a strictly stronger condition than pairwise independence.

Example 1.20 (An Application of Borel–Cantelli)**.** If the events A_n are independent with $\mathbb{P}(A_n) = 1/n^2$, then since $\sum 1/n^2 = \pi^2/6 < \infty$, with probability 1 only finitely many of the A_n occur. If $\mathbb{P}(A_n) = 1/n$, then $\sum 1/n = \infty$ and, under independence, the A_n occur infinitely often with probability 1. The difference between the two sequences decides whether "rare" events eventually die out.

1.4 Computational Applications

1.4.1 Verifying Continuity by Monte Carlo

For $\Omega = [0,1]$ with Lebesgue measure $\mathbb{P}$ and the increasing sequence $A_n = [0, 1 - 1/n]$, we have $A = \bigcup A_n = [0,1)$ and $\mathbb{P}(A) = 1$. By Theorem 1.10, $\mathbb{P}(A_n) = 1 - 1/n \to 1$.

code/ch01/c01_1_continuity_monte_carlo.py

```python
import numpy as np

rng = np.random.default_rng(42)
N_SAMPLES = 500_000
omega_samples = rng.uniform(0.0, 1.0, size=N_SAMPLES)

for n in [1, 5, 10, 25, 50]:
    threshold = 1.0 - (1.0 / n)
    p_sim = np.mean(omega_samples <= threshold)
    print(f"n = {n:2d} | Theoretical P(A_n): {threshold:.4f} | Monte Carlo: {p_sim:.4f}")
```

Output

```
n =  1 | Theoretical P(A_n): 0.0000 | Monte Carlo: 0.0000
n =  5 | Theoretical P(A_n): 0.8000 | Monte Carlo: 0.8005
n = 10 | Theoretical P(A_n): 0.9000 | Monte Carlo: 0.9000
n = 25 | Theoretical P(A_n): 0.9600 | Monte Carlo: 0.9601
n = 50 | Theoretical P(A_n): 0.9800 | Monte Carlo: 0.9799
```

1.4.2 The Birthday Problem: Exact Value and Simulation

code/ch01/c01_2_birthday_problem.py

```python
import numpy as np

rng = np.random.default_rng(42)
def exact(n):
    return 1.0 - np.prod((365 - np.arange(n)) / 365)

R = 100_000
for n in [10, 23, 30, 50]:
    days = rng.integers(0, 365, size=(R, n))
    days.sort(axis=1)
    collision = np.any(np.diff(days, axis=1) == 0, axis=1)
    approx = 1 - np.exp(-n * (n - 1) / 730)
    print(f"n={n:2d} | exact={exact(n):.4f} | exponential approx.={approx:.4f} | MC={
    collision.mean():.4f}")
```

Output

```
n=10 | exact=0.1169 | exponential approx.=0.1160 | MC=0.1179
n=23 | exact=0.5073 | exponential approx.=0.5000 | MC=0.5067
n=30 | exact=0.7063 | exponential approx.=0.6963 | MC=0.7086
n=50 | exact=0.9704 | exponential approx.=0.9651 | MC=0.9707
```

1.4.3 Bayes' Formula: Simulating a Diagnostic Test

code/ch01/c01_3_bayes_diagnostic_test.py

```python
import numpy as np

rng = np.random.default_rng(42)
N = 2_000_000
ill = rng.random(N) < 0.01
positive = np.where(ill, rng.random(N) < 0.99, rng.random(N) < 0.05)
print(f"P(D | +) theory    : {0.0099/0.0594:.4f}")
print(f"P(D | +) simulation: {ill[positive].mean():.4f}")
print(f"P(+)     theory    : {0.0594:.4f} | simulation: {positive.mean():.4f}")
```

Output

```
P(D | +) theory    : 0.1667
P(D | +) simulation: 0.1664
P(+)     theory    : 0.0594 | simulation: 0.0592
```

1.5 Comments and Notes

Remark 1.21 (Why Not Assign a Probability to Every Subset?). It is impossible to assign a translation-invariant, countably additive "length" to *all* subsets of $\Omega = [0,1]$ (the Vitali set; this requires the axiom of choice). We therefore define probability only on a σ-algebra of "sufficiently regular" sets. Every set met in practice (intervals, their countable unions and intersections, and so on) belongs to the Borel σ-algebra.

Remark 1.22 (Probability Zero Is Not Impossibility). In the uniform example above, $\mathbb{P}(\{\omega\}) = 0$ for every single point $\{\omega\}$, yet the outcome of the experiment is always some point. This is why events of probability 1 are said to occur "almost surely" (a.s.).

Remark 1.23 (Monte Carlo Error). When a probability p is estimated by the proportion in N independent trials, the standard error is $\sqrt{p(1-p)/N}$. For $N = 500\,000$ and $p = 0.5$ this is about 0.0007; the differences between the theoretical and simulated values in the outputs above are of this order. The proportion is guaranteed to converge by the Law of Large Numbers in Chapter 5.

1.6 Chapter Summary

- A probability model is a triple $(\Omega, \mathcal{F}, \mathbb{P})$: the events form a σ-algebra (Definition 1.2) and $\mathbb{P}$ satisfies Kolmogorov's axioms, in particular countable additivity (Definition 1.5).
- The complement rule, monotonicity, inclusion–exclusion and Boole's inequality are derived from the axioms (Proposition 1.9).
- A probability measure is continuous along monotone sequences of events (Theorem 1.10). Being able to interchange limits and probability underlies all later convergence results.
- Conditional probability, the law of total probability and Bayes' formula (Theorem 1.13) are the probabilistic counterpart of Bayesian inference (Chapter 19). Mutual independence is stronger than pairwise independence.
- Borel–Cantelli lemmas (Theorem 1.14): if $\sum \mathbb{P}(A_n) < \infty$, the A_n occur infinitely often with probability 0; for independent events with $\sum \mathbb{P}(A_n) = \infty$ this probability is 1. They are the main tool in proofs of almost sure convergence.

Further reading. For measure-theoretic foundations see [22, Chapter 1] and [9, Sections 1–4]; for a more introductory treatment see [16, Chapter 1] and [81, Chapter 1].

1.7 Exercises

Exercise 1.1. Show that if $\mathcal{F}_1$ and $\mathcal{F}_2$ are σ-algebras, then so is $\mathcal{F}_1 \cap \mathcal{F}_2$, but that $\mathcal{F}_1 \cup \mathcal{F}_2$ in general is not, by giving an example. → Solution

Exercise 1.2. Prove the inclusion–exclusion formula for three events and obtain the general formula for n events by induction. Use it to show that when n letters are placed in envelopes at random, the probability that no letter goes into its own envelope converges to e^{-1} as $n \to \infty$. → Solution

Exercise 1.3. Prove the Bonferroni inequality $\mathbb{P}(\bigcap_{i=1}^{n} A_i) \geq 1 - \sum_{i=1}^{n} \mathbb{P}(A_i^c)$. Discuss its use in multiple hypothesis testing (Chapter 13). → Solution

Exercise 1.4. Show that if A and B are independent, then so are A and B^c, and A^c and B^c. → Solution

Exercise 1.5. Write Python code that plots $\mathbb{P}(D \mid +)$ in Example 1.18 as the prevalence $\mathbb{P}(D)$ varies over $[0.001, 0.5]$. Find analytically the prevalence threshold above which the positive predictive value exceeds 50%. → Solution

Chapter 2

Random Variables and Distribution Functions

A probability space is an abstract structure: Ω may consist of sequences of coin tosses, particle positions or stock price paths. In statistics, however, we work with *numerical* summaries of these outcomes. A random variable is a function that carries the abstract Ω into the real numbers while preserving the probability structure. In this chapter we show that the distribution of a random variable is completely determined by a single function, the CDF; we study the characteristic properties of the CDF and the inverse transform method that underlies random number generation.

2.1 Definitions

Definition 2.1 (Borel σ-Algebra and Random Variable)**.** The smallest σ-algebra generated by all open intervals of $\mathbb{R}$ is called the Borel σ-algebra ($\mathcal{B}(\mathbb{R})$). A function from a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$ is called a *random variable* (RV).

Remark 2.2. $\mathcal{B}(\mathbb{R})$ is also generated by the family $\{(-\infty, x] : x \in \mathbb{R}\}$. Hence, for X to be a random variable it is *sufficient* that $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. (Idea of proof: the family $\{B : X^{-1}(B) \in \mathcal{F}\}$ is a σ-algebra containing the generating family.)

Definition 2.3 (Distribution)**.** If X is an RV, the measure $\mathbb{P}_X$ defined by $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B)$, $B \in \mathcal{B}(\mathbb{R})$, is called the *distribution* (or law) of X. $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_X)$ is a probability space in its own right. If $\mathbb{P}_X = \mathbb{P}_Y$, then X and Y are said to be *identically distributed*, written $X \stackrel{d}{=} Y$.

Definition 2.4 (Distribution Function (CDF))**.** The cumulative distribution function of X, $F_X : \mathbb{R} \to [0, 1]$, is defined by $F_X(x) = \mathbb{P}(X \leq x)$.

Definition 2.5 (Discrete and Continuous Random Variables)**.**

1. X is *discrete* if it takes values in a countable set $\{x_1, x_2, \dots\}$; the function $p_X(x) = \mathbb{P}(X = x)$ is called the *probability mass function* (pmf), and $F_X(x) = \sum_{x_i \leq x} p_X(x_i)$.
2. X is *(absolutely) continuous* if $F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$ for some non-negative integrable f_X; f_X is called the *probability density function* (pdf). At points where f_X is continuous, $F_X'(x) = f_X(x)$.

Definition 2.6 (Quantile (Generalised Inverse) Function)**.** For $u \in (0, 1)$, the function $F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}$ is called the *quantile function* of F. If F is continuous and strictly increasing, this is the ordinary inverse. $F^{-1}(1/2)$ is the median.

Definition 2.7 (Empirical Distribution Function)**.** For a sample $X_1, \ldots, X_n$, the function

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}\{X_i \leq x\}$$

is called the *empirical distribution function* (ECDF). For every fixed x, $n\hat{F}_n(x) \sim \text{Bin}(n, F(x))$.

2.2 Main Theorems and Proofs

Theorem 2.8 (Characteristic Properties of the CDF)**.** *Every distribution function F_X*

1. *is non-decreasing: $x < y \implies F_X(x) \leq F_X(y)$;*
2. *satisfies $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$;*
3. *is right-continuous.*

Conversely, every function F with these three properties is the CDF of some random variable on some probability space.

Proof. (1) If $x < y$ then $\{X \leq x\} \subseteq \{X \leq y\}$; apply the monotonicity in Proposition 1.9. (2) For $x_n \searrow -\infty$ the events $\{X \leq x_n\}$ decrease to $\emptyset$; by Corollary 1.11, $F_X(x_n) \to 0$. Similarly, for $x_n \nearrow \infty$ we have $\{X \leq x_n\} \nearrow \Omega$, and Theorem 1.10 gives $F_X(x_n) \to 1$. (3) This is Theorem 2.9 below. For the converse take $\Omega = (0,1)$, $\mathbb{P} =$ Lebesgue measure and $X(u) = F^{-1}(u)$; Theorem 2.13 shows that the CDF of this X is F. □

Theorem 2.9 (Right-Continuity of the CDF)**.** *The distribution function $F_X(x) = \mathbb{P}(X \leq x)$ is always right-continuous:*

$$\lim_{h \to 0^+} F_X(x + h) = F_X(x).$$

Proof. Since F_X is monotone, the right limit exists; it suffices to compute it along some sequence $h_n \searrow 0$. The events $A_n = \{\omega : X(\omega) \leq x + h_n\}$ form a decreasing chain $A_1 \supseteq A_2 \supseteq \ldots$ with $\bigcap_{n=1}^{\infty} A_n = \{\omega : X(\omega) \leq x\}$. By continuity of the measure from above (Corollary 1.11), $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = F_X(x)$. □

Corollary 2.10 (Left Limits and Jumps)**.** *$F_X(x^-) = \lim_{h \to 0^+} F_X(x - h) = \mathbb{P}(X < x)$, and therefore $\mathbb{P}(X = x) = F_X(x) - F_X(x^-)$. That is, F_X jumps exactly at the points to which X assigns positive probability; for a continuous RV, $\mathbb{P}(X = x) = 0$ for every x.*

Proof. For $x - h_n \nearrow x$ we have $\{X \leq x - h_n\} \nearrow \{X < x\}$; apply Theorem 1.10. □

Proposition 2.11. *The set of discontinuity (jump) points of any CDF is at most countable.*

Proof. Let $D_k = \{x : F(x) - F(x^-) > 1/k\}$. Since the jumps cannot add up to more than 1, $|D_k| < k$. The set of all discontinuities is $\bigcup_{k \geq 1} D_k$, a countable union of finite sets, hence countable. □

Lemma 2.12 (Basic Equivalence for the Quantile Function)**.** *For every $u \in (0,1)$ and $x \in \mathbb{R}$, $F^{-1}(u) \leq x \iff u \leq F(x)$.*

Proof. ($\Leftarrow$) If $u \leq F(x)$ then $x \in \{z : F(z) \geq u\}$, hence the infimum satisfies $F^{-1}(u) \leq x$. ($\Rightarrow$) Since F is non-decreasing, the set $\{z : F(z) \geq u\}$ is an interval extending to $+\infty$; since F is right-continuous, the infimum belongs to the set, i.e. $F(F^{-1}(u)) \geq u$. If $F^{-1}(u) \leq x$, monotonicity gives $F(x) \geq F(F^{-1}(u)) \geq u$. □

Theorem 2.13 (Inverse Transform Method)**.** *If $U \sim$ Uniform$(0,1)$ and F is any CDF, then the random variable $X = F^{-1}(U)$ has CDF F.*

Proof. By Lemma 2.12, $\mathbb{P}(X \le x) = \mathbb{P}(F^{-1}(U) \le x) = \mathbb{P}(U \le F(x)) = F(x)$. □

Theorem 2.14 (Probability Integral Transform)**.** *If the CDF F of X is* continuous*, then $F(X) \sim$ Uniform$(0,1)$.*

Proof. Let $u \in (0,1)$. Negating Lemma 2.12 gives $\{F(X) < u\} = \{X < F^{-1}(u)\}$, so $\mathbb{P}(F(X) < u) = F(F^{-1}(u)^-)$. Since F is continuous, the left limit equals $F(F^{-1}(u))$, and by the intermediate value theorem $F(F^{-1}(u)) = u$. Thus $\mathbb{P}(F(X) < u) = u$ for every u; taking right limits gives $\mathbb{P}(F(X) \le u) = u$. □

Remark 2.15. Continuity is essential in Theorem 2.14: for $X \sim$ Bernoulli$(1/2)$, $F(X)$ takes the values $\{1/2, 1\}$ and is not uniformly distributed. The theorem shows that for continuous test statistics, p-values are uniformly distributed under H_0 (Chapter 13).

Theorem 2.16 (Density Under a Monotone Transformation)**.** *Let X have density f_X, and let g be strictly monotone and differentiable on the support of X. The density of $Y = g(X)$ is*

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

(y in the range of g).

Proof. If g is increasing, $F_Y(y) = \mathbb{P}(g(X) \le y) = \mathbb{P}(X \le g^{-1}(y)) = F_X(g^{-1}(y))$; differentiate by the chain rule. If g is decreasing, $F_Y(y) = 1 - F_X(g^{-1}(y))$, and the minus sign of the derivative is absorbed by the absolute value. □

Theorem 2.17 (Glivenko–Cantelli [32, 15])**.** *If $X_1, X_2, \ldots \overset{i.i.d.}{\sim} F$, then*

$$D_n = \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \xrightarrow{a.s.} 0.$$

Moreover, the Dvoretzky–Kiefer–Wolfowitz (DKW) inequality [21, 56] states that $\mathbb{P}(D_n > \varepsilon) \le 2e^{-2n\varepsilon^2}$ for every $\varepsilon > 0$.

Proof (first part). For fixed x, $\hat{F}_n(x) = \frac{1}{n}\sum_i \mathbb{I}\{X_i \le x\}$ and $\hat{F}_n(x^-) = \frac{1}{n}\sum_i \mathbb{I}\{X_i < x\}$ are averages of i.i.d. Bernoulli variables with means $F(x)$ and $F(x^-)$; by the strong law of large numbers (Theorem 5.14) they converge a.s. to $F(x)$ and $F(x^-)$. The difficulty is taking the supremum over infinitely many x; we overcome it with a finite grid of quantiles.

Fix $k \in \mathbb{N}$. For $j = 1, \ldots, k-1$ let $x_j = F^{-1}(j/k) = \inf\{x : F(x) \ge j/k\}$, and set $x_0 = -\infty$, $x_k = +\infty$. By the definition of the quantile function, $F(x_j^-) \le j/k \le F(x_j)$, hence $F(x_j^-) - F(x_{j-1}) \le \frac{1}{k}$. For $x \in [x_{j-1}, x_j)$, monotonicity of $\hat{F}_n$ and F gives

$$\hat{F}_n(x) - F(x) \le \hat{F}_n(x_j^-) - F(x_{j-1}) \le \hat{F}_n(x_j^-) - F(x_j^-) + \tfrac{1}{k},$$

$$\hat{F}_n(x) - F(x) \ge \hat{F}_n(x_{j-1}) - F(x_j^-) \ge \hat{F}_n(x_{j-1}) - F(x_{j-1}) - \tfrac{1}{k}.$$

Since these intervals cover $\mathbb{R}$,

$$D_n \le \max_{0 \le j \le k} \left\{ |\hat{F}_n(x_j) - F(x_j)|,\ |\hat{F}_n(x_j^-) - F(x_j^-)| \right\} + \frac{1}{k}$$

($\hat{F}_n$ and F agree at the end points). The maximum on the right has finitely many terms, each tending to zero a.s.; hence on an event Ω_k of probability 1, $\limsup_n D_n \le 1/k$. On $\bigcap_k \Omega_k$ we have $D_n \to 0$, and this intersection has probability 1. The proof of the DKW inequality requires a finer combinatorial argument; for the sharp constant 2 see [56]. □

Remark 2.18. The Glivenko–Cantelli theorem is often called "the fundamental theorem of statistics": a large enough sample reveals the whole unknown distribution uniformly. The limit distribution of $\sqrt{n}D_n$ is the Kolmogorov distribution, which is the basis of the Kolmogorov–Smirnov goodness-of-fit test. Pointwise convergence ($\hat{F}_n(x) \xrightarrow{\text{a.s.}} F(x)$) follows at once from the law of large numbers in Chapter 5; the strength of the theorem lies in the *uniform* convergence.

2.3 Worked Examples

Example 2.19 (The Bernoulli CDF)**.** For $X \sim \text{Bernoulli}(p)$,

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1-p, & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$$

The function jumps at 0 and 1, with jump sizes $1-p$ and p respectively. At each jump F_X is right-continuous but not left-continuous.

Example 2.20 (A Mixed Distribution: Insurance Claims)**.** Suppose that a policy has no claim during the year with probability 0.7 and that, given a claim, the amount is $\text{Exp}(\lambda)$-distributed. The CDF of the claim amount X is

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 0.7 + 0.3(1 - e^{-\lambda x}), & x \ge 0 \end{cases}$$

X is neither discrete nor continuous: it has a jump of size 0.7 at 0 and density $0.3\lambda e^{-\lambda x}$ on $(0, \infty)$. Such distributions show why the CDF approach is more general than the pmf/pdf approach.

Example 2.21 (Deriving the Chi-Square Distribution)**.** Let $X \sim \mathcal{N}(0,1)$ and $Y = X^2$. Since $g(x) = x^2$ is not monotone, we use the CDF method: for $y > 0$,

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) = 2\Phi(\sqrt{y}) - 1, \qquad f_Y(y) = \frac{2\varphi(\sqrt{y})}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi y}} e^{-y/2}.$$

This is the χ^2_1 density. It will play a central role for confidence intervals in Chapter 16.

Example 2.22 (Generating Exponential and Cauchy Variates by Inversion)**.** For $F(x) = 1 - e^{-\lambda x}$ $(x \ge 0)$, $F^{-1}(u) = -\ln(1-u)/\lambda$; hence if $U \sim \text{Uniform}(0,1)$, then $X = -\ln(1-U)/\lambda \sim \text{Exp}(\lambda)$. (Since $1 - U \overset{d}{=} U$, $-\ln U/\lambda$ can also be used.) For the standard Cauchy, $F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan x$ and $F^{-1}(u) = \tan\left(\pi(u - \tfrac{1}{2})\right)$.

Example 2.23 (Inverse Transform for a Discrete Distribution)**.** If $\mathbb{P}(X = k) = p_k$, $k = 0, 1, 2, \ldots$, then $F^{-1}(u) = \min\{k : \sum_{j \le k} p_j \ge u\}$. That is, generate U and return the first index at which the cumulative probabilities exceed U. This is a concrete example showing that Theorem 2.13 also holds for discrete distributions, and it explains why the generalised inverse is defined with an inf.

2.4 Computational Applications

2.4.1 Empirical CDF and the Kolmogorov–Smirnov Distance

code/ch02/c02_1_ecdf_ks_distance.py

```python
import numpy as np
from scipy.stats import norm

rng = np.random.default_rng(42)
x_grid = np.linspace(-3.5, 3.5, 1000)
f_theo = norm.cdf(x_grid)

for n in [20, 100, 1000, 10000]:
    samples = rng.normal(0.0, 1.0, size=n)
    ecdf = np.array([np.mean(samples <= x) for x in x_grid])
    ks_dist = np.max(np.abs(ecdf - f_theo))
    print(f"n = {n:5d} | KS distance: {ks_dist:.4f}")
```

Output

```
n =    20 | KS distance: 0.1290
n =   100 | KS distance: 0.1215
n =  1000 | KS distance: 0.0291
n = 10000 | KS distance: 0.0087
```

Note that the distance shrinks at a rate of roughly $1/\sqrt{n}$. The supremum computed over a grid is a lower approximation to the true D_n; the exact value can be computed in $O(n \log n)$ time from the sorted sample using $D_n = \max_i \max\{i/n - F(X_{(i)}),\, F(X_{(i)}) - (i-1)/n\}$. The code below uses this exact formula, compares it with the DKW bound and plots the ECDF.

code/ch02/c02_2_exact_ks_dkw_ecdf_plot.py

```python
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
def ks_exact(x, cdf):
    x = np.sort(x); n = len(x); F = cdf(x); i = np.arange(1, n + 1)
    return max(np.max(i / n - F), np.max(F - (i - 1) / n))

eps, R = 0.05, 20_000
for n in [100, 400, 1000]:
    D = np.array([ks_exact(rng.normal(size=n), norm.cdf) for _ in range(R)])
    print(f"n={n:4d} | P(D_n > {eps}) MC = {np.mean(D > eps):.4f} | DKW bound = {min(1, 2*np.exp(-2*n*eps**2)):.4f}"
          f" | E[sqrt(n) D_n] = {np.sqrt(n)*D.mean():.3f}")

fig, ax = plt.subplots(figsize=(7, 3.6))
xg = np.linspace(-3.5, 3.5, 500)
ax.plot(xg, norm.cdf(xg), "k-", lw=2, label="True F (N(0,1))")
for n, c in [(20, "tab:red"), (200, "tab:blue")]:
    s = np.sort(rng.normal(size=n))
    ax.step(np.r_[-3.5, s, 3.5], np.r_[0, np.arange(1, n + 1) / n, 1], where="post", color=c,
     label=f"ECDF, n={n}")
ax.set_xlabel("x"); ax.set_ylabel("F(x)"); ax.legend(); fig.tight_layout()
fig.savefig("ecdf.pdf")
```

Output

```
n= 100 | P(D_n > 0.05) MC = 0.9544 | DKW bound = 1.0000 | E[sqrt(n) D_n] = 0.851
n= 400 | P(D_n > 0.05) MC = 0.2600 | DKW bound = 0.2707 | E[sqrt(n) D_n] = 0.861
n=1000 | P(D_n > 0.05) MC = 0.0131 | DKW bound = 0.0135 | E[sqrt(n) D_n] = 0.863
```

The output above shows that $\mathbb{E}[\sqrt{n} D_n]$ stays roughly constant in n; this is a numerical confirmation of the $D_n = O_P(n^{-1/2})$ rate and of the Kolmogorov limit distribution (whose mean is $\sqrt{\pi/2}\ln 2 \approx 0.869$).

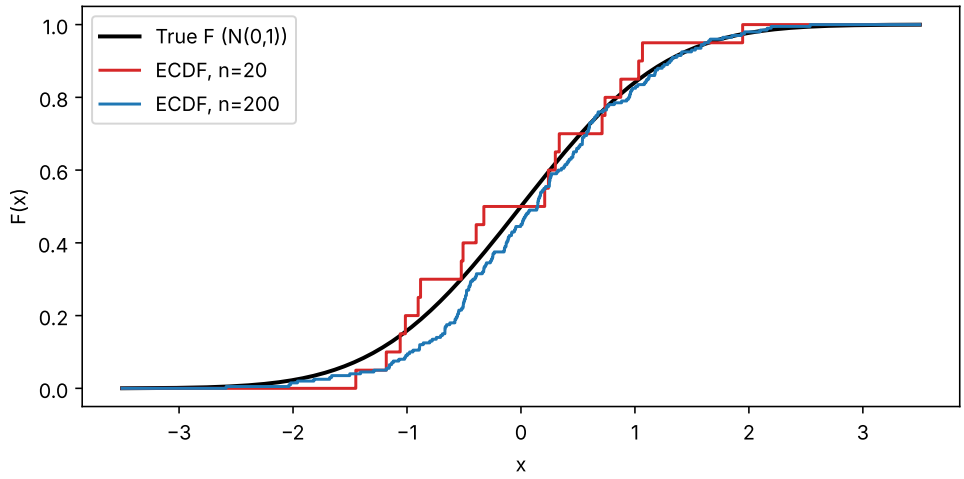


Figure 2.1: Empirical CDFs of standard normal samples. As n grows, the step function converges uniformly to the true CDF (Glivenko–Cantelli).

2.4.2 Inverse Transform and the Probability Integral Transform

code/ch02/c02_3_inverse_transform_pit.py

```python
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
U = rng.uniform(size=200_000)
lam = 2.0
X_exp = -np.log(1 - U) / lam                # Exp(2)
X_cau = np.tan(np.pi * (U - 0.5))           # standard Cauchy
X_poi = np.searchsorted(stats.poisson.cdf(np.arange(60), 3.0), U)  # Poisson(3)

print(f"Exponential: mean={X_exp.mean():.4f} (theory {1/lam}), KS p-value={stats.kstest(X_exp, 'expon', args=(0, 1/lam)).pvalue:.3f}")
print(f"Cauchy     : median={np.median(X_cau):.4f} (theory 0), KS p-value={stats.kstest(X_cau, 'cauchy').pvalue:.3f}")
print(f"Poisson    : mean={X_poi.mean():.4f}, variance={X_poi.var():.4f} (theory 3, 3)")

# Probability integral transform: F(X) ~ U(0,1) (for continuous F)
Z = rng.gamma(3.0, 2.0, size=200_000)      # Gamma(3, rate 1/2); numpy takes scale = 1/rate
V = stats.gamma.cdf(Z, a=3.0, scale=2.0)
print(f"PIT: mean of F(X)={V.mean():.4f} (1/2), variance={V.var():.4f} (1/12={1/12:.4f})")
```

Output

```
Exponential: mean=0.5002 (theory 0.5), KS p-value=0.964
Cauchy     : median=0.0004 (theory 0), KS p-value=0.964
Poisson    : mean=2.9998, variance=2.9995 (theory 3, 3)
PIT: mean of F(X)=0.4998 (1/2), variance=0.0833 (1/12=0.0833)
```

It is no coincidence that the KS p-values for the exponential and Cauchy samples are *exactly equal*: both samples were generated from the same vector U through their own F^{-1} and tested with their own F. Since $F(F^{-1}(U)) = U$, both tests actually test the same sample U against the uniform distribution. This is a direct illustration of the fact that the KS statistic is *distribution-free* for continuous F (Theorem 2.14).

2.5 Comments and Notes

Remark 2.24 (Why Is the CDF the Basic Object?). The distribution $\mathbb{P}_X$ of a random variable is a set function on $\mathcal{B}(\mathbb{R})$ and is hard to work with directly. By the Carathéodory extension theorem, $\mathbb{P}_X$ is *uniquely* determined by its values on the intervals $(-\infty, x]$, that is, by F_X. Hence the statement "two RVs have the same distribution" is equivalent to "their CDFs are equal".

Remark 2.25 (A Density Is Not a Probability). $f_X(x)$ is not a probability but a probability *density*; it may exceed 1. For example, for Uniform$(0, 1/2)$, $f_X \equiv 2$. Interpretation: for small h, $\mathbb{P}(x < X \le x + h) \approx f_X(x)h$.

Remark 2.26 (A Computational Note). The expression `[np.mean(samples <= x) for x in x_grid]` in the first code performs $O(n \cdot |\text{grid}|)$ operations. For large n, sorting the sample once and using `np.searchsorted` gives the ECDF in $O(n \log n)$ time.

2.6 Chapter Summary

- A random variable is a measurable function $X : \Omega \to \mathbb{R}$ (Definition 2.1). Its distribution is completely determined by its CDF.
- A function is a CDF if and only if it is non-decreasing and right-continuous with limits 0 and 1 (Theorem 2.8). Its jump points are at most countable, and $\mathbb{P}(X = x) = F(x) - F(x^-)$.
- The basic equivalence for the quantile function, $F^{-1}(u) \le x \iff u \le F(x)$ (Lemma 2.12), yields two results: the inverse transform method (Theorem 2.13) and, for continuous F, $F(X) \sim \text{Uniform}(0, 1)$ (Theorem 2.14).
- Under monotone transformations the density becomes $f_X(g^{-1}(y))\,|(g^{-1})'(y)|$ (Theorem 2.16).
- By the Glivenko–Cantelli theorem the empirical CDF converges uniformly to the true CDF (Theorem 2.17). The DKW inequality gives this convergence an exponential rate. The bootstrap and the Kolmogorov–Smirnov test rest on this result.

Further reading. [16, Chapters 1–2], [81, Chapter 2]; for the empirical CDF and the DKW inequality see [81, Chapter 7] and [78, Chapter 19].

2.7 Exercises

Exercise 2.1. If $X \sim \text{Uniform}(0, 1)$, find the distribution of $Y = -2 \ln X$ using Theorem 2.16 and show that it is χ^2_2. Relate this result to Fisher's method of combining p-values. → Solution

Exercise 2.2. Write the inverse transform algorithm for $F(x) = 1 - e^{-x^k}$ ($x \ge 0$, $k > 0$; Weibull) and compare the mean of your generated sample with $\Gamma(1 + 1/k)$. → Solution

Exercise 2.3. Let X and Y be independent with CDFs F and G. Find the CDFs of $\max(X, Y)$ and $\min(X, Y)$. Show that for $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ the CDF of $X_{(n)}$ is F^n (this will be used in Chapter 10; for general order statistics see Theorem 4.15). → Solution

Exercise 2.4. Using the DKW inequality, find the smallest n such that $\mathbb{P}(D_n > 0.02) \le 0.05$. From this, construct a 95% confidence *band* for F. → Solution

Chapter 3

Expectation, Variance and Cauchy–Schwarz

Knowing a whole distribution is often unnecessary or impossible; instead we work with summary numbers such as "centre" and "spread". The expectation is an integral with respect to the probability measure, and it is the common language of estimation theory (unbiasedness, MSE), limit theorems (LLN, CLT) and Bayesian decisions (posterior expected loss). In this chapter we discuss the measure-theoretic definition of expectation, the basic inequalities and the notion of conditional expectation, which will be used throughout the rest of the book.

3.1 Definitions

Definition 3.1 (Expectation as a Lebesgue Integral)**.** The expectation is defined in three steps:

1. *Simple RV:* for $X = \sum_{i=1}^{k} a_i \mathbb{I}_{A_i}$ ($A_i \in \mathcal{F}$ disjoint), $\mathbb{E}[X] = \sum_{i=1}^{k} a_i \mathbb{P}(A_i)$.
2. *Non-negative RV:* $\mathbb{E}[X] = \sup\{\mathbb{E}[Y] : Y \text{ simple},\ 0 \le Y \le X\}$ (the value may be $+\infty$).
3. *General RV:* write $X = X^+ - X^-$ ($X^+ = \max(X, 0)$, $X^- = \max(-X, 0)$); if at least one of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ is finite, $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$. If $\mathbb{E}|X| < \infty$, X is called *integrable* ($X \in L^1$).

The result of these three steps is denoted by $\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P}$.

Remark 3.2 (Computational Formulas). By the change-of-variables formula known as the Law of the Unconscious Statistician (LOTUS), for measurable g

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) \, d\mathbb{P}_X(x) = \begin{cases} \sum_x g(x) p_X(x), & X \text{ discrete} \\ \int_{-\infty}^{\infty} g(x) f_X(x) \, dx, & X \text{ continuous} \end{cases}$$

(provided the sum/integral converges absolutely). For $X \ge 0$ the tail formula $\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t) \, dt$ also holds.

Definition 3.3 (Moments, Variance, Covariance, Correlation)**.** For $k \in \mathbb{N}$, $\mu'_k = \mathbb{E}[X^k]$ is the *k-th raw moment* and $\mu_k = \mathbb{E}[(X - \mathbb{E}X)^k]$ the *k-th central moment.* Furthermore,

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2,$$
$$\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}X \, \mathbb{E}Y,$$

$$\operatorname{Corr}(X, Y) = \rho_{XY} = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)}}.$$

If $\operatorname{Cov}(X,Y)=0$, X and Y are called *uncorrelated*. The skewness is $\gamma_1=\mu_3/\sigma^3$ and the excess kurtosis is $\gamma_2=\mu_4/\sigma^4-3$.

Definition 3.4 (Moment Generating Function)**.** If $M_X(t)=\mathbb{E}[e^{tX}]$ is finite in a neighbourhood of 0, it is called the *moment generating function* (MGF) of X. In that case $\mathbb{E}[X^k]=M_X^{(k)}(0)$, and the MGF determines the distribution uniquely. (Not every distribution has an MGF; this gap will be filled by the characteristic function in Chapter 6.)

Definition 3.5 (Conditional Expectation — the Elementary Case)**.** Let (X,Y) be a random vector with joint density (or pmf) $f_{X,Y}$, and let $f_Y(y)>0$. The conditional density of X given $Y=y$ is $f_{X\mid Y}(x\mid y)=f_{X,Y}(x,y)/f_Y(y)$, and with

$$m(y)=\mathbb{E}[X\mid Y=y]=\int x\, f_{X\mid Y}(x\mid y)\,dx$$

the *random variable* $\mathbb{E}[X\mid Y]=m(Y)$ is called the conditional expectation of X given Y. Similarly, $\operatorname{Var}(X\mid Y)=\mathbb{E}[(X-\mathbb{E}[X\mid Y])^2\mid Y]$.

Remark 3.6 (The General Definition)*.* In measure-theoretic terms, $\mathbb{E}[X\mid\mathcal{G}]$ is the (a.s. unique) RV that is measurable with respect to a sub-σ-algebra $\mathcal{G}\subseteq\mathcal{F}$ and satisfies $\mathbb{E}[X\mathbb{I}_G]=\mathbb{E}[\mathbb{E}[X\mid\mathcal{G}]\mathbb{I}_G]$ for every $G\in\mathcal{G}$; its existence follows from the Radon–Nikodym theorem. $\mathbb{E}[X\mid Y]=\mathbb{E}[X\mid\sigma(Y)]$. In L^2, $\mathbb{E}[X\mid\mathcal{G}]$ is the *orthogonal projection* of X onto the subspace of $\mathcal{G}$-measurable RVs; this geometric interpretation is the essence of the Rao–Blackwell theorem (Chapter 9).

3.2 Main Theorems and Proofs

Proposition 3.7 (Properties of Expectation)**.** *Let $X,Y\in L^1$ and $a,b\in\mathbb{R}$.*

1. *(Linearity) $\mathbb{E}[aX+bY]=a\mathbb{E}X+b\mathbb{E}Y$.*
2. *(Monotonicity) If $X\le Y$ a.s., then $\mathbb{E}X\le\mathbb{E}Y$.*
3. *If $X\perp\!\!\!\perp Y$, then $\mathbb{E}[XY]=\mathbb{E}X\,\mathbb{E}Y$; hence independent RVs are uncorrelated.*
4. *$\operatorname{Var}(aX+b)=a^2\operatorname{Var}(X)$ and $\operatorname{Var}(X+Y)=\operatorname{Var}X+\operatorname{Var}Y+2\operatorname{Cov}(X,Y)$.*
5. *In general, $\operatorname{Var}\left(\sum_{i=1}^n X_i\right)=\sum_i\operatorname{Var}X_i+2\sum_{i<j}\operatorname{Cov}(X_i,X_j)$.*

Proof sketch. (1)–(2) follow directly for simple RVs and, in general, by passing to the limit with the Monotone Convergence Theorem. (3) First for indicators, $\mathbb{E}[\mathbb{I}_A\mathbb{I}_B]=\mathbb{P}(A\cap B)=\mathbb{P}(A)\mathbb{P}(B)$, then by linearity and a limiting argument. (4)–(5) follow by expanding the definition and using linearity. For measure-theoretic details see [22, Sections 1.4–1.6] or [9, Sections 15–16 and 21]. □

Theorem 3.8 (Cauchy–Schwarz Inequality)**.** *For any two random variables with $\mathbb{E}[X^2]<\infty$ and $\mathbb{E}[Y^2]<\infty$,*

$$(\mathbb{E}[XY])^2\le\mathbb{E}[X^2]\,\mathbb{E}[Y^2].$$

Equality holds if and only if $aX+bY=0$ a.s. for some $(a,b)\neq(0,0)$.

Proof. First, $\mathbb{E}[XY]$ is finite because $|XY|\le(X^2+Y^2)/2$. If $\mathbb{E}[X^2]=0$, then $X=0$ a.s. and both sides vanish. Let $\mathbb{E}[X^2]>0$. For every $t\in\mathbb{R}$,

$$g(t)=\mathbb{E}[(tX+Y)^2]=t^2\mathbb{E}[X^2]+2t\mathbb{E}[XY]+\mathbb{E}[Y^2]\ge 0.$$

A quadratic polynomial with positive leading coefficient that is never negative cannot have a positive discriminant: $\Delta=(2\mathbb{E}[XY])^2-4\mathbb{E}[X^2]\mathbb{E}[Y^2]\le 0\implies(\mathbb{E}[XY])^2\le\mathbb{E}[X^2]\mathbb{E}[Y^2]$. In the case of equality $\Delta=0$, so there is a t_0 with $g(t_0)=\mathbb{E}[(t_0X+Y)^2]=0$, i.e. $t_0X+Y=0$ a.s. □

Corollary 3.9 (Bounds on the Correlation)**.** $|\operatorname{Corr}(X,Y)| \le 1$*, with equality if and only if* $Y = a + bX$ *a.s. (*$b \neq 0$*).*

Proof. Apply Theorem 3.8 to the centred variables $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$. □

Theorem 3.10 (Jensen's Inequality)**.** *If* $\varphi : \mathbb{R} \to \mathbb{R}$ *is convex,* $X \in L^1$ *and* $\varphi(X) \in L^1$*, then*

$$\varphi(\mathbb{E}X) \le \mathbb{E}[\varphi(X)].$$

If φ *is strictly convex, equality holds only when* X *is a.s. constant.*

Proof. Let $\mu = \mathbb{E}X$. A convex function has a supporting line at every point: $\varphi(x) \ge \varphi(\mu) + c(x-\mu)$ for all x (for some $c \in \mathbb{R}$). Setting $x = X$ and taking expectations gives $\mathbb{E}\varphi(X) \ge \varphi(\mu) + c(\mathbb{E}X - \mu) = \varphi(\mu)$. □

Theorem 3.11 (Tower Rule (Law of Total Expectation))**.** *If* $X \in L^1$*, then* $\mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}[X]$*. Moreover, for every bounded measurable* h*,* $\mathbb{E}[h(Y)X \mid Y] = h(Y)\mathbb{E}[X \mid Y]$ *("taking out what is known").*

Proof (density case).

$$\mathbb{E}[m(Y)] = \int \Big(\int x f_{X|Y}(x \mid y) dx \Big) f_Y(y) dy = \iint x f_{X,Y}(x,y)\, dx\, dy = \mathbb{E}[X],$$

where the order of integration was exchanged by Fubini's theorem. The second property holds because $h(Y) = h(y)$ is constant given $Y = y$. □

Theorem 3.12 (Law of Total Variance)**.** *If* $X \in L^2$*, then*

$$\operatorname{Var}(X) = \mathbb{E}\big[\operatorname{Var}(X \mid Y)\big] + \operatorname{Var}\big(\mathbb{E}[X \mid Y]\big).$$

Proof. Let $m(Y) = \mathbb{E}[X \mid Y]$. Write $X - \mathbb{E}X = (X - m(Y)) + (m(Y) - \mathbb{E}X)$ and square. For the cross term the tower rule gives

$$\mathbb{E}\big[(X - m(Y))(m(Y) - \mathbb{E}X)\big] = \mathbb{E}\Big[(m(Y) - \mathbb{E}X)\, \mathbb{E}[X - m(Y) \mid Y]\Big] = 0,$$

because $\mathbb{E}[X - m(Y) \mid Y] = 0$. The remaining terms are $\mathbb{E}[(X - m(Y))^2] = \mathbb{E}[\operatorname{Var}(X \mid Y)]$ and $\mathbb{E}[(m(Y) - \mathbb{E}X)^2] = \operatorname{Var}(m(Y))$ (the latter uses $\mathbb{E}[m(Y)] = \mathbb{E}X$). □

Corollary 3.13 (Best Predictor)**.** *If* $X \in L^2$*, then among all measurable functions* g *of* Y*, the one minimising* $\mathbb{E}[(X - g(Y))^2]$ *is* $g(Y) = \mathbb{E}[X \mid Y]$*.*

Proof. The same decomposition as in the proof of the law of total variance gives $\mathbb{E}[(X - g(Y))^2] = \mathbb{E}[(X - m(Y))^2] + \mathbb{E}[(m(Y) - g(Y))^2]$; the second term vanishes for $g = m$. □

3.3 Worked Examples

Example 3.14 (The Binomial Distribution — the Indicator Trick)**.** $X \sim \operatorname{Bin}(n,p)$ can be written as $X = \sum_{i=1}^{n} Y_i$ with $Y_i \overset{\text{i.i.d.}}{\sim} \operatorname{Bernoulli}(p)$. By linearity $\mathbb{E}X = np$; by independence $\operatorname{Var} X = \sum \operatorname{Var} Y_i = np(1-p)$. This is much shorter than evaluating the sum $\sum_k k\binom{n}{k}p^k(1-p)^{n-k}$ directly.

Example 3.15 (Moments of the Exponential Distribution)**.** For $X \sim \operatorname{Exp}(\lambda)$, integration by parts gives $\mathbb{E}[X^k] = k!/\lambda^k$. Hence $\mathbb{E}X = 1/\lambda$ and $\operatorname{Var} X = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$. The MGF is $M_X(t) = \lambda/(\lambda - t)$, $t < \lambda$; its derivatives give the same moments.

Example 3.16 (A Distribution Without an Expectation: Cauchy)**.** For the standard Cauchy density $f(x) = \frac{1}{\pi(1+x^2)}$,

$$\mathbb{E}[X^+] = \int_0^\infty \frac{x}{\pi(1+x^2)}dx = \frac{1}{2\pi}\ln(1+x^2)\Big|_0^\infty = \infty,$$

and similarly $\mathbb{E}[X^-] = \infty$; that is, $\mathbb{E}X$ is *undefined*. Although the density is symmetric about zero, one cannot say "the mean is zero". In Chapters 5 and 6 this distribution will show why the assumptions of the law of large numbers and the central limit theorem are needed.

Example 3.17 (Uncorrelated but Dependent)**.** Let $X \sim \mathcal{N}(0,1)$ and $Y = X^2$. Then $\operatorname{Cov}(X,Y) = \mathbb{E}[X^3] - \mathbb{E}X\,\mathbb{E}[X^2] = 0$, yet Y is a deterministic function of X. Correlation measures only *linear* dependence. (Conversely, if (X,Y) is *jointly* normal, uncorrelatedness is equivalent to independence: Theorem 4.13. Normal marginals alone are not enough: Example 4.21.)

Example 3.18 (Applications of Jensen)**.**

- $\varphi(x) = x^2$: $(\mathbb{E}X)^2 \le \mathbb{E}X^2$, i.e. $\operatorname{Var} X \ge 0$.
- $\varphi(x) = 1/x$ $(x > 0)$: $\mathbb{E}[1/X] \ge 1/\mathbb{E}X$. For instance, computing an "average time" from an average speed systematically underestimates the time.
- $\varphi(x) = -\ln x$: $\mathbb{E}[\ln X] \le \ln \mathbb{E}X$, the probabilistic form of the arithmetic–geometric mean inequality. It will be used in Chapter 12 to prove the consistency of the MLE (the information inequality).
- For $S = \sqrt{S^2}$: $\mathbb{E}[S] = \mathbb{E}\sqrt{S^2} < \sqrt{\mathbb{E}S^2} = \sigma$, so S is a (downward) biased estimator of σ (Chapter 7).

Example 3.19 (Total Variance: a Poisson–Gamma Mixture)**.** Let the annual number of claims of a policyholder be $N \mid \Lambda \sim \text{Poisson}(\Lambda)$ with risk level $\Lambda \sim \text{Gamma}(\alpha, \beta)$ (shape α, rate β). By the tower rule and the law of total variance,

$$\mathbb{E}N = \mathbb{E}[\Lambda] = \frac{\alpha}{\beta}, \qquad \operatorname{Var} N = \mathbb{E}[\Lambda] + \operatorname{Var}(\Lambda) = \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2}.$$

Since $\operatorname{Var} N > \mathbb{E}N$, the mixture produces *overdispersion* relative to the Poisson. The marginal distribution of N is negative binomial. This structure is the probabilistic counterpart of the Gamma–Poisson conjugacy in Chapter 19.

3.4 Computational Applications

3.4.1 Verifying the Cauchy–Schwarz Inequality

code/ch03/c03_1_cauchy_schwarz_check.py

```python
import numpy as np

rng = np.random.default_rng(42)
cov = [[4.0, 0.85 * 2.0 * 3.0], [0.85 * 2.0 * 3.0, 9.0]]
samples = rng.multivariate_normal([2.0, -1.0], cov, size=500_000)
X, Y = samples[:, 0], samples[:, 1]
print(f"(E[XY])^2: {(np.mean(X*Y))**2:.4f} <= E[X^2]E[Y^2]: {np.mean(X**2)*np.mean(Y**2):.4f}
    ")

# Equality case: Y = a + bX  =>  |Corr| = 1 ; equality for the centred variables
Y_lin = 3.0 - 2.0 * X
Xc, Yc = X - X.mean(), Y_lin - Y_lin.mean()
print(f"Linear relation: (E[XcYc])^2 = {np.mean(Xc*Yc)**2:.4f} , E[Xc^2]E[Yc^2] = {np.mean(Xc
    **2)*np.mean(Yc**2):.4f}")
print(f"Sample correlation (for rho=0.85): {np.corrcoef(X, Y)[0,1]:.4f}")
```

Output

```
(E[XY])^2: 9.5445 <= E[X^2]E[Y^2]: 79.7080
Linear relation: (E[XcYc])^2 = 63.7241 , E[Xc^2]E[Yc^2] = 63.7241
Sample correlation (for rho=0.85): 0.8489
```

Theoretical values: $\mathbb{E}[XY] = \mathrm{Cov} + \mathbb{E}X\mathbb{E}Y = 5.1 - 2 = 3.1$, hence $(\mathbb{E}XY)^2 = 9.61$; $\mathbb{E}X^2 = 4 + 4 = 8$, $\mathbb{E}Y^2 = 9 + 1 = 10$, product 80. The inequality is rather loose because Cauchy–Schwarz was applied to *raw* moments; in centred form the ratio is $\rho^2 = 0.7225$.

3.4.2 Uncorrelatedness versus Independence; the Law of Total Variance

code/ch03/c03_2_uncorrelated_vs_independent_total_variance.py

```python
import numpy as np

rng = np.random.default_rng(42)
X = rng.normal(size=1_000_000)
Y = X**2
print(f"Corr(X, X^2) = {np.corrcoef(X, Y)[0,1]:+.4f}   (uncorrelated)")
print(f"P(Y>1 | |X|>1) = {np.mean(Y[np.abs(X) > 1] > 1):.1f}  vs  P(Y>1) = {np.mean(Y > 1):.4f}  (dependent)")

# Poisson-Gamma mixture: Var N = E[Var(N|L)] + Var(E[N|L])
alpha, beta = 3.0, 0.5
L = rng.gamma(alpha, 1/beta, size=1_000_000)   # numpy takes scale = 1/rate
N = rng.poisson(L)
print(f"E N   : MC={N.mean():.3f}  theory={alpha/beta:.3f}")
print(f"Var N : MC={N.var():.3f}  theory={alpha/beta + alpha/beta**2:.3f}  (would be {alpha/beta:.1f} for a Poisson)")
```

Output

```
Corr(X, X^2) = -0.0006   (uncorrelated)
P(Y>1 | |X|>1) = 1.0  vs  P(Y>1) = 0.3171  (dependent)
E N   : MC=6.005  theory=6.000
Var N : MC=18.034  theory=18.000  (would be 6.0 for a Poisson)
```

3.4.3 Jensen's Inequality: $\mathbb{E}[1/X]$ and $\mathbb{E}[S]$

code/ch03/c03_3_jensen_inverse_and_s.py

```python
import numpy as np
from scipy.special import gammaln

rng = np.random.default_rng(42)
X = rng.gamma(5.0, 1.0, size=1_000_000)       # E X = 5
print(f"E[1/X] = {np.mean(1/X):.4f} >= 1/E[X] = {1/X.mean():.4f}  (theory E[1/X]=1/4)")

n, sigma = 5, 2.0
S = np.std(rng.normal(0, sigma, size=(500_000, n)), axis=1, ddof=1)
c4 = np.sqrt(2/(n-1)) * np.exp(gammaln(n/2) - gammaln((n-1)/2))
print(f"E[S] MC = {S.mean():.4f} < sigma = {sigma}  | theory c4*sigma = {c4*sigma:.4f}")
```

Output

```
E[1/X] = 0.2501 >= 1/E[X] = 0.2000  (theory E[1/X]=1/4)
E[S] MC = 1.8787 < sigma = 2.0  | theory c4*sigma = 1.8800
```

Here $c_4(n) = \sqrt{2/(n-1)}\,\Gamma(n/2)/\Gamma((n-1)/2)$ is the correction factor giving $\mathbb{E}[S] = c_4\sigma$ for a normal sample ($c_4 \approx 0.940$ for $n = 5$).

3.5 Comments and Notes

Remark 3.20 (Numerical Stability When Computing a Variance). The formula $\operatorname{Var} X = \mathbb{E}X^2 - (\mathbb{E}X)^2$ is correct in theory but numerically dangerous: if the mean is large relative to the standard deviation, one subtracts two large numbers and floating-point errors are magnified (catastrophic cancellation). In practice a two-pass algorithm or Welford's one-pass algorithm should be used; `np.var` uses the two-pass method.

Remark 3.21 (Hierarchy of Moment Existence). For $1 \le p < q$, $\mathbb{E}|X|^q < \infty \implies \mathbb{E}|X|^p < \infty$ (Lyapunov's inequality $(\mathbb{E}|X|^p)^{1/p} \le (\mathbb{E}|X|^q)^{1/q}$, a consequence of Jensen). For example, the t_ν distribution has moments only of order less than ν; t_2 has a mean but infinite variance.

Remark 3.22 (Links Ahead). The Cauchy–Schwarz inequality is the key step for the Cramér–Rao lower bound in Chapter 8; the law of total variance for the Rao–Blackwell theorem in Chapter 9; and Jensen's inequality for the proof of the consistency of the MLE in Chapter 12.

3.6 Chapter Summary

- The expectation is a Lebesgue integral (Definition 3.1). It is linear and monotone, and multiplicative for independent variables (Proposition 3.7).
- The Cauchy–Schwarz inequality (Theorem 3.8) yields $|\operatorname{Corr}(X, Y)| \le 1$, with equality only for a linear relation (Corollary 3.9).
- Jensen's inequality: if φ is convex, $\varphi(\mathbb{E}X) \le \mathbb{E}\varphi(X)$ (Theorem 3.10). The information inequality and the Rao–Blackwell theorem are applications of it.
- The tower rule $\mathbb{E}[\mathbb{E}(X \mid Y)] = \mathbb{E}X$ (Theorem 3.11) and the law of total variance $\operatorname{Var} X = \mathbb{E}\operatorname{Var}(X \mid Y) + \operatorname{Var}\mathbb{E}(X \mid Y)$ (Theorem 3.12) simplify moment calculations in hierarchical models.
- $\mathbb{E}(X \mid Y)$ is the predictor that minimises the mean squared error among functions of Y (Corollary 3.13). This is the basic idea behind regression and Rao–Blackwell.
- The moment generating function produces moments through its derivatives. When it is not defined in a neighbourhood of zero, the characteristic function of Chapter 6 is used instead.

Further reading. [16, Chapter 2], [81, Chapter 3]; for expectation as a Lebesgue integral see [22, Sections 1.5–1.6].

3.7 Exercises

Exercise 3.1. Show using Fubini's theorem that $\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t)\,dt$ for $X \ge 0$. Use this to find $\mathbb{E}X = 1/p$ for $X \sim \text{Geometric}(p)$ (on $\{1, 2, \dots\}$). → Solution

Exercise 3.2. Let $X \sim \text{Uniform}(0, 1)$ and $Y \mid X = x \sim \text{Uniform}(0, x)$. Compute $\mathbb{E}Y$ and $\operatorname{Var} Y$ using the tower rule and the law of total variance, and verify the result by Monte Carlo. → Solution

Exercise 3.3. Show that $\operatorname{Var}(X) = \min_{c \in \mathbb{R}} \mathbb{E}[(X - c)^2]$ and $\mathbb{E}|X - m| = \min_c \mathbb{E}|X - c|$ (m a median). How do these two results relate to the Bayes estimators of Chapter 19? → Solution

Exercise 3.4. Let $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $X = Z_1$ and $Y = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$. Show that $\operatorname{Corr}(X, Y) = \rho$, $\mathbb{E}[Y \mid X] = \rho X$ and $\operatorname{Var}(Y \mid X) = 1 - \rho^2$. → Solution

Part II

Distributions and Limit Theorems

Chapter 4

Joint Distributions and Transformations

In statistics we rarely make do with a single random variable: a sample consists of n random variables, estimators are functions of them, and a regression model describes the joint behaviour of several variables. This chapter systematises tools that were used implicitly in the previous chapters: random vectors and joint distributions, the density criterion for independence, the distribution of sums (convolution), the multivariate change of variables (Jacobian), the multivariate normal distribution and order statistics. These results explain where sampling distributions such as χ^2, t, Beta and Gamma come from, and they form the basis of the normal sampling theory in Chapter 16.

4.1 Definitions

Definition 4.1 (Random Vector and Joint CDF)**.** A function $\mathbf{X} = (X_1, \dots, X_k)^\top : \Omega \to \mathbb{R}^k$ whose components are random variables is called a *random vector*. Its joint CDF is

$$F_{\mathbf{X}}(x_1, \dots, x_k) = \mathbb{P}(X_1 \le x_1, \dots, X_k \le x_k).$$

A function $f_{\mathbf{X}} \ge 0$ is called a *joint density* if $\mathbb{P}(\mathbf{X} \in B) = \int_B f_{\mathbf{X}}(\mathbf{x})\, d\mathbf{x}$ for every Borel set $B \subseteq \mathbb{R}^k$; in the discrete case the joint pmf is $p_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(\mathbf{X} = \mathbf{x})$.

Definition 4.2 (Marginal and Conditional Distributions)**.** If (X, Y) has joint density $f_{X,Y}$, the *marginal* density of X is $f_X(x) = \int f_{X,Y}(x, y)\, dy$. For $f_X(x) > 0$, the *conditional* density of Y given $X = x$ is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

This gives the *product rule* $f_{X,Y}(x, y) = f_X(x) f_{Y|X}(y \mid x)$, the density form of hierarchical models and of Bayes' formula (Chapter 19).

Definition 4.3 (Independence of Random Variables)**.** $X_1, \dots, X_k$ are *(mutually) independent* if

$$\mathbb{P}(X_1 \in B_1, \dots, X_k \in B_k) = \prod_{j=1}^{k} \mathbb{P}(X_j \in B_j)$$

for all $B_1, \dots, B_k \in \mathcal{B}(\mathbb{R})$. Equivalently, the joint CDF is the product of the marginal CDFs: $F_{\mathbf{X}}(\mathbf{x}) = \prod_j F_{X_j}(x_j)$ for every $\mathbf{x}$.

Definition 4.4 (Mean Vector and Covariance Matrix)**.** $\boldsymbol{\mu} = \mathbb{E}\mathbf{X} = (\mathbb{E}X_1, \dots, \mathbb{E}X_k)^\top$ and

$$\Sigma = \operatorname{Cov}(\mathbf{X}) = \mathbb{E}\big[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top\big], \qquad \Sigma_{jl} = \operatorname{Cov}(X_j, X_l).$$

Definition 4.5 (Multivariate Normal Distribution)**.** $\mathbf{X}$ is called *multivariate normal*, written $\mathbf{X} \sim \mathcal{N}_k(\boldsymbol{\mu}, \Sigma)$, if $\mathbf{a}^\top \mathbf{X}$ is (univariate) normal for every $\mathbf{a} \in \mathbb{R}^k$ (constants, with variance 0, included). If Σ is positive definite, the density is

$$f(\mathbf{x}) = (2\pi)^{-k/2} \det(\Sigma)^{-1/2} \exp\Big\{ -\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu}) \Big\}.$$

Definition 4.6 (Order Statistics)**.** The values $X_1, \dots, X_n$ arranged in increasing order, $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$, are called the *order statistics*. $X_{(1)}$ is the minimum, $X_{(n)}$ the maximum, and $R = X_{(n)} - X_{(1)}$ the range.

4.2 Main Theorems and Proofs

Theorem 4.7 (Density Criterion for Independence)**.** *Let (X, Y) have joint density $f_{X,Y}$. Then $X \perp\!\!\!\perp Y \iff$ there exist non-negative functions g, h such that $f_{X,Y}(x,y) = g(x)h(y)$ for almost every (x, y).*

Proof. ($\Rightarrow$) Independence gives $F_{X,Y} = F_X F_Y$; taking the mixed derivative yields $f_{X,Y} = f_X f_Y$. ($\Leftarrow$) Let $c_g = \int g$ and $c_h = \int h$; then $1 = \iint gh = c_g c_h$. Since the marginals are $f_X = c_h g$ and $f_Y = c_g h$, we get $f_X f_Y = c_g c_h\, gh = f_{X,Y}$; as the densities factorise, so do the probabilities. □

Remark 4.8. The factorisation criterion also involves the *support*: although $f(x,y) = 2\,\mathbb{I}\{0 < x < y < 1\}$ looks formally "constant", the factor $\mathbb{I}\{x < y\}$ cannot be written in the form $g(x)h(y)$; X and Y are dependent.

Theorem 4.9 (Convolution Formula)**.** *If $X \perp\!\!\!\perp Y$ with densities f_X, f_Y, the density of $Z = X + Y$ is*

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z-x)\, dx = (f_X * f_Y)(z).$$

In the discrete case, $p_Z(z) = \sum_x p_X(x) p_Y(z-x)$.

Proof. $F_Z(z) = \iint_{x+y \le z} f_X(x) f_Y(y)\, dy\, dx = \int f_X(x) F_Y(z-x)\, dx$; differentiating with respect to z (under the integral sign, by dominated convergence) gives the result. □

Theorem 4.10 (Multivariate Change of Variables)**.** *Let $\mathbf{X}$ have density $f_{\mathbf{X}}$ supported on an open set $A \subseteq \mathbb{R}^k$. Let $\mathbf{g} : A \to B$ be one-to-one, onto and continuously differentiable, and let the Jacobian determinant of its inverse $\mathbf{h} = \mathbf{g}^{-1}$,*

$$J(\mathbf{y}) = \det\Big[\frac{\partial h_j(\mathbf{y})}{\partial y_l}\Big]_{j,l=1}^{k},$$

be non-zero on B. Then the density of $\mathbf{Y} = \mathbf{g}(\mathbf{X})$ is

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}\big(\mathbf{h}(\mathbf{y})\big)\, |J(\mathbf{y})|, \qquad \mathbf{y} \in B.$$

Proof. For every Borel set $C \subseteq B$, the change-of-variables theorem for multiple integrals gives

$$\mathbb{P}(\mathbf{Y} \in C) = \mathbb{P}(\mathbf{X} \in \mathbf{h}(C)) = \int_{\mathbf{h}(C)} f_{\mathbf{X}}(\mathbf{x})\, d\mathbf{x} = \int_C f_{\mathbf{X}}(\mathbf{h}(\mathbf{y}))\, |J(\mathbf{y})|\, d\mathbf{y}.$$

Since this holds for every C, the integrand is the density of $\mathbf{Y}$. (The case $k = 1$ is Theorem 2.16.) □

Remark 4.11 (A Practical Recipe). When the distribution of $m < k$ functions of k variables is required, make the transformation one-to-one by adding $k - m$ "auxiliary" variables, apply Theorem 4.10, and then integrate the auxiliary variables out. Example 4.17 is a typical application of this recipe.

Proposition 4.12 (Properties of the Covariance Matrix)**.** *1. Σ is symmetric and positive semi-definite: $\mathbf{a}^\top \Sigma \mathbf{a} = \mathrm{Var}(\mathbf{a}^\top \mathbf{X}) \geq 0$ for every $\mathbf{a}$.*

2. For $A \in \mathbb{R}^{m\times k}$ and $\mathbf{b} \in \mathbb{R}^m$, $\mathbb{E}[A\mathbf{X} + \mathbf{b}] = A\boldsymbol{\mu} + \mathbf{b}$ and $\mathrm{Cov}(A\mathbf{X} + \mathbf{b}) = A\Sigma A^\top$.

3. If Σ is singular, $\mathbf{X}$ lies with probability 1 in a hyperplane of $\mathbb{R}^k$ (it has no joint density).

Proof. (1) $\mathbf{a}^\top \Sigma \mathbf{a} = \mathbb{E}[(\mathbf{a}^\top(\mathbf{X} - \boldsymbol{\mu}))^2]$. (2) Linearity of expectation. (3) If $\Sigma\mathbf{a} = \mathbf{0}$, then $\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = 0$, i.e. $\mathbf{a}^\top \mathbf{X} = \mathbf{a}^\top \boldsymbol{\mu}$ a.s. □

Theorem 4.13 (Properties of the Multivariate Normal)**.** *Let $\mathbf{X} \sim \mathcal{N}_k(\boldsymbol{\mu}, \Sigma)$.*

1. ***(Linear transformation)*** *$A\mathbf{X} + \mathbf{b} \sim \mathcal{N}_m(A\boldsymbol{\mu} + \mathbf{b},\, A\Sigma A^\top)$. In particular, every marginal (sub-vector) is normal.*

2. ***(Construction)*** *If $\Sigma = LL^\top$ (e.g. Cholesky) and $\mathbf{Z} \sim \mathcal{N}_k(\mathbf{0}, I)$, then $\boldsymbol{\mu} + L\mathbf{Z} \sim \mathcal{N}_k(\boldsymbol{\mu}, \Sigma)$.*

3. ***(Uncorrelated = independent)*** *For a partition $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$, if $\mathrm{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$ then $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2$.*

4. ***(Conditional distribution)*** *If $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \boldsymbol{\mu}_2)$, $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$ and Σ_{22} is invertible, then*
$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \;\sim\; \mathcal{N}\Big(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\;\; \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\Big).$$

5. ***(Quadratic form)*** *If Σ is positive definite, $(\mathbf{X} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_k$.*

Proof. (1) For every $\mathbf{c} \in \mathbb{R}^m$, $\mathbf{c}^\top(A\mathbf{X} + \mathbf{b}) = (A^\top\mathbf{c})^\top \mathbf{X} + \mathbf{c}^\top\mathbf{b}$ is normal; the mean and covariance come from Proposition 4.12(2). (2) From (1), since $\mathrm{Cov}(L\mathbf{Z}) = LL^\top = \Sigma$. (3) Via the characteristic function: the CF of $\mathcal{N}_k$ is $\exp\{i\mathbf{t}^\top\boldsymbol{\mu} - \frac{1}{2}\mathbf{t}^\top\Sigma\mathbf{t}\}$ (because every $\mathbf{t}^\top\mathbf{X}$ is normal). When $\Sigma_{12} = \mathbf{0}$, $\mathbf{t}^\top\Sigma\mathbf{t} = \mathbf{t}_1^\top\Sigma_{11}\mathbf{t}_1 + \mathbf{t}_2^\top\Sigma_{22}\mathbf{t}_2$ and the CF factorises; uniqueness of the CF (Theorem 6.6) gives independence. (4) Let $B = \Sigma_{12}\Sigma_{22}^{-1}$ and $\mathbf{W} = \mathbf{X}_1 - B\mathbf{X}_2$. Then $\mathrm{Cov}(\mathbf{W}, \mathbf{X}_2) = \Sigma_{12} - B\Sigma_{22} = \mathbf{0}$ and $(\mathbf{W}, \mathbf{X}_2)$ is jointly normal; by (3), $\mathbf{W} \perp\!\!\!\perp \mathbf{X}_2$. Hence the distribution of $\mathbf{X}_1 = \mathbf{W} + B\mathbf{X}_2$ given $\mathbf{X}_2 = \mathbf{x}_2$ is that of $\mathbf{W} + B\mathbf{x}_2$: mean $\boldsymbol{\mu}_1 - B\boldsymbol{\mu}_2 + B\mathbf{x}_2$ and covariance $\mathrm{Cov}(\mathbf{W}) = \Sigma_{11} - B\Sigma_{21}$. (5) With $\Sigma = LL^\top$, $\mathbf{Z} = L^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}_k(\mathbf{0}, I)$ and the quadratic form is $\mathbf{Z}^\top\mathbf{Z} = \sum Z_j^2 \sim \chi^2_k$. □

Remark 4.14 (Regression Interpretation). The conditional mean in (4) is a *linear* function of $\mathbf{x}_2$, and the conditional covariance does not depend on $\mathbf{x}_2$. In the bivariate case $\mathbb{E}[Y \mid X = x] = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X)$ and $\mathrm{Var}(Y \mid X) = \sigma_Y^2(1 - \rho^2)$: the mathematical source of linear regression and of "regression to the mean" (Galton). This shows that the best predictor of Corollary 3.13 is linear in the normal model.

Theorem 4.15 (Distribution of Order Statistics)**.** *Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} F$ with density f.*

1. *$F_{X_{(n)}}(x) = F(x)^n$ and $F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n$.*

2. *The density of $X_{(k)}$ is*
$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!\,(n-k)!} F(x)^{k-1}\big[1 - F(x)\big]^{n-k} f(x).$$

3. *The joint density of $(X_{(1)}, X_{(n)})$ is $n(n-1)\big[F(y) - F(x)\big]^{n-2} f(x) f(y)$ for $x < y$.*

Proof. (1) $\{X_{(n)} \le x\} = \bigcap_i \{X_i \le x\}$ and independence; for $X_{(1)}$ use the complement. (2) Heuristic derivation: the event $X_{(k)} \in [x, x+dx]$ means that $k-1$ observations fall below x, one in $[x, x+dx]$ and $n-k$ above. The multinomial probability is $\frac{n!}{(k-1)!1!(n-k)!} F^{k-1}(f\,dx)(1-F)^{n-k}$. (Rigorous proof: differentiate $\mathbb{P}(X_{(k)} \le x) = \sum_{j \ge k} \binom{n}{j} F^j (1-F)^{n-j}$; the terms telescope.) (3) $\mathbb{P}(X_{(1)} > x, X_{(n)} \le y) = [F(y) - F(x)]^n$; take the mixed derivative of $F_{X_{(1)},X_{(n)}}(x,y) = F(y)^n - [F(y)-F(x)]^n$. □

4.3 Worked Examples

Example 4.16 (A Sum of Exponentials Is Gamma)**.** If $X, Y \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$, then for $z > 0$ Theorem 4.9 gives

$$f_{X+Y}(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx = \lambda^2 z e^{-\lambda z},$$

which is the Gamma$(2, \lambda)$ density. By induction $\sum_{i=1}^n X_i \sim \text{Gamma}(n, \lambda)$; the result $2\lambda \sum X_i \sim \chi^2_{2n}$ used in Chapters 8 and 14 comes from here. Similar convolutions show that a sum of independent Poissons is Poisson, a sum of independent normals is normal, and a sum of independent χ^2's is χ^2 (the degrees of freedom add).

Example 4.17 (The Gamma–Beta Relation)**.** Let $X \sim \text{Gamma}(\alpha, 1)$ and $Y \sim \text{Gamma}(\beta, 1)$ be independent. The inverse of the transformation $U = X + Y$, $V = X/(X+Y)$ is $x = uv$, $y = u(1-v)$, with

$$J = \det \begin{pmatrix} v & u \\ 1-v & -u \end{pmatrix} = -u.$$

By Theorem 4.10, for $u > 0$, $0 < v < 1$,

$$f_{U,V}(u,v) = \frac{(uv)^{\alpha-1} e^{-uv}\,(u(1-v))^{\beta-1} e^{-u(1-v)}}{\Gamma(\alpha)\Gamma(\beta)}\, u = \underbrace{\frac{u^{\alpha+\beta-1}e^{-u}}{\Gamma(\alpha+\beta)}}_{\text{Gamma}(\alpha+\beta,1)} \cdot \underbrace{\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} v^{\alpha-1}(1-v)^{\beta-1}}_{\text{Beta}(\alpha,\beta)}.$$

By Theorem 4.7, $U \perp\!\!\!\perp V$, $U \sim \text{Gamma}(\alpha+\beta, 1)$ and $V \sim \text{Beta}(\alpha, \beta)$. This is the standard way of generating Beta variates from Gammas, and a direct check of the Basu exercise ($T \perp\!\!\!\perp X_1/T$) in Chapter 10.

Example 4.18 (The Box–Muller Transformation [12])**.** Let $U_1, U_2 \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$ and

$$Z_1 = \sqrt{-2\ln U_1}\cos(2\pi U_2), \qquad Z_2 = \sqrt{-2\ln U_1}\sin(2\pi U_2).$$

In polar coordinates, $S = R^2 = Z_1^2 + Z_2^2 = -2\ln U_1 \sim \text{Exp}(1/2) = \chi^2_2$ and the angle $\Theta = 2\pi U_2 \sim \text{Uniform}(0, 2\pi)$ are independent: $f_{S,\Theta}(s,\vartheta) = \frac{1}{2}e^{-s/2} \cdot \frac{1}{2\pi}$. The inverse map $s = z_1^2 + z_2^2$, $\vartheta = \text{atan2}(z_2, z_1)$ has Jacobian determinant $\left|\partial(s,\vartheta)/\partial(z_1,z_2)\right| = 2$, so by Theorem 4.10

$$f_{Z_1,Z_2}(z_1,z_2) = \frac{1}{2}e^{-(z_1^2+z_2^2)/2} \cdot \frac{1}{2\pi} \cdot 2 = \frac{1}{2\pi} e^{-(z_1^2+z_2^2)/2},$$

i.e. $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. (Whereas the inverse transform method needs Φ^{-1} for the normal, Box–Muller uses only ln, $\sqrt{\cdot}$ and trigonometric functions.)

Example 4.19 (The Ratio of Two Normals Is Cauchy)**.** Let $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ and $W = Z_1/Z_2$. With the auxiliary variable $V = Z_2$ we have $z_1 = wv$, $z_2 = v$, $|J| = |v|$:

$$f_W(w) = \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-v^2(1+w^2)/2} |v|\, dv = \frac{1}{\pi}\int_0^{\infty} v e^{-v^2(1+w^2)/2} dv = \frac{1}{\pi(1+w^2)}.$$

The standard Cauchy is also the t_1 distribution ($Z_1/\sqrt{Z_2^2}$, by symmetry). The distribution that we saw in Example 3.16 has no mean arises as the ratio of the two most "well-behaved" distributions.

Example 4.20 (Uniform Order Statistics Are Beta)**.** For $U_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$, Theorem 4.15(2) gives $f_{U_{(k)}}(u) = \frac{n!}{(k-1)!(n-k)!} u^{k-1}(1-u)^{n-k}$, i.e. $U_{(k)} \sim \text{Beta}(k, n-k+1)$ and $\mathbb{E}U_{(k)} = \frac{k}{n+1}$. By the probability integral transform (Theorem 2.14), $F(X_{(k)}) \sim \text{Beta}(k, n-k+1)$ for any continuous F: this is the source of quantile confidence intervals and of the plotting positions $k/(n+1)$ in Q–Q plots. For the range, Theorem 4.15(3) gives $R = U_{(n)} - U_{(1)} \sim \text{Beta}(n-1, 2)$.

Example 4.21 (A Vector with Normal Marginals That Is Not Jointly Normal)**.** Let $X \sim \mathcal{N}(0,1)$ and let S, independent of X, take the values $+1$ and -1 with equal probability $(\frac{1}{2})$; set $Y = SX$. By symmetry $Y \sim \mathcal{N}(0,1)$, and $\text{Cov}(X,Y) = \mathbb{E}[S]\mathbb{E}[X^2] = 0$. However, since $|Y| = |X|$, X and Y are not independent; moreover, $X+Y$ is exactly 0 with probability $\frac{1}{2}$, so it is not normal. "Uncorrelated = independent" in Theorem 4.13(3) holds only under *joint* normality; normal marginals are not enough.

Example 4.22 (Bivariate Normal: Height and Weight)**.** For height X (cm) and weight Y (kg) let $\mu_X = 170$, $\sigma_X = 10$, $\mu_Y = 70$, $\sigma_Y = 12$, $\rho = 0.6$. By Theorem 4.13(4),

$$\mathbb{E}[Y \mid X = 185] = 70 + 0.6 \cdot \frac{12}{10}(185 - 170) = 80.8, \qquad \sqrt{\text{Var}(Y \mid X)} = 12\sqrt{1 - 0.36} = 9.6.$$

The expected weight of a person of height 185 cm is only $0.9\sigma_Y$ above the mean, although the height is $1.5\sigma_X$ above its mean: regression to the mean.

4.4 Computational Applications

4.4.1 Box–Muller and Gamma–Beta Independence

code/ch04/c04_1_box_muller_gamma_beta.py

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
N = 500_000
U1, U2 = rng.random(N), rng.random(N)
Z1 = np.sqrt(-2*np.log(U1)) * np.cos(2*np.pi*U2)
Z2 = np.sqrt(-2*np.log(U1)) * np.sin(2*np.pi*U2)
print(f"Box-Muller: mean=({Z1.mean():+.4f}, {Z2.mean():+.4f}) | var=({Z1.var():.4f}, {Z2.var():.4f})"
      f" | Corr={np.corrcoef(Z1, Z2)[0,1]:+.4f} | Corr(Z1^2,Z2^2)={np.corrcoef(Z1**2, Z2**2)[0,1]:+.4f}")
print(f"KS p-value (Z1 ~ N(0,1)): {stats.kstest(Z1, 'norm').pvalue:.3f}")

a, b = 2.0, 5.0
X, Y = rng.gamma(a, 1, N), rng.gamma(b, 1, N)
U, V = X + Y, X / (X + Y)
print(f"Gamma-Beta: Corr(U, V)={np.corrcoef(U, V)[0,1]:+.4f} | Corr(U, V^2)={np.corrcoef(U, V**2)[0,1]:+.4f}")
print(f"  E[V]={V.mean():.4f} (Beta mean {a/(a+b):.4f}) | E[U]={U.mean():.4f} (Gamma mean {a+b})")
print(f"  KS: V~Beta({a},{b}) p={stats.kstest(V, 'beta', args=(a, b)).pvalue:.3f}")
# A stronger check of independence: is the mean of V constant across slices of U?
q = np.quantile(U, [0, .25, .5, .75, 1])
print("  E[V] within slices of U:", np.round([V[(U >= q[i]) & (U < q[i+1])].mean() for i in range(4)], 4))
```

Output

```
Box-Muller: mean=(-0.0006, -0.0021) | var=(1.0024, 1.0002) | Corr=-0.0022 | Corr(Z1^2,Z2^2)=-0.0038
```

```
KS p-value (Z1 ~ N(0,1)): 0.835
Gamma-Beta: Corr(U, V)=-0.0002 | Corr(U, V^2)=-0.0003
  E[V]=0.2857 (Beta mean 0.2857) | E[U]=7.0062 (Gamma mean 7.0)
  KS: V~Beta(2.0,5.0) p=0.498
  E[V] within slices of U: [0.2863 0.2853 0.2853 0.286 ]
```

4.4.2 Multivariate Normal: Cholesky Generation, Conditional Distribution, Mahalanobis

code/ch04/c04_2_multivariate_normal_cholesky.py

```python
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
mu = np.array([170.0, 70.0, 40.0])
sd = np.array([10.0, 12.0, 8.0])
R = np.array([[1.0, 0.6, 0.3], [0.6, 1.0, 0.5], [0.3, 0.5, 1.0]])
Sigma = np.outer(sd, sd) * R
L = np.linalg.cholesky(Sigma)
N = 1_000_000
X = mu + rng.normal(size=(N, 3)) @ L.T
print("Sample covariance - Sigma (largest absolute difference):", np.abs(np.cov(X.T) - Sigma)
    .max().round(3))

# Conditional distribution of X1 | (X2, X3): theoretical and empirical via "slicing"
x2 = np.array([80.0, 45.0])
S12, S22 = Sigma[0, 1:], Sigma[1:, 1:]
m_cond = mu[0] + S12 @ np.linalg.solve(S22, x2 - mu[1:])
v_cond = Sigma[0, 0] - S12 @ np.linalg.solve(S22, S12)
sel = (np.abs(X[:, 1] - x2[0]) < 1.0) & (np.abs(X[:, 2] - x2[1]) < 1.0)
print(f"E[X1|x2,x3]: theory={m_cond:.3f}, slice={X[sel, 0].mean():.3f} | sd: theory={np.sqrt(
    v_cond):.3f}, slice={X[sel, 0].std():.3f} (n={sel.sum()})")

# Squared Mahalanobis distance ~ chi2_3
D2 = np.einsum("ij,jk,ik->i", X - mu, np.linalg.inv(Sigma), X - mu)
print(f"Mahalanobis D^2: mean={D2.mean():.4f} (3) | P(D^2 > chi2_3,0.05)={np.mean(D2 > stats.
    chi2.ppf(0.95, 3)):.4f} (0.05)")
```

Output

```
Sample covariance - Sigma (largest absolute difference): 0.149
E[X1|x2,x3]: theory=175.000, slice=174.834 | sd: theory=8.000, slice=8.123 (n=5292)
Mahalanobis D^2: mean=2.9988 (3) | P(D^2 > chi2_3,0.05)=0.0499 (0.05)
```

4.4.3 Order Statistics and a Vector That Is Not Jointly Normal

code/ch04/c04_3_order_statistics_not_jointly_normal.py

```python
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
n, Rr = 9, 200_000
U = np.sort(rng.random((Rr, n)), axis=1)
for k in [1, 3, 5, 9]:
    print(f"U_({k}) : mean={U[:, k-1].mean():.4f} (k/(n+1)={k/(n+1):.4f}) | KS p, Beta({k},{n
    -k+1}): "
          f"{stats.kstest(U[:, k-1], 'beta', args=(k, n-k+1)).pvalue:.3f}")
R_ = U[:, -1] - U[:, 0]
```

```
print(f"Range: mean={R_.mean():.4f} (Beta({n-1},2) mean {(n-1)/(n+1):.4f})")

X = rng.normal(size=4000)
S = rng.choice([-1, 1], size=X.size)
Y = S * X
print(f"Y=SX: Corr(X,Y)={np.corrcoef(X, Y)[0,1]:+.4f} | Corr(X^2,Y^2)={np.corrcoef(X**2, Y
    **2)[0,1]:+.4f} | P(X+Y=0)={np.mean(X + Y == 0):.3f}")

rho = 0.6
Z = rng.normal(size=(4000, 2))
B = np.c_[Z[:, 0], rho*Z[:, 0] + np.sqrt(1 - rho**2)*Z[:, 1]]
fig, ax = plt.subplots(1, 2, figsize=(9, 3.8), sharex=True, sharey=True)
ax[0].scatter(B[:, 0], B[:, 1], s=2, alpha=0.3)
xx = np.linspace(-3.5, 3.5, 2)
ax[0].plot(xx, rho*xx, "r-", lw=2, label=r"$E[Y|X=x]=\rho x$")
ax[0].plot(xx, xx, "k:", lw=1, label="y = x")
ax[0].set_title(r"Jointly normal, $\rho=0.6$"); ax[0].legend(fontsize=8)
ax[1].scatter(X, Y, s=2, alpha=0.3, color="tab:orange")
ax[1].set_title("Normal marginals, not jointly normal (Y=SX)")
for a in ax: a.set_xlim(-3.5, 3.5); a.set_ylim(-3.5, 3.5); a.set_xlabel("x")
ax[0].set_ylabel("y"); fig.tight_layout(); fig.savefig("ortak_normal.pdf")
```

Output

```
U_(1) : mean=0.1003 (k/(n+1)=0.1000) | KS p, Beta(1,9): 0.545
U_(3) : mean=0.2999 (k/(n+1)=0.3000) | KS p, Beta(3,7): 0.391
U_(5) : mean=0.4998 (k/(n+1)=0.5000) | KS p, Beta(5,5): 0.344
U_(9) : mean=0.9000 (k/(n+1)=0.9000) | KS p, Beta(9,1): 0.902
Range: mean=0.7997 (Beta(8,2) mean 0.8000)
Y=SX: Corr(X,Y)=+0.0120 | Corr(X^2,Y^2)=+1.0000 | P(X+Y=0)=0.491
```

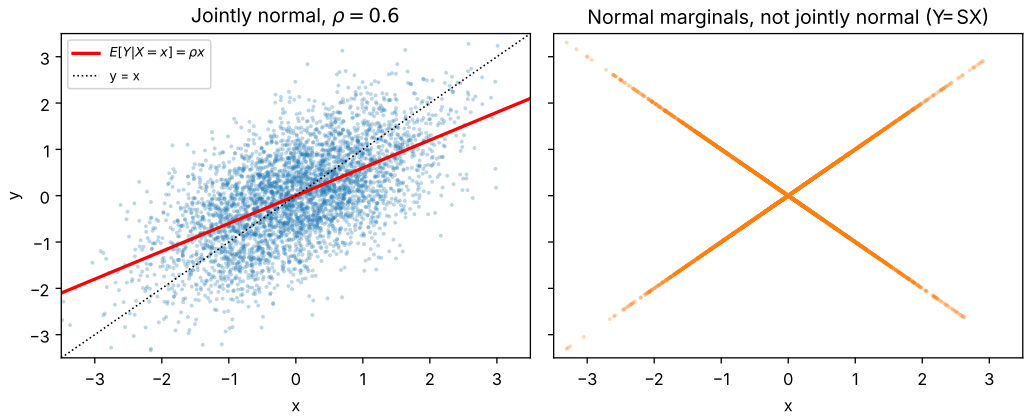


Figure 4.1: Left: a bivariate normal and the conditional mean line; since the slope is $\rho < 1$, it is flatter than the line $y = x$ (regression to the mean). Right: X and $Y = SX$ are each standard normal and uncorrelated, but their joint distribution concentrates on the two diagonals — it is not jointly normal and the variables are dependent.

4.5 Comments and Notes

Remark 4.23 (Copulas). Sklar's theorem [75]: every joint CDF can be written as $F(\mathbf{x}) = C(F_1(x_1), \ldots, F_k(x_k))$ with the marginal CDFs and a *copula* $C : [0,1]^k \to [0,1]$; if the marginals are continuous, C is unique. This way of separating the dependence structure from the marginals is widely used in financial risk and hydrology. Example 4.21 shows that the same (normal) marginals can be combined through very different copulas.

Remark 4.24 (The Central Role of the Multivariate Normal). Because of the multivariate CLT (Theorem 6.14), vectors of estimators are approximately multivariate normal in large samples. That is why the properties in Theorem 4.13 — linear transformations, conditional distributions, the χ^2 distribution of quadratic forms — are used throughout Wald tests (Chapter 15), confidence ellipsoids and linear models.

Remark 4.25 (A Computational Note). When generating multivariate normals, using the Cholesky factor L instead of $\Sigma^{1/2}$ is both faster ($O(k^3/3)$) and numerically more stable. For conditional distributions and Mahalanobis distances, solving a linear system (`np.linalg.solve`) is preferable to forming Σ^{-1} explicitly.

4.6 Chapter Summary

- The joint distribution determines the marginal and conditional distributions. With densities, independence is equivalent to factorisation of the joint density (Theorem 4.7).
- The density of an independent sum is found by convolution (Theorem 4.9), and that of a general one-to-one transformation by the Jacobian method (Theorem 4.10). The Box–Muller transformation is an application of the latter.
- The covariance matrix is symmetric and positive semi-definite, and $\operatorname{Cov}(A\mathbf{X}) = A\Sigma A^\top$ (Proposition 4.12).
- The multivariate normal distribution is closed under linear transformations, and its marginal and conditional distributions are normal (Theorem 4.13). Under joint normality, uncorrelatedness is equivalent to independence; normal marginals do not imply joint normality.
- The distributions of order statistics can be written explicitly in terms of F and f (Theorem 4.15). These results underlie the distribution-free intervals of Chapter 21.

Further reading. [16, Chapter 4], [81, Chapters 2–3]; for the multivariate normal see [69, Chapters 2–3].

4.7 Exercises

Exercise 4.1. For $X, Y \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$, find the density of $X+Y$ (the triangular distribution) by convolution. Write down the mean and variance of the sum of n independent uniforms (Irwin–Hall), and explain why for $n = 12$, $\sum U_i - 6$ was once used as an "approximately normal" generator. → Solution

Exercise 4.2. Let $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_k$ be independent. Derive the density of $T = Z/\sqrt{V/k}$ using Theorem 4.10 (auxiliary variable V) and obtain the t_k density. → Solution

Exercise 4.3. For $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$, show that $X_{(1)} \sim \text{Exp}(n\lambda)$ and that the "spacings" $D_k = (n-k+1)(X_{(k)} - X_{(k-1)})$ are i.i.d. $\text{Exp}(\lambda)$ (Rényi's representation). Verify this by simulation. → Solution

Exercise 4.4. For $(X,Y) \sim \mathcal{N}_2$ ($\mu = 0$, $\sigma = 1$, correlation ρ), show that $\mathbb{P}(X>0, Y>0) = \frac{1}{4} + \frac{\arcsin\rho}{2\pi}$ (Sheppard's formula). Hint: polar coordinates and Theorem 4.13(2). → Solution

Chapter 5

Basic Inequalities and the Law of Large Numbers

The most basic intuition of statistics is this: the average of independent observations approaches the true mean as the number of observations grows. To turn this intuition into a precise theorem we need two tools: (i) probability inequalities that bound the probability that a random variable deviates from its mean *in terms of its moments*, and (ii) precise notions of convergence for sequences of random variables. In this chapter we study the family of inequalities from Markov to Hoeffding, the hierarchy between the modes of convergence, and the Law of Large Numbers.

5.1 Definitions: Convergence of Random Variables

Let $X, X_1, X_2, \dots$ be random variables defined on the same space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 5.1 (Convergence in Probability)**.** If $\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0$ for every $\varepsilon > 0$, we write $X_n \xrightarrow{P} X$.

Definition 5.2 (Almost Sure Convergence)**.** If $\mathbb{P}(\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}) = 1$, we write $X_n \xrightarrow{\text{a.s.}} X$ ("convergence with probability 1").

Definition 5.3 (Convergence in L^p (in p-th Mean))**.** If $p \geq 1$, $\mathbb{E}|X_n|^p < \infty$ and $\lim_{n\to\infty} \mathbb{E}|X_n - X|^p = 0$, we write $X_n \xrightarrow{L^p} X$. The case $p = 2$ is *mean-square convergence*.

Remark 5.4. A fourth mode, *convergence in distribution* ($\xrightarrow{d}$), will be defined in Chapter 6. Almost sure convergence is a statement at the level of "each sample path", whereas convergence in probability only says that "for large n, the probability of a large deviation is small".

5.2 Probability Inequalities

Theorem 5.5 (Markov's Inequality)**.** *If $X \geq 0$ and $a > 0$, then* $\mathbb{P}(X \geq a) \leq \dfrac{\mathbb{E}X}{a}$.

Proof. $a\,\mathbb{I}\{X \geq a\} \leq X$ holds pointwise. Monotonicity of expectation gives $a\mathbb{P}(X \geq a) \leq \mathbb{E}X$. □

Theorem 5.6 (Chebyshev's Inequality)**.** *If $\mathbb{E}X = \mu$ and $\operatorname{Var} X < \infty$, then for every $\varepsilon > 0$*

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\operatorname{Var}(X)}{\varepsilon^2}.$$

Equivalently, with $\sigma = \sqrt{\operatorname{Var} X}$ and $k > 0$, $\mathbb{P}(|X - \mu| \geq k\sigma) \leq 1/k^2$.

Proof. Apply Markov's inequality to the variable $(X-\mu)^2 \geq 0$ with $a = \varepsilon^2$: $\mathbb{P}(|X-\mu| \geq \varepsilon) = \mathbb{P}((X-\mu)^2 \geq \varepsilon^2) \leq \mathbb{E}(X-\mu)^2/\varepsilon^2$. □

Example 5.7 (Chebyshev's Bound Is Sharp)**.** For $k > 1$ let X take the values $\{-k, 0, k\}$ with probabilities $\frac{1}{2k^2}$, $1 - \frac{1}{k^2}$, $\frac{1}{2k^2}$. Then $\mathbb{E}X = 0$, $\operatorname{Var} X = 1$ and $\mathbb{P}(|X| \geq k) = 1/k^2$. So without further assumptions Chebyshev's bound cannot be improved.

Theorem 5.8 (Chernoff Bound)**.** *For X with MGF $M_X(t)$ and every $a \in \mathbb{R}$,*

$$\mathbb{P}(X \geq a) \leq \inf_{t>0} e^{-ta} M_X(t).$$

Proof. For every $t > 0$ the map $x \mapsto e^{tx}$ is increasing, so by Markov $\mathbb{P}(X \geq a) = \mathbb{P}(e^{tX} \geq e^{ta}) \leq e^{-ta}\mathbb{E}e^{tX}$. The left-hand side does not depend on t, so we may take the infimum. □

Theorem 5.9 (Hoeffding's Inequality [37])**.** *Let $X_1, \ldots, X_n$ be independent with $X_i \in [a_i, b_i]$ a.s. For $S_n = \sum X_i$ and every $t > 0$,*

$$\mathbb{P}(|S_n - \mathbb{E}S_n| \geq t) \leq 2\exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

In particular, if the $X_i \in [a, b]$ are i.i.d., then $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq 2\exp\{-2n\varepsilon^2/(b-a)^2\}$.

Proof. Hoeffding's lemma: if $\mathbb{E}Y = 0$ and $Y \in [a, b]$, then $\mathbb{E}e^{\lambda Y} \leq e^{\lambda^2(b-a)^2/8}$ for every $\lambda \in \mathbb{R}$. Since $\mathbb{E}Y = 0$, $a \leq 0 \leq b$; the claim is obvious if $a = b$, so assume $a < b$. Since $y \mapsto e^{\lambda y}$ is convex, for $y \in [a, b]$

$$e^{\lambda y} \leq \frac{b-y}{b-a}e^{\lambda a} + \frac{y-a}{b-a}e^{\lambda b}.$$

Taking expectations and using $\mathbb{E}Y = 0$, with $p = -a/(b-a) \in [0, 1]$ and $u = \lambda(b-a)$,

$$\mathbb{E}e^{\lambda Y} \leq (1-p)e^{\lambda a} + pe^{\lambda b} = e^{L(u)}, \qquad L(u) = -pu + \ln\left(1 - p + pe^u\right).$$

$L(0) = 0$; $L'(u) = -p + \frac{pe^u}{1-p+pe^u}$ gives $L'(0) = 0$; and with $q = \frac{pe^u}{1-p+pe^u} \in [0, 1]$, $L''(u) = q(1-q) \leq \frac{1}{4}$. By Taylor's theorem, $L(u) \leq u^2/8 = \lambda^2(b-a)^2/8$.

The inequality. $Y_i = X_i - \mathbb{E}X_i \in [a_i - \mathbb{E}X_i,\ b_i - \mathbb{E}X_i]$, an interval of length $b_i - a_i$. For $\lambda > 0$, Markov's inequality and independence give

$$\mathbb{P}(S_n - \mathbb{E}S_n \geq t) \leq e^{-\lambda t}\prod_{i=1}^n \mathbb{E}e^{\lambda Y_i} \leq \exp\Big\{-\lambda t + \frac{\lambda^2}{8}\sum_i (b_i - a_i)^2\Big\}.$$

The exponent is minimised at $\lambda = 4t/\sum_i (b_i - a_i)^2$, where it equals $-2t^2/\sum_i (b_i - a_i)^2$. Applying the same bound to the $-X_i$ handles $\mathbb{P}(S_n - \mathbb{E}S_n \leq -t)$; add the two probabilities. The special case follows with $t = n\varepsilon$ and $b_i - a_i = b - a$. □

Remark 5.10 (Comparing the Inequalities). Chebyshev relies only on the second moment and bounds the deviation probability at rate $1/n$. Hoeffding (and Chernoff) give a bound decaying at an *exponential* rate in return for stronger assumptions such as boundedness or the existence of the MGF. Most generalisation bounds in machine learning (PAC learning) are of Hoeffding type.

5.3 Laws of Large Numbers

Theorem 5.11 (Weak Law of Large Numbers — WLLN)**.** *If $\{X_i\}$ is an i.i.d. sequence with mean μ and variance $\sigma^2 < \infty$, then $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{P} \mu$.*

Proof. By linearity and independence, $\mathbb{E}\bar{X}_n = \mu$ and $\operatorname{Var}(\bar{X}_n) = \frac{1}{n^2}\sum_{i=1}^n \operatorname{Var} X_i = \frac{\sigma^2}{n}$. By Chebyshev, for every $\varepsilon > 0$

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \xrightarrow{n\to\infty} 0. \qquad \square$$

Remark 5.12. The proof actually shows more: (i) it suffices that the X_i be *uncorrelated* with uniformly bounded variances; (ii) since $\mathbb{E}(\bar{X}_n - \mu)^2 = \sigma^2/n \to 0$, the convergence also holds in L^2.

Theorem 5.13 (Khinchin's WLLN)**.** *If $\{X_i\}$ are i.i.d. and only $\mathbb{E}|X_1| < \infty$, then $\bar{X}_n \xrightarrow{P} \mu = \mathbb{E}X_1$.*

Proof (truncation). For each n define the truncated variables $Y_{ni} = X_i \mathbb{I}\{|X_i| \leq n\}$, $i \leq n$, and $T_n = \frac{1}{n}\sum_{i\leq n} Y_{ni}$.

(i) *Truncation rarely matters:* $\mathbb{P}(\bar{X}_n \neq T_n) \leq \sum_{i\leq n}\mathbb{P}(|X_i| > n) = n\mathbb{P}(|X_1| > n) \leq \mathbb{E}\big[|X_1|\mathbb{I}\{|X_1| > n\}\big] \to 0$ (dominated convergence).

(ii) *Mean:* $\mathbb{E}T_n = \mathbb{E}[X_1\mathbb{I}\{|X_1| \leq n\}] \to \mu$ (dominated convergence, bounded by $|X_1|$).

(iii) *Variance:* $\operatorname{Var} T_n \leq \frac{1}{n}\mathbb{E}\big[X_1^2\mathbb{I}\{|X_1| \leq n\}\big] \leq \mathbb{E}\Big[|X_1|\min\Big(\frac{|X_1|}{n}, 1\Big)\Big] \to 0$, because the integrand tends to zero pointwise and is bounded by $|X_1|$.

By Chebyshev, (iii) gives $T_n - \mathbb{E}T_n \xrightarrow{P} 0$, and with (ii) $T_n \xrightarrow{P} \mu$. Finally, $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \mathbb{P}(\bar{X}_n \neq T_n) + \mathbb{P}(|T_n - \mu| > \varepsilon) \to 0$. (A short second proof using characteristic functions is given in Example 6.21.) $\square$

Theorem 5.14 (Kolmogorov's Strong Law of Large Numbers — SLLN [27])**.** *Let $\{X_i\}$ be i.i.d. Then $\bar{X}_n \xrightarrow{a.s.} \mu$ if and only if $\mathbb{E}|X_1| < \infty$ and $\mu = \mathbb{E}X_1$.*

A restricted proof: the case $\mathbb{E}X_1^4 < \infty$. Without loss of generality $\mu = 0$. For $S_n = \sum X_i$, $\mathbb{E}S_n^4 = n\mathbb{E}X_1^4 + 3n(n-1)\sigma^4 \leq Cn^2$ (terms with an odd power have zero expectation by independence). By Markov's inequality

$$\mathbb{P}(|\bar{X}_n| \geq \varepsilon) = \mathbb{P}(S_n^4 \geq n^4\varepsilon^4) \leq \frac{C}{n^2\varepsilon^4},$$

and $\sum_n 1/n^2 < \infty$. By Borel–Cantelli (Theorem 1.14), with probability 1, $|\bar{X}_n| \geq \varepsilon$ occurs for only finitely many n. Taking the countable intersection over $\varepsilon = 1/k$, $k \in \mathbb{N}$, gives $\bar{X}_n \xrightarrow{\text{a.s.}} 0$. $\square$

5.4 Relations Between the Modes of Convergence

Theorem 5.15 (Hierarchy of Convergence)**.** *1. $X_n \xrightarrow{a.s.} X \implies X_n \xrightarrow{P} X$.*

2. $X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{P} X$ ($p \geq 1$).

3. For $1 \leq r < p$, $X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{L^r} X$.

The converses are false in general.

Proof. (1) $A_n = \bigcup_{k\geq n}\{|X_k - X| > \varepsilon\}$ is a decreasing sequence and $\bigcap_n A_n \subseteq \{X_k \not\to X\}$, which has probability zero. By Corollary 1.11, $\mathbb{P}(|X_n - X| > \varepsilon) \leq \mathbb{P}(A_n) \to 0$. (2) Markov: $\mathbb{P}(|X_n - X| > \varepsilon) \leq \mathbb{E}|X_n - X|^p/\varepsilon^p \to 0$. (3) From Lyapunov's inequality $(\mathbb{E}|Y|^r)^{1/r} \leq (\mathbb{E}|Y|^p)^{1/p}$. $\square$

Example 5.16 (Convergence in Probability Without A.S. Convergence: the "Typewriter" Sequence)**.** Let $\Omega = [0,1]$ with Lebesgue measure. For $n = 2^k + j$ $(0 \le j < 2^k)$ let $X_n = \mathbb{I}_{[j/2^k,\,(j+1)/2^k]}$. Then $\mathbb{P}(X_n \neq 0) = 2^{-k} \to 0$, so $X_n \xrightarrow{P} 0$ (even in L^p). But for every ω, $X_n(\omega)$ takes the value 1 at least once in each block k, so $\limsup X_n(\omega) = 1$: the sequence converges at *no* point.

Example 5.17 (A.S. Convergence Without L^1 Convergence)**.** Let $U \sim \text{Uniform}(0,1)$ and $X_n = n\,\mathbb{I}\{U < 1/n\}$. For every ω (since $U(\omega) > 0$), $X_n(\omega) = 0$ for all large n; so $X_n \xrightarrow{\text{a.s.}} 0$. However $\mathbb{E}X_n = 1$ for every n, so $\mathbb{E}|X_n - 0| \not\to 0$. This example shows that convergence in probability does not guarantee convergence of moments: an estimator may be consistent while its MSE does not tend to zero (Chapter 7).

Theorem 5.18 (Continuous Mapping Theorem — in Probability)**.** *If $X_n \xrightarrow{P} c$ (a constant) and g is continuous at c, then $g(X_n) \xrightarrow{P} g(c)$.*

Proof. Given $\varepsilon > 0$, continuity gives a $\delta > 0$ with $|x - c| < \delta \implies |g(x) - g(c)| < \varepsilon$. Hence $\mathbb{P}(|g(X_n) - g(c)| \ge \varepsilon) \le \mathbb{P}(|X_n - c| \ge \delta) \to 0$. □

5.5 Worked Examples

Example 5.19 (Chebyshev versus the Normal Tail)**.** For $X \sim \mathcal{N}(\mu, \sigma^2)$, $\mathbb{P}(|X - \mu| \ge 2\sigma) \approx 0.0455$, whereas Chebyshev guarantees only ≤ 0.25. The looseness of the bound is the price of its validity for *all* distributions with finite variance (Example 5.7).

Example 5.20 (Choosing a Sample Size: Three Approaches)**.** We want to estimate a proportion p to within $\varepsilon = 0.05$ with at least 95% confidence: $\mathbb{P}(|\hat{p} - p| \ge 0.05) \le 0.05$.

- **Chebyshev:** since $p(1-p) \le 1/4$, $\frac{1}{4n\varepsilon^2} \le 0.05 \implies n \ge 2000$.
- **Hoeffding:** $X_i \in [0,1]$, $2e^{-2n\varepsilon^2} \le 0.05 \implies n \ge \frac{\ln 40}{2 \cdot 0.0025} \approx 737.8$, i.e. $n \ge 738$.
- **CLT approximation** (Chapter 6): $n \ge \left(\frac{z_{0.025}}{2\varepsilon}\right)^2 = \left(\frac{1.96}{0.1}\right)^2 \approx 384.2$, i.e. $n \ge 385$.

The first two are exact guarantees valid for every n; the third is an asymptotic approximation but gives the most economical result in practice.

Example 5.21 (Monte Carlo Integration)**.** The integral $I = \int_0^1 g(x)\,dx$ can be estimated by $\hat{I}_n = \frac{1}{n}\sum g(U_i)$ with $U_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$. Then $\mathbb{E}\hat{I}_n = I$ and, by the SLLN, $\hat{I}_n \xrightarrow{\text{a.s.}} I$. The error is of order $\sqrt{\text{Var}\,g(U)/n}$ and — unlike numerical quadrature rules — *independent of the dimension.* All simulations in the book rest on this principle.

5.6 Computational Applications

5.6.1 An Empirical Check of Chebyshev's Bound

For $X \sim \text{Gamma}(2, \frac{1}{2})$ (shape 2, rate $\frac{1}{2}$), $\mu = \alpha/\beta = 4$ and $\sigma^2 = \alpha/\beta^2 = 8$.

code/ch05/c05_1_chebyshev_empirical_check.py

```python
import numpy as np

rng = np.random.default_rng(42)
samples = rng.gamma(shape=2.0, scale=2.0, size=500_000)   # rate 1/2 <=> scale 2
sigma = np.sqrt(8.0)
for k in [1.5, 2.0, 3.0, 4.0]:
    print(f"k = {k:.1f} | Empirical: {np.mean(np.abs(samples - 4.0) >= k * sigma):.4f} <=
    Chebyshev: {1.0/(k**2):.4f}")
```

Output

```
k = 1.5 | Empirical: 0.0830 <= Chebyshev: 0.4444
k = 2.0 | Empirical: 0.0464 <= Chebyshev: 0.2500
k = 3.0 | Empirical: 0.0140 <= Chebyshev: 0.1111
k = 4.0 | Empirical: 0.0042 <= Chebyshev: 0.0625
```

5.6.2 How Well Do the Sample-Size Rules Really Perform?

code/ch05/c05_2_sample_size_rules.py

```python
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
eps, delta, R = 0.05, 0.05, 200_000
rules = {"Chebyshev": int(np.ceil(1/(4*eps**2*delta))),
            "Hoeffding": int(np.ceil(np.log(2/delta)/(2*eps**2))),
            "CLT":       int(np.ceil((stats.norm.ppf(1-delta/2)/(2*eps))**2))}
for p in [0.5, 0.2]:
    for name, n in rules.items():
        p_hat = rng.binomial(n, p, size=R) / n
        err = np.mean(np.abs(p_hat - p) >= eps)
        print(f"p={p} | {name:9s} n={n:4d} | P(|p_hat-p|>=0.05) = {err:.4f} (target <= {delta})")
```

Output

```
p=0.5 | Chebyshev n=2000 | P(|p_hat-p|>=0.05) = 0.0000 (target <= 0.05)
p=0.5 | Hoeffding n= 738 | P(|p_hat-p|>=0.05) = 0.0072 (target <= 0.05)
p=0.5 | CLT       n= 385 | P(|p_hat-p|>=0.05) = 0.0528 (target <= 0.05)
p=0.2 | Chebyshev n=2000 | P(|p_hat-p|>=0.05) = 0.0000 (target <= 0.05)
p=0.2 | Hoeffding n= 738 | P(|p_hat-p|>=0.05) = 0.0007 (target <= 0.05)
p=0.2 | CLT       n= 385 | P(|p_hat-p|>=0.05) = 0.0130 (target <= 0.05)
```

At $p = 0.5$ the CLT rule roughly hits the target (it may not be exactly 0.05 because of discreteness), whereas Chebyshev and Hoeffding are overly conservative. For $p = 0.2$ the variance is $p(1-p) = 0.16 < 0.25$, so all three rules recommend larger samples than necessary.

5.6.3 The Law of Large Numbers and the Cauchy Counterexample

code/ch05/c05_3_lln_and_cauchy.py

```python
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
n = np.arange(1, 10_001)
fig, ax = plt.subplots(1, 2, figsize=(9, 3.4), sharex=True)
for r in range(5):
    ax[0].plot(n, np.cumsum(rng.normal(size=n.size)) / n, lw=0.8)
    ax[1].plot(n, np.cumsum(rng.standard_cauchy(size=n.size)) / n, lw=0.8)
ax[0].set_title("Normal(0,1): the mean converges to 0"); ax[1].set_title("Cauchy: the mean does not converge")
for a in ax:
    a.set_xscale("log"); a.axhline(0, color="k", lw=0.6); a.set_xlabel("n")
ax[0].set_ylim(-1.5, 1.5); ax[0].set_ylabel(r"$\bar X_n$")
fig.tight_layout(); fig.savefig("lln_cauchy.pdf")

# The distribution of the Cauchy sample mean does not depend on n:
for m in [1, 10, 1000]:
    xbar = rng.standard_cauchy(size=(20_000, m)).mean(axis=1)
```

```
    q = np.percentile(xbar, [25, 75])
    print(f"n={m:5d} | interquartile width of the Cauchy mean: {q[1]-q[0]:.3f} (theory 2.000)
    ")
```

Output

```
n=    1 | interquartile width of the Cauchy mean: 1.948 (theory 2.000)
n=   10 | interquartile width of the Cauchy mean: 2.011 (theory 2.000)
n= 1000 | interquartile width of the Cauchy mean: 2.007 (theory 2.000)
```

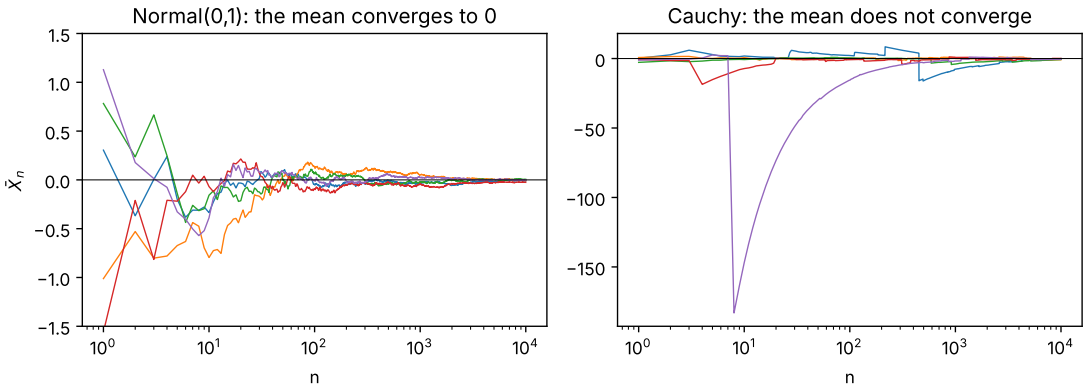


Figure 5.1: Running sample means along five independent paths. Left: by the SLLN every path converges to 0. Right: for the Cauchy distribution $\mathbb{E}|X| = \infty$, so occasional huge observations keep making the mean jump.

The output shows that the interquartile width of the Cauchy sample mean does *not* shrink at all with n; the reason is that $\bar{X}_n$ is again standard Cauchy (Chapter 6, Example 6.20).

5.7 Comments and Notes

Remark 5.22 (The Difference Between the Weak and Strong Laws). WLLN: "for a fixed large n, the probability that $\bar{X}_n$ is far from μ is small." SLLN: "for almost every sequence of observations, the sequence $\bar{X}_n$ converges to μ." Statistical consistency (Chapter 7) is usually defined via the weak version; but the "one long run" logic of Monte Carlo methods relies on the strong version.

Remark 5.23 (The Role of the Assumptions). The variance condition in the WLLN is for convenience of proof (Khinchin requires only $\mathbb{E}|X| < \infty$), but the *existence of the mean* is indispensable, as the Cauchy example clearly shows. This is the theoretical reason why, for very heavy-tailed data (financial returns, network traffic), robust estimators such as the median or a trimmed mean are preferred to the sample mean.

5.8 Chapter Summary

- Markov's and Chebyshev's inequalities (Theorems 5.5 and 5.6) bound tail probabilities by moments. The Chernoff and Hoeffding (Theorem 5.9) bounds give exponentially decaying bounds in return for an MGF or boundedness.
- Hierarchy of convergence (Theorem 5.15): almost sure and L^p convergence imply convergence in probability; the converses are false in general.
- The weak law of large numbers follows from Chebyshev under a finite variance (The-

orem 5.11). Khinchin's version requires only $\mathbb{E}|X_1| < \infty$ (Theorem 5.13). The strong law gives almost sure convergence (Theorem 5.14).

- Convergence in probability is preserved under continuous maps (Theorem 5.18); this is how consistency of estimators carries over to their functions.
- In sample-size planning, the Chebyshev and Hoeffding bounds are safe but conservative; the CLT-based rule is approximate.

Further reading. [16, Section 5.5], [81, Chapters 4–5]; for measure-theoretic proofs of the laws of large numbers see [22, Chapter 2].

5.9 Exercises

Exercise 5.1. Prove the one-sided Chebyshev (Cantelli) inequality: $\mathbb{P}(X - \mu \geq t) \leq \frac{\sigma^2}{\sigma^2 + t^2}$, $t > 0$. (Hint: for every $u \geq 0$, $\mathbb{P}(X - \mu \geq t) \leq \mathbb{P}((X - \mu + u)^2 \geq (t + u)^2)$; apply Markov and then minimise over u.) → Solution

Exercise 5.2. For $X \sim \text{Poisson}(\lambda)$, compute the Chernoff bound and show that $\mathbb{P}(X \geq a) \leq e^{-\lambda}(e\lambda/a)^a$ for $a > \lambda$. Compare Chebyshev, Chernoff and the exact value for $\lambda = 10$, $a = 20$. → Solution

Exercise 5.3. Show, using Borel–Cantelli, that if $X_n \xrightarrow{P} X$ then $\{X_n\}$ has a subsequence converging almost surely. → Solution

Exercise 5.4. Estimate π by Monte Carlo: for $(U_1, U_2) \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$, $\mathbb{P}(U_1^2 + U_2^2 \leq 1) = \pi/4$. Using Hoeffding's inequality, find the sample size needed to estimate π to within ± 0.01 with 99% confidence. → Solution

Chapter 6

Characteristic Functions and the Central Limit Theorem

The Law of Large Numbers says that $\bar{X}_n$ converges to μ, but it does not say *how*, i.e. what the distribution of the error $\bar{X}_n - \mu$ looks like. The Central Limit Theorem (CLT) states that this error, scaled by $\sqrt{n}$, converges to a normal distribution — whatever the initial distribution. Almost all approximate confidence intervals, z- and Wald tests and the asymptotic theory of the MLE are consequences of this theorem. The main tool of the proof is the characteristic function, which exists for every distribution and turns independent sums into products.

6.1 Definitions

Definition 6.1 (Convergence in Distribution)**.** Let F_n be the CDF of X_n and F that of X. If $\lim_{n\to\infty} F_n(x) = F(x)$ at *every point x where F is continuous*, we write $X_n \xrightarrow{d} X$ (or $F_n \Rightarrow F$).

Remark 6.2. The restriction to "continuity points" is necessary: for $X_n = 1/n$ (constant) we want $X_n \to 0$, but $F_n(0) = 0 \not\to 1 = F(0)$. Convergence in distribution is a statement about distributions only; the X_n need not even be defined on the same probability space.

Definition 6.3 (Characteristic Function)**.** The characteristic function (CF) of a random variable X, $\varphi_X : \mathbb{R} \to \mathbb{C}$, is defined by

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \mathbb{E}[\cos tX] + i\,\mathbb{E}[\sin tX].$$

Since $|e^{itX}| = 1$, the CF is defined for *every* distribution, whether or not moments exist. For a random vector $\mathbf{X} \in \mathbb{R}^k$, $\varphi_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}[e^{i\mathbf{t}^\top \mathbf{X}}]$.

6.2 Properties of the Characteristic Function

Proposition 6.4 (Basic Properties)**.**

1. *$\varphi_X(0) = 1$ and $|\varphi_X(t)| \le 1$ for every t.*
2. *φ_X is uniformly continuous on $\mathbb{R}$.*
3. *$\varphi_{aX+b}(t) = e^{ibt}\varphi_X(at)$.*
4. *If $X \perp\!\!\!\perp Y$, then $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$.*
5. *$\varphi_{-X}(t) = \overline{\varphi_X(t)}$; hence if X is symmetric about zero, φ_X is real-valued.*

Proof. (1) $|\mathbb{E}e^{itX}| \le \mathbb{E}|e^{itX}| = 1$. (2) $|\varphi(t+h)-\varphi(t)| \le \mathbb{E}|e^{ihX}-1|$; the right-hand side does not depend on t, and since $|e^{ihX}-1| \le 2$ it tends to zero as $h \to 0$ by dominated convergence. (3) $\mathbb{E}e^{it(aX+b)} = e^{itb}\mathbb{E}e^{i(at)X}$. (4) By independence, $\mathbb{E}[e^{itX}e^{itY}] = \mathbb{E}e^{itX}\mathbb{E}e^{itY}$ (the complex-valued version of Proposition 3.7). (5) Conjugation commutes with expectation. □

Lemma 6.5 (Portmanteau Lemma (Basic Form))**.** $X_n \xrightarrow{d} X \iff \mathbb{E}g(X_n) \to \mathbb{E}g(X)$ *for every bounded continuous* $g : \mathbb{R} \to \mathbb{R}$.

Proof. ($\Leftarrow$) Let x be a continuity point of F and $\delta > 0$. Let g_δ be the continuous function equal to 1 on $(-\infty, x]$, 0 on $[x+\delta, \infty)$ and linear in between. Since $\mathbb{I}_{(-\infty,x]} \le g_\delta \le \mathbb{I}_{(-\infty,x+\delta]}$,

$$\limsup_n F_n(x) \le \lim_n \mathbb{E}g_\delta(X_n) = \mathbb{E}g_\delta(X) \le F(x+\delta).$$

Similarly, with h_δ equal to 1 on $(-\infty, x-\delta]$ and 0 on $[x, \infty)$, $\liminf_n F_n(x) \ge F(x-\delta)$. Letting $\delta \downarrow 0$, continuity of F at x gives $F_n(x) \to F(x)$.

($\Rightarrow$) Let $|g| \le M$ and $\varepsilon > 0$. Since the discontinuity points of F are at most countable (Chapter 2), the continuity points are dense in $\mathbb{R}$; choose continuity points $a < b$ with $F(a) < \varepsilon$ and $1 - F(b) < \varepsilon$. Since g is uniformly continuous on $[a, b]$, continuity points $a = x_0 < x_1 < \cdots < x_m = b$ can be chosen with $|g(x) - g(x_j)| < \varepsilon$ on each $(x_{j-1}, x_j]$. For the step function $h = \sum_j g(x_j)\mathbb{I}_{(x_{j-1},x_j]}$,

$$\big|\mathbb{E}g(X_n) - \mathbb{E}h(X_n)\big| \le \varepsilon + M\,\mathbb{P}\big(X_n \notin (a,b]\big),$$
$$\mathbb{E}h(X_n) = \sum_{j=1}^{m} g(x_j)\big[F_n(x_j) - F_n(x_{j-1})\big] \to \mathbb{E}h(X).$$

Since $\mathbb{P}(X_n \notin (a,b]) = F_n(a) + 1 - F_n(b) \to F(a) + 1 - F(b) < 2\varepsilon$ and the same inequality holds for X, $\limsup_n |\mathbb{E}g(X_n) - \mathbb{E}g(X)| \le 2\varepsilon + 4M\varepsilon$. As ε is arbitrary, the result follows. □

Theorem 6.6 (Uniqueness and Inversion; see [9, Section 26])**.** *If* $\varphi_X = \varphi_Y$*, then* $X \overset{d}{=} Y$*. Moreover, if* $a < b$ *are continuity points of* F_X*,*

$$F_X(b) - F_X(a) = \lim_{T\to\infty} \frac{1}{2\pi}\int_{-T}^{T} \frac{e^{-ita} - e^{-itb}}{it}\varphi_X(t)\,dt.$$

If $\int |\varphi_X(t)|dt < \infty$*, then* X *has a continuous density:* $f_X(x) = \frac{1}{2\pi}\int e^{-itx}\varphi_X(t)\,dt$.

Proof. Inversion formula. Since $\big|\frac{e^{-ita}-e^{-itb}}{it}\big| = \big|\int_a^b e^{-itu}du\big| \le b-a$, the integrand is integrable on $[-T, T] \times \Omega$, and Fubini's theorem allows us to exchange the integral and the expectation:

$$I_T := \frac{1}{2\pi}\int_{-T}^{T} \frac{e^{-ita} - e^{-itb}}{it}\varphi_X(t)\,dt = \mathbb{E}\left[\frac{1}{2\pi}\int_{-T}^{T} \frac{e^{it(X-a)} - e^{it(X-b)}}{it}\,dt\right].$$

The cosine part is odd, so its integral vanishes, and $\int_{-T}^{T} \frac{e^{itc}}{it}dt = 2\int_0^T \frac{\sin tc}{t}dt = 2\,\mathrm{sgn}(c)\,S(T|c|)$, where $S(x) = \int_0^x \frac{\sin u}{u}du$. Hence

$$I_T = \mathbb{E}\Big[\tfrac{1}{\pi}\big\{\mathrm{sgn}(X-a)\,S(T|X-a|) - \mathrm{sgn}(X-b)\,S(T|X-b|)\big\}\Big].$$

S is bounded and $S(x) \to \pi/2$ as $x \to \infty$ (the Dirichlet integral). As $T \to \infty$, the quantity inside the expectation converges to 0 if $X < a$ or $X > b$, to $\frac{1}{2}$ if $X \in \{a, b\}$, and to 1 if $a < X < b$. By bounded convergence,

$$\lim_{T\to\infty} I_T = \mathbb{P}(a < X < b) + \tfrac{1}{2}\mathbb{P}(X = a) + \tfrac{1}{2}\mathbb{P}(X = b),$$

which equals $F_X(b) - F_X(a)$ when a, b are continuity points.

Uniqueness. If $\varphi_X = \varphi_Y$, the formula above gives $F_X(b) - F_X(a) = F_Y(b) - F_Y(a)$ for all common continuity points $a < b$ of F_X and F_Y. The common continuity points are dense (the complement of two countable sets); letting $a \to -\infty$ gives $F_X(b) = F_Y(b)$ at these points, and right-continuity gives $F_X = F_Y$ everywhere.

Density. If $\int |\varphi_X| < \infty$, by dominated convergence the limit equals the integral over $\mathbb{R}$, and $|F_X(b) - F_X(a)| \le \frac{b-a}{2\pi} \int |\varphi_X|$; hence F_X has no jumps and the formula holds for every $a < b$. Take $b = a + h$ and divide by h: $\frac{e^{-ita} - e^{-it(a+h)}}{ith} \to e^{-ita}$ with modulus bounded by 1; by dominated convergence $F_X'(a) = \frac{1}{2\pi} \int e^{-ita} \varphi_X(t)\, dt$. Continuity of the right-hand side in a again follows from dominated convergence. □

Theorem 6.7 (Lévy's Continuity Theorem; see [9, Section 26])**.** *$X_n \xrightarrow{d} X \iff \varphi_{X_n}(t) \to \varphi_X(t)$ for every $t \in \mathbb{R}$. More strongly: if $\varphi_{X_n}(t) \to \psi(t)$ pointwise and ψ is continuous at $t = 0$, then ψ is the CF of some random variable X and $X_n \xrightarrow{d} X$.*

Proof. ($\Rightarrow$) The maps $x \mapsto \cos tx$ and $x \mapsto \sin tx$ are bounded and continuous; apply Lemma 6.5.

($\Leftarrow$, strong form) *Step 1 (a tail inequality).* For every random variable and $u > 0$,

$$\mathbb{P}(|X| > 2/u) \le \frac{1}{u} \int_{-u}^{u} (1 - \varphi_X(t))\, dt.$$

Indeed, $\frac{1}{u} \int_{-u}^{u} (1 - e^{itx})\, dt = 2(1 - \frac{\sin ux}{ux}) \ge 0$, and this is ≥ 1 when $|ux| \ge 2$ because then $|\sin ux/(ux)| \le \frac{1}{2}$. By Fubini, the right-hand side equals $\mathbb{E}[2(1 - \frac{\sin uX}{uX})] \ge \mathbb{P}(|uX| \ge 2)$.

Step 2 (tightness). Since $\psi(0) = \lim \varphi_{X_n}(0) = 1$ and ψ is continuous at 0, for given $\varepsilon > 0$ there is a $u > 0$ with $\frac{1}{u} \int_{-u}^{u} |1 - \psi(t)|\, dt < \varepsilon$. Since $|1 - \varphi_{X_n}| \le 2$, dominated convergence gives $\frac{1}{u} \int_{-u}^{u} (1 - \varphi_{X_n}) \to \frac{1}{u} \int_{-u}^{u} (1 - \psi)$; hence $\mathbb{P}(|X_n| > 2/u) \le 2\varepsilon$ for large n. Choosing a larger bound for the finitely many initial terms: for every $\varepsilon > 0$ there is a K with $\sup_n \mathbb{P}(|X_n| > K) \le 2\varepsilon$, i.e. $\{X_n\}$ is *tight*.

Step 3 (identifying the limit). By Helly's selection theorem [9, Theorem 25.9], every sequence of CDFs has a subsequence converging at continuity points to a non-decreasing, right-continuous G; tightness ensures $G(-\infty) = 0$ and $G(\infty) = 1$, i.e. that G is a proper CDF. If $Y \sim G$, then $X_{n_k} \xrightarrow{d} Y$; by the first direction $\varphi_{X_{n_k}} \to \varphi_Y$, hence $\varphi_Y = \psi$: ψ is a CF.

Step 4 (convergence of the whole sequence). Let $X \sim G$. If $X_n \xrightarrow{d} X$ failed, there would be a continuity point x, a $\delta > 0$ and a subsequence with $|F_{n_k}(x) - F(x)| \ge \delta$. Applying Step 3 to this subsequence yields a further subsequence converging to a limit whose CF is again ψ, i.e. by Theorem 6.6 a limit with the same distribution as X; a contradiction. The first statement is the special case $\psi = \varphi_X$ (continuous). □

Proposition 6.8 (Moments and Taylor Expansion)**.** *If $\mathbb{E}|X|^k < \infty$, then φ_X is k times continuously differentiable, $\varphi_X^{(k)}(0) = i^k \mathbb{E}[X^k]$, and*

$$\varphi_X(t) = \sum_{j=0}^{k} \frac{(it)^j}{j!} \mathbb{E}[X^j] + o(|t|^k), \qquad t \to 0.$$

In particular, if $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$, then $\varphi_X(t) = 1 - \frac{t^2}{2} + o(t^2)$.

Proof. An inequality. For real x, integration by parts (induction) gives

$$e^{ix} - \sum_{j=0}^{k} \frac{(ix)^j}{j!} = \frac{i^{k+1}}{k!} \int_0^x (x - s)^k e^{is}\, ds = \frac{i^k}{(k-1)!} \int_0^x (x - s)^{k-1} (e^{is} - 1)\, ds \quad (k \ge 1).$$

The first representation with $|e^{is}| = 1$ gives the bound $|x|^{k+1}/(k+1)!$, and the second with $|e^{is} - 1| \le 2$ gives $2|x|^k/k!$:

$$\Big|e^{ix} - \sum_{j=0}^{k} \frac{(ix)^j}{j!}\Big| \le \min\Big(\frac{|x|^{k+1}}{(k+1)!},\ \frac{2|x|^k}{k!}\Big).$$

Differentiability. Since $\big|\frac{e^{i(t+h)X} - e^{itX}}{h}\big| \le |X|$ (the inequality with $k = 0$) and $\mathbb{E}|X| < \infty$, dominated convergence gives $\varphi_X'(t) = \mathbb{E}[iXe^{itX}]$. By induction, $\varphi_X^{(j)}(t) = \mathbb{E}[(iX)^j e^{itX}]$ for $j \le k$; these are continuous by dominated convergence and equal $i^j \mathbb{E}X^j$ at $t = 0$.

Expansion. Writing the inequality for $x = tX$ and taking expectations,

$$\Big|\varphi_X(t) - \sum_{j=0}^{k} \frac{(it)^j}{j!}\mathbb{E}X^j\Big| \le |t|^k\, \mathbb{E}\Big[\min\Big(\frac{|t||X|^{k+1}}{(k+1)!},\ \frac{2|X|^k}{k!}\Big)\Big].$$

The quantity in brackets tends to zero pointwise as $t \to 0$ and is bounded by the integrable $2|X|^k/k!$; by dominated convergence its expectation tends to zero, i.e. the remainder is $o(|t|^k)$. □

6.3 The Central Limit Theorem

Lemma 6.9. *If $c_n \in \mathbb{C}$ and $c_n \to c$, then $(1 + c_n/n)^n \to e^c$.*

Proof. For $|z| \le 1/2$, $|\log(1+z) - z| \le |z|^2$ (principal branch of the complex logarithm). For large n, $n\log(1 + c_n/n) = c_n + O(|c_n|^2/n) \to c$; continuity of the exponential gives the result. □

Theorem 6.10 (Lindeberg–Lévy CLT)**.** *Let $\{X_i\}$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Then*

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{X_i - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0,1).$$

Proof. Let $Y_i = (X_i - \mu)/\sigma$, i.i.d. with $\mathbb{E}Y_i = 0$, $\mathbb{E}Y_i^2 = 1$ and common CF φ. By Proposition 6.8, $\varphi(t) = 1 - \frac{t^2}{2} + o(t^2)$. By Proposition 6.4 (3)–(4),

$$\varphi_{Z_n}(t) = \Big[\varphi(t/\sqrt{n})\Big]^n = \left(1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right)^n = \left(1 + \frac{c_n}{n}\right)^n,$$

where $c_n = -t^2/2 + n \cdot o(t^2/n) \to -t^2/2$ (for fixed t). By Lemma 6.9, $\varphi_{Z_n}(t) \to e^{-t^2/2}$, the CF of the standard normal (Example 6.18). Lévy's continuity theorem (Theorem 6.7) gives the result. □

Theorem 6.11 (Lindeberg–Feller CLT [51])**.** *For each n, let $X_{n1}, \dots, X_{nn}$ be independent (not necessarily identically distributed) with $\mathbb{E}X_{ni} = 0$, and let $s_n^2 = \sum_i \operatorname{Var} X_{ni}$. If for every $\varepsilon > 0$*

$$\frac{1}{s_n^2} \sum_{i=1}^{n} \mathbb{E}\big[X_{ni}^2 \mathbb{I}\{|X_{ni}| > \varepsilon s_n\}\big] \to 0 \qquad \textit{(Lindeberg condition)},$$

then $\frac{1}{s_n}\sum_{i=1}^n X_{ni} \xrightarrow{d} \mathcal{N}(0,1)$. The Lindeberg condition says that no single term dominates the variance of the sum; this version is used for the asymptotic normality of regression coefficients.

Proof. Replacing X_{ni} by X_{ni}/s_n we may assume $s_n = 1$; write $\sigma_{ni}^2 = \operatorname{Var} X_{ni}$, so $\sum_i \sigma_{ni}^2 = 1$ and $L_n(\varepsilon) = \sum_i \mathbb{E}[X_{ni}^2 \mathbb{I}\{|X_{ni}| > \varepsilon\}] \to 0$. We use two basic inequalities:

(a) For complex numbers with $|z_i|, |w_i| \le 1$, $\big|\prod_{i=1}^n z_i - \prod_{i=1}^n w_i\big| \le \sum_{i=1}^n |z_i - w_i|$. (Induction: $\prod_{i\le n} z_i - \prod_{i\le n} w_i = z_n\big(\prod_{i<n} z_i - \prod_{i<n} w_i\big) + (z_n - w_n)\prod_{i<n} w_i$.)

(b) For real x, $|e^{ix} - 1 - ix + x^2/2| \le \min(|x|^3/6,\ x^2)$ (the inequality in the proof of Proposition 6.8, $k=2$); also $|e^{-c} - 1 + c| \le c^2/2$ for $c \ge 0$.

Step 1: $\max_i \sigma_{ni}^2 \to 0$. $\sigma_{ni}^2 \le \varepsilon^2 + \mathbb{E}[X_{ni}^2 \mathbb{I}\{|X_{ni}| > \varepsilon\}] \le \varepsilon^2 + L_n(\varepsilon)$; let first $n \to \infty$, then $\varepsilon \to 0$.

Step 2. Fix t and let φ_{ni} be the CF of X_{ni}. By (b) and $\mathbb{E}X_{ni} = 0$,

$$\Big|\varphi_{ni}(t) - \Big(1 - \frac{t^2\sigma_{ni}^2}{2}\Big)\Big| \le \mathbb{E}\Big[\frac{|t|^3|X_{ni}|^3}{6}\mathbb{I}\{|X_{ni}| \le \varepsilon\}\Big] + t^2\,\mathbb{E}\big[X_{ni}^2\mathbb{I}\{|X_{ni}| > \varepsilon\}\big]$$
$$\le \frac{\varepsilon|t|^3}{6}\sigma_{ni}^2 + t^2\,\mathbb{E}\big[X_{ni}^2\mathbb{I}\{|X_{ni}| > \varepsilon\}\big].$$

Summing over i, the right-hand side becomes $\varepsilon|t|^3/6 + t^2 L_n(\varepsilon)$. By Step 1, $w_i = 1 - t^2\sigma_{ni}^2/2 \in [0,1]$ for large n; by (a),

$$\limsup_n \Big|\prod_i \varphi_{ni}(t) - \prod_i \Big(1 - \frac{t^2\sigma_{ni}^2}{2}\Big)\Big| \le \frac{\varepsilon|t|^3}{6},$$

and since ε is arbitrary, this difference tends to zero.

Step 3. With $c_i = t^2\sigma_{ni}^2/2$, $\prod_i e^{-c_i} = e^{-t^2/2}$. By (a) and (b),

$$\Big|e^{-t^2/2} - \prod_i (1 - c_i)\Big| \le \sum_i \frac{c_i^2}{2} \le \frac{\max_i c_i}{2}\sum_i c_i = \frac{t^4}{8}\max_i \sigma_{ni}^2 \to 0.$$

Combining the two steps, $\prod_i \varphi_{ni}(t) \to e^{-t^2/2}$. By independence this product is the CF of $\sum_i X_{ni}$; Theorem 6.7 gives the result. □

Theorem 6.12 (Berry–Esseen [7, 26])**.** *If, in addition to the conditions of Theorem 6.10, $\rho = \mathbb{E}|X_1 - \mu|^3 < \infty$, then*

$$\sup_{x\in\mathbb{R}} |\mathbb{P}(Z_n \le x) - \Phi(x)| \le \frac{C\rho}{\sigma^3\sqrt{n}},$$

where C is a universal constant; one of the best known values is $C \le 0.4748$ [71].

Remark 6.13 (About the Proof). A full proof requires a long computation beyond the scope of this book; the main idea is as follows. Esseen's *smoothing inequality* states that if the derivative of G is bounded, $|G'| \le m$, then for every $T > 0$

$$\sup_x |F(x) - G(x)| \le \frac{1}{\pi}\int_{-T}^{T} \Big|\frac{\varphi_F(t) - \varphi_G(t)}{t}\Big|\,dt + \frac{24m}{\pi T}.$$

Take $F = F_{Z_n}$ and $G = \Phi$; the third-order remainder bound of Proposition 6.8 and inequality (a) in the proof of Theorem 6.11 give $|\varphi_{Z_n}(t) - e^{-t^2/2}| \le C'\frac{\rho}{\sigma^3\sqrt{n}}|t|^3 e^{-t^2/4}$ for $|t| \le \sigma^3\sqrt{n}/(4\rho)$. Choosing $T = \sigma^3\sqrt{n}/(4\rho)$ makes both terms of order $\rho/(\sigma^3\sqrt{n})$. For details see [22, Section 3.4] and the original papers [7, 26].

Theorem 6.14 (Multivariate CLT and Cramér–Wold)**.** *(Cramér–Wold) $\mathbf{X}_n \xrightarrow{d} \mathbf{X}$ (in $\mathbb{R}^k$) $\iff$ $\mathbf{a}^\top\mathbf{X}_n \xrightarrow{d} \mathbf{a}^\top\mathbf{X}$ for every $\mathbf{a} \in \mathbb{R}^k$. Consequently, if $\mathbf{X}_i \in \mathbb{R}^k$ are i.i.d. with mean $\boldsymbol{\mu}$ and covariance matrix Σ, then $\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}_k(\mathbf{0}, \Sigma)$.*

Proof. In $\mathbb{R}^k$, convergence in distribution means convergence of the joint CDFs at their continuity points. Lemma 6.5, Theorem 6.6 and Theorem 6.7 are proved in this setting by the same ideas [9, Section 29]; in particular, $\mathbf{X}_n \xrightarrow{d} \mathbf{X} \iff \varphi_{\mathbf{X}_n}(\mathbf{t}) \to \varphi_{\mathbf{X}}(\mathbf{t})$ for every $\mathbf{t} \in \mathbb{R}^k$.

Cramér–Wold. For $s \in \mathbb{R}$, $\varphi_{\mathbf{a}^\top \mathbf{X}_n}(s) = \mathbb{E}e^{is\mathbf{a}^\top \mathbf{X}_n} = \varphi_{\mathbf{X}_n}(s\mathbf{a})$. ($\Rightarrow$) $\varphi_{\mathbf{X}_n}(s\mathbf{a}) \to \varphi_{\mathbf{X}}(s\mathbf{a}) = \varphi_{\mathbf{a}^\top \mathbf{X}}(s)$. ($\Leftarrow$) Taking $s = 1$, for every $\mathbf{a}$, $\varphi_{\mathbf{X}_n}(\mathbf{a}) = \varphi_{\mathbf{a}^\top \mathbf{X}_n}(1) \to \varphi_{\mathbf{a}^\top \mathbf{X}}(1) = \varphi_{\mathbf{X}}(\mathbf{a})$; the multivariate Lévy theorem gives the result.

Multivariate CLT. For fixed $\mathbf{a}$, $Y_i = \mathbf{a}^\top \mathbf{X}_i$ are i.i.d. with mean $\mathbf{a}^\top \boldsymbol{\mu}$ and variance $\mathbf{a}^\top \Sigma \mathbf{a}$. If $\mathbf{a}^\top \Sigma \mathbf{a} > 0$, Theorem 6.10 gives $\mathbf{a}^\top \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) = \sqrt{n}(\bar{Y}_n - \mathbf{a}^\top \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}(0, \mathbf{a}^\top \Sigma \mathbf{a})$; if $\mathbf{a}^\top \Sigma \mathbf{a} = 0$, the left-hand side is a.s. zero. In both cases the limit is the distribution of $\mathbf{a}^\top \mathbf{Z}$ with $\mathbf{Z} \sim \mathcal{N}_k(\mathbf{0}, \Sigma)$; Cramér–Wold gives the result. □

6.4 Relations Between Modes of Convergence and Slutsky

Proposition 6.15. *(1) $X_n \xrightarrow{P} X \implies X_n \xrightarrow{d} X$. (2) If c is a constant, $X_n \xrightarrow{d} c \implies X_n \xrightarrow{P} c$.*

Proof. (1) For $\varepsilon > 0$, $F_n(x) \le F(x + \varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon)$ and $F(x - \varepsilon) \le F_n(x) + \mathbb{P}(|X_n - X| > \varepsilon)$. Let $n \to \infty$ and then $\varepsilon \to 0$ (at a continuity point). (2) $\mathbb{P}(|X_n - c| > \varepsilon) \le F_n(c - \varepsilon) + 1 - F_n(c + \varepsilon/2) \to 0 + 1 - 1 = 0$, because $c \pm \varepsilon$ are continuity points of the CDF of the constant. □

Theorem 6.16 (Slutsky's Theorem)**.** *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} c$ (a constant), then*

$$X_n + Y_n \xrightarrow{d} X + c, \qquad X_n Y_n \xrightarrow{d} cX, \qquad X_n / Y_n \xrightarrow{d} X/c \ (c \neq 0).$$

Proof. Joint convergence. For $(s, t) \in \mathbb{R}^2$,

$$\begin{aligned} \left|\mathbb{E}e^{i(sX_n + tY_n)} - \mathbb{E}e^{i(sX + tc)}\right| &\le \mathbb{E}\left|e^{isX_n}\left(e^{itY_n} - e^{itc}\right)\right| + \left|e^{itc}\right| \left|\varphi_{X_n}(s) - \varphi_X(s)\right| \\ &\le \mathbb{E}\left|e^{it(Y_n - c)} - 1\right| + \left|\varphi_{X_n}(s) - \varphi_X(s)\right|. \end{aligned}$$

The second term tends to zero by Theorem 6.7. For the first, with $\delta > 0$ and $|e^{iu} - 1| \le \min(|u|, 2)$, $\mathbb{E}|e^{it(Y_n - c)} - 1| \le |t|\delta + 2\mathbb{P}(|Y_n - c| > \delta) \to |t|\delta$; δ is arbitrary. By the multivariate Lévy theorem (proof of Theorem 6.14), $(X_n, Y_n) \xrightarrow{d} (X, c)$.

Conclusion. $g(x, y) = x + y$ and $g(x, y) = xy$ are continuous; Theorem 6.17 (continuous g) gives the first two statements. If $c \neq 0$, Theorem 5.18 gives $1/Y_n \xrightarrow{P} 1/c$, and the product rule gives the third statement. □

Theorem 6.17 (Continuous Mapping Theorem)**.** *If $X_n \xrightarrow{d} X$ and g is measurable with $\mathbb{P}(X \in D_g) = 0$ (D_g: the set of discontinuity points), then $g(X_n) \xrightarrow{d} g(X)$. The same statement holds for $\xrightarrow{P}$ and $\xrightarrow{a.s.}$.*

Proof. Almost sure case. The event $\{X_n \to X\} \cap \{X \notin D_g\}$ has probability 1, and on it $g(X_n) \to g(X)$ by continuity of g at X.

In distribution, continuous g. If h is bounded and continuous, so is $h \circ g$; by Lemma 6.5 (or its multivariate version) $\mathbb{E}h(g(X_n)) \to \mathbb{E}h(g(X))$, i.e. $g(X_n) \xrightarrow{d} g(X)$.

In probability, continuous g. Given $\varepsilon, \eta > 0$, choose K with $\mathbb{P}(|X| > K) < \eta$. Since g is uniformly continuous on the compact set $[-K - 1, K + 1]$, there is $\delta \in (0, 1)$ such that $|g(y) - g(x)| < \varepsilon$ whenever $|x| \le K$ and $|y - x| < \delta$. Hence

$$\mathbb{P}\left(|g(X_n) - g(X)| \ge \varepsilon\right) \le \mathbb{P}(|X| > K) + \mathbb{P}(|X_n - X| \ge \delta) \to \text{ at most } \eta.$$

General g. The case of convergence in distribution under $\mathbb{P}(X \in D_g) = 0$ is proved with the form of the portmanteau lemma "$\mathbb{P}(X_n \in A) \to \mathbb{P}(X \in A)$ for every A whose boundary has probability zero"; see [78, Theorem 2.3]. The case of convergence in probability then follows from the principle "every subsequence has a further subsequence converging a.s.". □

6.5 Worked Examples

Example 6.18 (The CF of the Normal Distribution)**.** For $Z \sim \mathcal{N}(0,1)$, $\varphi_Z(t) = \int \cos(tx)\varphi(x)dx$ (by symmetry). Differentiating and integrating by parts gives $\varphi_Z'(t) = -t\varphi_Z(t)$, $\varphi_Z(0) = 1$; the solution of this differential equation is $\varphi_Z(t) = e^{-t^2/2}$. By Proposition 6.4(3), for $X \sim \mathcal{N}(\mu, \sigma^2)$, $\varphi_X(t) = e^{i\mu t - \sigma^2 t^2/2}$. Consequence: a sum of independent normals is normal, because the product of such CFs keeps the same form.

Example 6.19 (Normal Approximation to the Poisson)**.** For $X \sim \text{Poisson}(\lambda)$, $\varphi_X(t) = \sum_k e^{itk} e^{-\lambda}\lambda^k/k! = \exp\{\lambda(e^{it}-1)\}$. For $Z_\lambda = (X-\lambda)/\sqrt{\lambda}$,

$$\log \varphi_{Z_\lambda}(t) = \lambda(e^{it/\sqrt{\lambda}} - 1) - it\sqrt{\lambda} = \lambda\Big(\tfrac{it}{\sqrt{\lambda}} - \tfrac{t^2}{2\lambda} + O(\lambda^{-3/2})\Big) - it\sqrt{\lambda} \to -\tfrac{t^2}{2},$$

i.e. $Z_\lambda \xrightarrow{d} \mathcal{N}(0,1)$ as $\lambda \to \infty$.

Example 6.20 (Cauchy: Where the CLT Fails)**.** The CF of the standard Cauchy distribution is $\varphi(t) = e^{-|t|}$ (this can be verified with the inversion formula of Theorem 6.6). If the X_i are i.i.d. Cauchy,

$$\varphi_{\bar{X}_n}(t) = \big[\varphi(t/n)\big]^n = e^{-n|t|/n} = e^{-|t|},$$

i.e. $\bar{X}_n$ is again standard Cauchy! Averaging gains no information at all. φ is not differentiable at $t = 0$; this is the reflection in the CF of the fact that $\mathbb{E}|X| = \infty$.

Example 6.21 (The Weak Law of Large Numbers via CFs)**.** Let the X_i be i.i.d. with $\mathbb{E}|X_1| < \infty$. By Proposition 6.8 ($k = 1$), $\varphi(t) = 1 + i\mu t + o(t)$; hence

$$\varphi_{\bar{X}_n}(t) = \big(1 + i\mu t/n + o(1/n)\big)^n \to e^{i\mu t},$$

the CF of the constant μ. By Lévy's theorem $\bar{X}_n \xrightarrow{d} \mu$, and by Proposition 6.15(2) $\bar{X}_n \xrightarrow{P} \mu$. This is a proof of Khinchin's WLLN (Theorem 5.13) that requires no variance.

Example 6.22 (An Application of Slutsky: the Studentised Mean)**.** Let the X_i be i.i.d. with $\sigma^2 < \infty$ (not necessarily normal). By the CLT, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0,1)$; by the WLLN and continuous mapping, $S_n \xrightarrow{P} \sigma$. By Slutsky,

$$T_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} = \frac{\sigma}{S_n} \cdot \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0,1).$$

Thus $\bar{X}_n \pm z_{\alpha/2} S_n/\sqrt{n}$ is an asymptotic $1-\alpha$ confidence interval for any distribution with finite variance (Chapter 16).

Example 6.23 (Continuity Correction for the Binomial)**.** Let $X \sim \text{Bin}(100, 0.3)$, $\mathbb{E}X = 30$, $\sigma = \sqrt{21} \approx 4.583$. For $\mathbb{P}(X \le 25)$ the direct approximation is $\Phi((25-30)/4.583) = \Phi(-1.091) \approx 0.138$, whereas the continuity-corrected approximation is $\Phi((25.5-30)/4.583) = \Phi(-0.982) \approx 0.163$. The exact value is ≈ 0.163 (see the code below). The correction comes from assuming that each discrete mass is spread over the interval $[k - \tfrac{1}{2}, k + \tfrac{1}{2}]$.

6.6 Computational Applications

6.6.1 Convergence to the CLT: Skewness and Kurtosis

For $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda = 0.5)$, $\mu = \sigma = 2$. A cumulant calculation shows that the skewness of Z_n is $2/\sqrt{n}$ and its excess kurtosis $6/n$; the simulated values are compared with these theoretical values.

code/ch06/c06_1_clt_skewness_kurtosis.py

```python
import numpy as np
import scipy.stats as stats

rng = np.random.default_rng(42)
lam = 0.5
for n in [1, 5, 30, 200]:
    samples = rng.exponential(scale=1.0/lam, size=(100_000, n))
    z = (np.mean(samples, axis=1) - 2.0) / (2.0 / np.sqrt(n))
    print(f"n = {n:3d} | Skewness: {stats.skew(z):6.3f} (theory {2/np.sqrt(n):5.3f}) |
    Kurtosis: {stats.kurtosis(z):6.3f} (theory {6/n:5.3f})")
```

Output

```
n =   1 | Skewness:  2.023 (theory 2.000) | Kurtosis:  6.238 (theory 6.000)
n =   5 | Skewness:  0.903 (theory 0.894) | Kurtosis:  1.281 (theory 1.200)
n =  30 | Skewness:  0.356 (theory 0.365) | Kurtosis:  0.204 (theory 0.200)
n = 200 | Skewness:  0.148 (theory 0.141) | Kurtosis:  0.040 (theory 0.030)
```

The slow $1/\sqrt{n}$ decay of the skewness shows that for skewed distributions the normal approximation needs a larger n than for symmetric ones. The rule "$n \geq 30$ is enough" is a rule of thumb for moderate skewness; it is not part of the theorem.

6.6.2 Checking the Berry–Esseen Bound by Exact Computation

Since a sum of Exp(1) variables is Gamma$(n, 1)$, the CDF of Z_n can be computed exactly, without simulation: $\mathbb{P}(Z_n \leq x) = G_n(n + x\sqrt{n})$.

code/ch06/c06_2_berry_esseen_exact.py

```python
import numpy as np
from scipy import stats, integrate

rho = integrate.quad(lambda x: abs(x - 1)**3 * np.exp(-x), 0, np.inf)[0]   # E|X-1|^3
x = np.linspace(-6, 6, 20_001)
print(f"rho = E|X-mu|^3 = {rho:.4f}")
for n in [1, 10, 100, 1000]:
    Fn = stats.gamma.cdf(n + x*np.sqrt(n), a=n)
    sup = np.max(np.abs(Fn - stats.norm.cdf(x)))
    print(f"n={n:5d} | sup|F_n - Phi| = {sup:.5f} | Berry-Esseen bound = {0.4748*rho/np.sqrt(
    n):.5f} | sqrt(n)*sup = {np.sqrt(n)*sup:.4f}")
```

Output

```
rho = E|X-mu|^3 = 2.4146
n=    1 | sup|F_n - Phi| = 0.15861 | Berry-Esseen bound = 1.14643 | sqrt(n)*sup = 0.1586
n=   10 | sup|F_n - Phi| = 0.04211 | Berry-Esseen bound = 0.36253 | sqrt(n)*sup = 0.1332
n=  100 | sup|F_n - Phi| = 0.01330 | Berry-Esseen bound = 0.11464 | sqrt(n)*sup = 0.1330
n= 1000 | sup|F_n - Phi| = 0.00421 | Berry-Esseen bound = 0.03625 | sqrt(n)*sup = 0.1330
```

That $\sqrt{n} \cdot \sup |F_n - \Phi|$ converges to a constant shows that the $O(n^{-1/2})$ rate is sharp. The Berry–Esseen bound gets the rate right, but its constant is conservative.

6.6.3 Visualising the CLT and the Continuity Correction

code/ch06/c06_3_clt_visualisation_continuity_correction.py

```python
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
```

```python
fig, axes = plt.subplots(1, 4, figsize=(11, 2.8), sharey=True)
zz = np.linspace(-4, 4, 300)
for ax, n in zip(axes, [1, 3, 10, 50]):
    z = (rng.exponential(size=(200_000, n)).mean(axis=1) - 1) * np.sqrt(n)
    ax.hist(z, bins=120, range=(-4, 6), density=True, color="tab:blue", alpha=0.6)
    ax.plot(zz, stats.norm.pdf(zz), "k-", lw=1.5)
    ax.set_title(f"n = {n}"); ax.set_xlim(-4, 6)
fig.tight_layout(); fig.savefig("clt_ustel.pdf")

n, p, k = 100, 0.3, 25
mu, sd = n*p, np.sqrt(n*p*(1-p))
print(f"Exact P(X<=25)                    = {stats.binom.cdf(k, n, p):.4f}")
print(f"Normal approx. (no correction)    = {stats.norm.cdf((k - mu)/sd):.4f}")
print(f"Normal approx. (with correction)  = {stats.norm.cdf((k + 0.5 - mu)/sd):.4f}")
```

Output

```
Exact P(X<=25)                    = 0.1631
Normal approx. (no correction)    = 0.1376
Normal approx. (with correction)  = 0.1631
```

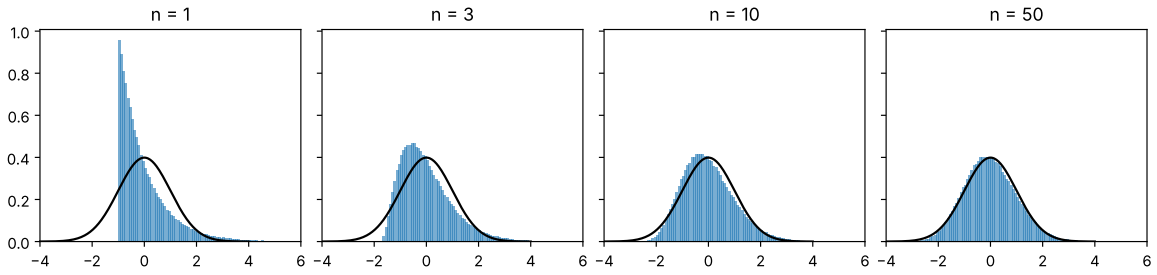


Figure 6.1: Standardised histograms of the sample mean of Exp(1) variables and the $\mathcal{N}(0,1)$ density. The right skewness disappears as n grows.

6.7 Comments and Notes

Remark 6.24 (What the CLT Does Not Say). The CLT says that the *distribution* of Z_n converges to the normal; it does not say that Z_n itself converges anywhere (in fact Z_n does not converge in probability). Moreover, the convergence is pointwise: for tail probabilities (such as $\mathbb{P}(Z_n > 5)$) the relative error may be large; such questions are the subject of large deviations theory.

Remark 6.25 (How Critical the Assumptions Are). If the finite-variance condition is dropped, the limit may be a *stable* distribution instead of the normal (the Cauchy is a special case), and the scaling becomes $n^{1/\alpha}$ instead of $\sqrt{n}$. If independence is dropped — for example in time series — the variance σ^2 must be replaced by the long-run variance $\sum_{h} \gamma(h)$.

Remark 6.26 (Link with Chapter 12). The asymptotic normality of the MLE comes from the fact that the score is an i.i.d. sum to which the CLT applies; the Slutsky and continuous mapping theorems justify replacing the Hessian by a consistent estimate.

6.8 Chapter Summary

- Convergence in distribution is convergence of the CDFs at their continuity points (Definition 6.1). It is equivalent to convergence of the expectations of bounded continuous functions (Lemma 6.5).
- The characteristic function is defined for every distribution and determines it uniquely (Theorem 6.6). Lévy's continuity theorem (Theorem 6.7) reduces convergence in distribution to pointwise convergence of CFs.
- The central limit theorem (Theorem 6.10) says that i.i.d. sums with finite variance converge to the normal. The Lindeberg–Feller version (Theorem 6.11) drops the identical-distribution assumption. Berry–Esseen (Theorem 6.12) bounds the approximation error by $O(n^{-1/2})$.
- The Cramér–Wold device reduces multivariate convergence to univariate linear combinations; this yields the multivariate CLT (Theorem 6.14).
- Slutsky's theorem (Theorem 6.16) and the continuous mapping theorem (Theorem 6.17) are the two basic tools of asymptotic inference. For example, replacing σ by S does not change the asymptotic distribution.

Further reading. [22, Chapter 3], [9, Sections 25–27]; from a statistical point of view, [78, Chapters 2–3] and [28].

6.9 Exercises

Exercise 6.1. Show that for $X \sim \text{Gamma}(\alpha, \beta)$ (rate parametrisation) $\varphi_X(t) = (1 - it/\beta)^{-\alpha}$. Use this to find the distribution of a sum of independent Gamma variables (with the same β). → Solution

Exercise 6.2. Show using CFs that if $X_n \sim \text{Bin}(n, \lambda/n)$, then $X_n \xrightarrow{d} \text{Poisson}(\lambda)$. → Solution

Exercise 6.3. Let the X_i be i.i.d. Uniform$(0, 1)$ and $Y_n = n(1 - X_{(n)})$. Show via the CDF that $Y_n \xrightarrow{d} \text{Exp}(1)$. This result shows that the CLT does not apply to *extreme-value* statistics and that a different scaling (n instead of $\sqrt{n}$) is needed; relate it to the German tank problem in Chapter 10. → Solution

Exercise 6.4. Show by a counterexample that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ do *not* imply $X_n + Y_n \xrightarrow{d} X + Y$. (Hint: $Y_n = -X_n$.) Discuss why the limit of Y_n in Slutsky's theorem must be a constant. → Solution

Part III

Point Estimation Theory

Chapter 7

Unbiasedness, Consistency and the MSE Decomposition

With this chapter we move from probability theory to statistical inference. The distribution is now unknown; we assume that it belongs to a parametric family $\{f(\cdot;\theta) : \theta \in \Theta\}$ and make inferences about θ (or a function $g(\theta)$ of it) from the observed sample $\mathbf{X} = (X_1, \dots, X_n)$. The first question is: what is a "good" estimator? In this chapter we discuss bias, variance and mean squared error (MSE); we see that unbiasedness, while a virtue, is not the only criterion; and we introduce consistency, which measures large-sample behaviour.

7.1 Definitions

Definition 7.1 (Statistic and Estimator)**.** A measurable function $T = T(\mathbf{X})$ of the sample that does not depend on the unknown parameter is called a *statistic*. A statistic used to estimate $g(\theta)$ is called an *estimator* of $g(\theta)$, and its observed value $T(\mathbf{x})$ an *estimate*.

Definition 7.2 (Bias and MSE)**.** Let $\hat{\theta}$ be an estimator of θ:

$$\operatorname{Bias}_\theta(\hat{\theta}) = \mathbb{E}_\theta[\hat{\theta}] - \theta, \qquad \operatorname{MSE}_\theta(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2].$$

If $\operatorname{Bias}_\theta(\hat{\theta}) = 0$ for every $\theta \in \Theta$, $\hat{\theta}$ is *unbiased*. If $\lim_{n\to\infty} \operatorname{Bias}_\theta(\hat{\theta}_n) = 0$, it is *asymptotically unbiased*.

Definition 7.3 (Consistency)**.** A sequence of estimators $\{\hat{\theta}_n\}$ is *(weakly) consistent* if $\hat{\theta}_n \xrightarrow{P} \theta$ (under $\mathbb{P}_\theta$) for every $\theta \in \Theta$. It is *strongly consistent* if $\hat{\theta}_n \xrightarrow{\text{a.s.}} \theta$, and *MSE-consistent* if $\operatorname{MSE}_\theta(\hat{\theta}_n) \to 0$.

Definition 7.4 (Loss, Risk and Admissibility)**.** A *loss function* $L : \Theta \times \mathbb{R} \to [0, \infty)$ measures the cost of the estimate a when the true value is θ. The *risk function* of $\hat{\theta}$ is $R(\theta, \hat{\theta}) = \mathbb{E}_\theta[L(\theta, \hat{\theta}(\mathbf{X}))]$. Under squared loss $L(\theta, a) = (a-\theta)^2$ the risk is the MSE. If $R(\theta, \hat{\theta}_1) \le R(\theta, \hat{\theta}_2)$ for every θ, with strict inequality for at least one θ, then $\hat{\theta}_1$ *dominates* $\hat{\theta}_2$. An estimator dominated by no other estimator is called *admissible*.

Definition 7.5 (Method of Moments)**.** Let $\theta = (\theta_1, \dots, \theta_k)$ and $\mu'_j(\theta) = \mathbb{E}_\theta[X^j]$. The *method of moments* (MM) estimator is the solution in θ of the system

$$\mu'_j(\theta) = m'_j = \frac{1}{n}\sum_{i=1}^{n} X_i^j, \qquad j = 1, \dots, k.$$

7.2 Main Theorems and Proofs

Theorem 7.6 (Bias–Variance Decomposition of the MSE)**.** *If* $\mathbb{E}_\theta[\hat\theta^2] < \infty$*, then*

$$\mathrm{MSE}_\theta(\hat\theta) = \mathrm{Var}_\theta(\hat\theta) + [\mathrm{Bias}_\theta(\hat\theta)]^2.$$

Proof. Write $\hat\theta - \theta = (\hat\theta - \mathbb{E}_\theta[\hat\theta]) + (\mathbb{E}_\theta[\hat\theta] - \theta)$. The second term is a constant. Squaring and taking expectations,

$$\mathrm{MSE}_\theta(\hat\theta) = \mathbb{E}_\theta(\hat\theta - \mathbb{E}_\theta\hat\theta)^2 + 2(\mathbb{E}_\theta\hat\theta - \theta)\underbrace{\mathbb{E}_\theta[\hat\theta - \mathbb{E}_\theta\hat\theta]}_{=0} + (\mathbb{E}_\theta\hat\theta - \theta)^2,$$

i.e. the cross term vanishes: $2\mathbb{E}[(\hat\theta - \mathbb{E}[\hat\theta])(\mathbb{E}[\hat\theta] - \theta)] = 0$. □

Remark 7.7. This decomposition is the formula behind the *bias–variance trade-off* of statistics and machine learning: if accepting some bias reduces the variance by more, the MSE goes down. Shrinkage, regularisation (ridge, lasso) and Bayes estimators rest on this principle.

Theorem 7.8 (MSE-Consistency Implies Consistency)**.** *If* $\mathrm{MSE}_\theta(\hat\theta_n) \to 0$*, then* $\hat\theta_n \xrightarrow{P} \theta$*. Equivalently,* $\mathrm{Bias}_\theta(\hat\theta_n) \to 0$ *and* $\mathrm{Var}_\theta(\hat\theta_n) \to 0$ *together form a* sufficient *(but not necessary) condition for consistency.*

Proof. Markov's inequality: $\mathbb{P}_\theta(|\hat\theta_n - \theta| \ge \varepsilon) \le \mathrm{MSE}_\theta(\hat\theta_n)/\varepsilon^2 \to 0$. □

Theorem 7.9 (Unbiasedness and Consistency of the Sample Variance)**.** *Let* $X_1, \dots, X_n$ *be i.i.d. with* $\mathbb{E}X = \mu$ *and* $\mathrm{Var}\,X = \sigma^2 < \infty$*. For* $S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar X)^2$*,*

1. $\mathbb{E}[S^2] = \sigma^2$ *(unbiased), whereas* $\hat\sigma^2 = \frac{n-1}{n}S^2$ *has bias* $-\sigma^2/n$*;*
2. $S^2 \xrightarrow{a.s.} \sigma^2$ *(strongly consistent);*
3. *if* $\mu_4 = \mathbb{E}(X-\mu)^4 < \infty$*, then* $\mathrm{Var}(S^2) = \frac{1}{n}\Big(\mu_4 - \frac{n-3}{n-1}\sigma^4\Big)$*. In particular, for the normal distribution* ($\mu_4 = 3\sigma^4$)*,* $\mathrm{Var}(S^2) = \frac{2\sigma^4}{n-1}$*.*

Proof. (1) From the algebraic identity $\sum(X_i - \bar X)^2 = \sum(X_i - \mu)^2 - n(\bar X - \mu)^2$ and $\mathbb{E}(\bar X - \mu)^2 = \sigma^2/n$, $\mathbb{E}\sum(X_i - \bar X)^2 = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$. (2) $S^2 = \frac{n}{n-1}\Big[\frac{1}{n}\sum X_i^2 - \bar X^2\Big]$; by the SLLN, $\frac{1}{n}\sum X_i^2 \xrightarrow{\text{a.s.}} \sigma^2 + \mu^2$ and $\bar X \xrightarrow{\text{a.s.}} \mu$; by continuous mapping, $S^2 \xrightarrow{\text{a.s.}} \sigma^2$. (3) By expanding fourth moments (a long but direct computation); the normal case also follows from $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ in Chapter 16 and $\mathrm{Var}(\chi^2_k) = 2k$. □

Theorem 7.10 (Consistency of the Method of Moments)**.** *If the map* $\theta \mapsto \mu'(\theta) = (\mu'_1(\theta), \dots, \mu'_k(\theta))$ *is one-to-one with continuous inverse* $h = \mu'^{-1}$ *and* $\mathbb{E}|X|^k < \infty$*, then the MM estimator* $\tilde\theta_n = h(m'_1, \dots, m'_k)$ *is consistent.*

Proof. By the WLLN, $m'_j \xrightarrow{P} \mu'_j(\theta)$ for every $j \le k$; by the continuous mapping theorem (the vector version of Theorem 5.18), $\tilde\theta_n \xrightarrow{P} h(\mu'(\theta)) = \theta$. □

7.3 Worked Examples

Example 7.11 (The Best Divisor in the Family cS^2)**.** Consider the estimators $\hat\sigma^2_c = cS^2$ for a normal sample. By Theorems 7.9(3) and 7.6,

$$\mathrm{MSE}(c) = c^2\frac{2\sigma^4}{n-1} + (c-1)^2\sigma^4.$$

Setting the derivative in c to zero gives $c^* = \frac{n-1}{n+1}$; i.e. the smallest MSE is attained by $\frac{1}{n+1}\sum(X_i - \bar X)^2$. For three divisors:

Divisor	Bias	Variance	MSE
$n-1$ (S^2, unbiased)	0	$\frac{2\sigma^4}{n-1}$	$\frac{2\sigma^4}{n-1}$
n (MLE, Chapter 11)	$-\frac{\sigma^2}{n}$	$\frac{2(n-1)\sigma^4}{n^2}$	$\frac{(2n-1)\sigma^4}{n^2}$
$n+1$ (smallest MSE)	$-\frac{2\sigma^2}{n+1}$	$\frac{2(n-1)\sigma^4}{(n+1)^2}$	$\frac{2\sigma^4}{n+1}$

For $n = 5$ and $\sigma^2 = 4$ the MSE values are 8, 5.76 and 5.33 respectively. The unbiased estimator is the worst in terms of MSE!

Example 7.12 (Uniform$(0,\theta)$: Three Estimators)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,\theta)$. Since $\mathbb{E}X = \theta/2$, the MM estimator is $\tilde\theta = 2\bar X$; it is unbiased with $\text{MSE} = 4\cdot\frac{\theta^2}{12n} = \frac{\theta^2}{3n}$. $X_{(n)} = \max X_i$ has density $f(x) = nx^{n-1}/\theta^n$, $0 < x < \theta$; $\mathbb{E}X_{(n)} = \frac{n}{n+1}\theta$ and $\mathbb{E}X_{(n)}^2 = \frac{n}{n+2}\theta^2$. Hence

$$\text{MSE}(X_{(n)}) = \frac{2\theta^2}{(n+1)(n+2)}, \qquad \text{MSE}\left(\tfrac{n+1}{n}X_{(n)}\right) = \frac{\theta^2}{n(n+2)}.$$

The estimators based on the order statistic have MSE $O(1/n^2)$, while the MM estimator has $O(1/n)$. The reason will be explained in Chapter 9: $X_{(n)}$ is sufficient, $\bar X$ is not.

Example 7.13 (Method of Moments for the Gamma Distribution)**.** For $X \sim \text{Gamma}(\alpha,\beta)$ (shape α, rate β): $\mathbb{E}X = \alpha/\beta$ and $\text{Var}\,X = \alpha/\beta^2$. With $m_1' = \bar X$ and $\hat\sigma^2 = m_2' - \bar X^2$, the ratio of the two equations gives $\beta = \mathbb{E}X/\,\text{Var}\,X$:

$$\tilde\alpha = \frac{\bar X^2}{\hat\sigma^2}, \qquad \tilde\beta = \frac{\bar X}{\hat\sigma^2}.$$

It is closed-form and fast, but (as we shall see in Chapter 12) less efficient than the MLE; in practice it serves as a good starting value for the numerical optimisation of the MLE.

Example 7.14 (When No Unbiased Estimator Exists)**.** Let $X \sim \text{Bin}(n,p)$, $p \in (0,1)$, and $g(p) = 1/p$. For any T,

$$\mathbb{E}_p[T(X)] = \sum_{k=0}^{n} T(k)\binom{n}{k}p^k(1-p)^{n-k}$$

is a polynomial in p of degree at most n, which stays bounded as $p \to 0$, whereas $1/p \to \infty$. Hence $1/p$ has *no* unbiased estimator. Unbiasedness is not always an attainable goal.

Example 7.15 (Consistent but Biased; Unbiased but Inconsistent)**.** $\hat\sigma^2 = \frac{1}{n}\sum(X_i - \bar X)^2$ is biased but consistent. On the other hand, $T_n = X_1$ (the first observation only) is an unbiased but inconsistent estimator of μ: its distribution does not depend on n. Unbiasedness is a finite-sample property and consistency a large-sample property; neither implies the other.

7.4 Computational Applications

7.4.1 Comparing the MSE of S^2 and the MLE

code/ch07/c07_1_s2_vs_mle_mse.py

```
import numpy as np

rng = np.random.default_rng(42)
n, s2_true = 5, 4.0
samples = rng.normal(10.0, np.sqrt(s2_true), size=(200_000, n))
s2 = np.var(samples, axis=1, ddof=1)
```

```
mle = np.var(samples, axis=1, ddof=0)
print(f"S^2 MSE: {np.mean((s2 - s2_true)**2):.4f} | MLE MSE: {np.mean((mle - s2_true)**2):.4f}")

# Family of divisors: SS / d , d = n-1, n, n+1 -- bias, variance, MSE
SS = np.sum((samples - samples.mean(axis=1, keepdims=True))**2, axis=1)
for d in [n - 1, n, n + 1]:
    est = SS / d
    print(f"divisor={d} | bias={est.mean()-s2_true:+.4f} | var={est.var():.4f} | MSE={np.mean((est-s2_true)**2):.4f}"
          f" | theory MSE={((d-n+1)**2 + 2*(n-1)) * s2_true**2 / d**2:.4f}")
```

Output

```
S^2 MSE: 8.0765 | MLE MSE: 5.8004
divisor=4 | bias=+0.0067 | var=8.0765 | MSE=8.0765 | theory MSE=8.0000
divisor=5 | bias=-0.7946 | var=5.1689 | MSE=5.8004 | theory MSE=5.7600
divisor=6 | bias=-1.3289 | var=3.5895 | MSE=5.3554 | theory MSE=5.3333
```

Theoretical MSE formula: for $\hat{\sigma}^2_d = SS/d$, $\mathrm{MSE} = \sigma^4\left[(n-1-d)^2+2(n-1)\right]/d^2$; $d = n+1$ minimises this expression.

7.4.2 Estimators for Uniform$(0,\theta)$

code/ch07/c07_2_uniform_estimators.py

```
import numpy as np

rng = np.random.default_rng(42)
theta, R = 10.0, 100_000
print(" n   | MSE(2*Xbar)  theory   | MSE(X_max)  theory   | MSE((n+1)/n X_max) theory")
for n in [5, 20, 100]:
    X = rng.uniform(0, theta, size=(R, n))
    t1, t2 = 2*X.mean(axis=1), X.max(axis=1)
    t3 = (n + 1) / n * t2
    m = lambda t: np.mean((t - theta)**2)
    print(f"{n:4d} | {m(t1):8.4f} {theta**2/(3*n):8.4f} | {m(t2):8.4f} {2*theta**2/((n+1)*(n+2)):8.4f}"
          f" | {m(t3):8.4f} {theta**2/(n*(n+2)):8.4f}")
```

Output

```
 n   | MSE(2*Xbar)  theory   | MSE(X_max)  theory   | MSE((n+1)/n X_max) theory
   5 |   6.6706   6.6667 |   4.7873   4.7619 |   2.8688   2.8571
  20 |   1.6550   1.6667 |   0.4315   0.4329 |   0.2262   0.2273
 100 |   0.3319   0.3333 |   0.0194   0.0194 |   0.0098   0.0098
```

7.4.3 Inadmissibility: the James–Stein Estimator

For a single observation $\mathbf{X} \sim \mathcal{N}_k(\boldsymbol{\theta}, I_k)$ under total squared loss $\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}\|^2$, the natural estimator $\hat{\boldsymbol{\theta}} = \mathbf{X}$ has risk k. Stein [76] showed that for $k \geq 3$, $\mathbf{X}$ is *inadmissible*: the James–Stein estimator [40]

$$\hat{\boldsymbol{\theta}}_{JS} = \Big(1 - \frac{k-2}{\|\mathbf{X}\|^2}\Big)\mathbf{X}$$

has smaller risk for *every* $\boldsymbol{\theta}$.

code/ch07/c07_3_james_stein.py

```
import numpy as np

rng = np.random.default_rng(42)
```

```
k, R = 10, 200_000
for norm_theta in [0.0, 1.0, 3.0, 10.0]:
    theta = np.full(k, norm_theta / np.sqrt(k))
    X = theta + rng.normal(size=(R, k))
    js = (1 - (k - 2) / np.sum(X**2, axis=1, keepdims=True)) * X
    r_mle = np.mean(np.sum((X - theta)**2, axis=1))
    r_js = np.mean(np.sum((js - theta)**2, axis=1))
    print(f"||theta||={norm_theta:5.1f} | Risk(X)={r_mle:6.3f} | Risk(JS)={r_js:6.3f}")
```

Output

```
||theta||=  0.0 | Risk(X)= 9.997 | Risk(JS)= 1.998
||theta||=  1.0 | Risk(X)= 9.998 | Risk(JS)= 2.732
||theta||=  3.0 | Risk(X)= 9.995 | Risk(JS)= 5.979
||theta||= 10.0 | Risk(X)= 9.992 | Risk(JS)= 9.394
```

The gain is very large near $\boldsymbol{\theta} = \mathbf{0}$ (the risk falls from $k = 10$ to about 2) and decreases as $\|\boldsymbol{\theta}\|$ grows, but it never becomes negative. This result shows that estimating unrelated problems jointly ("borrowing strength") can pay off at the cost of unbiasedness, and it is the starting point of the empirical Bayes approach in Chapter 19.

7.5 Comments and Notes

Remark 7.16 (Why Unbiasedness Alone Is Not Enough). (i) As in Example 7.11, an unbiased estimator may be poor in terms of MSE. (ii) Unbiasedness is not preserved under transformations: S^2 is unbiased for σ^2, whereas S is biased for σ. (iii) An unbiased estimator may not exist (Example 7.14) or may be absurd (Exercise 7.2). Still, unbiasedness is valuable as a constraint that makes estimators comparable: within the class of unbiased estimators the notion of a "best" estimator (UMVUE) makes sense (Chapter 10), whereas in the class of all estimators there is in general no estimator with the smallest MSE at every θ (the constant $\hat{\theta} \equiv \theta_0$ has zero MSE at θ_0).

Remark 7.17 (The Asymptotic View). In large samples the bias is usually of order $O(1/n)$ and the standard deviation of order $O(1/\sqrt{n})$; hence the variance dominates the MSE, and asymptotic variances suffice to compare estimators. This is the rationale for the notion of asymptotic efficiency in Chapter 12.

7.6 Chapter Summary

- An estimator is a function of the data that does not depend on the parameter. Its quality is measured by bias and MSE, and the decomposition $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2$ holds (Theorem 7.6).
- $\mathrm{MSE} \to 0$ implies consistency (Theorem 7.8); the converse is false.
- S^2 is unbiased and consistent (Theorem 7.9). Yet in the normal model the divisor minimising the MSE is $n + 1$, not $n - 1$ (Example 7.11): unbiasedness alone is not an optimality criterion.
- The method of moments (Definition 7.5) is simple and consistent (Theorem 7.10), but usually not efficient.
- Loss and risk functions provide a general framework for comparing estimators (Definition 7.4). By Stein's paradox, in dimension $k \geq 3$ the natural estimator $\mathbf{X}$ is inadmissible; the James–Stein estimator beats it everywhere.

Further reading. [16, Chapter 7], [46, Chapters 1–2]; for inadmissibility and James–Stein see [46, Chapter 5].

7.7 Exercises

Exercise 7.1. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$. Show that both $\bar{X}$ and S^2 are unbiased for λ. Which has the smaller variance? (Hint: for the Poisson, $\mu_4 = \lambda + 3\lambda^2$.) → Solution

Exercise 7.2. Let $X \sim \text{Poisson}(\lambda)$ be a single observation. Show that the only unbiased estimator of $g(\lambda) = e^{-2\lambda}$ is $T(X) = (-1)^X$. Why is this estimator absurd? Compare its MSE numerically with that of $\tilde{T} = e^{-2X}$. → Solution

Exercise 7.3. Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(\theta - \frac{1}{2}, \theta + \frac{1}{2})$. Show that $\bar{X}$ and the midrange $\frac{X_{(1)} + X_{(n)}}{2}$ are unbiased, and compare their MSEs by simulation. Which converges faster? → Solution

Exercise 7.4. Derive the method of moments estimators for the $\text{Beta}(\alpha, \beta)$ distribution and compute their bias and MSE by simulation for $\alpha = 2, \beta = 5$, $n = 50$. → Solution

Chapter 8

The Cramér–Rao Lower Bound and Fisher Information

In the previous chapter we compared estimators by their MSE. A natural question is: how small can the variance be among unbiased estimators? The Cramér–Rao inequality answers with a lower bound that comes *from the model itself* and that no unbiased estimator can beat. At the heart of this bound is the Fisher information: the amount of information an observation carries about the parameter is measured by how "sharply" the log-likelihood changes with the parameter. Fisher information will reappear as the asymptotic variance of the MLE (Chapter 12), in Wald and score tests (Chapter 15) and in the Jeffreys prior (Chapter 19).

8.1 Definitions

Definition 8.1 (Regularity Conditions)**.** Let $\{f(x;\theta) : \theta \in \Theta\}$, with $\Theta \subseteq \mathbb{R}$ an open interval, be a family of densities (or pmfs). We call the following the *(Cramér) regularity conditions*:

(R1) The support $\{x : f(x;\theta) > 0\}$ does not depend on θ.

(R2) For every x, $\theta \mapsto f(x;\theta)$ is twice differentiable.

(R3) The integral $\int f(x;\theta)\,dx$ can be differentiated twice with respect to θ under the integral sign.

(R4) $0 < I_1(\theta) < \infty$ (defined below).

Definition 8.2 (Score Function)**.** For a single observation the *score* is $s(\theta; X) = \frac{\partial}{\partial\theta} \ln f(X;\theta)$; for an i.i.d. sample,

$$S_n(\theta) = \frac{\partial}{\partial\theta} \ln L_n(\theta; \mathbf{X}) = \sum_{i=1}^{n} s(\theta; X_i).$$

Definition 8.3 (Fisher Information [29])**.** The Fisher information of a single observation is

$$I_1(\theta) = \mathbb{E}_\theta\Big[\big(s(\theta;X)\big)^2\Big] = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2} \ln f(X;\theta)\right]$$

(the second equality holds under Lemma 8.6). The information of an i.i.d. sample of size n is

$$I_n(\theta) = -n\,\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2} \ln f(X;\theta)\right] = n\,I_1(\theta).$$

Definition 8.4 (Efficient Estimator)**.** An unbiased estimator of $g(\theta)$ whose variance equals the Cramér–Rao lower bound (below) for every θ is called *efficient*. The *efficiency* of an unbiased T is $e(T) = \mathrm{CRLB}/\operatorname{Var}_\theta(T) \in (0, 1]$.

Definition 8.5 (Fisher Information Matrix)**.** For $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^k$, the $k \times k$ Fisher information matrix is

$$[I_1(\boldsymbol{\theta})]_{jl} = \mathbb{E}_{\boldsymbol{\theta}}\Big[\frac{\partial \ln f}{\partial \theta_j}\frac{\partial \ln f}{\partial \theta_l}\Big] = -\mathbb{E}_{\boldsymbol{\theta}}\Big[\frac{\partial^2 \ln f}{\partial \theta_j \partial \theta_l}\Big].$$

8.2 Main Theorems and Proofs

Lemma 8.6 (Moments of the Score and the Information Identity)**.** *Under (R1)–(R3),*

$$\mathbb{E}_\theta[s(\theta; X)] = 0, \qquad \operatorname{Var}_\theta[s(\theta; X)] = I_1(\theta) = -\mathbb{E}_\theta\big[\partial_\theta^2 \ln f(X; \theta)\big].$$

Proof. Differentiate the identity $\int f(x;\theta)dx = 1$ with respect to θ (R3):

$$0 = \int \partial_\theta f \, dx = \int \frac{\partial_\theta f}{f} f \, dx = \mathbb{E}_\theta[s(\theta; X)].$$

Differentiating once more: $0 = \int \partial_\theta(s\, f)dx = \int (\partial_\theta s) f \, dx + \int s\, \partial_\theta f \, dx = \mathbb{E}[\partial_\theta s] + \mathbb{E}[s^2]$, because $\partial_\theta f = s f$. Hence $\mathbb{E}[s^2] = -\mathbb{E}[\partial_\theta^2 \ln f]$, and since $\mathbb{E}s = 0$, $\operatorname{Var}(s) = \mathbb{E}s^2$. □

Theorem 8.7 (Cramér–Rao Lower Bound — CRLB [65, 18])**.** *Assume (R1)–(R4), and let $T = T(\mathbf{X})$ be a statistic with $\mathbb{E}_\theta[T] = g(\theta)$, $\operatorname{Var}_\theta T < \infty$ and $\frac{d}{d\theta}\int T(\mathbf{x}) L(\theta; \mathbf{x}) d\mathbf{x} = \int T(\mathbf{x}) \partial_\theta L(\theta; \mathbf{x}) d\mathbf{x}$. Then*

$$\operatorname{Var}_\theta(T) \geq \frac{[g'(\theta)]^2}{n I_1(\theta)} \qquad \text{for every } \theta \in \Theta.$$

Proof. For the score $S_n = \sum s(\theta; X_i)$, Lemma 8.6 and independence give $\mathbb{E}S_n = 0$ and $\operatorname{Var}(S_n) = n I_1(\theta)$. Moreover,

$$g'(\theta) = \frac{d}{d\theta}\int T\, L \, d\mathbf{x} = \int T \, \frac{\partial_\theta L}{L} \, L \, d\mathbf{x} = \mathbb{E}_\theta[T S_n] = \operatorname{Cov}_\theta(T, S_n),$$

where the last step used $\mathbb{E}S_n = 0$. By the Cauchy–Schwarz inequality (Corollary 3.9),

$$[\operatorname{Cov}_\theta(T, S_n)]^2 \leq \operatorname{Var}_\theta(T) \operatorname{Var}_\theta(S_n), \quad \text{i.e.} \quad [g'(\theta)]^2 \leq \operatorname{Var}_\theta(T) \cdot n I_1(\theta).$$ □

Corollary 8.8 (The Equality Condition and Exponential Families)**.** *Equality in the CRLB holds if and only if, for some function $a(\theta)$,*

$$S_n(\theta) = a(\theta)\big(T(\mathbf{X}) - g(\theta)\big) \quad \text{a.s.}$$

Integrating this equation in θ gives $\ln L = A(\theta)T(\mathbf{x}) + B(\theta) + H(\mathbf{x})$*; that is, an efficient estimator exists only if the model is an* exponential family *(Definition 9.4) and the estimand is a* linear *function of the expectation of the sufficient statistic T.*

Proof. Equality in Cauchy–Schwarz is equivalent to S_n and $T - g(\theta)$ being a.s. linearly dependent (Theorem 3.8); since $\mathbb{E}S_n = 0$, there is no constant term. □

Proposition 8.9 (Reparametrisation)**.** *If $\theta = h(\eta)$ with h differentiable and one-to-one, the information in the η parametrisation is $I^{(\eta)}(\eta) = I^{(\theta)}(h(\eta))\,[h'(\eta)]^2$.*

Proof. Chain rule: $\partial_\eta \ln f = \partial_\theta \ln f \cdot h'(\eta)$; take the expectation of its square. □

Theorem 8.10 (Multiparameter CRLB)**.** *If $\boldsymbol{\theta} \in \mathbb{R}^k$, T is scalar and $\mathbb{E}_{\boldsymbol{\theta}} T = g(\boldsymbol{\theta})$, then under suitable regularity conditions*

$$\operatorname{Var}_{\boldsymbol{\theta}}(T) \geq \nabla g(\boldsymbol{\theta})^{\top} \left[n I_1(\boldsymbol{\theta}) \right]^{-1} \nabla g(\boldsymbol{\theta}).$$

In particular, the variance of an unbiased estimator of θ_1 is $\geq [I_n^{-1}]_{11} \geq 1/[I_n]_{11}$: unknown nuisance parameters reduce information *(or at best leave it unchanged).*

Proof. Let $I_n = n I_1(\boldsymbol{\theta})$ be positive definite and $\mathbf{S} = \nabla_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}; \mathbf{X})$ the score vector. Assume the regularity conditions of the one-parameter case (interchanging derivative and integral) hold for each component. Then $\mathbb{E}\mathbf{S} = \mathbf{0}$, $\operatorname{Cov}(\mathbf{S}) = I_n$ and, differentiating $\mathbb{E}T = g(\boldsymbol{\theta})$,

$$\operatorname{Cov}(T, \mathbf{S}) = \mathbb{E}[T\,\mathbf{S}] = \nabla g(\boldsymbol{\theta}).$$

For every $\mathbf{c} \in \mathbb{R}^k$,

$$0 \leq \operatorname{Var}\left(T - \mathbf{c}^{\top}\mathbf{S}\right) = \operatorname{Var} T - 2\,\mathbf{c}^{\top} \nabla g + \mathbf{c}^{\top} I_n \mathbf{c}.$$

Choosing $\mathbf{c} = I_n^{-1} \nabla g$ gives $0 \leq \operatorname{Var} T - \nabla g^{\top} I_n^{-1} \nabla g$, the first claim. ($\mathbf{c}$ is the minimising choice; it corresponds to the L^2 projection of T onto the score vector.)

For $g(\boldsymbol{\theta}) = \theta_1$, $\nabla g = \mathbf{e}_1$ and the bound is $[I_n^{-1}]_{11}$. By Cauchy–Schwarz,

$$1 = \left(\mathbf{e}_1^{\top} I_n^{-1/2}\, I_n^{1/2} \mathbf{e}_1\right)^2 \leq \left(\mathbf{e}_1^{\top} I_n^{-1} \mathbf{e}_1\right)\left(\mathbf{e}_1^{\top} I_n \mathbf{e}_1\right) = [I_n^{-1}]_{11}\,[I_n]_{11},$$

i.e. $[I_n^{-1}]_{11} \geq 1/[I_n]_{11}$. The block inverse formula $[I_n^{-1}]_{11} = \left(I_{11} - I_{12} I_{22}^{-1} I_{21}\right)^{-1}$ (Schur complement) shows that equality holds if and only if $I_{12} = \mathbf{0}$, i.e. when the score for θ_1 is uncorrelated with the other scores. □

8.3 Worked Examples

Example 8.11 (Poisson)**.** $\ln f(x;\lambda) = -\lambda + x \ln \lambda - \ln x!$, $s = -1 + x/\lambda$, $\partial_\lambda^2 \ln f = -x/\lambda^2$. Hence $I_1(\lambda) = \mathbb{E}X/\lambda^2 = 1/\lambda$ and the CRLB is λ/n. Since $\operatorname{Var}(\bar{X}) = \lambda/n$, $\bar{X}$ is efficient; indeed $S_n = \frac{n}{\lambda}(\bar{X} - \lambda)$ has the form of Corollary 8.8.

Example 8.12 (Bernoulli and the Normal Mean)**.** Bernoulli(p): $I_1(p) = \frac{1}{p(1-p)}$; $\hat{p} = \bar{X}$ is efficient. The information is smallest at $p = 1/2$: when a coin toss is most "uncertain", a single observation carries the least information. $\mathcal{N}(\mu, \sigma^2)$ with σ^2 known: $s = (x-\mu)/\sigma^2$, $I_1(\mu) = 1/\sigma^2$; $\bar{X}$ is efficient.

Example 8.13 (Normal: Two Parameters and an Inefficient UMVUE)**.** For $\boldsymbol{\theta} = (\mu, \sigma^2)$,

$$\ln f = -\tfrac{1}{2}\ln(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}, \qquad I_1(\mu, \sigma^2) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 1/(2\sigma^4) \end{pmatrix}.$$

The CRLB for unbiased estimators of σ^2 is $2\sigma^4/n$, whereas $\operatorname{Var}(S^2) = 2\sigma^4/(n-1)$. So S^2 is not efficient (efficiency $(n-1)/n$). Nevertheless, in Chapter 10 we shall see that S^2 is the UMVUE: *the CRLB is not always attainable.* (The diagonal form of the matrix shows that not knowing μ does not reduce the information about σ^2; this is the case of "orthogonal parameters".)

Example 8.14 (Exponential: λ Cannot Be Estimated Efficiently)**.** $f(x;\lambda) = \lambda e^{-\lambda x}$ and $I_1(\lambda) = 1/\lambda^2$. For the mean $g(\lambda) = 1/\lambda$, $\bar{X}$ is unbiased with $\operatorname{Var}\bar{X} = \frac{1}{n\lambda^2} = \frac{[g'(\lambda)]^2}{n I_1} = \frac{\lambda^{-4}}{n\lambda^{-2}}$; i.e. it is efficient. But for λ itself, with $T = \sum X_i \sim \text{Gamma}(n, \lambda)$, $\mathbb{E}[1/T] = \frac{\lambda}{n-1}$ and $\mathbb{E}[1/T^2] = \frac{\lambda^2}{(n-1)(n-2)}$, the estimator $\tilde{\lambda} = \frac{n-1}{T}$ is unbiased and

$$\operatorname{Var}(\tilde{\lambda}) = \frac{\lambda^2}{n-2} > \frac{\lambda^2}{n} = \text{CRLB}.$$

Since λ is not a linear function of $\mathbb{E}T = n/\lambda$, by Corollary 8.8 no efficient estimator exists; but the efficiency $(n-2)/n \to 1$.

Example 8.15 (Violated Regularity: Uniform$(0,\theta)$)**.** The support depends on θ; (R1) is violated. Applying the "formula" anyway gives $\partial_\theta \ln(1/\theta) = -1/\theta$, $I_1 = 1/\theta^2$ and a pseudo-bound θ^2/n. Yet $\frac{n+1}{n}X_{(n)}$ is unbiased with variance $\frac{\theta^2}{n(n+2)}$ — $n+2$ times smaller than the pseudo-bound! Moreover $\mathbb{E}[s] = -1/\theta \neq 0$: Lemma 8.6 also breaks down. The regularity conditions are not decoration.

Example 8.16 (The Cauchy Location Family)**.** For $f(x;\theta) = \frac{1}{\pi(1+(x-\theta)^2)}$, $I_1(\theta) = \frac{1}{2}$. Hence the variance of any unbiased estimator is $\geq 2/n$. The sample mean has infinite variance (indeed $\bar{X}$ is not even consistent); the sample median has asymptotic variance $\frac{\pi^2}{4n} \approx \frac{2.47}{n}$, and the MLE $\frac{2}{n}$ (Chapter 12).

8.4 Computational Applications

8.4.1 Poisson: $\bar{X}$ Attains the CRLB

code/ch08/c08_1_poisson_attains_crlb.py

```python
import numpy as np

rng = np.random.default_rng(42)
lam = 3.0
for n in [5, 15, 30, 100]:
    samples = rng.poisson(lam=lam, size=(150_000, n))
    var_x = np.var(np.mean(samples, axis=1), ddof=1)
    print(f"n = {n:3d} | CRLB: {lam/n:.6f} | Empirical Var: {var_x:.6f}")
```

Output

```
n =   5 | CRLB: 0.600000 | Empirical Var: 0.597129
n =  15 | CRLB: 0.200000 | Empirical Var: 0.199763
n =  30 | CRLB: 0.100000 | Empirical Var: 0.099790
n = 100 | CRLB: 0.030000 | Empirical Var: 0.030091
```

8.4.2 Moments of the Score: Checking Lemma 8.6

code/ch08/c08_2_score_moments.py

```python
import numpy as np

rng = np.random.default_rng(42)
N = 2_000_000
# Gamma(known shape a, rate b): ln f = a ln b - ln G(a) + (a-1) ln x - b x
a, b = 3.0, 2.0
X = rng.gamma(a, 1/b, size=N)
score = a / b - X              # d/db ln f
hess = -a / b**2 * np.ones(N) # d2/db2 ln f
print(f"Gamma : E[s]={score.mean():+.4f} | Var(s)={score.var():.4f} | -E[s']={-hess.mean():.4
    f} | theory I={a/b**2:.4f}")

# Cauchy location: s = 2(x-t)/(1+(x-t)^2), I = 1/2
Y = rng.standard_cauchy(N)
s = 2*Y / (1 + Y**2)
h = (2*Y**2 - 2) / (1 + Y**2)**2     # d/dt s
print(f"Cauchy: E[s]={s.mean():+.4f} | Var(s)={s.var():.4f} | -E[s']={-h.mean():.4f} | theory
     I=0.5000")

# Uniform(0, theta): regularity violated, E[s] != 0
theta = 2.0
print(f"Uniform: E[s] = -1/theta = {-1/theta:.3f}  (not zero!)")
```

Output

```
Gamma : E[s]=-0.0009 | Var(s)=0.7509 | -E[s']=0.7500 | theory I=0.7500
Cauchy: E[s]=+0.0007 | Var(s)=0.5001 | -E[s']=0.5005 | theory I=0.5000
Uniform: E[s] = -1/theta = -0.500  (not zero!)
```

8.4.3 When the CRLB Is Not Attained and When It Is Violated

code/ch08/c08_3_crlb_not_attained_or_violated.py

```python
import numpy as np

rng = np.random.default_rng(42)
R = 300_000
lam = 2.0
print("Exp(lambda): lambda_tilde = (n-1)/sum X")
for n in [5, 20, 100]:
    T = rng.exponential(1/lam, size=(R, n)).sum(axis=1)
    lt = (n - 1) / T
    print(f"  n={n:3d} | mean={lt.mean():.4f} | Var={lt.var():.5f} | theory {lam**2/(n-2):.5f} | CRLB {lam**2/n:.5f}")

theta = 1.0
print("Uniform(0,theta): (n+1)/n * max  vs  'pseudo-CRLB' theta^2/n")
for n in [5, 20, 100]:
    t = (n + 1) / n * rng.uniform(0, theta, size=(R, n)).max(axis=1)
    print(f"  n={n:3d} | Var={t.var():.6f} | theory {theta**2/(n*(n+2)):.6f} | pseudo-CRLB {theta**2/n:.6f}")
```

Output

```
Exp(lambda): lambda_tilde = (n-1)/sum X
  n=  5 | mean=2.0012 | Var=1.33392 | theory 1.33333 | CRLB 0.80000
  n= 20 | mean=1.9998 | Var=0.22107 | theory 0.22222 | CRLB 0.20000
  n=100 | mean=2.0003 | Var=0.04089 | theory 0.04082 | CRLB 0.04000
Uniform(0,theta): (n+1)/n * max  vs  'pseudo-CRLB' theta^2/n
  n=  5 | Var=0.028442 | theory 0.028571 | pseudo-CRLB 0.200000
  n= 20 | Var=0.002279 | theory 0.002273 | pseudo-CRLB 0.050000
  n=100 | Var=0.000099 | theory 0.000098 | pseudo-CRLB 0.010000
```

8.5 Comments and Notes

Remark 8.17 (A Geometric Interpretation of Fisher Information). $-\partial_\theta^2 \ln L$ is the curvature of the log-likelihood curve at its peak. A sharp peak means that the parameter is well determined by the data; a flat peak expresses uncertainty. The observed information $J_n(\hat\theta) = -\partial_\theta^2 \ell(\hat\theta)$ is the data-based counterpart of the expected information $I_n(\theta)$ and is often preferred in practice for computing standard errors [23].

Remark 8.18 (Information Is Additive). The information of independent observations adds up: $I_n = nI_1$. This is the source of the $1/n$ rate of the CRLB and of the rule "standard error $\propto 1/\sqrt{n}$". Information is also the local curvature of the Kullback–Leibler divergence between $f(\cdot;\theta)$ and $f(\cdot;\theta+\delta)$: $\mathrm{KL}(\theta \,\|\, \theta+\delta) \approx \frac{1}{2} I_1(\theta)\delta^2$.

Remark 8.19 (Limits of the CRLB). The CRLB is a bound for *unbiased* estimators only. For biased estimators the general form is $\mathrm{Var}_\theta T \geq [1 + b'(\theta)]^2 / I_n(\theta)$ (b: the bias function); hence a biased estimator can go below the bound in MSE (James–Stein, shrinkage estimators). Furthermore, when regularity fails as in Example 8.15, convergence faster than $1/n$ is possible ("superefficiency").

8.6 Chapter Summary

- The score function has mean 0 and variance equal to the Fisher information. The information identity reads $I(\theta) = -\mathbb{E}\,\partial_\theta^2 \ln f$ (Lemma 8.6).
- Cramér–Rao lower bound (Theorem 8.7): under regularity conditions, every T with $\mathbb{E}T = g(\theta)$ satisfies $\operatorname{Var} T \geq g'(\theta)^2/(nI_1(\theta))$.
- The bound is attained only in exponential families and only by linear functions of the natural sufficient statistic (Corollary 8.8). Hence in most problems the CRLB is not attainable.
- Under reparametrisation, Fisher information scales with the square of the derivative (Proposition 8.9). In the multiparameter case the bound becomes $\nabla g^\top I_n^{-1} \nabla g$; unknown nuisance parameters raise the bound (Theorem 8.10).
- In models whose support depends on the parameter, regularity fails and the bound is invalid (Example 8.15).

Further reading. [16, Section 7.3], [46, Section 2.5], [70, Chapter 3].

8.7 Exercises

Exercise 8.1. Compute $I_1(p)$ for the Geometric(p) distribution ($\mathbb{P}(X = k) = p(1-p)^{k-1}$, $k \geq 1$). Does $1/p$ have an efficient estimator? → Solution

Exercise 8.2. In $\mathcal{N}(\mu, \sigma^2)$, find the Fisher information for σ (the standard deviation, not the variance) using Proposition 8.9, and compare the variance of $c_4^{-1} S$ (Chapter 3) with the CRLB by simulation. → Solution

Exercise 8.3. Find the 2×2 Fisher information matrix of Gamma(α, β) (shape–rate) in terms of $\psi'(\alpha)$ (the trigamma function). For $\alpha = 2$, compute how much larger the CRLB for α is when β is unknown than when β is known. → Solution

Exercise 8.4. Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, 1)$ and $g(\theta) = \theta^2$. Show that $\bar{X}^2 - 1/n$ is unbiased with variance $\frac{4\theta^2}{n} + \frac{2}{n^2}$. What is the CRLB? Why is the difference $O(1/n^2)$? → Solution

Chapter 9

Sufficiency and the Rao–Blackwell Theorem

A sample of n observations contains many details that carry no information about θ (the order of the observations, a particular rearrangement, and so on). A sufficient statistic is a summary that compresses the sample while retaining *all* the information about θ: once it is known, the rest of the raw data says nothing about θ. The Rao–Blackwell theorem links this idea to estimation theory: any estimator can be improved, *never made worse*, by conditioning on a sufficient statistic. This chapter is the first half of the UMVUE theory of Chapter 10.

9.1 Definitions

Definition 9.1 (Sufficient Statistic)**.** Let $T = T(\mathbf{X})$ be a statistic. If the conditional distribution of $\mathbf{X}$ given T does not depend on θ, i.e. if $\mathbb{P}_\theta(\mathbf{X} \in A \mid T = t)$ is free of θ for every A and every t, then T is called *sufficient* for θ.

Remark 9.2 (Interpretation). If T is sufficient, a statistician who knows $T = t$ can, without knowing θ, generate an $\mathbf{X}^*$ from the known conditional distribution; $\mathbf{X}^*$ has the same distribution as $\mathbf{X}$. Hence every inference that can be made with $\mathbf{X}$ can be made with the same performance using T alone. This is the rationale for the *sufficiency principle*.

Definition 9.3 (Minimal Sufficient Statistic)**.** T is called *minimal sufficient* if it is sufficient and, for every other sufficient statistic U, there is a function h with $T = h(U)$. A minimal sufficient statistic is the greatest data compression achievable without loss of information.

Definition 9.4 (Exponential Family)**.** A family of densities (or pmfs) is called a *k-parameter exponential family* if it can be written as

$$f(x;\boldsymbol{\theta}) = h(x)\, c(\boldsymbol{\theta}) \exp\Big\{ \sum_{j=1}^{k} w_j(\boldsymbol{\theta})\, t_j(x) \Big\}.$$

The form $f(x;\boldsymbol{\eta}) = h(x)\exp\{\sum \eta_j t_j(x) - A(\boldsymbol{\eta})\}$ obtained through $\eta_j = w_j(\boldsymbol{\theta})$ is called the *natural (canonical) form*; $\boldsymbol{\eta}$ is the natural parameter and A the log-partition function.

9.2 Main Theorems and Proofs

Theorem 9.5 (Neyman–Fisher Factorisation [34])**.** *$T(\mathbf{X})$ is sufficient for θ $\iff$ there exist non-negative functions g and h such that for every $\mathbf{x}$ and θ*

$$L_n(\theta;\mathbf{x}) = g(T(\mathbf{x}),\theta)\, h(\mathbf{x}).$$

Proof (discrete case). ($\Rightarrow$) Let T be sufficient and $t = T(\mathbf{x})$. Then $L(\theta;\mathbf{x}) = \mathbb{P}_\theta(\mathbf{X}=\mathbf{x}) = \mathbb{P}_\theta(\mathbf{X}=\mathbf{x}, T=t) = \mathbb{P}_\theta(T=t)\,\mathbb{P}(\mathbf{X}=\mathbf{x}\mid T=t)$. The second factor is free of θ by sufficiency; take $g(t,\theta)=\mathbb{P}_\theta(T=t)$ and $h(\mathbf{x})=\mathbb{P}(\mathbf{X}=\mathbf{x}\mid T=T(\mathbf{x}))$.

($\Leftarrow$) Suppose the factorisation holds. For $\mathbf{x}$ with $T(\mathbf{x})=t$,

$$\mathbb{P}_\theta(\mathbf{X}=\mathbf{x}\mid T=t) = \frac{\mathbb{P}_\theta(\mathbf{X}=\mathbf{x})}{\mathbb{P}_\theta(T=t)} = \frac{g(t,\theta)h(\mathbf{x})}{\sum_{\mathbf{y}:T(\mathbf{y})=t} g(t,\theta)h(\mathbf{y})} = \frac{h(\mathbf{x})}{\sum_{\mathbf{y}:T(\mathbf{y})=t} h(\mathbf{y})},$$

which is free of θ. (If $T(\mathbf{x})\neq t$, the conditional probability is zero anyway.) In the continuous case the proof requires measure-theoretic tools (Halmos–Savage). □

Theorem 9.6 (Criterion for Minimal Sufficiency [48])**.** *If for every pair of sample points* $\mathbf{x},\mathbf{y}$

$$\textit{the ratio } \frac{L(\theta;\mathbf{x})}{L(\theta;\mathbf{y})} \textit{ is free of } \theta \iff T(\mathbf{x})=T(\mathbf{y}),$$

then T is minimal sufficient.

Proof sketch. Sufficiency: For each value t choose a representative $\mathbf{x}_t$ with $T(\mathbf{x}_t)=t$. Since $T(\mathbf{x}) = T(\mathbf{x}_{T(\mathbf{x})})$, the ratio $h(\mathbf{x}) = L(\theta;\mathbf{x})/L(\theta;\mathbf{x}_{T(\mathbf{x})})$ is free of θ; with $g(t,\theta)=L(\theta;\mathbf{x}_t)$ we obtain a factorisation. *Minimality:* Let U be another sufficient statistic with $U(\mathbf{x})=U(\mathbf{y})$. By factorisation, $L(\theta;\mathbf{x})/L(\theta;\mathbf{y}) = h_U(\mathbf{x})/h_U(\mathbf{y})$ is free of θ; by assumption $T(\mathbf{x})=T(\mathbf{y})$. So T is a function of U. (This proof is given for discrete or density models, ignoring "almost sure" details; see [16, Theorem 6.2.13], and for the general case [46, Section 1.6].) □

Theorem 9.7 (Sufficiency in Exponential Families)**.** *If $X_1,\dots,X_n$ are i.i.d. from the k-parameter exponential family of Definition 9.4, then*

$$\mathbf{T}(\mathbf{X}) = \Big(\sum_{i=1}^n t_1(X_i),\dots,\sum_{i=1}^n t_k(X_i)\Big)$$

is sufficient. If the set $\{(w_1(\boldsymbol\theta),\dots,w_k(\boldsymbol\theta)) : \boldsymbol\theta\in\Theta\}$ contains a non-empty open set in $\mathbb{R}^k$ ("full rank"), $\mathbf{T}$ is minimal sufficient.

Proof. $L = \prod h(x_i)\cdot c(\boldsymbol\theta)^n \exp\{\sum_j w_j(\boldsymbol\theta)\sum_i t_j(x_i)\}$; the factorisation can be read off directly. For minimality use Theorem 9.6: $L(\mathbf{x})/L(\mathbf{y}) = \frac{\prod h(x_i)}{\prod h(y_i)}\exp\{\sum_j w_j(\boldsymbol\theta)[T_j(\mathbf{x})-T_j(\mathbf{y})]\}$; under the open-set condition this is free of θ if and only if $\mathbf{T}(\mathbf{x})=\mathbf{T}(\mathbf{y})$. □

Theorem 9.8 (Rao–Blackwell Theorem [65, 10])**.** *Let W be an unbiased estimator of $g(\theta)$ ($\mathbb{E}_\theta[W]=g(\theta)$, $\operatorname{Var}_\theta W<\infty$) and T a sufficient statistic. Define $W^*=\mathbb{E}[W\mid T]$. Then for every θ:*

1. *W^* is a statistic (i.e. it does not depend on θ);*
2. *$\mathbb{E}_\theta[W^*]=g(\theta)$;*
3. *$\operatorname{Var}_\theta(W^*)\le\operatorname{Var}_\theta(W)$, with equality only if W is already a function of T ($W=W^*$ a.s.).*

Proof. (1) Since T is sufficient, the conditional distribution of $\mathbf{X}\mid T$, and hence of $W(\mathbf{X})\mid T$, does not depend on θ; so $\mathbb{E}[W\mid T]$ is a known function of T alone. *This is where sufficiency is crucial*: conditioning on a non-sufficient T produces an "estimator" that depends on θ and cannot be used. (2) Tower rule (Theorem 3.11): $\mathbb{E}_\theta[W^*]=\mathbb{E}_\theta[\mathbb{E}(W\mid T)]=\mathbb{E}_\theta W=g(\theta)$. (3) Law of total variance (Theorem 3.12):

$$\operatorname{Var}(W)=\operatorname{Var}(\mathbb{E}[W\mid T])+\mathbb{E}[\operatorname{Var}(W\mid T)]=\operatorname{Var}(W^*)+\mathbb{E}[\operatorname{Var}(W\mid T)]\ge\operatorname{Var}(W^*).$$

Equality holds when $\mathbb{E}[\operatorname{Var}(W\mid T)]=0$, i.e. $W=\mathbb{E}[W\mid T]$ a.s. □

Remark 9.9 (Generalisation). Even without unbiasedness, the conditional Jensen inequality gives $R(\theta, W^*) \le R(\theta, W)$ for every convex loss $L(\theta, \cdot)$. So under convex loss it always suffices to restrict attention to estimators that are functions of the sufficient statistic.

9.3 Worked Examples

Example 9.10 (Bernoulli: Sufficiency from the Definition)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $T = \sum X_i \sim \text{Bin}(n, p)$. For every $\mathbf{x}$ with $\sum x_i = t$,

$$\mathbb{P}_p(\mathbf{X} = \mathbf{x} \mid T = t) = \frac{p^t(1-p)^{n-t}}{\binom{n}{t} p^t (1-p)^{n-t}} = \frac{1}{\binom{n}{t}},$$

which is free of p: given the number of successes, *which* trials were successes carries no information about p.

Example 9.11 (Normal: $(\bar{X}, S^2)$)**.** For $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, the identity $\sum (x_i - \mu)^2 = \sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$ gives

$$L(\mu, \sigma^2; \mathbf{x}) = (2\pi\sigma^2)^{-n/2} \exp\Big\{ - \frac{(n-1)s^2 + n(\bar{x} - \mu)^2}{2\sigma^2} \Big\}.$$

With $h(\mathbf{x}) = 1$, factorisation shows that $(\bar{X}, S^2)$ is sufficient; since this is a two-parameter exponential family, it is minimal sufficient. If σ^2 is known, $\bar{X}$ alone is sufficient.

Example 9.12 (Uniform$(0, \theta)$: the Importance of the Indicator)**.**

$$L(\theta; \mathbf{x}) = \prod_{i=1}^{n} \frac{1}{\theta} \mathbb{I}\{0 < x_i < \theta\} = \theta^{-n}\, \mathbb{I}\{x_{(n)} < \theta\} \cdot \mathbb{I}\{x_{(1)} > 0\}.$$

With $g(t, \theta) = \theta^{-n} \mathbb{I}\{t < \theta\}$ and $h(\mathbf{x}) = \mathbb{I}\{x_{(1)} > 0\}$, $T = X_{(n)}$ is sufficient. Forgetting the indicators and writing $L = \theta^{-n}$ is a common mistake that leads to wrong conclusions such as "nothing is sufficient".

Example 9.13 (Members and Non-Members of the Exponential Family)**.**

Distribution	Natural parameter η	$t(x)$	Sufficient statistic
Bin(m, p)	$\ln \frac{p}{1-p}$	x	$\sum X_i$
Poisson(λ)	$\ln \lambda$	x	$\sum X_i$
Exp(λ)	$-\lambda$	x	$\sum X_i$
$\mathcal{N}(\mu, \sigma^2)$	$(\mu/\sigma^2,\ -1/(2\sigma^2))$	(x, x^2)	$(\sum X_i, \sum X_i^2)$
Gamma(α, β)	$(\alpha - 1,\ -\beta)$	$(\ln x, x)$	$(\sum \ln X_i, \sum X_i)$
Beta(a, b)	$(a - 1,\ b - 1)$	$(\ln x, \ln(1-x))$	$(\sum \ln X_i, \sum \ln(1 - X_i))$

Distributions whose support depends on the parameter (Uniform$(0, \theta)$, the shifted exponential) and location families such as the Cauchy, t and logistic are not exponential families. For example, in the Cauchy location family the minimal sufficient statistic is the full set of order statistics $(X_{(1)}, \ldots, X_{(n)})$: no dimension reduction is possible.

Example 9.14 (Rao–Blackwell: Estimating $e^{-\lambda}$ for the Poisson)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$ and $g(\lambda) = \mathbb{P}(X = 0) = e^{-\lambda}$ (for example, the probability that no customer arrives in an hour). A crude unbiased estimator is $W = \mathbb{I}\{X_1 = 0\}$. $T = \sum X_i$ is sufficient, and $X_1 \mid T = t \sim \text{Bin}(t, 1/n)$ (the conditional distribution of independent Poissons). Hence

$$W^* = \mathbb{E}[W \mid T] = \mathbb{P}(X_1 = 0 \mid T) = \Big(1 - \frac{1}{n}\Big)^T.$$

Using the probability generating function of $T \sim \text{Poisson}(n\lambda)$, $\mathbb{E}[(1-1/n)^T] = e^{n\lambda(-1/n)} = e^{-\lambda}$ (unbiased), and

$$\text{Var}(W^*) = e^{-2\lambda}(e^{\lambda/n} - 1) \quad \ll \quad \text{Var}(W) = e^{-\lambda}(1 - e^{-\lambda}).$$

The intuitive "plug-in" estimator $e^{-\bar{X}}$ is biased, whereas W^* is unbiased and will be shown to be the UMVUE in Chapter 10.

Example 9.15 (Rao–Blackwell: Uniform$(0,\theta)$)**.** $W = 2X_1$ is unbiased. Given $X_{(n)} = m$, X_1 equals m with probability $1/n$ and is otherwise Uniform$(0, m)$. Hence

$$\mathbb{E}[2X_1 \mid X_{(n)} = m] = 2\Big[\frac{m}{n} + \frac{n-1}{n} \cdot \frac{m}{2}\Big] = \frac{n+1}{n} m.$$

The Rao–Blackwell procedure mechanically produces the "intuitive" estimator $\frac{n+1}{n} X_{(n)}$ of Chapter 7.

9.4 Computational Applications

9.4.1 Bernoulli: $X_1 \to \bar{X}$

code/ch09/c09_1_bernoulli.py

```
import numpy as np

rng = np.random.default_rng(42)
samples = rng.binomial(1, 0.4, size=(200_000, 10))
print(f"Var(W=X_1): {np.var(samples[:,0], ddof=1):.4f} | Var(W*=X_bar): {np.var(np.mean(
    samples, axis=1), ddof=1):.4f}")
print(f"Theory    : {0.4*0.6:.4f}                | {0.4*0.6/10:.4f}")
```

Output

```
Var(W=X_1): 0.2400 | Var(W*=X_bar): 0.0240
Theory    : 0.2400                | 0.0240
```

$\mathbb{E}[X_1 \mid \sum X_i = t] = t/n$ (by symmetry), i.e. $W^* = \bar{X}$ and the variance falls by a factor of n.

9.4.2 Poisson: Three Estimators of $\mathbb{P}(X = 0)$

code/ch09/c09_2_poisson_three_estimators.py

```
import numpy as np

rng = np.random.default_rng(42)
lam, n, R = 2.0, 10, 400_000
X = rng.poisson(lam, size=(R, n))
T = X.sum(axis=1)
target = np.exp(-lam)
estimators = {"W  = I(X1=0)        ": (X[:, 0] == 0).astype(float),
              "W* = (1-1/n)^T (RB) ": (1 - 1/n)**T,
              "exp(-Xbar) (plug-in)": np.exp(-T/n)}
print(f"Target e^-lambda = {target:.5f} | CRLB = {np.exp(-2*lam)*lam/n:.3e}")
for name, t in estimators.items():
    print(f"{name} | mean={t.mean():.5f} | bias={t.mean()-target:+.5f} | Var={t.var():.3e} |
    MSE={np.mean((t-target)**2):.3e}")
print(f"Theory Var(W*) = {np.exp(-2*lam)*(np.exp(lam/n)-1):.3e}")
```

Output

```
Target e^-lambda = 0.13534 | CRLB = 3.663e-03
W  = I(X1=0)          | mean=0.13553 | bias=+0.00020 | Var=1.172e-01 | MSE=1.172e-01
W* = (1-1/n)^T (RB)   | mean=0.13539 | bias=+0.00005 | Var=4.042e-03 | MSE=4.042e-03
exp(-Xbar) (plug-in) | mean=0.14914 | bias=+0.01380 | Var=4.399e-03 | MSE=4.590e-03
Theory Var(W*) = 4.055e-03
```

The Rao–Blackwell estimator reduces the variance by a factor of about 29. Its variance is slightly above the CRLB: since $e^{-\lambda}$ is not a linear function of $\mathbb{E}T$, no efficient estimator exists (Corollary 8.8). The intuitive plug-in estimator $e^{-\bar{X}}$ is biased upwards by Jensen's inequality (e^{-x} is convex), and in this example both its variance and its MSE exceed those of W^*: here the Rao–Blackwell procedure beats intuition not only in unbiasedness but also in MSE. (This need not always be so; in Example 7.11 the biased estimator was better.)

9.4.3 Uniform$(0, \theta)$: Rao–Blackwell by Simulation

code/ch09/c09_3_rao_blackwell_simulation.py

```python
import numpy as np

rng = np.random.default_rng(42)
theta, n, R = 5.0, 8, 300_000
X = rng.uniform(0, theta, size=(R, n))
W = 2 * X[:, 0]
M = X.max(axis=1)
W_star = (n + 1) / n * M
# Check the function E[W | M] empirically: split M into slices
bins = np.quantile(M, np.linspace(0, 1, 6))
idx = np.digitize(M, bins[1:-1])
for k in range(5):
    sel = idx == k
    print(f"M slice {k}: mean(W | M) = {W[sel].mean():.3f}  vs  (n+1)/n * mean(M) = {(n+1)/n*M[sel].mean():.3f}")
print(f"Var(W) = {W.var():.4f} (theory {theta**2/3:.4f}) | Var(W*) = {W_star.var():.4f} (theory {theta**2/(n*(n+2)):.4f})")
```

Output

```
M slice 0: mean(W | M) = 4.078  vs  (n+1)/n * mean(M) = 4.089
M slice 1: mean(W | M) = 4.822  vs  (n+1)/n * mean(M) = 4.827
M slice 2: mean(W | M) = 5.143  vs  (n+1)/n * mean(M) = 5.153
M slice 3: mean(W | M) = 5.381  vs  (n+1)/n * mean(M) = 5.377
M slice 4: mean(W | M) = 5.553  vs  (n+1)/n * mean(M) = 5.550
Var(W) = 8.3277 (theory 8.3333) | Var(W*) = 0.3123 (theory 0.3125)
```

9.5 Comments and Notes

Remark 9.16 (Sufficiency Depends on the Model). $\bar{X}$ is sufficient for μ in the normal model (σ^2 known), but not in the Laplace or Cauchy location models. If the model is wrong, reducing to the "sufficient" statistic can lose information; that is why the raw data are always needed for model checking (residual analysis).

Remark 9.17 (The Pitman–Koopman–Darmois Theorem). Among regular families whose support does not depend on the parameter, only exponential families admit a sufficient statistic whose dimension does not grow with n. That is, "lossless compression of the data into a fixed-dimensional summary" is essentially a property of exponential families. This explains their central place in statistics (GLMs, conjugate priors).

Remark 9.18 (Rao–Blackwell in Practice). The Rao–Blackwell principle is also used in Monte Carlo: when estimating an expectation by simulation, integrating out analytically the part whose conditional expectation can be computed reduces the variance ("Rao–Blackwellised" MCMC estimators, conditional Monte Carlo).

9.6 Chapter Summary

- If T is sufficient, the conditional distribution of the data given T is free of θ (Definition 9.1). T carries all the information in the data about θ.
- The factorisation theorem is the practical criterion for sufficiency: $f(\mathbf{x};\theta) = g(T(\mathbf{x});\theta)h(\mathbf{x})$ (Theorem 9.5).
- A minimal sufficient statistic is the coarsest sufficient reduction. It is characterised by likelihood ratios being free of θ (Theorem 9.6).
- In an exponential family, $\left(\sum_i t_1(X_i), \dots, \sum_i t_k(X_i)\right)$ is sufficient (Theorem 9.7); its dimension does not depend on n.
- Rao–Blackwell theorem (Theorem 9.8): conditioning any estimator on a sufficient statistic preserves its bias and does not increase its MSE. Good estimators should be sought among functions of the sufficient statistic.

Further reading. [16, Chapter 6], [46, Section 1.6], [70, Chapter 2].

9.7 Exercises

Exercise 9.1. Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Gamma}(\alpha, \beta)$ with both parameters unknown. Find a minimal sufficient statistic. What changes if α is known? → Solution

Exercise 9.2. For $X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(\theta, \theta+1)$, show using Theorem 9.6 that $(X_{(1)}, X_{(n)})$ is minimal sufficient. What does the existence of a two-dimensional minimal sufficient statistic in a one-parameter model indicate? (See Chapter 10.) → Solution

Exercise 9.3. Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $g(p) = p^2$. $W = X_1 X_2$ is unbiased. Compute $W^* = \mathbb{E}[W \mid T]$ ($T = \sum X_i$) and show that $W^* = \frac{T(T-1)}{n(n-1)}$. → Solution

Exercise 9.4. Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, 1)$ and $g(\mu) = \mathbb{P}(X \le c) = \Phi(c - \mu)$. Rao–Blackwellise $W = \mathbb{I}\{X_1 \le c\}$ with respect to $\bar{X}$. (Hint: $X_1 \mid \bar{X} \sim \mathcal{N}(\bar{X}, 1 - 1/n)$.) Compare the result with the plug-in estimator $\Phi(c - \bar{X})$ by simulation. → Solution

Chapter 10

Completeness, Lehmann–Scheffé and the UMVUE

The Rao–Blackwell theorem lets us improve any unbiased estimator by conditioning on a sufficient statistic. Two questions remain open, however: (i) can different starting estimators lead to different results? (ii) is the resulting estimator really the "best"? The notion of completeness answers both at once: if the sufficient statistic is also complete, the unbiased estimator that is a function of it is *unique* and has uniformly minimum variance among all unbiased estimators. In this chapter we study this summit of point estimation theory and Basu's theorem, an unexpected by-product of completeness.

10.1 Definitions

Definition 10.1 (Complete Statistic)**.** Let $\{\mathbb{P}_\theta^T : \theta \in \Theta\}$ be the family of distributions of a statistic T. If for every measurable g

$$\mathbb{E}_\theta[g(T)] = 0 \quad \forall \theta \in \Theta \implies \mathbb{P}_\theta(g(T) = 0) = 1 \quad \forall \theta \in \Theta,$$

then T (or its family) is called *complete*. If the condition is required only for bounded g, it is called *boundedly complete*.

Remark 10.2 (Interpretation). Completeness says that the family of distributions of T is "rich enough": as θ varies, the distributions of T are so varied that no non-zero $g(T)$ can have mean zero for every θ. In linear-algebra terms, the densities $\{f_T(\cdot;\theta)\}_\theta$ "span" the function space; the only function orthogonal to all of them is zero.

Definition 10.3 (Ancillary Statistic)**.** A statistic $A(\mathbf{X})$ whose distribution does not depend on θ is called *ancillary*. An ancillary statistic by itself carries no information about θ.

Definition 10.4 (UMVUE)**.** An estimator W° of $g(\theta)$ is called a *uniformly minimum variance unbiased estimator* (UMVUE) if (i) it is unbiased and (ii) $\mathrm{Var}_\theta(W^\circ) \le \mathrm{Var}_\theta(W)$ for every unbiased W and *every* $\theta \in \Theta$.

10.2 Main Theorems and Proofs

Theorem 10.5 (Lehmann–Scheffé Theorem [48])**.** *Let T be a complete sufficient statistic. If there is a $\phi(T)$ with $\mathbb{E}_\theta[\phi(T)] = g(\theta)$, then $\phi(T)$ is the* unique *UMVUE of $g(\theta)$.*

Proof. Uniqueness: if $\phi_1(T)$ and $\phi_2(T)$ are unbiased, then $\mathbb{E}_\theta[\phi_1(T) - \phi_2(T)] = 0$ for every θ; by completeness $\phi_1(T) = \phi_2(T)$ a.s. So the unbiased estimator that is a function of T is unique (in the a.s. sense).

Optimality: let W be any unbiased estimator of $g(\theta)$. By Rao–Blackwell (Theorem 9.8), $W^* = \mathbb{E}[W \mid T]$ is an unbiased function of T with $\operatorname{Var}_\theta W^* \le \operatorname{Var}_\theta W$. By uniqueness, $W^* = \phi(T)$ a.s. Hence $\operatorname{Var}_\theta \phi(T) \le \operatorname{Var}_\theta W$ for every θ. □

Remark 10.6 (Two Ways to Find a UMVUE). The theorem gives two practical recipes: (a) guess a function of the complete sufficient T whose expectation is $g(\theta)$; (b) take any simple unbiased W and compute $\mathbb{E}[W \mid T]$ (Rao–Blackwellisation). The estimator $(1 - 1/n)^T$ in Example 9.14 was found by route (b), and we now know that it is the UMVUE.

Theorem 10.7 (Completeness in Exponential Families; see [47, Section 4.3])**.** *Let $X_1, \dots, X_n$ be i.i.d. from a k-parameter exponential family (Definition 9.4) whose natural parameter space $\{(w_1(\boldsymbol{\theta}), \dots, w_k(\boldsymbol{\theta}))\}$ contains a non-empty open set in $\mathbb{R}^k$. Then*

$$\mathbf{T}(\mathbf{X}) = \Big(\sum_i t_1(X_i), \dots, \sum_i t_k(X_i) \Big)$$

is a complete (and sufficient) statistic.

Idea of proof. $\mathbb{E}_{\boldsymbol{\eta}}[g(\mathbf{T})] = \int g(\mathbf{t}) e^{\boldsymbol{\eta}^\top \mathbf{t}} \, d\nu(\mathbf{t}) \cdot e^{-nA(\boldsymbol{\eta})}$; this means that the Laplace transforms of g^+ and g^- agree on an open set. By uniqueness of the Laplace transform (analytic continuation), $g^+ = g^-$ ν-a.e. For details on uniqueness and analytic continuation see [47, Theorem 2.7.1 and Section 4.3]. □

Theorem 10.8 (Basu's Theorem [3])**.** *If T is complete and (minimal) sufficient and A is ancillary, then T and A are independent for every θ.*

Proof. Fix a set B. Since A is ancillary, $p_B = \mathbb{P}_\theta(A \in B)$ does not depend on θ. Since T is sufficient, $\eta_B(T) = \mathbb{P}(A \in B \mid T)$ is also a statistic free of θ. By the tower rule, $\mathbb{E}_\theta[\eta_B(T) - p_B] = 0$ for every θ; since $\eta_B - p_B$ is bounded, bounded completeness suffices and $\eta_B(T) = p_B$ a.s. That is, $\mathbb{P}(A \in B \mid T) = \mathbb{P}(A \in B)$: independence. □

Theorem 10.9 (Characterisation of the UMVUE)**.** *Let W be an unbiased estimator of $g(\theta)$ with finite variance. Then W is the UMVUE $\iff$ W is uncorrelated with every unbiased estimator of zero with finite variance: $\mathbb{E}_\theta U = 0 \ \forall \theta \implies \operatorname{Cov}_\theta(W, U) = 0 \ \forall \theta$.*

Proof. ($\Rightarrow$) Let W be the UMVUE and U an unbiased estimator of zero. For every $a \in \mathbb{R}$, $W + aU$ is unbiased and $\operatorname{Var}(W + aU) = \operatorname{Var} W + 2a \operatorname{Cov}(W, U) + a^2 \operatorname{Var} U \ge \operatorname{Var} W$. This is possible for all a only if $\operatorname{Cov}(W, U) = 0$ (otherwise $a = -\operatorname{Cov}(W, U)/\operatorname{Var} U$ reduces the variance). ($\Leftarrow$) Let W' be another unbiased estimator; $U = W' - W$ is an unbiased estimator of zero. Then $\operatorname{Var} W' = \operatorname{Var} W + \operatorname{Var} U + 2 \operatorname{Cov}(W, U) = \operatorname{Var} W + \operatorname{Var} U \ge \operatorname{Var} W$. □

Remark 10.10. Bahadur's theorem: if a minimal sufficient statistic exists, every boundedly complete sufficient statistic is minimal sufficient. The converse is false: a minimal sufficient statistic need not be complete (Example 10.15).

10.3 Worked Examples

Example 10.11 (Completeness of the Binomial)**.** Let $T \sim \operatorname{Bin}(n, p)$, $p \in (0, 1)$. With $r = p/(1 - p) \in (0, \infty)$,

$$\mathbb{E}_p[g(T)] = (1 - p)^n \sum_{t=0}^{n} g(t) \binom{n}{t} r^t = 0 \quad \forall p \iff \sum_{t=0}^{n} g(t) \binom{n}{t} r^t = 0 \quad \forall r > 0.$$

A polynomial that vanishes identically on an open interval has all coefficients zero: $g(t) = 0$, $t = 0, \dots, n$. Hence T is complete and $\bar{X} = T/n$ is the UMVUE of p.

Example 10.12 (Bernoulli: the UMVUE of $p(1-p)$)**.** Since $\mathbb{E}[T] = np$ and $\mathbb{E}[T^2] = np(1-p) + n^2p^2$, $\mathbb{E}[T(n-T)] = n(n-1)p(1-p)$. Hence

$$\widehat{p(1-p)} = \frac{T(n-T)}{n(n-1)}$$

is unbiased and, by Lehmann–Scheffé, the UMVUE. This is nothing but S^2 of 0/1 data: $S^2 = \frac{T(n-T)}{n(n-1)}$. The plug-in estimator $\bar{X}(1-\bar{X})$ is biased by the factor $\frac{n-1}{n}$.

Example 10.13 (Uniform$(0,\theta)$ and the German Tank Problem)**.** $T = X_{(n)}$ has density $f_T(t) = nt^{n-1}/\theta^n$, $0 < t < \theta$. If $\mathbb{E}_\theta g(T) = 0$ for every $\theta > 0$, then

$$\int_0^\theta g(t)t^{n-1}dt = 0 \quad \forall \theta > 0;$$

differentiating with respect to θ (Lebesgue's differentiation theorem) gives $g(\theta)\theta^{n-1} = 0$ for almost every θ, i.e. $g = 0$ a.e. T is complete. Since $\mathbb{E}T = \frac{n}{n+1}\theta$, $\frac{n+1}{n}X_{(n)}$ is the UMVUE, and

$$\operatorname{Var} X_{(n)} = \frac{n}{n+2}\theta^2 - \frac{n^2}{(n+1)^2}\theta^2 = \frac{n\theta^2}{(n+1)^2(n+2)} \implies \operatorname{Var}\left(\tfrac{n+1}{n}X_{(n)}\right) = \frac{\theta^2}{n(n+2)}.$$

Historical note: in the Second World War the Allies estimated monthly production from the serial numbers of captured German tanks with estimators of this kind. In the discrete version (k draws without replacement from serial numbers $1,\dots,N$, with maximum m) the UMVUE is $\hat{N} = m(1+1/k) - 1$. While intelligence estimates often overstated monthly production several times over, the statistical estimates turned out to be quite close to the production records captured after the war.

Example 10.14 (Normal: UMVUEs and Basu)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$. $(\sum X_i, \sum X_i^2)$, hence $(\bar{X}, S^2)$, is complete sufficient (Theorem 10.7). Accordingly, $\bar{X}$ is the UMVUE of μ, S^2 that of σ^2, and $S/c_4(n)$ that of σ. We saw in Example 8.13 that S^2 does not attain the CRLB: a UMVUE need not be efficient.

An application of Basu: hold σ^2 fixed. In the family indexed by μ, $\bar{X}$ is complete sufficient and S^2 is ancillary (the distribution of the $X_i - \bar{X}$ does not depend on μ). By Basu's theorem, $\bar{X} \perp\!\!\!\perp S^2$. This is the key to the derivation of the t distribution in Chapter 16, and it requires no density computations at all.

Example 10.15 (A Minimal Sufficient Statistic That Is Not Complete)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(\theta, \theta+1)$. $(X_{(1)}, X_{(n)})$ is minimal sufficient (Chapter 9, Exercise). However, the distribution of the range $R = X_{(n)} - X_{(1)}$ does not depend on θ, and $\mathbb{E}R = \frac{n-1}{n+1}$. Hence $g(X_{(1)}, X_{(n)}) = R - \frac{n-1}{n+1}$ has mean zero for every θ without being zero: the statistic is not complete. In this model θ has *no* UMVUE; different estimators attain the smallest variance at different values of θ.

Example 10.16 (Poisson: the UMVUE of $\mathbb{P}(X = k)$)**.** $T = \sum X_i \sim \text{Poisson}(n\lambda)$ is complete sufficient. $W = \mathbb{I}\{X_1 = k\}$ is unbiased, and since $X_1 \mid T = t \sim \text{Bin}(t, 1/n)$,

$$\widehat{\mathbb{P}(X=k)} = \binom{T}{k}\left(\frac{1}{n}\right)^k\left(1-\frac{1}{n}\right)^{T-k}$$

is the UMVUE. The case $k = 0$ is $(1-1/n)^T$ of Example 9.14.

10.4 Computational Applications

10.4.1 The German Tank Problem

code/ch10/c10_1_german_tank.py

```python
import numpy as np

rng = np.random.default_rng(42)
theta, n = 100.0, 10
samples = rng.uniform(0.0, theta, size=(200_000, n))
umvue = ((n + 1.0) / n) * np.max(samples, axis=1)
print(f"UMVUE variance: {np.var(umvue, ddof=1):.4f} (theory: {theta**2/(n*(n+2)):.4f})")
mm = 2 * samples.mean(axis=1)
print(f"MM (2*Xbar) variance: {np.var(mm, ddof=1):.4f} (theory: {theta**2/(3*n):.4f})")
```

Output

```
UMVUE variance: 83.0691 (theory: 83.3333)
MM (2*Xbar) variance: 332.5476 (theory: 333.3333)
```

Discrete version: suppose $k = 8$ tanks have been captured without replacement from $N = 250$ serial numbers.

code/ch10/c10_2_german_tank_discrete.py

```python
import numpy as np

rng = np.random.default_rng(42)
N, k, R = 250, 8, 200_000
m = np.array([rng.choice(N, k, replace=False).max() + 1 for _ in range(R)])
umvue = m * (1 + 1/k) - 1
print(f"m(1+1/k)-1 : mean={umvue.mean():.2f} (target {N}) | std={umvue.std():.2f}")
print(f"2*mean - 1 : ", end="")
o = np.array([rng.choice(N, k, replace=False).mean() + 1 for _ in range(R)])
print(f"mean={(2*o - 1).mean():.2f} | std={(2*o - 1).std():.2f}")
print(f"Theory std(UMVUE) = {np.sqrt((N - k)*(N + 1)/(k*(k + 2))):.2f}")
```

Output

```
m(1+1/k)-1 : mean=249.91 (target 250) | std=27.65
2*mean - 1 : mean=250.12 | std=50.36
Theory std(UMVUE) = 27.55
```

10.4.2 Basu's Theorem: $\bar{X} \perp\!\!\!\perp S^2$ Only for the Normal

code/ch10/c10_3_basu_only_normal.py

```python
import numpy as np

rng = np.random.default_rng(42)
n, R = 10, 300_000
dists = {"Normal": lambda: rng.normal(0, 1, (R, n)),
              "Expon. ": lambda: rng.exponential(1, (R, n)),
              "Uniform": lambda: rng.uniform(0, 1, (R, n))}
for name, gen in dists.items():
    X = gen()
    xb, s2 = X.mean(axis=1), X.var(axis=1, ddof=1)
    print(f"{name:7s} | Corr(Xbar, S^2) = {np.corrcoef(xb, s2)[0,1]:+.4f} | Corr(|Xbar-med|,
    S^2) = "
          f"{np.corrcoef(np.abs(xb - np.median(xb)), s2)[0,1]:+.4f}")
```

Output

```
Normal  | Corr(Xbar, S^2) = +0.0042 | Corr(|Xbar-med|, S^2) = +0.0016
Expon.  | Corr(Xbar, S^2) = +0.6962 | Corr(|Xbar-med|, S^2) = +0.3975
Uniform | Corr(Xbar, S^2) = -0.0000 | Corr(|Xbar-med|, S^2) = -0.2594
```

For the exponential distribution $\mathrm{Cov}(\bar{X}, S^2) = \mu_3/n > 0$, hence the strong positive correlation. For the uniform the first correlation is zero (symmetry, $\mu_3 = 0$), but the second column reveals dependence: uncorrelatedness is not independence. The normal is the *only* distribution for which $\bar{X}$ and S^2 are independent (the Geary–Lukacs characterisation).

10.4.3 Checking the Characterisation of the UMVUE

In the Bernoulli model, $U = X_1 - X_2$ is an unbiased estimator of zero. By Theorem 10.9, the UMVUE $\bar{X}$ must be uncorrelated with U, whereas the non-UMVUE $W = X_1$ is not.

code/ch10/c10_4_umvue_characterisation_check.py

```python
import numpy as np

rng = np.random.default_rng(42)
p, n, R = 0.3, 6, 1_000_000
X = rng.binomial(1, p, size=(R, n))
U = X[:, 0] - X[:, 1]
T = X.sum(axis=1)
print(f"Cov(Xbar, U) = {np.cov(X.mean(axis=1), U)[0,1]:+.5f}  (UMVUE -> 0)")
print(f"Cov(X_1,  U) = {np.cov(X[:, 0], U)[0,1]:+.5f}  (theory p(1-p) = {p*(1-p):.5f})")
est = T*(n - T)/(n*(n - 1))
print(f"T(n-T)/(n(n-1)) mean = {est.mean():.5f} | target p(1-p) = {p*(1-p):.5f}")
print(f"Xbar(1-Xbar)    mean = {(T/n*(1 - T/n)).mean():.5f} | theory {(n-1)/n*p*(1-p):.5f}")
```

Output

```
Cov(Xbar, U) = -0.00017  (UMVUE -> 0)
Cov(X_1,  U) = +0.21002  (theory p(1-p) = 0.21000)
T(n-T)/(n(n-1)) mean = 0.20996 | target p(1-p) = 0.21000
Xbar(1-Xbar)    mean = 0.17497 | theory 0.17500
```

10.5 Comments and Notes

Remark 10.17 (Limits of the UMVUE). The UMVUE is a strong notion of optimality, but: (i) because of the unbiasedness constraint it can be beaten in MSE (in Example 7.11, S^2 is the UMVUE but $\frac{1}{n+1}SS$ has smaller MSE); (ii) it can be absurd (Chapter 7, Exercise 7.2: $(-1)^X$ is the only unbiased function of the complete sufficient X, hence the UMVUE!); (iii) it may not exist (Example 10.15).

Remark 10.18 (Completeness and Inference). Completeness is used not only in estimation but also in testing: in constructing similar tests and UMPU tests, one conditions on a complete sufficient statistic for the nuisance parameter (Neyman structure). Fisher's exact test is an example.

10.6 Chapter Summary

- If T is complete, $\mathbb{E}_\theta g(T) = 0$ $(\forall\theta)$ implies $g(T) = 0$ a.s. (Definition 10.1). Completeness makes unbiased functions of T unique.
- Lehmann–Scheffé theorem (Theorem 10.5): an unbiased function of a complete sufficient statistic is the unique UMVUE. The practical recipe: either find this function

directly, or improve any unbiased estimator by Rao–Blackwell.

- In exponential families whose natural parameter space contains an open set, the natural sufficient statistic is complete (Theorem 10.7).
- Basu's theorem (Theorem 10.8): a complete sufficient statistic is independent of every ancillary statistic. The independence of $\bar{X}$ and S^2 in normal samples follows in a few lines.
- An estimator is the UMVUE if and only if it is uncorrelated with every unbiased estimator of zero (Theorem 10.9). A UMVUE may not exist and may be beaten in MSE by biased estimators.

Further reading. [16, Sections 6.2 and 7.3], [46, Sections 1.6 and 2.1], [47, Section 4.3].

10.7 Exercises

Exercise 10.1. Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$ (rate). Find the UMVUEs of λ and $1/\lambda$. Show that the UMVUE of $\mathbb{P}(X > c) = e^{-\lambda c}$ is $\big(1 - \frac{c}{T}\big)_+^{n-1}$ $(T = \sum X_i)$. → Solution

Exercise 10.2. Show that for $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, 1)$ the UMVUE of $\Phi(c - \mu) = \mathbb{P}(X \le c)$ is $\Phi\Big(\frac{c - \bar{X}}{\sqrt{1 - 1/n}}\Big)$. → Solution

Exercise 10.3. Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, \theta^2)$, $\theta > 0$ (a "curved exponential family"). Show that $(\bar{X}, S^2)$ is minimal sufficient but not complete. (Hint: write $\mathbb{E}[\bar{X}^2]$ and $\mathbb{E}[S^2]$ in terms of θ^2.) Which condition of Theorem 10.7 is violated? → Solution

Exercise 10.4. Use Basu's theorem to show that for $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$, $T = \sum X_i$ and the vector $\mathbf{U} = (X_1/T, \dots, X_n/T)$ are independent. Use this to compute $\mathbb{E}[X_1/T]$. → Solution

Chapter 11

Maximum Likelihood and the Invariance Principle

The unbiased estimation theory of the previous chapters is elegant but not applicable to every problem: a complete sufficient statistic may not exist, and an unbiased estimator may not exist. Fisher's maximum likelihood (ML) principle, by contrast, offers an almost universal recipe: choose the parameter value that makes the observed data "most likely". ML estimators are usually biased, but in large samples they are consistent, asymptotically normal and efficient (Chapter 12). In this chapter we discuss the definition of the MLE, its computation, its invariance property and its pitfalls.

11.1 Definitions

Definition 11.1 (Likelihood and Log-Likelihood)**.** After $\mathbf{x}$ has been observed, the function $\theta \mapsto L(\theta; \mathbf{x}) = f(\mathbf{x}; \theta)$ is called the *likelihood function* and $\ell(\theta; \mathbf{x}) = \ln L(\theta; \mathbf{x})$ the *log-likelihood*. For an i.i.d. sample, $L(\theta; \mathbf{x}) = \prod_{i=1}^{n} f(x_i; \theta)$ and $\ell(\theta; \mathbf{x}) = \sum_{i=1}^{n} \ln f(x_i; \theta)$.

Remark 11.2. The likelihood is a function of θ and is *not* a probability distribution on θ: $\int L(\theta; \mathbf{x}) d\theta$ is generally not 1. Only its ratios are meaningful; $L(\theta_1; \mathbf{x})/L(\theta_2; \mathbf{x})$ measures how much the data support θ_1 relative to θ_2 (the likelihood principle).

Definition 11.3 (Maximum Likelihood Estimator — MLE)**.** A statistic $\hat{\theta}(\mathbf{X})$ satisfying, for every $\mathbf{x}$,

$$L(\hat{\theta}(\mathbf{x}); \mathbf{x}) = \sup_{\theta \in \Theta} L(\theta; \mathbf{x})$$

is called the *maximum likelihood estimator* (MLE) of θ. If ℓ is differentiable and the maximum is at an interior point of Θ, the MLE is a solution of the *likelihood equations* $\nabla_\theta \ell(\theta; \mathbf{x}) = \mathbf{0}$.

Definition 11.4 (Induced (Profile) Likelihood)**.** Let $g : \Theta \to \Lambda$ be a function, not necessarily one-to-one. For $\eta \in \Lambda$ the function

$$L^*(\eta; \mathbf{x}) = \sup_{\{\theta:\, g(\theta) = \eta\}} L(\theta; \mathbf{x})$$

is called the *induced likelihood*. The $\hat{\eta}$ maximising L^* is called the MLE of $g(\theta)$.

11.2 Main Theorems and Proofs

Theorem 11.5 (Zehna's Invariance Principle [90])**.** *If $\hat{\theta}$ is the MLE of θ, then for* any *function g the MLE of $g(\theta)$ is $g(\hat{\theta})$:*

$$\widehat{g(\theta)}_{MLE} = g(\hat{\theta}_{MLE}).$$

Proof. Let $\hat{\eta} = g(\hat{\theta})$. For every $\eta \in \Lambda$,

$$L^*(\eta; \mathbf{x}) = \sup_{\theta : g(\theta) = \eta} L(\theta; \mathbf{x}) \le \sup_{\theta \in \Theta} L(\theta; \mathbf{x}) = L(\hat{\theta}; \mathbf{x}).$$

On the other hand, since $\hat{\theta}$ belongs to $\{\theta : g(\theta) = \hat{\eta}\}$, $L(\hat{\theta}; \mathbf{x}) \le L^*(\hat{\eta}; \mathbf{x})$. Combining the two inequalities, $L^*(\eta; \mathbf{x}) \le L^*(\hat{\eta}; \mathbf{x})$ for every η; i.e. $\hat{\eta} = g(\hat{\theta})$ maximises L^*. □

Remark 11.6. Invariance is a property that unbiasedness *lacks*: S^2 is unbiased for σ^2 while S is biased for σ; but if $\hat{\sigma}^2$ is the MLE of σ^2, then $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ is the MLE of σ. Practical consequence: the parametrisation in which you set up a model does not change the MLE; in numerical optimisation, unconstrained parametrisations (e.g. $\ln \sigma$, logit) can be used freely.

Proposition 11.7 (The MLE Is a Function of the Sufficient Statistic)**.** *If T is sufficient and the MLE is unique, then $\hat{\theta}$ depends only on T.*

Proof. By factorisation, $L(\theta; \mathbf{x}) = g(T(\mathbf{x}), \theta) h(\mathbf{x})$; the constant factor $h(\mathbf{x}) > 0$ does not affect the maximiser, so $\hat{\theta} = \arg\max_\theta g(T(\mathbf{x}), \theta)$. □

Theorem 11.8 (The MLE in a Natural Exponential Family)**.** *Let the X_i be i.i.d. from the natural-form family $f(x; \boldsymbol{\eta}) = h(x) \exp\{\boldsymbol{\eta}^\top \mathbf{t}(x) - A(\boldsymbol{\eta})\}$ with an open natural parameter space. Then $\ell(\boldsymbol{\eta}) = \boldsymbol{\eta}^\top \sum \mathbf{t}(x_i) - nA(\boldsymbol{\eta}) + \text{const}$ is strictly concave and the MLE (if it exists) is unique; the likelihood equation*

$$\nabla A(\hat{\boldsymbol{\eta}}) = \mathbb{E}_{\hat{\boldsymbol{\eta}}}[\mathbf{t}(X)] = \frac{1}{n} \sum_{i=1}^{n} \mathbf{t}(X_i)$$

takes the form "equate theoretical moments with sample moments".

Proof. Differentiating $e^{A(\boldsymbol{\eta})} = \int h(x) e^{\boldsymbol{\eta}^\top \mathbf{t}(x)} dx$ gives $\nabla A = \mathbb{E}[\mathbf{t}(X)]$ and $\nabla^2 A = \operatorname{Cov}(\mathbf{t}(X))$. Since the covariance matrix is positive definite (for a full-rank family), A is strictly convex and ℓ strictly concave; a strictly concave function has at most one critical point, and that point is the global maximum. □

11.3 Worked Examples

Example 11.9 (The Normal Distribution)**.** $\ell(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$. Partial derivatives:

$$\begin{aligned} \partial_\mu \ell &= \frac{1}{\sigma^2} \sum (x_i - \mu) = 0 && \Longrightarrow \quad \hat{\mu} = \bar{x}, \\ \partial_{\sigma^2} \ell &= -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2 = 0 && \Longrightarrow \quad \hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2. \end{aligned}$$

The Hessian at $(\hat{\mu}, \hat{\sigma}^2)$ is $\operatorname{diag}(-n/\hat{\sigma}^2, -n/(2\hat{\sigma}^4))$, negative definite, so this is a maximum. $\hat{\sigma}^2$ has bias $-\sigma^2/n$ (Theorem 7.9).

Example 11.10 (Uniform$(0, \theta)$: When Differentiation Does Not Help)**.** $L(\theta; \mathbf{x}) = \theta^{-n} \mathbb{I}\{\theta \ge x_{(n)}\}$. For $\theta < x_{(n)}$, $L = 0$; for $\theta \ge x_{(n)}$, L is strictly decreasing. Hence $\hat{\theta} = X_{(n)}$. The likelihood equation $-n/\theta = 0$ has no solution: the maximum is at the boundary point where the likelihood is discontinuous. The MLE is biased ($\mathbb{E} X_{(n)} = \frac{n}{n+1}\theta$); the UMVUE rescales it by $\frac{n+1}{n}$.

Example 11.11 (Bernoulli and the Boundary Problem)**.** $\ell(p) = t \ln p + (n - t) \ln(1 - p)$ and $\hat{p} = t/n$. But if $t = 0$ is observed, $\hat{p} = 0$; the open interval $\Theta = (0, 1)$ does not contain this value, so the MLE (on the open interval) does not exist. Taking $\Theta = [0, 1]$ gives $\hat{p} = 0$; but then the Wald standard error $\sqrt{\hat{p}(1 - \hat{p})/n} = 0$ and the Wald confidence interval collapses to a point (to be fixed with the Wilson interval in Chapter 16).

Example 11.12 (The Laplace Location Family: the Median)**.** For $f(x;\theta)=\tfrac{1}{2}e^{-|x-\theta|}$, $\ell(\theta)=-n\ln 2-\sum|x_i-\theta|$. $\sum|x_i-\theta|$ is a piecewise linear convex function of θ minimised at the sample median. Hence the MLE is the sample median (if n is even, every point between the two middle values is an MLE: the MLE need not be unique).

Example 11.13 (An Application of Invariance)**.** In the normal model, by Theorem 11.5, the MLE of $\mathbb{P}(X>c)=1-\Phi\big((c-\mu)/\sigma\big)$ is simply $1-\Phi\big((c-\bar X)/\hat\sigma\big)$. Similarly, for the Bernoulli the MLE of the odds $p/(1-p)$ is $\hat p/(1-\hat p)$ and that of the log-odds is $\ln\frac{\hat p}{1-\hat p}$.

Example 11.14 (The Gamma Distribution: an MLE Without Closed Form)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Gamma}(\alpha,\beta)$ (shape α, rate β). $\ell=n\alpha\ln\beta-n\ln\Gamma(\alpha)+(\alpha-1)\sum\ln x_i-\beta\sum x_i$. From $\partial_\beta\ell=n\alpha/\beta-\sum x_i=0$, $\hat\beta=\hat\alpha/\bar x$; substituting into $\partial_\alpha\ell=0$ gives the one-dimensional equation

$$\ln\hat\alpha-\psi(\hat\alpha)=\ln\bar x-\overline{\ln x}\quad(=:s>0)$$

(ψ: digamma). The left-hand side is strictly decreasing on $(0,\infty)$, so there is a unique solution, found by Newton–Raphson:

$$\alpha_{k+1}=\alpha_k-\frac{\ln\alpha_k-\psi(\alpha_k)-s}{1/\alpha_k-\psi'(\alpha_k)}.$$

The method of moments estimate (Example 7.13) is a good starting value.

Example 11.15 (Cauchy: a Multimodal Likelihood)**.** For $f(x;\theta)=\frac{1}{\pi(1+(x-\theta)^2)}$, the likelihood equation $\sum\frac{2(x_i-\theta)}{1+(x_i-\theta)^2}=0$ is equivalent to a polynomial equation of degree $2n-1$ in θ and may have several local maxima. Numerical optimisation may converge to the wrong peak depending on the starting point. In large samples the sample median (being consistent) is a good start that falls in the basin of attraction of the global peak; in small samples there is no such guarantee. In the five-observation example below, even BFGS started at the median *misses* the global maximum. Safe practice: first evaluate ℓ on a coarse grid and start a local optimisation from the best grid point, or try many starting points.

11.4 Computational Applications

11.4.1 Analytic and Numerical MLE in the Normal Model; Invariance

code/ch11/c11_1_normal_analytic_numeric_mle.py

```python
import numpy as np
from scipy.optimize import minimize

rng = np.random.default_rng(42)
data = rng.normal(10.0, 3.0, size=250)
mu_an, var_an = np.mean(data), np.var(data, ddof=0)
print(f"Analytic MLE: mu={mu_an:.4f}, s2={var_an:.4f}, sd={np.sqrt(var_an):.4f}")

def negll(par, param):
    mu, s = par
    sd = {"sd": abs(s), "var": np.sqrt(abs(s)), "logsd": np.exp(s)}[param]
    return -np.sum(-np.log(sd) - 0.5*np.log(2*np.pi) - (data - mu)**2 / (2*sd**2))

for param, x0 in [("sd", [5, 1]), ("var", [5, 1]), ("logsd", [5, 1])]:
    r = minimize(negll, x0, args=(param,), method="Nelder-Mead", options={"xatol": 1e-8, "
    fatol": 1e-10})
    mu, s = r.x
    sd = {"sd": abs(s), "var": np.sqrt(abs(s)), "logsd": np.exp(s)}[param]
    print(f"Numerical ({param:5s} param.): mu={mu:.4f}, sd={sd:.4f}, -logL={r.fun:.4f}")
```

Output

```
Analytic MLE: mu=9.8540, s2=7.9348, sd=2.8169
Numerical (sd    param.): mu=9.8540, sd=2.8169, -logL=613.6412
Numerical (var   param.): mu=9.8540, sd=2.8169, -logL=613.6412
Numerical (logsd param.): mu=9.8540, sd=2.8169, -logL=613.6412
```

Three different parametrisations give the same $(\hat{\mu}, \hat{\sigma})$ and the same maximal log-likelihood: the numerical counterpart of the invariance principle.

11.4.2 Gamma MLE: Newton–Raphson

code/ch11/c11_2_gamma_mle_newton_raphson.py

```python
import numpy as np
from scipy.special import digamma, polygamma
from scipy import stats

rng = np.random.default_rng(42)
x = rng.gamma(2.5, 1/0.625, size=400)      # true: alpha=2.5, beta(rate)=0.625; numpy takes scale=1/rate
s = np.log(x.mean()) - np.mean(np.log(x))
a = x.mean()**2 / x.var()                          # MM starting value
print(f"MM start: alpha={a:.5f}")
for k in range(6):
    a -= (np.log(a) - digamma(a) - s) / (1/a - polygamma(1, a))
    print(f"iter {k+1}: alpha={a:.8f}")
b = a / x.mean()                                   # beta_hat = alpha_hat / xbar (rate)
a_sp, _, scale_sp = stats.gamma.fit(x, floc=0)     # scipy also returns the scale
print(f"Newton MLE : alpha={a:.5f}, beta(rate)={b:.5f}")
print(f"scipy.fit  : alpha={a_sp:.5f}, beta(rate)={1/scale_sp:.5f}")
```

Output

```
MM start: alpha=2.52321
iter 1: alpha=2.54125944
iter 2: alpha=2.54139646
iter 3: alpha=2.54139646
iter 4: alpha=2.54139646
iter 5: alpha=2.54139646
iter 6: alpha=2.54139646
Newton MLE : alpha=2.54140, beta(rate)=0.63481
scipy.fit  : alpha=2.54140, beta(rate)=0.63481
```

Newton–Raphson converges quadratically: the number of correct digits roughly doubles at each iteration.

11.4.3 Multimodality of the Cauchy Likelihood

code/ch11/c11_3_cauchy_multimodal_likelihood.py

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize_scalar, minimize

x = np.array([-4.2, -3.1, 0.4, 5.6, 6.3])
ll = lambda t: -np.sum(np.log1p((x - t)**2))
grid = np.linspace(-10, 12, 4001)
L = np.array([ll(t) for t in grid])
peaks = grid[1:-1][(L[1:-1] > L[:-2]) & (L[1:-1] > L[2:])]
print("Local maxima:", np.round(peaks, 3))
for t0 in [-5.0, 0.0, 6.0, np.median(x), grid[np.argmax(L)]]:   # the last two: the median and the best grid point
```

```
    r = minimize(lambda t: -ll(t[0]), [t0], method="BFGS")
    print(f"start={t0:5.2f} -> theta_hat={r.x[0]:7.4f}, logL={-r.fun:.4f}")

fig, ax = plt.subplots(figsize=(7, 3))
ax.plot(grid, L, "k-")
ax.plot(x, np.full_like(x, L.min()), "r|", ms=15, label="observations")
ax.set_xlabel(r"$\theta$"); ax.set_ylabel(r"$\ell(\theta)$ (+ const.)"); ax.legend()
fig.tight_layout(); fig.savefig("cauchy_ll.pdf")
```

Output

```
Local maxima: [-3.12   0.252  5.658]
start=-5.00 -> theta_hat=-3.1171, logL=-12.2092
start= 0.00 -> theta_hat= 0.2503, logL=-12.5762
start= 6.00 -> theta_hat= 5.6581, logL=-12.6432
start= 0.40 -> theta_hat= 0.2503, logL=-12.5762
start=-3.12 -> theta_hat=-3.1171, logL=-12.2092
```

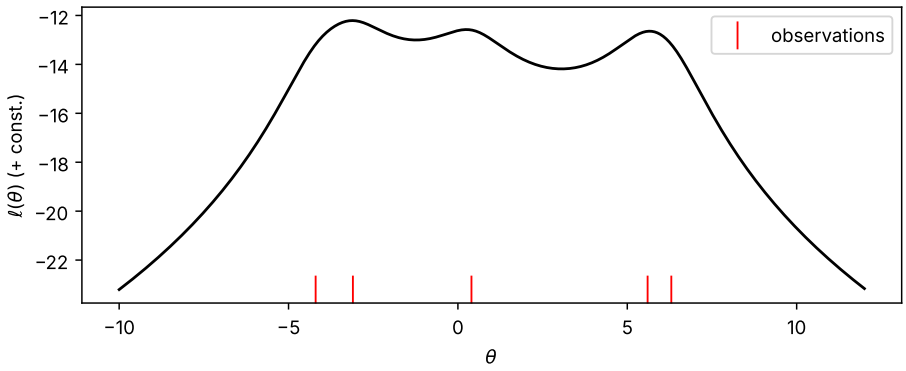


Figure 11.1: Log-likelihood for a Cauchy sample of five observations. Data split into two clusters produce three local maxima (the two cluster centres and a compromise point between them); gradient methods converge to different peaks depending on the starting point. The global maximum is in the left cluster ($\hat\theta \approx -3.12$).

11.5 Comments and Notes

Remark 11.16 (When the MLE Can Fail).

- **Unbounded likelihood:** in a two-component normal mixture, if the mean of one component is set equal to an observation and its variance is sent to 0, then $L \to \infty$; the global MLE is meaningless and one looks for "the best local maximum" (the EM algorithm).
- **Number of parameters growing with** n (the Neyman–Scott problem [62]): if $X_{ij} \sim \mathcal{N}(\mu_i, \sigma^2)$, $j = 1, 2$, $i = 1, \dots, n$, then $\hat\sigma^2 \xrightarrow{P} \sigma^2/2$: the MLE is inconsistent.
- **True parameter on the boundary:** asymptotic normality breaks down (in Chapter 15, the counterpart of Wilks' theorem is a mixture of chi-squares).

Remark 11.17 (Likelihood and the KL Divergence). $\frac{1}{n}\ell(\theta) = \frac{1}{n}\sum \ln f(X_i;\theta) \xrightarrow{\text{a.s.}} \mathbb{E}_{\theta_0} \ln f(X;\theta)$ and $\mathbb{E}_{\theta_0} \ln f(X;\theta_0) - \mathbb{E}_{\theta_0} \ln f(X;\theta) = \mathrm{KL}(f_{\theta_0} \| f_\theta) \geq 0$. That is, the MLE minimises the KL divergence between the empirical distribution and the model. If the model is misspecified, the MLE converges to the member of the model closest in KL to the true distribution (the quasi-likelihood / QMLE interpretation).

11.6 Chapter Summary

- The MLE is the parameter value that maximises the likelihood. It is usually found as a root of the likelihood equation; when there is no closed form, numerical methods such as Newton–Raphson are used.
- Zehna's invariance principle (Theorem 11.5): even when g is not one-to-one, $g(\hat\theta)$ is the MLE of $g(\theta)$ in the sense of the induced likelihood (Definition 11.4).
- The MLE is a function of the sufficient statistic (Proposition 11.7). In a natural exponential family the likelihood equation becomes a moment equation: $\mathbb{E}_\eta T = T(\mathbf{x})$ (Theorem 11.8).
- The MLE does not always behave well. It may not exist, may not be unique, and the likelihood may have several local maxima (Cauchy). When the number of parameters grows with n, the MLE may even be inconsistent (Neyman–Scott).

Further reading. [16, Section 7.2], [81, Chapter 9]; for misbehaviour of the MLE see [46, Chapter 6].

11.7 Exercises

Exercise 11.1. Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$. Find the MLE of $\mathbb{P}(X=0)=e^{-\lambda}$ and compare it with the UMVUE in Example 9.14. Which is biased? → Solution

Exercise 11.2. Find the MLE of (μ,λ) for the shifted exponential distribution $f(x;\mu,\lambda)=\lambda e^{-\lambda(x-\mu)}\mathbb{I}\{x\ge\mu\}$. Compute the bias of $\hat\mu$ and propose an unbiased correction. → Solution

Exercise 11.3. Under Hardy–Weinberg equilibrium the genotype probabilities are $(\theta^2, 2\theta(1-\theta), (1-\theta)^2)$. Given observed counts (n_1,n_2,n_3), show that $\hat\theta = \frac{2n_1+n_2}{2n}$ and compute $I_n(\theta)$. → Solution

Exercise 11.4. Study the Neyman–Scott problem by simulation: show that for $n=1000$ pairs, $\hat\sigma^2$ approaches $\sigma^2/2$. Construct a consistent estimator by differencing $(X_{i1}-X_{i2})$. → Solution

Chapter 12

Asymptotic Theory of the MLE and the Delta Method

The finite-sample properties of the MLE vary from model to model: sometimes it is biased, sometimes it has no closed form at all. In large samples, however, under regularity conditions, the MLE shows universal and very strong behaviour: it is consistent, asymptotically normal, and its asymptotic variance attains the Cramér–Rao bound. This chapter covers these three results, the information inequality that makes them possible, and the delta method, which gives the asymptotic distribution of transformations of an estimator. These results are the theoretical basis of almost all standard errors and Wald confidence intervals reported in applied statistics.

12.1 Definitions

Definition 12.1 (Asymptotic Normality and Asymptotic Variance)**.** If $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, v(\theta))$, then T_n is called *asymptotically normal* (AN) and $v(\theta)$ its *asymptotic variance*; we write $T_n \approx \mathcal{N}(\theta, v(\theta)/n)$. The asymptotic variance need not equal $\lim n \operatorname{Var}(T_n)$ (the latter may not exist).

Definition 12.2 (Asymptotic Efficiency and Asymptotic Relative Efficiency)**.** In a regular model, T_n is called *asymptotically efficient* if $v(\theta) = 1/I_1(\theta)$. For two AN estimators $T_n^{(1)}, T_n^{(2)}$,

$$\operatorname{ARE}(T^{(1)}, T^{(2)}) = \frac{v_2(\theta)}{v_1(\theta)}$$

is the *asymptotic relative efficiency*. Interpretation: $T^{(2)}$ needs roughly n/ARE observations to reach the precision that $T^{(1)}$ achieves with n.

Definition 12.3 (Observed Information)**.** The random quantity $J_n(\theta) = -\ell''(\theta; \mathbf{X}) = -\sum_{i=1}^n \partial_\theta^2 \ln f(X_i; \theta)$ is called the *observed Fisher information*. $\mathbb{E}_\theta J_n(\theta) = I_n(\theta)$ and, by the WLLN, $J_n(\theta)/n \xrightarrow{P} I_1(\theta)$.

12.2 Consistency

Lemma 12.4 (The Information Inequality)**.** *If $\theta \neq \theta_0$ and $\mathbb{P}_\theta \neq \mathbb{P}_{\theta_0}$ (identifiability), then*

$$\mathbb{E}_{\theta_0}\big[\ln f(X;\theta)\big] < \mathbb{E}_{\theta_0}\big[\ln f(X;\theta_0)\big].$$

Proof. Since ln is strictly concave, Jensen's inequality (Theorem 3.10) gives

$$\mathbb{E}_{\theta_0} \ln \frac{f(X;\theta)}{f(X;\theta_0)} < \ln \mathbb{E}_{\theta_0} \frac{f(X;\theta)}{f(X;\theta_0)} = \ln \int f(x;\theta)\,dx = \ln 1 = 0.$$

The strict inequality comes from the ratio not being a.s. constant (identifiability). The left-hand side is $-\mathrm{KL}(f_{\theta_0}\|f_\theta)$. □

Theorem 12.5 (Consistency of the MLE [79])**.** *Let $X_i \overset{i.i.d.}{\sim} f(\cdot;\theta_0)$ with an identifiable model. Suppose moreover that (i) Θ is compact, (ii) $\theta \mapsto \ln f(x;\theta)$ is continuous for every x, and (iii) $\mathbb{E}_{\theta_0}\sup_{\theta\in\Theta}|\ln f(X;\theta)| < \infty$. Then $\hat\theta_n \xrightarrow{a.s.} \theta_0$.*

Proof. Let $M_n(\theta) = \frac{1}{n}\ell(\theta)$ and $M(\theta) = \mathbb{E}_{\theta_0}\ln f(X;\theta)$; by (iii), M is finite everywhere. By Lemma 12.4, $M(\theta) < M(\theta_0)$ for $\theta \neq \theta_0$.

Fix $\varepsilon > 0$ and let $K = \{\theta \in \Theta : |\theta - \theta_0| \geq \varepsilon\}$; K is compact. For $\theta \in K$ and $r > 0$ let $B(\theta, r) = \{\theta' \in \Theta : |\theta' - \theta| < r\}$ and

$$m_r(x;\theta) = \sup_{\theta' \in B(\theta,r)} \ln f(x;\theta').$$

By (ii), $m_r(x;\theta) \to \ln f(x;\theta)$ as $r \downarrow 0$, and by (iii) $|m_r| \leq \sup_\Theta |\ln f(x;\cdot)|$ is integrable. By dominated convergence, $\mathbb{E}_{\theta_0} m_r(X;\theta) \to M(\theta) < M(\theta_0)$; hence there is an $r_\theta > 0$ with $\mathbb{E}_{\theta_0} m_{r_\theta}(X;\theta) < M(\theta_0)$. Let $B_1, \ldots, B_m$ (centres $\theta^{(1)}, \ldots, \theta^{(m)}$) be a finite subcover of the open cover $\{B(\theta, r_\theta) : \theta \in K\}$.

By the strong law of large numbers, for each of the finitely many j and for θ_0, almost surely

$$\frac{1}{n}\sum_{i=1}^{n} m_{r_j}(X_i;\theta^{(j)}) \to \mathbb{E}_{\theta_0} m_{r_j}(X;\theta^{(j)}) < M(\theta_0), \qquad M_n(\theta_0) \to M(\theta_0).$$

On this event, for all sufficiently large n,

$$\sup_{\theta\in K} M_n(\theta) \leq \max_{j\leq m} \frac{1}{n}\sum_{i=1}^{n} m_{r_j}(X_i;\theta^{(j)}) < M_n(\theta_0) \leq M_n(\hat\theta_n),$$

i.e. $\hat\theta_n \notin K$: $|\hat\theta_n - \theta_0| < \varepsilon$. Intersecting over $\varepsilon = 1/k$, $k \in \mathbb{N}$, gives $\hat\theta_n \xrightarrow{\text{a.s.}} \theta_0$. (For relaxing compactness and for the multidimensional case see [78, Theorems 5.7 and 5.14] and [28, Chapter 17].) □

12.3 Asymptotic Normality

Theorem 12.6 (Asymptotic Normality of the MLE [18])**.** *In addition to conditions (R1)–(R4) of Chapter 8, let θ_0 be an interior point, let $\hat\theta_n$ be a consistent root of the likelihood equation, and suppose $|\partial_\theta^3 \ln f(x;\theta)| \leq M(x)$ in a neighbourhood of θ_0 with $\mathbb{E}_{\theta_0} M(X) < \infty$. Then*

$$\sqrt{n}(\hat\theta_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I_1(\theta_0)}\right).$$

Proof. Expand the equation $\ell'(\hat\theta_n) = 0$ around θ_0: for some θ_n^* between $\hat\theta_n$ and θ_0,

$$0 = \ell'(\theta_0) + (\hat\theta_n - \theta_0)\ell''(\theta_0) + \tfrac{1}{2}(\hat\theta_n - \theta_0)^2 \ell'''(\theta_n^*).$$

Rearranging,

$$\sqrt{n}(\hat\theta_n - \theta_0) = \frac{\frac{1}{\sqrt{n}}\ell'(\theta_0)}{-\frac{1}{n}\ell''(\theta_0) - \frac{1}{2n}(\hat\theta_n - \theta_0)\ell'''(\theta_n^*)}.$$

- *Numerator:* $\ell'(\theta_0) = \sum s(\theta_0; X_i)$ is a sum of i.i.d. terms with mean 0 and variance $I_1(\theta_0)$ (Lemma 8.6). By the CLT, $\frac{1}{\sqrt{n}}\ell'(\theta_0) \xrightarrow{d} \mathcal{N}(0, I_1(\theta_0))$.

- *Denominator, first term:* by the WLLN, $-\frac{1}{n}\ell''(\theta_0) \xrightarrow{P} I_1(\theta_0)$ (information identity).

- *Denominator, second term:* $|\frac{1}{n}\ell'''(\theta_n^*)| \le \frac{1}{n}\sum M(X_i) \xrightarrow{P} \mathbb{E}M(X) < \infty$ and $\hat\theta_n - \theta_0 \xrightarrow{P} 0$; the product $\xrightarrow{P} 0$.

By Slutsky's theorem (Theorem 6.16), $\sqrt{n}(\hat\theta_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I_1)/I_1 = \mathcal{N}(0, 1/I_1(\theta_0))$. □

Corollary 12.7 (The Wald Confidence Interval)**.** *If I_1 is continuous, then $I_1(\hat\theta_n) \xrightarrow{P} I_1(\theta_0)$, and by Slutsky $\sqrt{nI_1(\hat\theta_n)}(\hat\theta_n - \theta_0) \xrightarrow{d} \mathcal{N}(0,1)$. Hence*

$$\hat\theta_n \pm z_{\alpha/2}/\sqrt{nI_1(\hat\theta_n)} \qquad \text{or} \qquad \hat\theta_n \pm z_{\alpha/2}/\sqrt{J_n(\hat\theta_n)}$$

are asymptotic $1-\alpha$ confidence intervals. In the multiparameter case, $\sqrt{n}(\hat{\boldsymbol\theta}_n - \boldsymbol\theta_0) \xrightarrow{d} \mathcal{N}_k(\mathbf{0}, I_1(\boldsymbol\theta_0)^{-1})$.

Proof. Since $\hat\theta_n \xrightarrow{P} \theta_0$ and I_1 is continuous, Theorem 5.18 gives $I_1(\hat\theta_n) \xrightarrow{P} I_1(\theta_0) > 0$. Since $Z_n = \sqrt{nI_1(\hat\theta_n)}(\hat\theta_n - \theta_0) = \sqrt{I_1(\hat\theta_n)} \cdot \sqrt{n}(\hat\theta_n - \theta_0)$, Slutsky gives $Z_n \xrightarrow{d} \sqrt{I_1}\,\mathcal{N}(0, 1/I_1) = \mathcal{N}(0,1)$, and

$$\mathbb{P}_{\theta_0}\Big(\theta_0 \in \hat\theta_n \pm z_{\alpha/2}/\sqrt{nI_1(\hat\theta_n)}\Big) = \mathbb{P}(|Z_n| \le z_{\alpha/2}) \to 1-\alpha$$

(because $\pm z_{\alpha/2}$ are continuity points of Φ). For the observed information write $\frac{1}{n}J_n(\hat\theta_n) = -\frac{1}{n}\ell''(\hat\theta_n)$; as in the proof of Theorem 12.6, $|\frac{1}{n}\ell''(\hat\theta_n) - \frac{1}{n}\ell''(\theta_0)| \le |\hat\theta_n - \theta_0|\frac{1}{n}\sum M(X_i) \xrightarrow{P} 0$ and $-\frac{1}{n}\ell''(\theta_0) \xrightarrow{P} I_1(\theta_0)$; the same argument applies. The multiparameter statement is the vector form of the proof of Theorem 12.6 (multivariate CLT for the score vector, Theorem 6.14). □

12.4 The Delta Method

Theorem 12.8 (The Delta Method)**.** *Let $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ and let g be differentiable at θ with $g'(\theta) \neq 0$. Then*

$$\sqrt{n}\big(g(T_n) - g(\theta)\big) \xrightarrow{d} \mathcal{N}\big(0, [g'(\theta)]^2\sigma^2\big).$$

Proof. Since $\sqrt{n}(T_n - \theta) = O_P(1)$, $T_n \xrightarrow{P} \theta$. By the definition of the derivative, $g(t) - g(\theta) = [g'(\theta) + r(t)](t - \theta)$, where $r(t) \to 0$ as $t \to \theta$ and $r(\theta) = 0$. Hence

$$\sqrt{n}\big(g(T_n) - g(\theta)\big) = \big[g'(\theta) + r(T_n)\big] \cdot \sqrt{n}(T_n - \theta).$$

By the continuous mapping theorem $r(T_n) \xrightarrow{P} 0$; by Slutsky the limit is $g'(\theta)\cdot\mathcal{N}(0,\sigma^2)$. □

Theorem 12.9 (Second-Order Delta Method)**.** *Under the conditions of Theorem 12.8, if $g'(\theta) = 0$ and $g''(\theta) \neq 0$, then*

$$n\big(g(T_n) - g(\theta)\big) \xrightarrow{d} \frac{\sigma^2 g''(\theta)}{2}\,\chi^2_1.$$

Proof. Second-order Taylor: $n(g(T_n) - g(\theta)) = \frac{g''(\theta)}{2}\big[\sqrt{n}(T_n - \theta)\big]^2 + o_P(1)$; and $[\sqrt{n}(T_n - \theta)/\sigma]^2 \xrightarrow{d} \chi^2_1$ (continuous mapping). □

Theorem 12.10 (Multivariate Delta Method)**.** *If $\sqrt{n}(\mathbf{T}_n - \boldsymbol\theta) \xrightarrow{d} \mathcal{N}_k(\mathbf{0}, \Sigma)$ and $g : \mathbb{R}^k \to \mathbb{R}$ is differentiable at $\boldsymbol\theta$, then $\sqrt{n}(g(\mathbf{T}_n) - g(\boldsymbol\theta)) \xrightarrow{d} \mathcal{N}\big(0, \nabla g(\boldsymbol\theta)^\top \Sigma\, \nabla g(\boldsymbol\theta)\big)$.*

Proof. By the definition of differentiability, $g(\mathbf{t}) - g(\boldsymbol{\theta}) = \nabla g(\boldsymbol{\theta})^\top (\mathbf{t} - \boldsymbol{\theta}) + \|\mathbf{t} - \boldsymbol{\theta}\|\, r(\mathbf{t})$, where $r(\boldsymbol{\theta}) = 0$ and r is continuous at $\boldsymbol{\theta}$. Hence

$$\sqrt{n}(g(\mathbf{T}_n) - g(\boldsymbol{\theta})) = \nabla g(\boldsymbol{\theta})^\top \sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta}) + \big\|\sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta})\big\|\, r(\mathbf{T}_n).$$

By the continuous mapping theorem (Theorem 6.17, linear g), the first term converges to $\nabla g^\top \mathbf{Z}$ with $\mathbf{Z} \sim \mathcal{N}_k(\mathbf{0}, \Sigma)$, which is $\mathcal{N}(0, \nabla g^\top \Sigma \nabla g)$. In the second term, $\|\sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta})\| \xrightarrow{d} \|\mathbf{Z}\|$, and hence $\mathbf{T}_n - \boldsymbol{\theta} = n^{-1/2} \cdot \sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta}) \xrightarrow{P} \mathbf{0}$ (Slutsky); by continuity of r, $r(\mathbf{T}_n) \xrightarrow{P} 0$, and by Slutsky the product $\xrightarrow{P} 0$. Apply Slutsky once more. □

Definition 12.11 (Variance-Stabilising Transformation)**.** If $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2(\theta))$, then for the transformation $g(\theta) = \int \frac{c}{\sigma(\theta)} d\theta$, which satisfies $g'(\theta) = c/\sigma(\theta)$, we have $\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, c^2)$: the asymptotic variance no longer depends on θ.

12.5 Worked Examples

Example 12.12 (The Odds)**.** $\hat{p} = \bar{X}$, $\sqrt{n}(\hat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1-p))$. For the odds $\tau = g(p) = \frac{p}{1-p}$, $g'(p) = \frac{1}{(1-p)^2}$. By the delta method,

$$\sqrt{n}(\hat{\tau} - \tau) \xrightarrow{d} \mathcal{N}\Big(0, \frac{p(1-p)}{(1-p)^4}\Big) = \mathcal{N}\Big(0, \frac{p}{(1-p)^3}\Big).$$

For the log-odds $\eta = \ln \frac{p}{1-p}$, $g'(p) = \frac{1}{p(1-p)}$ and the asymptotic variance is $\frac{1}{p(1-p)}$; this is the source of standard errors in logistic regression.

Example 12.13 (Variance-Stabilising Transformations)**.**

- **Poisson:** $\sigma^2(\lambda) = \lambda$; with $g(\lambda) = 2\sqrt{\lambda}$, $\sqrt{n}(2\sqrt{\bar{X}} - 2\sqrt{\lambda}) \xrightarrow{d} \mathcal{N}(0, 1)$.
- **Binomial:** $\sigma^2(p) = p(1-p)$; with $g(p) = \arcsin\sqrt{p}$ the asymptotic variance is $1/4$.
- **Correlation coefficient (Fisher's z):** for the bivariate normal, $\sqrt{n}(r - \rho) \xrightarrow{d} \mathcal{N}(0, (1-\rho^2)^2)$; with $z = \tanh^{-1} r = \frac{1}{2}\ln\frac{1+r}{1-r}$ the asymptotic variance is 1.

These transformations both improve the coverage of confidence intervals and speed up the normal approximation.

Example 12.14 (Asymptotic Relative Efficiency: Mean or Median?)**.** In a location family with symmetric density f, the asymptotic variance of the sample median is $\frac{1}{4f(\theta)^2}$ and that of the mean is σ^2.

Distribution	$v(\bar{X})$	$v(\text{median})$	$\text{ARE}(\text{median}, \bar{X})$
$\mathcal{N}(\theta, 1)$	1	$\pi/2$	$2/\pi \approx 0.637$
$\text{Laplace}(\theta, 1)$	2	1	2
$\text{Cauchy}(\theta, 1)$	∞	$\pi^2/4$	∞

For normal data the median needs about 57% more observations to match the efficiency of the mean; for the Laplace the median is twice as efficient (and is the MLE). For the Cauchy, the efficiency of the median relative to the MLE is $8/\pi^2 \approx 0.81$.

Example 12.15 (Second-Order Delta: $p(1-p)$ at $p = 1/2$)**.** $g(p) = p(1-p)$, $g'(1/2) = 0$, $g'' = -2$. Since $\sigma^2 = 1/4$ at $p = 1/2$, $n(\bar{X}(1-\bar{X}) - \frac{1}{4}) \xrightarrow{d} -\frac{1}{4}\chi^2_1$. That is, at this point the estimator $\bar{X}(1-\bar{X})$ converges at rate n rather than $\sqrt{n}$, and its limit distribution is not normal but a (negatively scaled) chi-square.

12.6 Computational Applications

12.6.1 The Delta Method for the Odds

code/ch12/c12_1_delta_method_odds.py

```python
import numpy as np

rng = np.random.default_rng(42)
p, n = 0.3, 500
samples = rng.binomial(1, p, size=(100_000, n))
tau_hat = np.mean(samples, axis=1) / (1.0 - np.mean(samples, axis=1))
print(f"Delta theoretical variance: {(p/((1-p)**3))/n:.6f} | Empirical: {np.var(tau_hat, ddof=1):.6f}")
eta_hat = np.log(tau_hat)
print(f"Log-odds delta variance: {1/(p*(1-p)*n):.6f} | Empirical: {np.var(eta_hat, ddof=1):.6f}")
```

Output

```
Delta theoretical variance: 0.001749 | Empirical: 0.001764
Log-odds delta variance: 0.009524 | Empirical: 0.009570
```

12.6.2 Asymptotic Normality of the MLE and Coverage of the Wald Interval

For the Gamma shape parameter α (rate known, $\beta = 1$), $I_1(\alpha) = \psi'(\alpha)$ (trigamma). The MLE solves $\psi(\hat{\alpha}) = \overline{\ln X}$.

code/ch12/c12_2_mle_asymptotic_normality_wald.py

```python
import numpy as np
from scipy.special import digamma, polygamma
from scipy import stats

rng = np.random.default_rng(42)
alpha0, R = 2.0, 50_000
def mle_alpha(lx_bar):
    a = np.full_like(lx_bar, 1.0)
    for _ in range(50):                                    # vectorised Newton
        a = np.maximum(a - (digamma(a) - lx_bar) / polygamma(1, a), 1e-6)
    return a
for n in [10, 50, 400]:
    X = rng.gamma(alpha0, 1.0, size=(R, n))
    a_hat = mle_alpha(np.log(X).mean(axis=1))
    z = np.sqrt(n * polygamma(1, alpha0)) * (a_hat - alpha0)
    se = 1 / np.sqrt(n * polygamma(1, a_hat))
    coverage = np.mean(np.abs(a_hat - alpha0) <= 1.96 * se)
    print(f"n={n:4d} | bias={a_hat.mean()-alpha0:+.4f} | n*Var={n*a_hat.var():.4f} (1/I={1/polygamma(1,alpha0):.4f})"
          f" | skewness(z)={stats.skew(z):+.3f} | Wald 95% coverage={coverage:.4f}")
```

Output

```
n=  10 | bias=+0.0471 | n*Var=1.6058 (1/I=1.5505) | skewness(z)=+0.486 | Wald 95% coverage=0.9476
n=  50 | bias=+0.0113 | n*Var=1.5671 (1/I=1.5505) | skewness(z)=+0.219 | Wald 95% coverage=0.9501
n= 400 | bias=+0.0012 | n*Var=1.5378 (1/I=1.5505) | skewness(z)=+0.075 | Wald 95% coverage=0.9522
```

For small n the positive bias and right skewness are clear, and $n\,\mathrm{Var}(\hat{\alpha})$ is above its asymptotic value; as n grows, the bias vanishes at rate $O(1/n)$ and the skewness at rate $O(1/\sqrt{n})$. In this model the *total* coverage of the Wald interval is close to the target even at $n = 10$, but this is because the errors in the two tails offset each other: because of the skewness, the interval misses too often on one side and too rarely on the other (Exercise: compute the two tail errors separately).

12.6.3 ARE, Variance Stabilisation and the Second-Order Delta Method

code/ch12/c12_3_are_variance_stabilisation.py

```python
import numpy as np

rng = np.random.default_rng(42)
n, R = 101, 100_000
for name, X in [("Normal ", rng.normal(size=(R, n))),
                ("Laplace", rng.laplace(size=(R, n))),
                ("t_3    ", rng.standard_t(3, size=(R, n)))]:
    vm, vmed = X.mean(axis=1).var(), np.median(X, axis=1).var()
    print(f"{name} | n*Var(mean)={n*vm:.3f} | n*Var(median)={n*vmed:.3f} | ARE(median, mean)
    ={vm/vmed:.3f}")

print("Poisson: Var(2*sqrt(Xbar)) * n  (target 1, independent of lambda)")
for lam in [1, 5, 50]:
    Xb = rng.poisson(lam, size=(R, 200)).mean(axis=1)
    print(f"  lambda={lam:3d} | n*Var(Xbar)={200*Xb.var():7.3f} | n*Var(2 sqrt(Xbar))={200*np
    .var(2*np.sqrt(Xb)):.3f}")

n = 400
pb = rng.binomial(n, 0.5, size=R) / n
W = n * (pb*(1 - pb) - 0.25)
print(f"Second-order delta: E[W]={W.mean():.4f} (theory -0.25), Var[W]={W.var():.4f} (theory
    2/16={2/16:.4f})")
```

Output

```
Normal  | n*Var(mean)=0.999 | n*Var(median)=1.561 | ARE(median, mean)=0.640
Laplace | n*Var(mean)=2.003 | n*Var(median)=1.156 | ARE(median, mean)=1.733
t_3     | n*Var(mean)=2.980 | n*Var(median)=1.862 | ARE(median, mean)=1.601
Poisson: Var(2*sqrt(Xbar)) * n  (target 1, independent of lambda)
  lambda=  1 | n*Var(Xbar)=  1.004 | n*Var(2 sqrt(Xbar))=1.006
  lambda=  5 | n*Var(Xbar)=  5.022 | n*Var(2 sqrt(Xbar))=1.005
  lambda= 50 | n*Var(Xbar)= 50.119 | n*Var(2 sqrt(Xbar))=1.002
Second-order delta: E[W]=-0.2487 (theory -0.25), Var[W]=0.1237 (theory 2/16=0.1250)
```

The normal and t_3 rows are very close to the theoretical ARE values (0.637 and $3{\cdot}4f(0)^2 \approx 1.62$): for the t_3 distribution (finite variance $= 3$, but heavy-tailed) the median is more efficient than the mean. For the Laplace the empirical ARE (≈ 1.73) is below the theoretical value 2; since the Laplace density has a sharp (non-differentiable) peak, $n\,\mathrm{Var}(\text{median})$ approaches its asymptotic value 1 only slowly, at rate $O(n^{-1/2})$. How well asymptotic results hold in finite samples should always be checked by simulation. The superiority of robust estimators for heavy-tailed data is a consequence of this table.

12.7 Comments and Notes

Remark 12.16 (Hodges' Superefficient Estimator). The estimator $T_n = \bar{X}\,\mathbb{I}\{|\bar{X}| \geq n^{-1/4}\}$ behaves like $\bar{X}$ in the $\mathcal{N}(\theta, 1)$ model for $\theta \neq 0$, while at $\theta = 0$ its asymptotic variance is 0: it seemingly "beats" the CRLB. However, its risk deteriorates in a neighbourhood of $\theta = 0$;

Le Cam's theorem [45] shows that superefficiency is possible only on a set of measure zero. This is why asymptotic efficiency is defined together with local uniformity (regularity) (the Hájek–Le Cam convolution theorem).

Remark 12.17 (Expected or Observed Information?). Either $I_n(\hat\theta)$ or $J_n(\hat\theta)$ may be used in the Wald interval; both are asymptotically valid. The observed information is usually more accurate from the point of view of conditional inference, because it reflects ancillary information in the data, and it comes for free as a by-product of the Newton–Raphson algorithm.

Remark 12.18 (Limits of the Delta Method). The delta method is a local linearisation; if g changes rapidly or the variance of T_n is large, the approximation deteriorates. For example, $\hat\tau = \hat p/(1-\hat p)$ has infinite mean in finite samples because $\hat p = 1$ has positive probability; yet its asymptotic distribution is well defined. In such cases the bootstrap (Chapter 18) or profile likelihood intervals are preferred.

12.8 Chapter Summary

- By the information inequality (Lemma 12.4), the expected log-likelihood has its unique maximum at the true parameter. Consistency of the MLE (Theorem 12.5) follows from this.
- In regular models, $\sqrt{n}(\hat\theta_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, 1/I_1(\theta_0))$ (Theorem 12.6): the MLE attains the CRLB asymptotically.
- The Wald interval is built from the expected or the observed information (Corollary 12.7). In small samples its coverage is sensitive to the parametrisation.
- The delta method (Theorem 12.8) carries asymptotic normality over to differentiable transformations. When $g'(\theta) = 0$ the second-order version (Theorem 12.9) is used, and in several dimensions the gradient version (Theorem 12.10). A variance-stabilising transformation makes the asymptotic variance constant.
- Hodges' superefficient estimator shows that asymptotic efficiency must be defined together with a local uniformity condition.

Further reading. [16, Chapter 10], [78, Chapters 3, 5 and 8], [28, Chapters 17–18].

12.9 Exercises

Exercise 12.1. Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$. Find the asymptotic distribution of $\hat\lambda = 1/\bar X$ both by Theorem 12.6 and by the delta method. Show that the asymptotic variance of $\ln\hat\lambda$ does not depend on λ. → Solution

Exercise 12.2. Let (X_i, Y_i) be i.i.d. with $\mathbb{E}X = \mu_X \neq 0$. Find the asymptotic variance of $\bar Y/\bar X$, the estimator of the ratio μ_Y/μ_X, by the multivariate delta method. → Solution

Exercise 12.3. In the Cauchy location model, show (by simulation) that the one-step Newton–Raphson estimator starting from the median, $\tilde\theta = \hat m + \ell'(\hat m)/(nI_1)$ ($I_1 = 1/2$), is asymptotically efficient. Look up the general theory of such "one-step" estimators. → Solution

Exercise 12.4. Using Fisher's z transformation, construct an approximate 95% confidence interval for ρ from bivariate normal data. For $\rho = 0.8$, $n = 20$, compare its coverage by simulation with that of the direct interval $r \pm 1.96(1-r^2)/\sqrt{n}$. → Solution

Part IV

Hypothesis Testing and Interval Estimation

Chapter 13

The Neyman–Pearson Lemma and Most Powerful Tests

Estimation answers the question "what is θ?"; hypothesis testing answers the question "are the data consistent with a particular claim about θ?". The contribution of Neyman and Pearson in 1933 [61] was to set up the testing problem as an optimisation problem: bound the probability of a type I error at a prespecified level α and, subject to this constraint, minimise the type II error (maximise the power). For two simple hypotheses the solution is surprisingly simple: reject H_0 where the likelihood ratio is large. This chapter lays the foundation for the next three (UMP tests, likelihood ratio tests, confidence intervals).

13.1 Definitions

Definition 13.1 (Hypotheses)**.** Let $\Theta = \Theta_0 \cup \Theta_1$ be a disjoint decomposition. $H_0 : \theta \in \Theta_0$ is the *null hypothesis* and $H_1 : \theta \in \Theta_1$ the *alternative hypothesis*. If Θ_j consists of a single point the hypothesis is *simple*, otherwise it is *composite*.

Definition 13.2 (Test Function)**.** A *(randomised) test* is a measurable function $\phi : \mathcal{X}^n \to [0,1]$: when $\mathbf{x}$ is observed, H_0 is rejected with probability $\phi(\mathbf{x})$. If $\phi \in \{0,1\}$ the test is *non-randomised* and $\mathcal{R} = \{\mathbf{x} : \phi(\mathbf{x}) = 1\}$ is its *rejection region*.

Definition 13.3 (Error Types and the Power Function)**.**

$$\beta_\phi(\theta) = \mathbb{E}_\theta[\phi(\mathbf{X})] \qquad \text{(power function)}.$$

For $\theta \in \Theta_0$, $\beta_\phi(\theta)$ is the probability of a *type I error*; for $\theta \in \Theta_1$, $1 - \beta_\phi(\theta)$ is the probability of a *type II error* and $\beta_\phi(\theta)$ is the *power* of the test at θ.

Definition 13.4 (Level and Size)**.** $\sup_{\theta \in \Theta_0} \beta_\phi(\theta)$ is called the *size* of the test. A test with size $\leq \alpha$ is called a *level-α* test.

Definition 13.5 (Most Powerful (MP) Test)**.** For simple hypotheses $H_0 : \theta = \theta_0$, $H_1 : \theta = \theta_1$, a level-α test ϕ^* is *most powerful* (MP) if $\beta_{\phi^*}(\theta_1) \geq \beta_\phi(\theta_1)$ for every level-α test ϕ.

Definition 13.6 (p-Value)**.** Let $\{\mathcal{R}_\alpha\}_{\alpha \in (0,1)}$ be nested rejection regions satisfying $\alpha' < \alpha \implies \mathcal{R}_{\alpha'} \subseteq \mathcal{R}_\alpha$. The value $p(\mathbf{x}) = \inf\{\alpha : \mathbf{x} \in \mathcal{R}_\alpha\}$ is called the *p-value*: the smallest level at which H_0 can be rejected with the observed data. For tests of the form "reject for large T", $p(\mathbf{x}) = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(T(\mathbf{X}) \geq T(\mathbf{x}))$.

13.2 The Neyman–Pearson Lemma

Theorem 13.7 (Neyman–Pearson Lemma [61])**.** *Let $H_0 : \theta = \theta_0$, $H_1 : \theta = \theta_1$ and $f_j(\mathbf{x}) = L(\theta_j; \mathbf{x})$.*

1. ***(Sufficiency)*** *If for some $k \geq 0$ and $\gamma \in [0, 1]$*

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & f_1(\mathbf{x}) > k f_0(\mathbf{x}) \\ \gamma, & f_1(\mathbf{x}) = k f_0(\mathbf{x}) \\ 0, & f_1(\mathbf{x}) < k f_0(\mathbf{x}) \end{cases} \qquad \text{and} \qquad \mathbb{E}_{\theta_0}[\phi^*(\mathbf{X})] = \alpha,$$

   then ϕ^* is a most powerful (MP) level-α test.

2. ***(Existence)*** *For every $\alpha \in (0, 1)$ there is a test of this form with size exactly α.*

3. ***(Necessity)*** *If ϕ is any MP level-α test, then $\phi = \phi^*$ a.e. on the set $\{f_1 \neq k f_0\}$.*

Randomisation is needed only for discrete distributions; for continuous distributions $\mathbb{P}_{\theta_0}(f_1 = k f_0) = 0$ and the test reduces to $\phi^*(\mathbf{x}) = \mathbb{I}\{L(\theta_1)/L(\theta_0) > k\}$.

Proof. (1) Let ϕ be any level-α test. For every $\mathbf{x}$,

$$\big(\phi^*(\mathbf{x}) - \phi(\mathbf{x})\big)\big(f_1(\mathbf{x}) - k f_0(\mathbf{x})\big) \geq 0,$$

because when $f_1 > k f_0$, $\phi^* = 1 \geq \phi$; when $f_1 < k f_0$, $\phi^* = 0 \leq \phi$; and in the case of equality the second factor is zero. Integrating,

$$\beta_{\phi^*}(\theta_1) - \beta_{\phi}(\theta_1) \geq k\big(\beta_{\phi^*}(\theta_0) - \beta_{\phi}(\theta_0)\big) = k\big(\alpha - \beta_{\phi}(\theta_0)\big) \geq 0.$$

(2) Let $R(\mathbf{x}) = f_1(\mathbf{x})/f_0(\mathbf{x})$ (on $f_0 > 0$) and $a(c) = \mathbb{P}_{\theta_0}(R > c)$. a is a non-increasing, right-continuous function of c (the complement of a CDF). Choosing $k = \inf\{c : a(c) \leq \alpha\}$ gives $a(k) \leq \alpha \leq a(k^-) = \mathbb{P}_{\theta_0}(R \geq k)$. If $\mathbb{P}_{\theta_0}(R = k) > 0$ take $\gamma = \frac{\alpha - a(k)}{\mathbb{P}_{\theta_0}(R = k)} \in [0, 1]$, otherwise $\gamma = 0$; the size is then exactly α. (3) If ϕ is MP, the inequality in (1) must be an equality; a non-negative integrand with zero integral is zero a.e., i.e. $\phi = \phi^*$ wherever $f_1 \neq k f_0$. □

Corollary 13.8 (An MP Test Is Unbiased)**.** *If ϕ^* is an MP test of level $\alpha \in (0, 1)$, then $\beta_{\phi^*}(\theta_1) \geq \alpha$; if $f_0 \neq f_1$, the inequality is strict.*

Proof. The test $\phi \equiv \alpha$, which does not use the data at all, has level α and power α; the MP test must be at least as powerful. □

Corollary 13.9 (Expression via a Sufficient Statistic)**.** *If T is sufficient and $g(t; \theta_j)$ is the factor in the factorisation, then the MP test has the form $g(T; \theta_1)/g(T; \theta_0) > k$; that is, it depends only on T.*

Proof. By factorisation, $f(\mathbf{x}; \theta_j) = g(T(\mathbf{x}); \theta_j) h(\mathbf{x})$. For every $\mathbf{x}$ with $h(\mathbf{x}) > 0$, $\frac{f(\mathbf{x};\theta_1)}{f(\mathbf{x};\theta_0)} = \frac{g(T(\mathbf{x});\theta_1)}{g(T(\mathbf{x});\theta_0)}$; the points with $h(\mathbf{x}) = 0$ have probability zero under both hypotheses, and how the test is defined there affects neither its level nor its power. Apply Theorem 13.7. □

Theorem 13.10 (Distribution of the p-Value)**.** *If the distribution of T under (simple) H_0 is continuous and the test has the form "reject for large T", then $p(\mathbf{X}) = 1 - F_0(T(\mathbf{X})) \sim$ Uniform$(0, 1)$ under H_0. For discrete T, $\mathbb{P}_{\theta_0}(p(\mathbf{X}) \leq u) \leq u$ (stochastically larger than uniform).*

Proof. By the probability integral transform (Theorem 2.14), $F_0(T) \sim \text{Uniform}(0, 1)$, and therefore so is $1 - F_0(T)$. □

13.3 Worked Examples

Example 13.11 (Normal Mean: MP Test and the Power Formula)**.** $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ known; $H_0 : \mu = \mu_0$, $H_1 : \mu = \mu_1 > \mu_0$. The likelihood ratio

$$\frac{L(\mu_1)}{L(\mu_0)} = \exp\Big\{\frac{\mu_1 - \mu_0}{\sigma^2}\sum x_i - \frac{n(\mu_1^2 - \mu_0^2)}{2\sigma^2}\Big\}$$

is an increasing function of $\bar{x}$. Hence the MP test has the form $\bar{X} > c$, and $\mathbb{P}_{\mu_0}(\bar{X} > c) = \alpha$ gives $c = \mu_0 + z_\alpha \sigma/\sqrt{n}$. Power:

$$\beta(\mu_1) = \mathbb{P}_{\mu_1}(\bar{X} > c) = 1 - \Phi\Big(z_\alpha - \frac{\sqrt{n}(\mu_1 - \mu_0)}{\sigma}\Big).$$

With $\mu_0 = 0$, $\mu_1 = 1$, $\sigma = 2$, $n = 16$, $\alpha = 0.05$: $c = 1.645 \cdot 0.5 = 0.822$ and $\beta = 1 - \Phi(1.645 - 2) = \Phi(0.355) \approx 0.639$.

Note: the critical value c *does not depend on* μ_1. This observation is the key to the UMP tests of Chapter 14.

Example 13.12 (Sample Size Calculation)**.** In the same problem, the n required for a level-α test with power $1 - \beta$ is

$$z_\alpha - \frac{\sqrt{n}\,\Delta}{\sigma} = -z_\beta \implies n = \Big(\frac{(z_\alpha + z_\beta)\sigma}{\Delta}\Big)^2, \qquad \Delta = \mu_1 - \mu_0.$$

For $\sigma = 2$, $\Delta = 1$, $\alpha = 0.05$ and power 0.80 ($z_{0.20} = 0.842$), $n = (2.487 \cdot 2)^2 \approx 24.7$, i.e. $n = 25$.

Example 13.13 (Poisson: A Randomised MP Test)**.** $X_1, \dots, X_{10} \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$, $H_0 : \lambda = 1$, $H_1 : \lambda = 2$, $\alpha = 0.05$. $T = \sum X_i$ is sufficient and the likelihood ratio $(2/1)^T e^{-10}$ is increasing in T. Under H_0, $T \sim \text{Poisson}(10)$. Since $\mathbb{P}_0(T \geq 16) \approx 0.0487 < 0.05 < \mathbb{P}_0(T \geq 15) \approx 0.0835$ (values verified with the code below), no non-randomised test can have size exactly 0.05. The MP test is

$$\phi^*(T) = \begin{cases} 1, & T \geq 16 \\ \gamma, & T = 15 \\ 0, & T \leq 14 \end{cases}, \qquad \gamma = \frac{0.05 - \mathbb{P}_0(T \geq 16)}{\mathbb{P}_0(T = 15)}.$$

Example 13.14 (Normal Variance)**.** $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, $H_0 : \sigma^2 = \sigma_0^2$, $H_1 : \sigma^2 = \sigma_1^2 > \sigma_0^2$. The likelihood ratio is increasing in $\sum x_i^2$; since $\sum X_i^2/\sigma_0^2 \sim \chi^2_n$ under H_0, the MP test rejects when $\sum X_i^2 > \sigma_0^2 \chi^2_{n,\alpha}$.

13.4 Computational Applications

13.4.1 Power of the Test for a Normal Mean

code/ch13/c13_1_power_normal_mean_test.py

```python
import numpy as np
import scipy.stats as stats

rng = np.random.default_rng(42)
s_h0 = rng.normal(0.0, 2.0, size=(150_000, 16))
s_h1 = rng.normal(1.0, 2.0, size=(150_000, 16))
c = stats.norm.ppf(0.95) * (2.0 / np.sqrt(16))
print(f"Critical value c = {c:.4f}")
```

```
print(f"Type I error (MC): {np.mean(np.mean(s_h0, axis=1) > c):.4f}  (target 0.05)")
print(f"Neyman-Pearson power: {np.mean(np.mean(s_h1, axis=1) > c):.4f}")
print(f"Analytic power       : {1 - stats.norm.cdf(stats.norm.ppf(0.95) - np.sqrt(16)
    *1.0/2.0):.4f}")
za, zb = stats.norm.ppf(0.95), stats.norm.ppf(0.80)
print(f"Required n for power 0.80 = {((za + zb) * 2.0 / 1.0)**2:.2f}")
```

Output

```
Critical value c = 0.8224
Type I error (MC): 0.0497  (target 0.05)
Neyman-Pearson power: 0.6397
Analytic power       : 0.6388
Required n for power 0.80 = 24.73
```

13.4.2 A Randomised Poisson Test

code/ch13/c13_2_randomised_poisson_test.py

```
import numpy as np
from scipy import stats

alpha, n, l0, l1 = 0.05, 10, 1.0, 2.0
T0, T1 = stats.poisson(n*l0), stats.poisson(n*l1)
c = int(T0.ppf(1 - alpha))            # smallest c: P0(T > c) <= alpha
while T0.sf(c) > alpha: c += 1
gamma = (alpha - T0.sf(c)) / T0.pmf(c)
print(f"Reject: T > {c} ; if T = {c} reject with probability gamma = {gamma:.4f}")
print(f"P0(T >= {c+1}) = {T0.sf(c):.4f} | P0(T >= {c}) = {T0.sf(c-1):.4f}")
print(f"Size = {T0.sf(c) + gamma*T0.pmf(c):.4f}")
print(f"Power (randomised)        = {T1.sf(c) + gamma*T1.pmf(c):.4f}")
print(f"Power (T > {c}, conservative) = {T1.sf(c):.4f}")

rng = np.random.default_rng(42)
T = rng.poisson(n*l1, size=1_000_000)
rej = (T > c) | ((T == c) & (rng.random(T.size) < gamma))
print(f"Power (simulation)        = {rej.mean():.4f}")
```

Output

```
Reject: T > 15 ; if T = 15 reject with probability gamma = 0.0363
P0(T >= 16) = 0.0487 | P0(T >= 15) = 0.0835
Size = 0.0500
Power (randomised)        = 0.8454
Power (T > 15, conservative) = 0.8435
Power (simulation)        = 0.8452
```

13.4.3 Power Curves and the Distribution of p-Values

code/ch13/c13_3_power_curves_pvalue_distribution.py

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
sigma, za = 2.0, stats.norm.ppf(0.95)
mu = np.linspace(-0.5, 2.5, 300)
fig, ax = plt.subplots(1, 2, figsize=(10, 3.4))
for n in [4, 16, 64]:
    ax[0].plot(mu, 1 - stats.norm.cdf(za - np.sqrt(n)*mu/sigma), label=f"n={n}")
```

```
ax[0].axhline(0.05, color="gray", ls=":"); ax[0].set_xlabel(r"$\mu$"); ax[0].set_ylabel("
    power")
ax[0].set_title(r"$H_0:\mu=0$, $\alpha=0.05$"); ax[0].legend()

for mu1, col in [(0.0, "tab:blue"), (0.5, "tab:red")]:
    xb = rng.normal(mu1, sigma/4, size=100_000)
    p = 1 - stats.norm.cdf(xb / (sigma/4))
    ax[1].hist(p, bins=40, range=(0, 1), density=True, alpha=0.6, color=col, label=rf"$\mu={
    mu1}$")
    print(f"mu={mu1}: P(p <= 0.05) = {np.mean(p <= 0.05):.4f}")
ax[1].set_xlabel("p-value"); ax[1].set_title("distribution of p-values (n=16)"); ax[1].legend
    ()
fig.tight_layout(); fig.savefig("guc_pdeger.pdf")
```

Output

```
mu=0.0: P(p <= 0.05) = 0.0503
mu=0.5: P(p <= 0.05) = 0.2606
```

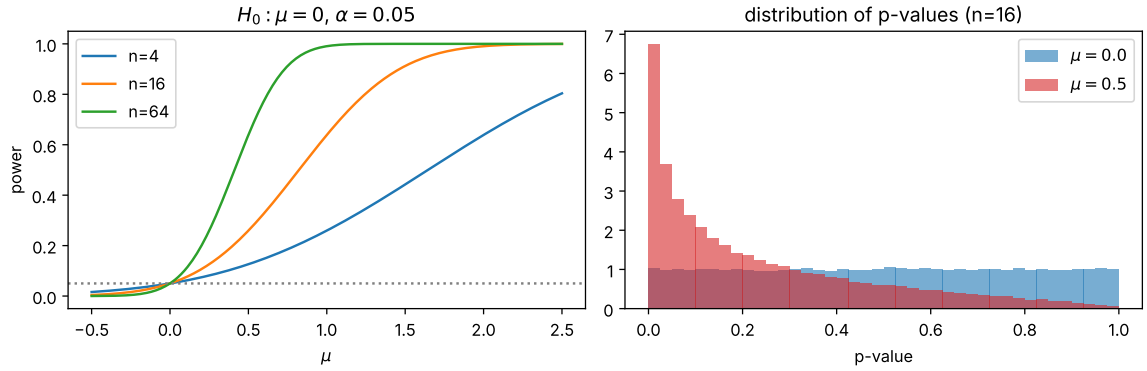


Figure 13.1: Left: power functions of the one-sided z-test; all curves pass through α at $\mu = 0$ and become steeper as n grows. Right: when H_0 is true the p-values are uniformly distributed (Theorem 13.10); under H_1 they pile up near zero.

13.5 Comments and Notes

Remark 13.15 (An Asymmetric Approach). The Neyman–Pearson framework does not treat H_0 and H_1 symmetrically: the type I error is kept under control, while the type II error is only minimised. Therefore "H_0 could not be rejected" does not mean "H_0 is true"; a low-power test frequently fails to reject H_0. This is why power analysis is an integral part of experimental design.

Remark 13.16 (Randomisation in Practice). Randomised tests are needed in theory (for the existence of an MP test), but they are rarely used in practice, because the same data can lead to different decisions. Practitioners report either the conservative test ($\gamma = 0$) or the p-value; the "mid-p" value $\mathbb{P}(T > t) + \frac{1}{2}\mathbb{P}(T = t)$ is a compromise.

Remark 13.17 (What Does a p-Value Not Measure?). The p-value is *not* the probability that H_0 is true; it is the probability, when H_0 is true, of obtaining a result at least as extreme as the one observed. Nor does it measure effect size: for very large n, even negligibly small differences produce very small p-values. The 2016 statement of the American Statistical Association [83] discusses these misinterpretations in detail. The relationship between p-values and posterior probabilities is taken up in Chapter 19.

13.6 Chapter Summary

- A test is a decision rule. The probability of a type I error is bounded by the level, and the power is maximised subject to this constraint.
- The Neyman–Pearson lemma (Theorem 13.7): for simple hypotheses the most powerful test is the likelihood ratio test. For discrete distributions, randomisation may be needed to attain the level exactly.
- An MP test is unbiased (Corollary 13.8) and depends only on a sufficient statistic (Corollary 13.9).
- The p-value (Definition 13.6) is the probability under H_0 of a result at least as extreme as the one observed; for a continuous test statistic it is uniformly distributed under H_0 (Theorem 13.10). The p-value is not the probability that H_0 is true, and it does not measure effect size.

Further reading. [16, Chapter 8], [47, Chapter 3], [81, Chapter 10].

13.7 Exercises

Exercise 13.1. $X \sim \text{Uniform}(0, \theta)$, a single observation. Find the MP level-α test of $H_0 : \theta = 1$ against $H_1 : \theta = 2$. Show that the likelihood ratio (on the region where $f_0 > 0$ or $f_1 > 0$) takes only two values ($1/2$ and ∞) and that the MP test is not unique. → Solution

Exercise 13.2. $X_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$, $n = 20$, $H_0 : p = 0.5$, $H_1 : p = 0.7$. Construct the randomised MP test at level $\alpha = 0.05$ and compute its power. → Solution

Exercise 13.3. $H_0 : X \sim \mathcal{N}(0, 1)$, $H_1 : X \sim \text{Cauchy}(0, 1)$ (a single observation). Show that the rejection region of the MP test has the form $\{|x| > c\}$ and compute the power for $\alpha = 0.05$. (Is the likelihood ratio a monotone function of x^2?) → Solution

Exercise 13.4. Using Theorem 13.10, show that for the p-values of m independent tests the Bonferroni correction (performing each test at level α/m) bounds the family-wise error rate by α (Chapter 1, Boole's inequality). → Solution

Chapter 14

The Karlin–Rubin Theorem and UMP Tests

The Neyman–Pearson lemma gives the optimal test for two *simple* hypotheses. In real problems, however, the alternative is almost always composite, as in "the new drug is more effective" ($\theta > \theta_0$). A test that is most powerful simultaneously at every point of a composite alternative is called a *uniformly most powerful* (UMP) test. The existence of such a test is a strong requirement; the Karlin–Rubin theorem shows that it is met for one-sided hypotheses in families whose likelihood ratio is monotone in a statistic. In this chapter we also discuss why no UMP test exists in two-sided problems and how the constraint of unbiasedness fills this gap.

14.1 Definitions

Definition 14.1 (Uniformly Most Powerful (UMP) Test)**.** A level-α test ϕ^* is called a level-α *UMP test* of $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ if $\beta_{\phi^*}(\theta) \geq \beta_\phi(\theta)$ for every level-α test ϕ and *every* $\theta \in \Theta_1$.

Definition 14.2 (Monotone Likelihood Ratio (MLR))**.** Let $\Theta \subseteq \mathbb{R}$. If for every $\theta_1 < \theta_2$ the likelihood ratio

$$\frac{L(\theta_2; \mathbf{x})}{L(\theta_1; \mathbf{x})}$$

(on $\{\mathbf{x} : L(\theta_1; \mathbf{x}) > 0 \text{ or } L(\theta_2; \mathbf{x}) > 0\}$) is a *non-decreasing* function of a statistic $T(\mathbf{x})$, the family is said to have a *monotone likelihood ratio* (MLR) in T. If the ratio is a non-increasing function of T, the family has MLR in $-T$.

Definition 14.3 (Unbiased Test)**.** If $\sup_{\theta \in \Theta_0} \beta_\phi(\theta) \leq \alpha \leq \inf_{\theta \in \Theta_1} \beta_\phi(\theta)$, then ϕ is an *unbiased* level-α test: it rejects H_0 more often when it is false than when it is true. A test that is UMP among unbiased tests is called a *UMPU* test.

14.2 Main Theorems and Proofs

Proposition 14.4 (MLR in the Exponential Family)**.** *Let $f(x;\theta) = h(x)c(\theta)\exp\{w(\theta)t(x)\}$ be a one-parameter exponential family and $T = \sum t(X_i)$. If w is non-decreasing the family has MLR in T; if it is non-increasing, in $-T$.*

Proof. For $\theta_1 < \theta_2$, $\frac{L(\theta_2)}{L(\theta_1)} = \left(\frac{c(\theta_2)}{c(\theta_1)}\right)^n \exp\{[w(\theta_2) - w(\theta_1)]T\}$; the coefficient in the exponent is ≥ 0 if w is non-decreasing. $\square$

Theorem 14.5 (Karlin–Rubin Theorem [42])**.** *Let the family have MLR in T, and test $H_0 : \theta \le \theta_0$ against $H_1 : \theta > \theta_0$. For $\alpha \in (0,1)$ the test*

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & T(\mathbf{x}) > c \\ \gamma, & T(\mathbf{x}) = c \\ 0, & T(\mathbf{x}) < c \end{cases}, \qquad \mathbb{E}_{\theta_0}[\phi^*] = \alpha$$

(for continuous T, briefly "reject if $T(\mathbf{X}) > c$") is a UMP level-α test. Moreover the power function $\beta_{\phi^*}(\theta)$ is non-decreasing, and strictly increasing at every point where $\beta_{\phi^*}(\theta) < 1$.

If the family has MLR in $-T$ (or if the hypotheses are $H_0 : \theta \ge \theta_0$, $H_1 : \theta < \theta_0$), the inequalities are reversed: "reject if $T(\mathbf{X}) < c$".

Proof. Step 1 (MP against each simple alternative). Fix $\theta_1 > \theta_0$. By MLR, $L(\theta_1)/L(\theta_0) = g(T)$ with g non-decreasing. Taking $k = g(c)$, $\{T > c\} \subseteq \{g(T) \ge k\}$ and $\{g(T) > k\} \subseteq \{T > c\}$ or $\{T \ge c\}$; hence ϕ^* has the Neyman–Pearson form (Theorem 13.7(1)) and is MP at level α for θ_0 against θ_1. The crucial point: c and γ are determined solely by the condition $\mathbb{E}_{\theta_0}\phi^* = \alpha$ and *do not depend on* θ_1.

Step 2 (the power is monotone). Take $\theta' < \theta''$. By the reasoning of Step 1, ϕ^* is MP at level $\beta_{\phi^*}(\theta')$ for θ' against θ''. By Corollary 13.8, $\beta_{\phi^*}(\theta'') \ge \beta_{\phi^*}(\theta')$.

Step 3 (level α). By Step 2, $\sup_{\theta \le \theta_0} \beta_{\phi^*}(\theta) = \beta_{\phi^*}(\theta_0) = \alpha$.

Step 4 (UMP). Let ϕ be any level-α test for the composite H_0. In particular $\beta_\phi(\theta_0) \le \alpha$, i.e. ϕ also has level α for the simple $H_0 : \theta = \theta_0$. By Step 1, $\beta_{\phi^*}(\theta_1) \ge \beta_\phi(\theta_1)$ for every $\theta_1 > \theta_0$. □

Proposition 14.6 (No UMP Test in the Two-Sided Problem)**.** *Let $X_i \overset{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ known, $H_0 : \mu = \mu_0$, $H_1 : \mu \ne \mu_0$, $\alpha \in (0,1)$. There is no UMP test for this problem.*

Proof. If a UMP ϕ existed, it would have to be MP for every $\mu_1 > \mu_0$; by the necessity part of Neyman–Pearson (Theorem 13.7(3)), $\phi = \mathbb{I}\{\bar{X} > \mu_0 + z_\alpha \sigma/\sqrt{n}\}$ a.e. Likewise it would have to be MP for every $\mu_1 < \mu_0$, i.e. $\phi = \mathbb{I}\{\bar{X} < \mu_0 - z_\alpha \sigma/\sqrt{n}\}$ a.e. These two tests differ on a set of positive probability; contradiction. □

Theorem 14.7 (Two-Sided UMPU Test (Exponential Family); see [47, Section 4.4])**.** *In a one-parameter exponential family (natural parameter θ, sufficient statistic T), for $H_0 : \theta = \theta_0$, $H_1 : \theta \ne \theta_0$ the test of the form*

$$\phi(T) = \begin{cases} 1, & T < c_1 \text{ or } T > c_2 \\ \gamma_i, & T = c_i \\ 0, & c_1 < T < c_2 \end{cases}$$

is UMPU under the conditions $\mathbb{E}_{\theta_0}\phi = \alpha$ and $\mathbb{E}_{\theta_0}[T\phi(T)] = \alpha \mathbb{E}_{\theta_0} T$. In symmetric cases (e.g. the normal mean) these conditions reduce to the equal-tailed test: $|\bar{X} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}$.

Main steps of the proof. 1. Unbiasedness reduces to two constraints. In the exponential family the power function $\beta_\phi(\theta) = \mathbb{E}_\theta \phi(T)$ of every test ϕ is differentiable, and $\beta'_\phi(\theta) = \mathbb{E}_\theta[T\phi] - \mathbb{E}_\theta T \cdot \mathbb{E}_\theta \phi$ (differentiation under the integral sign, permitted in the interior of the natural parameter space). If ϕ is unbiased, then $\beta_\phi(\theta) \ge \alpha$ ($\theta \ne \theta_0$) and by continuity $\beta_\phi(\theta_0) = \alpha$; hence β_ϕ has a minimum at θ_0 and $\beta'_\phi(\theta_0) = 0$. These two conditions are exactly the constraints in the theorem. So every unbiased level-α test satisfies these constraints.

2. The solution of the constrained problem does not depend on θ_1. For fixed $\theta_1 \ne \theta_0$, by the generalised Neyman–Pearson lemma [47, Theorem 3.6.1], the test maximising $\mathbb{E}_{\theta_1}\phi$ under the two constraints rejects, for some constants k_1, k_2, when

$$f(\mathbf{x};\theta_1) > (k_1 + k_2 T(\mathbf{x})) f(\mathbf{x};\theta_0) \iff e^{(\theta_1 - \theta_0)T} > a + bT.$$

Since the left-hand side is a strictly convex function of T, the set where it lies above a line is the complement of an interval: $\{T < c_1\} \cup \{T > c_2\}$. c_1, c_2 and γ_i are determined by the two constraints, which depend only on θ_0; hence the same test is most powerful for every θ_1.

3. The test itself is unbiased. The test $\phi \equiv \alpha$ also satisfies both constraints; by Step 2 the power of the test found is at least α at every θ_1. Consequently the test is uniformly most powerful among the tests satisfying the constraints, this class contains all unbiased tests, and the test itself is unbiased: it is UMPU. For details (in particular the existence of k_1, k_2) see [47, Sections 4.2–4.4].

The symmetric case. For the normal mean, $Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma$ is symmetric under H_0; with $c_1 = -c_2$ the second constraint $\mathbb{E}_{\theta_0}[Z\phi(Z)] = 0$ holds automatically by symmetry, and the first constraint gives $c_2 = z_{\alpha/2}$. □

14.3 Worked Examples

Example 14.8 (The Rate Parameter of the Exponential Distribution)**.** $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\theta)$, $f(x;\theta) = \theta e^{-\theta x}$, $x > 0$. Here $w(\theta) = -\theta$ is *decreasing*, so the family has MLR in $-T$ ($T = \sum X_i$). For $H_0 : \theta \le \theta_0$, $H_1 : \theta > \theta_0$ the UMP test rejects, in line with intuition, for *small* T: a large rate means short waiting times. Since $2\theta T \sim \chi^2_{2n}$,

$$T < c = \frac{\chi^2_{2n,\,1-\alpha}}{2\theta_0} \quad (\chi^2_{2n,1-\alpha} : \text{ lower } \alpha \text{ quantile}), \qquad \beta(\theta) = \mathbb{P}(\chi^2_{2n} < \theta\,\chi^2_{2n,1-\alpha}/\theta_0).$$

For $n = 10$, $\theta_0 = 1$, $\alpha = 0.05$, the lower 5% quantile of χ^2_{20} is 10.851 and $c \approx 5.425$; the power at $\theta = 1.5$ is $\mathbb{P}(\chi^2_{20} < 16.28)$ (computed with the code below).

In terms of the mean $\mu = 1/\theta$, the same test takes the form "$H_0 : \mu \ge \mu_0$ against $H_1 : \mu < \mu_0$; reject if $\bar{X}$ is small" — the test does not change even though the parametrisation does.

Example 14.9 (Binomial Proportion)**.** $X \sim \text{Bin}(n, p)$: $w(p) = \ln \frac{p}{1-p}$ is increasing, and the MLR is in X. For $H_0 : p \le p_0$, $H_1 : p > p_0$ the UMP test rejects for large X; because X is discrete, randomisation at $X = c$ is needed for exact size.

Example 14.10 (Uniform$(0, \theta)$)**.** For $\theta_1 < \theta_2$,

$$\frac{L(\theta_2)}{L(\theta_1)} = \Big(\frac{\theta_1}{\theta_2}\Big)^n \frac{\mathbb{I}\{x_{(n)} < \theta_2\}}{\mathbb{I}\{x_{(n)} < \theta_1\}} = \begin{cases} (\theta_1/\theta_2)^n, & x_{(n)} < \theta_1 \\ \infty, & \theta_1 \le x_{(n)} < \theta_2 \end{cases}$$

is non-decreasing in $X_{(n)}$ (there is MLR although this is not an exponential family). For $H_0 : \theta \le \theta_0$ the UMP test rejects when $X_{(n)} > \theta_0(1-\alpha)^{1/n}$; its power for $\theta > \theta_0$ is $1 - (1-\alpha)(\theta_0/\theta)^n$.

Example 14.11 (No MLR in the Cauchy Location Family)**.** $f(x;\theta) = \frac{1}{\pi(1+(x-\theta)^2)}$, a single observation. $\frac{f(x;\theta_2)}{f(x;\theta_1)} = \frac{1+(x-\theta_1)^2}{1+(x-\theta_2)^2} \to 1$ as $x \to \pm\infty$, while in between it takes values larger than 1; hence it is not monotone in x. In this model there is no UMP test even for one-sided hypotheses: the MP test against θ_1 depends on θ_1.

14.4 Computational Applications

14.4.1 The Exponential Rate Test: Simulation and Exact Power

code/ch14/c14_1_exponential_rate_test_exact_power.py

```python
import numpy as np
import scipy.stats as stats

rng = np.random.default_rng(42)
c = stats.chi2.ppf(0.05, df=20) / 2.0
samples = rng.exponential(scale=1.0/1.5, size=(100_000, 10))
print(f"theta=1.5 power: {np.mean(np.sum(samples, axis=1) < c):.4f}")
print(f"Critical value c = {c:.4f} | exact power = P(chi2_20 < 3c) = {stats.chi2.cdf(1.5*2*c,
    20):.4f}")
h0 = rng.exponential(scale=1.0, size=(100_000, 10))
print(f"rejection rate at theta=1.0 (size): {np.mean(h0.sum(axis=1) < c):.4f}")
for th in [0.5, 0.8, 1.0, 1.5, 2.0, 3.0]:
    print(f"  beta({th:.1f}) = {stats.chi2.cdf(2*th*c, 20):.4f}")
```

Output

```
theta=1.5 power: 0.3003
Critical value c = 5.4254 | exact power = P(chi2_20 < 3c) = 0.3007
rejection rate at theta=1.0 (size): 0.0512
  beta(0.5) = 0.0005
  beta(0.8) = 0.0137
  beta(1.0) = 0.0500
  beta(1.5) = 0.3007
  beta(2.0) = 0.6431
  beta(3.0) = 0.9622
```

The power function stays below $\alpha = 0.05$ on the region $\theta \leq 1$ and increases with θ — Steps 2 and 3 of Theorem 14.5.

14.4.2 The Two-Sided Problem: One-Sided Tests Cannot Be UMP

code/ch14/c14_2_two_sided_problem_one_sided_tests.py

```python
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

sigma, n, a = 1.0, 25, 0.05
d = np.linspace(-1, 1, 401) * np.sqrt(n) / sigma        # sqrt(n)(mu - mu0)/sigma
za, za2 = stats.norm.ppf(1 - a), stats.norm.ppf(1 - a/2)
upp  = 1 - stats.norm.cdf(za - d)
low  = stats.norm.cdf(-za - d)
two  = 1 - stats.norm.cdf(za2 - d) + stats.norm.cdf(-za2 - d)
fig, ax = plt.subplots(figsize=(7, 3.4))
mu = d * sigma / np.sqrt(n)
ax.plot(mu, upp, label="Upper one-sided ($\\bar X > c$)")
ax.plot(mu, low, label="Lower one-sided ($\\bar X < -c$)")
ax.plot(mu, two, "k", lw=2, label="Two-sided (UMPU)")
ax.axhline(a, color="gray", ls=":"); ax.set_xlabel(r"$\mu-\mu_0$"); ax.set_ylabel("power");
    ax.legend(fontsize=8)
fig.tight_layout(); fig.savefig("iki_tarafli.pdf")
for m in [-0.4, 0.2, 0.4]:
    k = np.argmin(np.abs(mu - m))
    print(f"mu-mu0={m:+.1f} | upper={upp[k]:.3f} | lower={low[k]:.3f} | two-sided={two[k]:.3f
    }")
```

Output

```
mu-mu0=-0.4 | upper=0.000 | lower=0.639 | two-sided=0.516
mu-mu0=+0.2 | upper=0.260 | lower=0.004 | two-sided=0.170
mu-mu0=+0.4 | upper=0.639 | lower=0.000 | two-sided=0.516
```

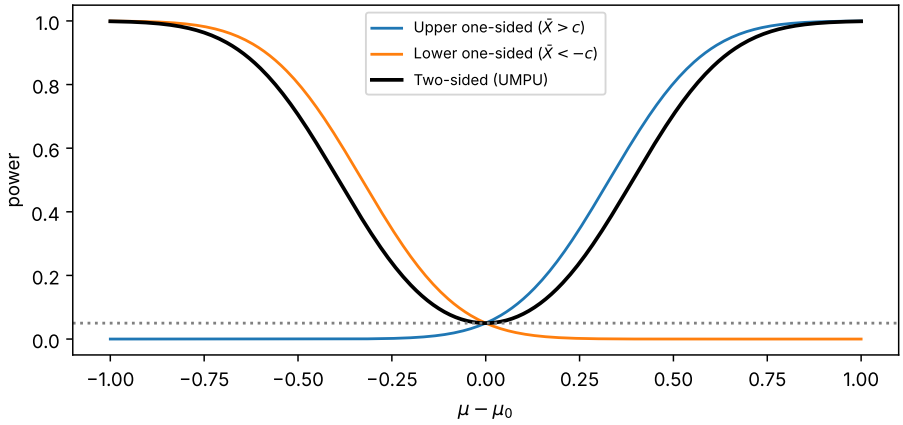


Figure 14.1: Power functions of three tests for $H_1 : \mu \neq \mu_0$ ($n = 25$, $\sigma = 1$). Each one-sided test is more powerful than the two-sided test in its own direction, but in the other direction its power falls even below α (it is biased). No test is most powerful at every $\mu \neq \mu_0$; the two-sided test is most powerful among unbiased tests.

14.4.3 Uniform: MLR and the Power of the UMP Test

code/ch14/c14_3_uniform_mlr_ump_power.py

```
import numpy as np

rng = np.random.default_rng(42)
theta0, n, a, R = 1.0, 8, 0.05, 200_000
c = theta0 * (1 - a)**(1/n)
for th in [0.9, 1.0, 1.1, 1.3]:
    M = rng.uniform(0, th, size=(R, n)).max(axis=1)
    theory = max(0.0, 1 - (1 - a)*(theta0/th)**n) if th > theta0 else (1 - (c/th)**n if c < th else 0.0)
    print(f"theta={th:.1f} | rejection rate={np.mean(M > c):.4f} | theory={theory:.4f}")
```

Output

```
theta=0.9 | rejection rate=0.0000 | theory=0.0000
theta=1.0 | rejection rate=0.0505 | theory=0.0500
theta=1.1 | rejection rate=0.5587 | theory=0.5568
theta=1.3 | rejection rate=0.8836 | theory=0.8835
```

14.5 Comments and Notes

Remark 14.12 (The Rarity of UMP Tests). UMP tests exist essentially in one-parameter families with MLR and for one-sided hypotheses. In multiparameter problems (e.g. testing a normal mean with σ unknown) the presence of a nuisance parameter usually rules out a UMP test; one then uses either (a) optimality under additional constraints such as unbiasedness or invariance (UMPU, UMPI; the t-test is optimal in this sense) or (b) the general likelihood ratio test of Chapter 15.

Remark 14.13 (One-Sided or Two-Sided?). A one-sided test is more powerful when its direction is scientifically justified in advance (before seeing the data). Choosing the direction after looking at the data effectively raises the type I error to 2α. For this reason regulators (e.g. for clinical trials) usually require two-sided tests or one-sided tests at level $\alpha/2$.

14.6 Chapter Summary

- For composite alternatives, a test that is most powerful simultaneously for all parameter values is called UMP. Such a test exists only in special structures.
- A monotone likelihood ratio (Definition 14.2) provides this structure. In the exponential family there is MLR if $w(\theta)$ is monotone (Proposition 14.4).
- The Karlin–Rubin theorem (Theorem 14.5): in MLR families the test "$T > c$" is UMP for one-sided hypotheses.
- There is no UMP test in the two-sided problem (Proposition 14.6). Adding the constraint of unbiasedness yields a UMPU test in the exponential family (Theorem 14.7).

Further reading. [16, Section 8.3], [47, Chapters 3–4].

14.7 Exercises

Exercise 14.1. $X_i \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$, $H_0 : \lambda \le 1$, $H_1 : \lambda > 1$, $n = 20$, $\alpha = 0.05$. Construct the UMP randomised test and plot its power function on the interval $\lambda \in [0.5,\, 2]$. → Solution

Exercise 14.2. For $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, find the UMP test against $H_0 : \sigma \le \sigma_0$. Show numerically that for $H_0 : \sigma = \sigma_0$, $H_1 : \sigma \ne \sigma_0$ the equal-tailed chi-square test is *biased*. → Solution

Exercise 14.3. Show that the logistic location family $f(x;\theta) = \frac{e^{-(x-\theta)}}{(1+e^{-(x-\theta)})^2}$ has MLR in x (a single observation). Discuss why for $n > 1$ there is no MLR in $\sum X_i$, so that Karlin–Rubin cannot be applied directly. → Solution

Exercise 14.4. Using Step 2 of the proof of Theorem 14.5, show that in a family with MLR the p-value is $p(\mathbf{x}) = \mathbb{P}_{\theta_0}(T \ge T(\mathbf{x}))$ and that, for the composite $H_0 : \theta \le \theta_0$, the supremum is attained at θ_0. → Solution

Chapter 15

The Likelihood Ratio Test and Wilks' Theorem

UMP tests are elegant but rare. For most hypotheses met in practice — composite null hypotheses, nuisance parameters, multidimensional constraints — a general method is needed. The generalised likelihood ratio test (LRT) extends the Neyman–Pearson ratio naturally to composite hypotheses: it compares the best fits of the restricted and unrestricted models. Wilks' theorem says that, under H_0, this test statistic converges to a chi-square distribution independently of the model. Together with the Wald and score tests, the LRT forms the "holy trinity" of asymptotic tests; almost all tests in regression, GLMs and goodness of fit are special cases of this trinity.

15.1 Definitions

Definition 15.1 (Generalised Likelihood Ratio)**.** For $H_0 : \boldsymbol{\theta} \in \Theta_0$, $H_1 : \boldsymbol{\theta} \in \Theta \setminus \Theta_0$,

$$\Lambda(\mathbf{x}) = \frac{\sup_{\boldsymbol{\theta}\in\Theta_0} L(\boldsymbol{\theta};\mathbf{x})}{\sup_{\boldsymbol{\theta}\in\Theta} L(\boldsymbol{\theta};\mathbf{x})} = \frac{L(\hat{\boldsymbol{\theta}}_0;\mathbf{x})}{L(\hat{\boldsymbol{\theta}};\mathbf{x})} \in [0,1]$$

is the *generalised likelihood ratio* ($\hat{\boldsymbol{\theta}}_0$: restricted MLE, $\hat{\boldsymbol{\theta}}$: unrestricted MLE). The *likelihood ratio test* (LRT) rejects H_0 when $\Lambda(\mathbf{x}) \le k$. Equivalently, it rejects when

$$G^2 = -2\ln\Lambda = 2\big[\ell(\hat{\boldsymbol{\theta}}) - \ell(\hat{\boldsymbol{\theta}}_0)\big] \ge 0$$

is large.

Definition 15.2 (Wald and Score (Rao) Tests)**.** For the simple $H_0 : \theta = \theta_0$ (one parameter),

$$W = (\hat{\theta} - \theta_0)^2\, I_n(\hat{\theta}) \qquad \text{(Wald)}, \qquad S = \frac{\big[\ell'(\theta_0)\big]^2}{I_n(\theta_0)} \qquad \text{(score / Lagrange multiplier)}.$$

In the multidimensional case $W = (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)^\top I_n(\hat{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ and $S = \nabla\ell(\boldsymbol{\theta}_0)^\top I_n(\boldsymbol{\theta}_0)^{-1}\nabla\ell(\boldsymbol{\theta}_0)$. The Wald test requires fitting only the unrestricted model, the score test only the restricted one; the LRT requires both.

15.2 Main Theorems and Proofs

Proposition 15.3 (The LRT Is Based on a Sufficient Statistic)**.** *If T is sufficient, then $\Lambda(\mathbf{x})$ is a function of $T(\mathbf{x})$ alone.*

Proof. By factorisation $L(\boldsymbol{\theta};\mathbf{x}) = g(T(\mathbf{x}),\boldsymbol{\theta})h(\mathbf{x})$; $h(\mathbf{x})$ cancels between numerator and denominator. □

Proposition 15.4 (For Simple Hypotheses, LRT = NP Test)**.** *If $\Theta = \{\theta_0, \theta_1\}$, then*

$$\Lambda = \min\Big\{1,\, \frac{L(\theta_0)}{L(\theta_1)}\Big\},$$

and $\Lambda \le k < 1$ and $L(\theta_1)/L(\theta_0) \ge 1/k$ give the same rejection region. The LRT is the natural generalisation of the Neyman–Pearson test.

Proof. Since $\Theta_0 = \{\theta_0\}$ and $\Theta = \{\theta_0, \theta_1\}$,

$$\Lambda = \frac{L(\theta_0)}{\max\{L(\theta_0), L(\theta_1)\}} = \min\Big\{1, \frac{L(\theta_0)}{L(\theta_1)}\Big\}.$$

For $k < 1$, $\Lambda \le k$ can occur only when the minimum comes from the second term, which means $L(\theta_0)/L(\theta_1) \le k$, i.e. $L(\theta_1)/L(\theta_0) \ge 1/k$. □

Theorem 15.5 (Wilks' Theorem [87])**.** *Let $X_i \overset{i.i.d.}{\sim} f(\cdot;\boldsymbol{\theta})$ with $\Theta \subseteq \mathbb{R}^k$ open, and suppose the conditions for asymptotic normality of the MLE of Chapter 12 hold. Let Θ_0 be a smooth $(k-r)$-dimensional submanifold of Θ (e.g. a subspace in which r components are fixed), and let the true $\boldsymbol{\theta}_0$ be an interior point of Θ_0. Then under H_0*

$$G_n^2 = -2\ln\Lambda_n(\mathbf{X}) \xrightarrow{d} \chi^2_r, \qquad r = \dim(\Theta) - \dim(\Theta_0).$$

Proof (simple H_0, $k = r = 1$). Expand $\ell(\theta_0)$ in a Taylor series around $\hat\theta$; since $\ell'(\hat\theta) = 0$, for some θ^* between the two,

$$\ell(\theta_0) = \ell(\hat\theta) + \tfrac{1}{2}(\theta_0 - \hat\theta)^2\ell''(\theta^*).$$

Hence

$$G^2 = 2[\ell(\hat\theta) - \ell(\theta_0)] = \Big[\sqrt{n}(\hat\theta - \theta_0)\Big]^2 \cdot \Big(-\frac{1}{n}\ell''(\theta^*)\Big).$$

By Theorem 12.6, $\sqrt{n}(\hat\theta - \theta_0)\sqrt{I_1(\theta_0)} \xrightarrow{d} \mathcal{N}(0,1)$; since the MLE is consistent, $\theta^* \xrightarrow{P} \theta_0$ and (by the third-derivative condition) $-\frac{1}{n}\ell''(\theta^*) \xrightarrow{P} I_1(\theta_0)$. By Slutsky and the continuous mapping theorem, $G^2 = \big[\sqrt{nI_1}(\hat\theta - \theta_0)\big]^2 \cdot \frac{-\ell''(\theta^*)/n}{I_1} \xrightarrow{d} Z^2 \cdot 1 = \chi^2_1$.

In the general case the same expansion gives $G^2 \approx \sqrt{n}(\hat{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}_0)^\top I_1 \sqrt{n}(\hat{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}_0)$; this is the squared norm of the r-dimensional component of an $\mathcal{N}_k(\mathbf{0}, I_1^{-1})$ vector orthogonal (in the Fisher metric) to the tangent space of Θ_0, and it has a χ^2_r distribution. □

Theorem 15.6 (Asymptotic Equivalence: LRT, Wald, Score)**.** *Under the conditions of Wilks' theorem and when H_0 is true, $W_n - G_n^2 \xrightarrow{P} 0$ and $S_n - G_n^2 \xrightarrow{P} 0$; hence all three statistics have the limit distribution χ^2_r. Moreover, under $\sqrt{n}$-local alternatives $\boldsymbol{\theta}_n = \boldsymbol{\theta}_0 + \mathbf{h}/\sqrt{n}$ all three converge to the same non-central $\chi^2_r(\delta)$ distribution; that is, their asymptotic powers are equal.*

Idea of proof ($r = k = 1$). From the proof of Wilks' theorem, $G^2 = n(\hat\theta - \theta_0)^2 I_1 + o_P(1)$. Since $W = n(\hat\theta - \theta_0)^2 I_1(\hat\theta)$ and $I_1(\hat\theta) \xrightarrow{P} I_1(\theta_0)$, $W - G^2 \xrightarrow{P} 0$. For the score, $\ell'(\theta_0) = -\ell''(\theta^{**})(\hat\theta - \theta_0) \approx nI_1(\hat\theta - \theta_0)$, so $S = \ell'(\theta_0)^2/(nI_1) \approx nI_1(\hat\theta - \theta_0)^2$. □

Remark 15.7 (The Assumptions of Wilks' Theorem).

- **Interior point:** if $\boldsymbol{\theta}_0$ lies on the boundary of Θ (e.g. $H_0 : \sigma^2_{\text{random effect}} = 0$), the limit is the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$; using χ^2_1 gives a conservative test.
- **Identifiability:** if some parameters become unidentified under H_0 (e.g. testing the number of components in a mixture model), Wilks' theorem fails.
- **Fixed dimension:** if the number of parameters grows with n, the chi-square approximation may break down.

15.3 Worked Examples

Example 15.8 (Poisson Mean). $X_i \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$, $H_0 : \lambda = \lambda_0$. Since $\hat\lambda = \bar x$ and $\ell(\lambda) = n\bar x \ln\lambda - n\lambda + \text{const}$,

$$G^2 = 2n\Big[\bar x \ln\frac{\bar x}{\lambda_0} - (\bar x - \lambda_0)\Big], \qquad W = \frac{n(\bar x - \lambda_0)^2}{\bar x}, \qquad S = \frac{n(\bar x - \lambda_0)^2}{\lambda_0}.$$

(If $\bar x = 0$, the convention $0 \ln 0 = 0$ gives $G^2 = 2n\lambda_0$; the Wald statistic is then undefined.) By Wilks, $G^2 \xrightarrow{d} \chi^2_1$. The score statistic is the counterpart of the one-cell Pearson chi-square statistic.

Example 15.9 (Normal with Unknown Variance: LRT = t-Test). $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, $H_0 : \mu = \mu_0$, with σ^2 a nuisance parameter. The unrestricted MLE is $(\bar x, \hat\sigma^2)$, the restricted MLE $(\mu_0, \hat\sigma_0^2)$ with $\hat\sigma_0^2 = \frac{1}{n}\sum(x_i - \mu_0)^2 = \hat\sigma^2 + (\bar x - \mu_0)^2$. Since the exponential term at the maximum of both likelihoods is $e^{-n/2}$,

$$\Lambda = \Big(\frac{\hat\sigma^2}{\hat\sigma_0^2}\Big)^{n/2} = \Big(1 + \frac{(\bar x - \mu_0)^2}{\hat\sigma^2}\Big)^{-n/2} = \Big(1 + \frac{t^2}{n-1}\Big)^{-n/2}, \qquad t = \frac{\sqrt{n}(\bar x - \mu_0)}{S}.$$

Since Λ is a decreasing function of $|t|$, the LRT is the two-sided t-test, and its distribution under H_0 is known *exactly* (t_{n-1}; Chapter 16). As $n \to \infty$, $G^2 = n\ln(1 + t^2/(n-1)) \approx t^2 \xrightarrow{d} \chi^2_1$, consistent with Wilks.

Example 15.10 (Multinomial Goodness of Fit: G^2 and Pearson's X^2 [64]). Let the cell counts $(N_1, \dots, N_k)$ have a Multinom$(n; p_1, \dots, p_k)$ distribution and test $H_0 : p_j = p_{j0}$ $(j = 1, \dots, k)$. The unrestricted MLE is $\hat p_j = N_j/n$; $\dim\Theta = k - 1$, $\dim\Theta_0 = 0$. Thus

$$G^2 = 2\sum_{j=1}^{k} N_j \ln\frac{N_j}{np_{j0}} \xrightarrow{d} \chi^2_{k-1}, \qquad X^2 = \sum_{j=1}^{k}\frac{(N_j - np_{j0})^2}{np_{j0}} \xrightarrow{d} \chi^2_{k-1}.$$

X^2 is the score test for this problem; using the expansion $\ln(1+u) \approx u - u^2/2$ one can show that $G^2 = X^2 + o_P(1)$. If H_0 involves estimating s parameters, the degrees of freedom become $k - 1 - s$.

Example 15.11 (Equality of Two Poisson Means). $X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda_1)$, $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda_2)$, $H_0 : \lambda_1 = \lambda_2$. $\dim\Theta = 2$, $\dim\Theta_0 = 1$, $r = 1$. The restricted MLE is $\hat\lambda = \frac{\sum X_i + \sum Y_j}{m+n}$ and

$$G^2 = 2\Big[S_X \ln\frac{\bar x}{\hat\lambda} + S_Y \ln\frac{\bar y}{\hat\lambda}\Big] \xrightarrow{d} \chi^2_1, \qquad S_X = \textstyle\sum X_i,\ S_Y = \sum Y_j.$$

15.4 Computational Applications

15.4.1 Verifying Wilks' Theorem (Poisson)

code/ch15/c15_1_verifying_wilks_theorem_poisson.py

```
import numpy as np
import scipy.stats as stats

rng = np.random.default_rng(42)
samples = rng.poisson(lam=4.0, size=(150_000, 100))
xb = np.mean(samples, axis=1)
g2 = 2.0 * 100 * (xb * np.log(xb / 4.0) - (xb - 4.0))
print(f"E[G^2]: {np.mean(g2):.4f} (theory: 1.0) | type I error: {np.mean(g2 > stats.chi2.ppf(0.95, 1)):.4f}")
print(f"Var[G^2]: {np.var(g2):.4f} (theory: 2.0) | KS p-value (against chi2_1): {stats.kstest(g2, 'chi2', args=(1,)).pvalue:.3f}")
```

Output

```
E[G^2]: 0.9998 (theory: 1.0) | type I error: 0.0517
Var[G^2]: 2.0141 (theory: 2.0) | KS p-value (against chi2_1): 0.000
```

The very small p-value of the KS test arises because, in an enormous simulation with 150 000 replications, it detects even the small deviations caused by the discreteness of G^2 (a Poisson sum is an integer); the mean, the variance and the type I error agree with χ^2_1. This is a nice example of the difference between "statistical significance" and "practical importance" in large-sample simulations.

15.4.2 The Holy Trinity: Exact Sizes in Small Samples

Since the Poisson sum $T = n\bar{X} \sim \text{Poisson}(n\lambda_0)$, the true sizes of the three tests under H_0 can be computed exactly, without simulation.

code/ch15/c15_2_holy_trinity_exact_sizes.py

```python
import numpy as np
from scipy import stats

def sizes(n, lam0, alpha=0.05):
    t = np.arange(0, int(n*lam0 + 20*np.sqrt(n*lam0) + 30))
    w = stats.poisson.pmf(t, n*lam0)
    xb = t / n
    with np.errstate(divide="ignore", invalid="ignore"):
        G = 2*n*(np.where(xb > 0, xb*np.log(xb/lam0), 0.0) - (xb - lam0))
        W = np.where(xb > 0, n*(xb - lam0)**2 / xb, np.inf)
    S = n*(xb - lam0)**2 / lam0
    k = stats.chi2.ppf(1 - alpha, 1)
    return [np.sum(w[s > k]) for s in (G, W, S)]

print(" n   lambda0 |  LRT    Wald   Score  (target 0.05)")
for n, l0 in [(5, 1.0), (10, 1.0), (10, 0.3), (50, 1.0), (200, 1.0)]:
    g, w, s = sizes(n, l0)
    print(f"{n:4d}  {l0:5.1f}   | {g:.4f} {w:.4f} {s:.4f}")
```

Output

```
 n   lambda0 |  LRT    Wald   Score  (target 0.05)
   5    1.0   | 0.0723 0.1301 0.0386
  10    1.0   | 0.0563 0.0743 0.0374
  10    0.3   | 0.0833 0.2030 0.0335
  50    1.0   | 0.0474 0.0512 0.0556
 200    1.0   | 0.0476 0.0537 0.0517
```

For small values of $n\lambda_0$ the Wald test badly exceeds the size (20% instead of 5% for $n\lambda_0 = 3$!): the probability of $\bar{x} = 0$ is high, and in that case $W = \infty$. The LRT also exceeds the nominal level, but much more mildly; the score test is slightly conservative in this example. The practical lesson: in small samples the Wald test should be avoided and the score test or the LRT preferred. As n grows, all three approach the nominal level (Theorem 15.6). Because of the discreteness, the sizes do not vary monotonically with n but fluctuate in a saw-tooth pattern.

15.4.3 Multinomial Goodness of Fit

code/ch15/c15_3_multinomial_goodness_of_fit.py

```python
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
```

```
p0 = np.array([0.1, 0.2, 0.3, 0.4])
n, R = 40, 200_000
N = rng.multinomial(n, p0, size=R)
E = n * p0
with np.errstate(divide="ignore", invalid="ignore"):
    G2 = 2 * np.sum(np.where(N > 0, N * np.log(N / E), 0.0), axis=1)
X2 = np.sum((N - E)**2 / E, axis=1)
k = stats.chi2.ppf(0.95, df=3)
print(f"n={n}: P(G^2 > chi2_3,0.05) = {np.mean(G2 > k):.4f} | P(X^2 > chi2_3,0.05) = {np.mean
    (X2 > k):.4f}")
print(f"means: G^2={G2.mean():.3f}, X^2={X2.mean():.3f} (theory 3) | Corr(G^2, X^2)={np.
    corrcoef(G2, X2)[0,1]:.4f}")
# An example data set
observed = np.array([7, 5, 13, 15])
r1 = stats.chisquare(observed, E)
r2 = stats.power_divergence(observed, E, lambda_="log-likelihood")
print(f"Observed {observed}, expected {E}")
print(f"Pearson X^2 = {r1.statistic:.4f}, p = {r1.pvalue:.4f} | G^2 = {r2.statistic:.4f}, p =
     {r2.pvalue:.4f}")
```

Output

```
n=40: P(G^2 > chi2_3,0.05) = 0.0600 | P(X^2 > chi2_3,0.05) = 0.0474
means: G^2=3.104, X^2=2.996 (theory 3) | Corr(G^2, X^2)=0.9666
Observed [ 7  5 13 15], expected [ 4.  8. 12. 16.]
Pearson X^2 = 3.5208, p = 0.3181 | G^2 = 3.2795, p = 0.3505
```

With $n = 40$ and a smallest expected cell frequency of 4, the size of Pearson's X^2 is closer to the nominal level than that of G^2; for small expected frequencies the chi-square approximation for G^2 improves more slowly. The rule of thumb "expected frequency ≥ 5 in every cell" is based on this observation.

15.5 Comments and Notes

Remark 15.12 (Which Test?).

- The **LRT** is invariant under reparametrisation (Zehna invariance) and usually gives the best chi-square approximation in small samples; with the Bartlett correction its error can be reduced to $O(n^{-2})$.
- The **Wald** test requires only the unrestricted model and is readily available in standard output (coefficient/SE); however, it is sensitive to the parametrisation (e.g. the result changes if $\ln\theta$ is tested instead of θ) and behaves badly near the boundary (the Hauck–Donner effect).
- The **score** test requires only the model under H_0; this makes it ideal for model-extension tests (e.g. should an extra variable be added?) and diagnostic tests. Pearson's X^2, the log-rank test and the Cochran–Armitage trend test are score tests.

Remark 15.13 (Connection with Confidence Regions). All three tests can be inverted into confidence regions (Chapter 16, Theorem 16.7): inverting the LRT gives the *profile likelihood interval* $\{\theta : 2[\ell(\hat\theta) - \ell(\theta)] \leq \chi^2_{1,\alpha}\}$, inverting Wald gives the symmetric interval $\hat\theta \pm z_{\alpha/2}\mathrm{SE}$, and inverting the score test gives (for the binomial) the Wilson interval.

15.6 Chapter Summary

- The generalised likelihood ratio $\Lambda = \sup_{\Theta_0} L / \sup_{\Theta} L$ depends only on a sufficient statistic. For simple hypotheses it reduces to the Neyman–Pearson test.
- Wilks' theorem (Theorem 15.5): in regular models, under H_0, $-2\ln\Lambda \xrightarrow{d} \chi^2_r$, where r

is the difference in the numbers of free parameters.
- The likelihood ratio, Wald and score tests are asymptotically equivalent (Theorem 15.6). In finite samples they may give different results; the Wald test is not invariant under reparametrisation.
- If the true parameter lies on the boundary or the model is not identifiable (e.g. the number of mixture components), the Wilks approximation breaks down.
- In the multinomial model the goodness-of-fit statistics G^2 and Pearson's X^2 are special cases of the likelihood ratio approach.

Further reading. [16, Section 10.3], [78, Chapters 16–17], [28, Chapter 22].

15.7 Exercises

Exercise 15.1. $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, $H_0 : \sigma^2 = \sigma_0^2$ (μ unknown). Write G^2 in terms of S^2/σ_0^2 and show that the LRT is a chi-square test that is *not equal-tailed*. → Solution

Exercise 15.2. Write down the G^2 and Pearson X^2 statistics for the hypothesis of independence in a 2×2 contingency table. Verify that the number of degrees of freedom is $(2-1)(2-1) = 1$ by computing $\dim \Theta - \dim \Theta_0$. → Solution

Exercise 15.3. For the exponential distribution and $H_0 : \lambda = \lambda_0$, construct the Wald test (a) on the λ scale and (b) on the $\ln \lambda$ scale. For $n = 10$, compute and compare the exact sizes of the two tests (using $2\lambda_0 \sum X_i \sim \chi^2_{2n}$). Compare them with the LRT. → Solution

Exercise 15.4. A boundary parameter: $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, 1)$, $\Theta = [0, \infty)$, $H_0 : \mu = 0$. Show that $G^2 = n\bar{X}^2 \mathbb{I}\{\bar{X} > 0\}$ and that under H_0, $G^2 \xrightarrow{d} \frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$. What is the true size of the test that uses the χ^2_1 critical value? → Solution

Chapter 16

Confidence Intervals and the Pivotal Quantity Method

A point estimate on its own is an incomplete answer: the value $\hat{\theta} = 3.2$ does not say whether the uncertainty is ± 0.1 or ± 5. A confidence interval is the frequentist way of reporting an estimate together with its uncertainty: in the long run, 95% of the intervals constructed by the same method cover the true parameter. In this chapter we discuss the two general methods for constructing confidence intervals (pivotal quantities and the inversion of tests), the basic results of normal sampling theory, and why the "textbook" Wald interval for a binomial proportion is surprisingly bad.

16.1 Definitions

Definition 16.1 (Confidence Set and Confidence Coefficient)**.** Let $C(\mathbf{X}) \subseteq \Theta$ be a set depending on the data. $\mathbb{P}_\theta(\theta \in C(\mathbf{X}))$ is the *coverage probability* at θ. If

$$\inf_{\theta \in \Theta} \mathbb{P}_\theta(\theta \in C(\mathbf{X})) \geq 1 - \alpha,$$

then C is called a *confidence set* of level $1 - \alpha$; the infimum on the left is the *confidence coefficient*. If $\Theta \subseteq \mathbb{R}$ and $C(\mathbf{X}) = [L(\mathbf{X}), U(\mathbf{X})]$, it is called a *confidence interval*.

Remark 16.2 (The Correct Interpretation)*.* It is the interval that is random, not the parameter. The statement "the probability that the interval $[2.9,\ 3.5]$ contains θ is 0.95" is — in the frequentist sense — false: the observed interval either contains θ or it does not. The correct statement is: "the method that produced this interval covers θ in 95% of repeated samples." A probability statement about the parameter is given by a Bayesian credible interval (Chapter 19).

Definition 16.3 (Pivotal Quantity)**.** A function $Q(\mathbf{X}, \theta)$ whose distribution does not depend on θ is called a *pivot*. (It need not be a statistic; it involves θ.) If its asymptotic distribution does not depend on θ, it is called an *asymptotic pivot*.

Definition 16.4 (The Chi-Square, Student t and F Distributions)**.** Let $Z, Z_1, \ldots, Z_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$.

1. $\sum_{j=1}^{k} Z_j^2 \sim \chi^2_k$ (chi-square with k degrees of freedom); $\chi^2_k = \text{Gamma}(k/2, \frac{1}{2})$ (rate $\frac{1}{2}$), with mean k and variance $2k$.
2. If $V \sim \chi^2_k$ and $Z \perp\!\!\!\perp V$, then $T = Z/\sqrt{V/k} \sim t_k$ (Student t).
3. If $V_1 \sim \chi^2_{k_1}$ and $V_2 \sim \chi^2_{k_2}$ are independent, then $\frac{V_1/k_1}{V_2/k_2} \sim F_{k_1, k_2}$.

The densities of these distributions are derived by the Jacobian method of Chapter 4 (see Example 4.17 and the exercises there).

16.2 Normal Sampling Theory

Theorem 16.5 (The Normal Sample Theorem [77])**.** *If* $X_1, \dots, X_n \overset{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$*, then*

1. $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$*;*
2. $\bar{X}$ *and* S^2 *are independent;*
3. $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$*;*
4. $T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$.

Proof. (1) A linear combination of independent normals is normal (Example 6.18). (2) Basu's theorem (Example 10.14). (3) $\sum_{i=1}^n \left(\frac{X_i-\mu}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2} + \left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right)^2$. The left-hand side is χ^2_n, the second term on the right is χ^2_1, and by (2) the two terms are independent. In terms of MGFs, $(1-2t)^{-n/2} = M_{(n-1)S^2/\sigma^2}(t) \cdot (1-2t)^{-1/2}$, so $M_{(n-1)S^2/\sigma^2}(t) = (1-2t)^{-(n-1)/2}$; this is the MGF of χ^2_{n-1}. (4) $T = \frac{(\bar{X}-\mu)/(\sigma/\sqrt{n})}{\sqrt{[(n-1)S^2/\sigma^2]/(n-1)}}$; the numerator is $\mathcal{N}(0,1)$, and by (3) and the independence (2) the denominator has the form in the definition. □

16.3 Methods for Constructing Confidence Intervals

Theorem 16.6 (The Pivot Method)**.** *Let* $Q(\mathbf{X}, \theta)$ *be a pivot and* $a < b$ *constants with* $\mathbb{P}(a \le Q \le b) = 1 - \alpha$*. Then* $C(\mathbf{X}) = \{\theta : a \le Q(\mathbf{X}, \theta) \le b\}$ *is an (exact) confidence set of level* $1 - \alpha$*. If* Q *is monotone in* θ*,* C *is an interval.*

Proof. $\theta \in C(\mathbf{X}) \iff a \le Q(\mathbf{X}, \theta) \le b$; the probability of the right-hand side is $1 - \alpha$ for every θ. □

Theorem 16.7 (Test–Interval Duality)**.**

1. *For each* $\theta_0 \in \Theta$ *let* $A(\theta_0)$ *be the acceptance region of a level-*α *test of* $H_0 : \theta = \theta_0$*. Then*

$$C(\mathbf{X}) = \{\theta : \mathbf{X} \in A(\theta)\}$$

   is a confidence set of level $1 - \alpha$*.*

2. *Conversely, if* $C(\mathbf{X})$ *is a confidence set of level* $1 - \alpha$*, then* $A(\theta_0) = \{\mathbf{x} : \theta_0 \in C(\mathbf{x})\}$ *is the acceptance region of a level-*α *test of* $H_0 : \theta = \theta_0$*.*

Proof. Both directions rest on the identity $\theta \in C(\mathbf{x}) \iff \mathbf{x} \in A(\theta)$. Hence

$$\mathbb{P}_\theta(\theta \in C(\mathbf{X})) = \mathbb{P}_\theta(\mathbf{X} \in A(\theta)) = 1 - \mathbb{P}_\theta(H_0 : \theta \text{ is rejected}) \ge 1 - \alpha. \qquad \square$$

Remark 16.8. Duality carries the optimality notions of test theory over to intervals: inverting UMP tests gives *uniformly most accurate* confidence bounds, i.e. bounds with the smallest probability of covering false values θ'. One-sided tests correspond to one-sided confidence bounds and two-sided tests to intervals.

Theorem 16.9 (The Shortest Pivotal Interval)**.** *Let the density* f *of* Q *be unimodal, and let the length of the interval be proportional to* $b - a$*. The interval minimising* $b - a$ *subject to* $\int_a^b f = 1 - \alpha$ *satisfies* $f(a) = f(b) > 0$ *and* $a \le x^* \le b$ *(*x^**: the mode). For symmetric* f *this is the equal-tailed interval.*

Proof. Unimodality: f is non-decreasing on $(-\infty, x^*]$ and non-increasing on $[x^*, \infty)$. Let $[a, b]$ satisfy the conditions ($\int_a^b f = 1-\alpha$, $f(a) = f(b) > 0$, $a \le x^* \le b$); by unimodality $f \ge f(a)$ on $[a, b]$ and $f \le f(a)$ outside it. Suppose that some $[a', b']$ with $\int_{a'}^{b'} f \ge 1-\alpha$ has $b'-a' < b-a$, and derive a contradiction.

Case 1: $a' \le a$. Then $b' < b$. (i) If $b' \le a$, then $f \le f(a)$ on $[a', b'] \subseteq (-\infty, a]$ and

$$\int_{a'}^{b'} f \le f(a)(b'-a') < f(a)(b-a) \le \int_a^b f = 1-\alpha.$$

(ii) If $a < b' < b$, then

$$\begin{aligned}\int_{a'}^{b'} f = \int_a^b f + \int_{a'}^{a} f - \int_{b'}^{b} f &\le (1-\alpha) + f(a)(a-a') - f(b)(b-b') \\ &= (1-\alpha) + f(a)\big[(b'-a') - (b-a)\big] < 1-\alpha,\end{aligned}$$

because $f \le f(a)$ on $[a', a]$, $f \ge f(b) = f(a)$ on $[b', b]$, and $f(a) > 0$. In both subcases $\int_{a'}^{b'} f < 1-\alpha$: a contradiction.

Case 2: $a' > a$. The symmetric argument (Case 1 under the reflection $x \mapsto -x$) gives the same contradiction.

Hence $[a, b]$ is the shortest interval. If f is symmetric about 0, the choice $a = -b$ satisfies $f(a) = f(b)$ and the interval is equal-tailed. □

16.4 Worked Examples

Example 16.10 (Normal Mean: The t Interval)**.** The T of Theorem 16.5(4) is a pivot. Since t_{n-1} is symmetric and unimodal, the shortest interval is

$$\bar{X} \pm t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}.$$

If σ were known, $z_{\alpha/2}$ would be used; the difference $t_{n-1,\alpha/2} > z_{\alpha/2}$ is the price of estimating σ (2.145 vs 1.960 for $n = 15$).

Example 16.11 (Normal Variance)**.** From the pivot $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$,

$$\Big[\frac{(n-1)S^2}{\chi^2_{n-1,\alpha/2}},\ \frac{(n-1)S^2}{\chi^2_{n-1,1-\alpha/2}}\Big].$$

Since the chi-square distribution is skewed, the equal-tailed interval is not the shortest (Theorem 16.9). Moreover, this interval is very sensitive to normality: if the excess kurtosis $\gamma_2 \ne 0$, even the asymptotic coverage is not $1-\alpha$ (because, by Theorem 7.9(3), $\mathrm{Var}(S^2)$ depends on μ_4).

Example 16.12 (Exponential Distribution)**.** $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(\lambda)$: $2\lambda\sum X_i \sim \chi^2_{2n}$ is a pivot. Hence

$$\Big[\frac{\chi^2_{2n,1-\alpha/2}}{2\sum X_i},\ \frac{\chi^2_{2n,\alpha/2}}{2\sum X_i}\Big]$$

is an exact $1-\alpha$ confidence interval for λ.

Example 16.13 (Uniform$(0, \theta)$)**.** The CDF of $Q = X_{(n)}/\theta$ is q^n $(0 < q < 1)$; it does not depend on θ. Since the density nq^{n-1} is increasing, the shortest interval is obtained with the choice $Q \in [\alpha^{1/n}, 1]$:

$$[X_{(n)},\ X_{(n)}\alpha^{-1/n}].$$

Its length is $O(1/n)$, faster than the $O(1/\sqrt{n})$ of normal-based intervals.

Example 16.14 (Binomial Proportion: Four Intervals)**.** $X \sim \mathrm{Bin}(n, p)$, $\hat{p} = X/n$, $z = z_{\alpha/2}$.

1. **Wald** (inverting the Wald test): $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$.
2. **Wilson** [88] (inverting the score test): $\{p : |\hat{p} - p| \le z\sqrt{p(1-p)/n}\}$. Solving the quadratic inequality in p gives
$$\frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}.$$
3. **Clopper–Pearson** [17] (inverting exact binomial tests): the solutions of the equations $\mathbb{P}_p(X \ge x) = \alpha/2$ and $\mathbb{P}_p(X \le x) = \alpha/2$; in terms of Beta quantiles, $[B_{\alpha/2}(x, n-x+1),\ B_{1-\alpha/2}(x+1, n-x)]$.
4. **Agresti–Coull** [1]: with $\tilde{n} = n + z^2$ and $\tilde{p} = (X + z^2/2)/\tilde{n}$, $\tilde{p} \pm z\sqrt{\tilde{p}(1-\tilde{p})/\tilde{n}}$ ("add two successes and two failures").

These four intervals show how duality produces different intervals when different tests are applied to the same data. In the simulation below we shall see that the coverage of the Wald interval falls far below the nominal level at some values of p, even when n is large [13].

16.5 Computational Applications

16.5.1 Coverage of the t Interval

code/ch16/c16_1_coverage_of_t_interval.py

```
import numpy as np
import scipy.stats as stats

rng = np.random.default_rng(42)
s = rng.normal(50.0, 10.0, size=(100_000, 15))
xb, se = np.mean(s, axis=1), np.std(s, axis=1, ddof=1) / np.sqrt(15)
tc = stats.t.ppf(0.975, df=14)
print(f"Coverage: {np.mean((xb - tc*se <= 50.0) & (50.0 <= xb + tc*se)) * 100:.2f}%")
zc = stats.norm.ppf(0.975)
print(f"with z instead (wrong): {np.mean(np.abs(xb - 50.0) <= zc*se) * 100:.2f}%")

# Departure from normality: the t interval for the mean is robust, the variance interval is not
for name, gen, mu, var in [("Normal", lambda: rng.normal(0, 1, (100_000, 15)), 0, 1),
                           ("Expon.", lambda: rng.exponential(1, (100_000, 15)), 1, 1)]:
    X = gen(); xb = X.mean(1); S2 = X.var(1, ddof=1)
    ct = np.mean(np.abs(xb - mu) <= tc*np.sqrt(S2/15))
    lo, hi = 14*S2/stats.chi2.ppf(0.975, 14), 14*S2/stats.chi2.ppf(0.025, 14)
    cv = np.mean((lo <= var) & (var <= hi))
    print(f"{name} | t-interval coverage={ct:.4f} | chi-square variance interval coverage={cv:.4f}  (target 0.95)")
```

Output

```
Coverage: 95.12%
with z instead (wrong): 93.12%
Normal | t-interval coverage=0.9501 | chi-square variance interval coverage=0.9506  (target 0.95)
Expon. | t-interval coverage=0.9118 | chi-square variance interval coverage=0.7430  (target 0.95)
```

For exponential data the t interval for the mean gives reasonable coverage even at $n = 15$ (CLT and Slutsky), but the chi-square interval for the variance collapses: the excess kurtosis of the exponential distribution is 6.

16.5.2 Binomial Proportion: Wald, Wilson, Clopper–Pearson, Agresti–Coull

The coverage probabilities were computed exactly (by weighting with binomial probabilities).

code/ch16/c16_2_binomial_wald_wilson_clopper_pearson.py

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

def intervals(x, n, a=0.05):
    z = stats.norm.ppf(1 - a/2); p = x / n
    wald = (p - z*np.sqrt(p*(1-p)/n), p + z*np.sqrt(p*(1-p)/n))
    m = (p + z**2/(2*n)) / (1 + z**2/n); h = z/(1 + z**2/n)*np.sqrt(p*(1-p)/n + z**2/(4*n**2))
    wilson = (m - h, m + h)
    cp = (np.nan_to_num(stats.beta.ppf(a/2, x, n - x + 1), nan=0.0),
          np.nan_to_num(stats.beta.ppf(1 - a/2, x + 1, n - x), nan=1.0))
    nt = n + z**2; pt = (x + z**2/2)/nt
    ac = (pt - z*np.sqrt(pt*(1-pt)/nt), pt + z*np.sqrt(pt*(1-pt)/nt))
    return {"Wald": wald, "Wilson": wilson, "Clopper-Pearson": cp, "Agresti-Coull": ac}

def coverage(n, ps):
    x = np.arange(n + 1); A = intervals(x, n)
    return {k: np.array([np.sum(stats.binom.pmf(x, n, p) * ((L <= p) & (p <= U))) for p in ps])
            for k, (L, U) in A.items()}

ps = np.linspace(0.005, 0.995, 991)
fig, axes = plt.subplots(1, 2, figsize=(10, 3.4), sharey=True)
for ax, n in zip(axes, [30, 100]):
    K = coverage(n, ps)
    for k, v in K.items():
        ax.plot(ps, v, lw=0.8, label=k)
        print(f"n={n:3d} {k:16s} | mean coverage={v.mean():.4f} | min coverage={v.min():.4f}")
    ax.axhline(0.95, color="k", ls=":"); ax.set_ylim(0.75, 1.0); ax.set_xlabel("p"); ax.set_title(f"n = {n}")
axes[0].set_ylabel("coverage probability"); axes[1].legend(fontsize=8, loc="lower center")
fig.tight_layout(); fig.savefig("binom_kapsama.pdf")
for p in [0.01, 0.05, 0.2, 0.5]:
    K = coverage(100, [p])
    print(f"n=100, p={p:.2f} | Wald={K['Wald'][0]:.4f} | Wilson={K['Wilson'][0]:.4f} | CP={K['Clopper-Pearson'][0]:.4f}")
```

Output

```
n= 30 Wald             | mean coverage=0.8822 | min coverage=0.1396
n= 30 Wilson           | mean coverage=0.9527 | min coverage=0.8604
n= 30 Clopper-Pearson  | mean coverage=0.9731 | min coverage=0.9537
n= 30 Agresti-Coull    | mean coverage=0.9598 | min coverage=0.9339
n=100 Wald             | mean coverage=0.9288 | min coverage=0.3941
n=100 Wilson           | mean coverage=0.9511 | min coverage=0.9102
n=100 Clopper-Pearson  | mean coverage=0.9642 | min coverage=0.9504
n=100 Agresti-Coull    | mean coverage=0.9552 | min coverage=0.9391
n=100, p=0.01 | Wald=0.6334 | Wilson=0.9206 | CP=0.9816
n=100, p=0.05 | Wald=0.8775 | Wilson=0.9659 | CP=0.9826
n=100, p=0.20 | Wald=0.9331 | Wilson=0.9405 | CP=0.9674
n=100, p=0.50 | Wald=0.9431 | Wilson=0.9431 | CP=0.9648
```

16.5.3 The Pivotal Interval for Uniform$(0, \theta)$

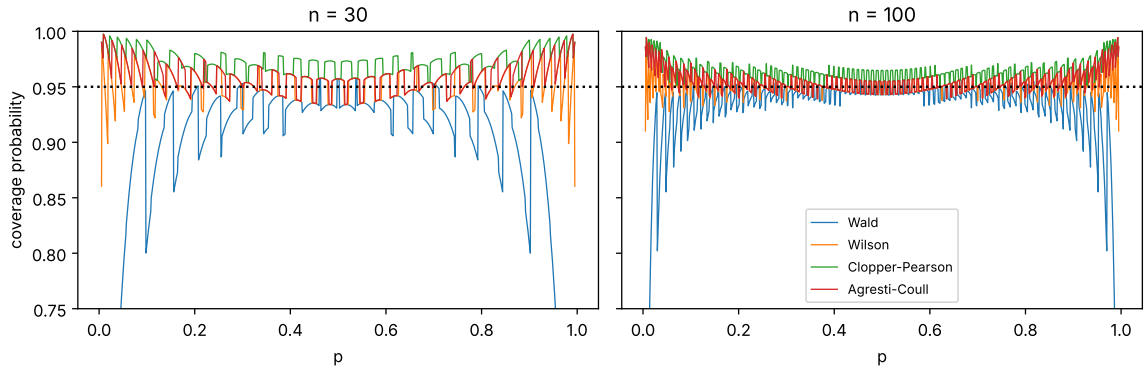


Figure 16.1: Exact coverage probabilities of 95% intervals for a binomial proportion. The Wald interval (blue) gives dramatically low coverage, especially near the ends, and is erratic even at $n = 100$. Clopper–Pearson is ≥ 0.95 everywhere (conservative); Wilson and Agresti–Coull are close to the nominal level on average.

code/ch16/c16_3_uniform_pivotal_interval.py

```
import numpy as np

rng = np.random.default_rng(42)
theta, a, R = 7.0, 0.05, 200_000
for n in [5, 20, 100]:
    M = rng.uniform(0, theta, size=(R, n)).max(axis=1)
    L, U = M, M * a**(-1/n)
    print(f"n={n:3d} | coverage={np.mean((L <= theta) & (theta <= U)):.4f} | mean length={np.mean(U - L):.4f}"
          f" | theory={theta*n/(n+1)*(a**(-1/n) - 1):.4f}")
```

Output

```
n=  5 | coverage=0.9502 | mean length=4.7865 | theory=4.7866
n= 20 | coverage=0.9495 | mean length=1.0773 | theory=1.0772
n=100 | coverage=0.9500 | mean length=0.2108 | theory=0.2108
```

16.6 Comments and Notes

Remark 16.15 (Why Is the Wald Interval Bad?). The Wald interval combines two approximations: the normal approximation *and* evaluating the standard error at $\hat{p}$. When p is near the ends, the distribution of $\hat{p}$ is skewed and $\hat{p}(1-\hat{p})$ systematically underestimates the variance; moreover, when $\hat{p} \in \{0, 1\}$ the interval collapses to a single point. The Wilson interval computes the variance at the value of p under H_0 (the score approach) and avoids both problems. The general lesson: if the standard error of a parameter depends on that parameter, test inversion (score or LRT) is more reliable than Wald.

Remark 16.16 (Why Are Exact Intervals Conservative?). For discrete distributions, coverage cannot be made exactly $1-\alpha$ without randomisation; Clopper–Pearson guarantees $\geq 1-\alpha$ for every p at the cost of intervals that are wider than necessary on average. This is the interval counterpart of the conservativeness of the non-randomised tests of Chapter 13.

Remark 16.17 (Bootstrap Intervals). When no pivot can be found and the asymptotic variance is complicated, the distribution of $\hat{\theta}$ can be imitated by resampling from the sample with replacement [25]. The percentile, BCa and bootstrap-t intervals are data-driven versions of the pivot idea of this chapter; detailed theory and comparative simulations are in Chapter 18.

16.7 Chapter Summary

- In a confidence set it is the set, not the parameter, that is random. The coverage probability is the probability that the set contains the true parameter.
- The normal sample theorem (Theorem 16.5): $\bar{X}$ and S^2 are independent, $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, and the studentised mean has a t_{n-1} distribution.
- Intervals are constructed in two ways: the pivot method (Theorem 16.6) and test–interval duality (Theorem 16.7), i.e. inverting acceptance regions.
- For a unimodal pivot density, the shortest interval satisfies $f(a) = f(b)$ (Theorem 16.9). For symmetric pivots this interval is equal-tailed.
- For a binomial proportion the Wald interval gives very low coverage near the ends. The Wilson, Agresti–Coull and Clopper–Pearson intervals are more reliable.

Further reading. [16, Chapter 9], [47, Section 3.5 and Chapter 5]; for a comparison of binomial intervals, [13].

16.8 Exercises

Exercise 16.1. For two independent normal samples ($\sigma_1 = \sigma_2$ unknown), derive the pooled t interval for $\mu_1 - \mu_2$. If the variances are not equal, compare the coverage of the Welch [84] approach by simulation. → Solution

Exercise 16.2. For the interval for σ^2 based on the χ^2_{n-1} pivot, find the shortest interval by solving numerically the condition of Theorem 16.9 (careful: the length is proportional to $(n-1)S^2(1/a - 1/b)$!) and compare it with the equal-tailed interval for $n = 10$. → Solution

Exercise 16.3. For the Poisson mean λ, plot for $\lambda \in (0, 5]$ the coverage of (a) the Wald interval, (b) the (inverted) score interval, and (c) the exact interval (Garwood: $\frac{1}{2}\chi^2_{2x,1-\alpha/2} \le \lambda n \le \frac{1}{2}\chi^2_{2x+2,\alpha/2}$). → Solution

Exercise 16.4. $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$. When the t interval for μ and the chi-square interval for σ^2 are each constructed at level 95%, compute by simulation the probability that both are correct *simultaneously*. Why is the result not $0.95^2 = 0.9025$, even though $\bar{X} \perp\!\!\!\perp S^2$? (Hint: the coverage event of the t interval also depends on S.) Construct a joint region with confidence at least 95% using Bonferroni. → Solution

Chapter 17

The Linear Model

Linear regression is where all the tools of statistical theory come together: estimation (least squares, Gauss–Markov), distribution theory (the multivariate normal and quadratic forms), hypothesis testing (t and F tests, likelihood ratio) and interval estimation (confidence and prediction intervals). In this chapter we study the geometry of the linear model, the finite-sample and asymptotic properties of the least squares estimator, and exact inference under the assumption of normal errors. The multivariate normal results of Chapter 4 (Theorem 4.13) will be used throughout.

17.1 Definitions

Definition 17.1 (Linear Model)**.** Let $\mathbf{Y} \in \mathbb{R}^n$ be the response vector, $X \in \mathbb{R}^{n \times p}$ a known (fixed) *design matrix*, $\boldsymbol{\beta} \in \mathbb{R}^p$ an unknown coefficient vector and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ the error vector. The model

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

is called a *linear model*. Unless stated otherwise, X is assumed to have full column rank ($\operatorname{rank} X = p < n$) and the following *Gauss–Markov assumptions* are assumed to hold:

$$\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}, \qquad \operatorname{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n.$$

If in addition $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 I_n)$, the model is called the *normal linear model*. "Linear" refers to $\boldsymbol{\beta}$: the columns of X may be x, x^2, $\ln x$ or indicator variables.

Definition 17.2 (Least Squares Estimator)**.** The $\hat{\boldsymbol{\beta}}$ minimising $S(\boldsymbol{\beta}) = \|\mathbf{Y} - X\boldsymbol{\beta}\|^2 = \sum_{i=1}^n (Y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2$ is called the *least squares (LS) estimator*. $\hat{\mathbf{Y}} = X\hat{\boldsymbol{\beta}}$ are the *fitted values*, $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ the *residuals*, and $\mathrm{SSE} = \mathbf{e}^\top \mathbf{e}$ the *residual sum of squares*.

Definition 17.3 (Hat Matrix and Leverage)**.** The matrix $H = X(X^\top X)^{-1} X^\top$ is called the *hat matrix* ($\hat{\mathbf{Y}} = H\mathbf{Y}$: it "puts the hat on $\mathbf{Y}$"). Its diagonal elements h_{ii} are the *leverages* of the observations.

Definition 17.4 (Best Linear Unbiased Estimator (BLUE))**.** Among the (linear) unbiased estimators of $\mathbf{c}^\top \boldsymbol{\beta}$ of the form $\mathbf{a}^\top \mathbf{Y}$, the one with the smallest variance is called the *best linear unbiased estimator* (BLUE).

Definition 17.5 (Coefficient of Determination)**.** If the model contains an intercept, then with $\mathrm{SST} = \sum (Y_i - \bar{Y})^2$ and $\mathrm{SSR} = \sum (\hat{Y}_i - \bar{Y})^2$ we have $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ and

$$R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}, \qquad R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{SSE}/(n-p)}{\mathrm{SST}/(n-1)}.$$

17.2 Least Squares: Geometry and Moments

Theorem 17.6 (Normal Equations and Projection)**.** *1. The LS estimator satisfies the* normal equations $X^\top X\hat{\boldsymbol{\beta}} = X^\top \mathbf{Y}$ *and is unique:* $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top \mathbf{Y}$.

2. H is symmetric and idempotent ($H^2 = H$) with $\operatorname{rank} H = \operatorname{tr} H = p$*; H is the orthogonal projection of $\mathbb{R}^n$ onto the column space $\mathcal{C}(X)$ of X. $I - H$ is the projection onto $\mathcal{C}(X)^\perp$ and has rank $n - p$.*

3. The residuals are orthogonal to the columns of X: $X^\top \mathbf{e} = \mathbf{0}$. If there is an intercept, $\sum e_i = 0$.

Proof. (1) For any $\mathbf{b}$ write $\mathbf{Y} - X\mathbf{b} = (\mathbf{Y} - X\hat{\boldsymbol{\beta}}) + X(\hat{\boldsymbol{\beta}} - \mathbf{b})$. Since $X^\top(\mathbf{Y} - X\hat{\boldsymbol{\beta}}) = \mathbf{0}$, the cross term vanishes and

$$S(\mathbf{b}) = S(\hat{\boldsymbol{\beta}}) + \|X(\hat{\boldsymbol{\beta}} - \mathbf{b})\|^2 \geq S(\hat{\boldsymbol{\beta}}),$$

with equality only when $X(\hat{\boldsymbol{\beta}} - \mathbf{b}) = \mathbf{0}$, i.e. (by full rank) when $\mathbf{b} = \hat{\boldsymbol{\beta}}$. (2) $H^\top = H$ is clear; $H^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = H$. The eigenvalues of an idempotent matrix are 0 or 1, so rank = trace = $\operatorname{tr}\left((X^\top X)^{-1}X^\top X\right) = \operatorname{tr} I_p = p$. That $H\mathbf{v} = \mathbf{v}$ ($\mathbf{v} \in \mathcal{C}(X)$) and $H\mathbf{w} = \mathbf{0}$ ($\mathbf{w} \perp \mathcal{C}(X)$) is seen directly. (3) $X^\top \mathbf{e} = X^\top(I - H)\mathbf{Y} = \mathbf{0}$ because $X^\top H = X^\top$. The intercept is the column $\mathbf{1}$, so $\mathbf{1}^\top \mathbf{e} = 0$. □

Theorem 17.7 (Moments of the LS Estimator)**.** *Under the Gauss–Markov assumptions (normality is not required):*

$$\mathbb{E}\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}, \qquad \operatorname{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(X^\top X)^{-1}, \qquad \mathbb{E}[\mathrm{SSE}] = (n - p)\sigma^2.$$

Hence $s^2 = \mathrm{SSE}/(n - p)$ is an unbiased estimator of σ^2.

Proof. $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top(X\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (X^\top X)^{-1}X^\top\boldsymbol{\varepsilon}$; the mean and covariance follow at once from Proposition 4.12(2). Also $\mathbf{e} = (I - H)\mathbf{Y} = (I - H)\boldsymbol{\varepsilon}$ (because $(I - H)X = 0$), and by the cyclic property of the trace

$$\mathbb{E}[\boldsymbol{\varepsilon}^\top(I - H)\boldsymbol{\varepsilon}] = \mathbb{E}\operatorname{tr}\left((I - H)\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top\right) = \sigma^2 \operatorname{tr}(I - H) = \sigma^2(n - p).$$ □

Remark 17.8. The divisor $n - p$ generalises the divisor $n - 1$ of S^2 in Chapter 7: in the model with an intercept and no explanatory variables ($p = 1$, $X = \mathbf{1}$), $s^2 = S^2$. Each estimated parameter "uses up" one degree of freedom.

Theorem 17.9 (Gauss–Markov Theorem; see [69, Chapter 3])**.** *Under the Gauss–Markov assumptions, for every $\mathbf{c} \in \mathbb{R}^p$, $\mathbf{c}^\top\hat{\boldsymbol{\beta}}$ is the BLUE of $\mathbf{c}^\top\boldsymbol{\beta}$. Equivalently, for every linear unbiased estimator $\tilde{\boldsymbol{\beta}} = A\mathbf{Y}$ of $\boldsymbol{\beta}$,* $\operatorname{Cov}(\tilde{\boldsymbol{\beta}}) - \operatorname{Cov}(\hat{\boldsymbol{\beta}})$ *is positive semidefinite.*

Proof. Let $\mathbf{a}^\top\mathbf{Y}$ be unbiased for $\mathbf{c}^\top\boldsymbol{\beta}$: $\mathbf{a}^\top X\boldsymbol{\beta} = \mathbf{c}^\top\boldsymbol{\beta}$ for every $\boldsymbol{\beta}$, i.e. $X^\top\mathbf{a} = \mathbf{c}$. Let $\mathbf{a}_0 = X(X^\top X)^{-1}\mathbf{c}$ (the coefficients corresponding to LS) and $\mathbf{d} = \mathbf{a} - \mathbf{a}_0$. Then $X^\top\mathbf{d} = \mathbf{c} - \mathbf{c} = \mathbf{0}$, so $\mathbf{a}_0^\top\mathbf{d} = \mathbf{c}^\top(X^\top X)^{-1}X^\top\mathbf{d} = 0$. Thus

$$\operatorname{Var}(\mathbf{a}^\top\mathbf{Y}) = \sigma^2\|\mathbf{a}\|^2 = \sigma^2(\|\mathbf{a}_0\|^2 + \|\mathbf{d}\|^2) \geq \sigma^2\|\mathbf{a}_0\|^2 = \operatorname{Var}(\mathbf{c}^\top\hat{\boldsymbol{\beta}}),$$

with equality only when $\mathbf{d} = \mathbf{0}$. The matrix statement is equivalent to this result holding for every $\mathbf{c}$. □

Remark 17.10 (The Limits of Gauss–Markov). The theorem gives optimality only within the class of *linear* and *unbiased* estimators. With normal errors, LS is in fact the UMVUE among all unbiased estimators (Lehmann–Scheffé; $(X^\top\mathbf{Y}, \mathbf{Y}^\top\mathbf{Y})$ is complete sufficient). With heavy-tailed errors, however, non-linear robust estimators (e.g. least absolute deviations) can be much better than LS; and if the unbiasedness constraint is dropped, shrinkage estimators such as ridge regression can reduce the MSE (compare with James–Stein, Chapter 7).

17.3 Exact Inference in the Normal Linear Model

Lemma 17.11 (Idempotent Quadratic Forms)**.** *If $\mathbf{Z} \sim \mathcal{N}_n(\mathbf{0}, I_n)$ and A is a symmetric idempotent matrix of rank r, then $\mathbf{Z}^\top A\mathbf{Z} \sim \chi^2_r$. Moreover, if A and B are two such matrices with $AB = 0$, then $\mathbf{Z}^\top A\mathbf{Z}$ and $\mathbf{Z}^\top B\mathbf{Z}$ are independent.*

Proof. Spectral decomposition: $A = Q\Lambda Q^\top$ with Q orthogonal and Λ having r ones and $n-r$ zeros on its diagonal. $\mathbf{W} = Q^\top\mathbf{Z} \sim \mathcal{N}_n(\mathbf{0}, Q^\top Q) = \mathcal{N}_n(\mathbf{0}, I)$ (Theorem 4.13(1)) and $\mathbf{Z}^\top A\mathbf{Z} = \mathbf{W}^\top\Lambda\mathbf{W} = \sum_{j=1}^{r} W_j^2 \sim \chi^2_r$. Second part: $\mathbf{Z}^\top A\mathbf{Z} = \|A\mathbf{Z}\|^2$, $\mathbf{Z}^\top B\mathbf{Z} = \|B\mathbf{Z}\|^2$ and $\operatorname{Cov}(A\mathbf{Z}, B\mathbf{Z}) = AB^\top = AB = 0$; for jointly normal vectors uncorrelatedness is independence (Theorem 4.13(3)). □

Theorem 17.12 (Sampling Theory of the Normal Linear Model)**.** *If $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 I)$, then*

1. *$\hat{\boldsymbol{\beta}}$ is the MLE of $\boldsymbol{\beta}$ and $\hat{\boldsymbol{\beta}} \sim \mathcal{N}_p(\boldsymbol{\beta}, \sigma^2(X^\top X)^{-1})$; the MLE of σ^2 is $\hat\sigma^2 = \mathrm{SSE}/n$.*
2. $\dfrac{\mathrm{SSE}}{\sigma^2} = \dfrac{(n-p)s^2}{\sigma^2} \sim \chi^2_{n-p}$.
3. *$\hat{\boldsymbol{\beta}}$ and* SSE *are independent.*
4. *For every $\mathbf{c} \neq \mathbf{0}$, $T = \dfrac{\mathbf{c}^\top\hat{\boldsymbol{\beta}} - \mathbf{c}^\top\boldsymbol{\beta}}{s\sqrt{\mathbf{c}^\top(X^\top X)^{-1}\mathbf{c}}} \sim t_{n-p}$.*

Proof. (1) The log-likelihood is $-\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{Y} - X\boldsymbol{\beta}\|^2$; maximising over $\boldsymbol{\beta}$ means minimising $S(\boldsymbol{\beta})$. The derivative with respect to σ^2 gives $\hat\sigma^2 = S(\hat{\boldsymbol{\beta}})/n$. The distribution follows from the linear transformation $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1}X^\top\boldsymbol{\varepsilon}$ (Theorem 4.13(1)). (2) $\mathrm{SSE}/\sigma^2 = \mathbf{Z}^\top(I-H)\mathbf{Z}$ with $\mathbf{Z} = \boldsymbol{\varepsilon}/\sigma$; by Lemma 17.11 it is χ^2_{n-p}. (3) $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1}X^\top\boldsymbol{\varepsilon}$ and $\mathbf{e} = (I-H)\boldsymbol{\varepsilon}$ are jointly normal with $\operatorname{Cov} = \sigma^2(X^\top X)^{-1}X^\top(I-H) = 0$. Hence $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \mathbf{e}$, and since $\mathrm{SSE} = \mathbf{e}^\top\mathbf{e}$, $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \mathrm{SSE}$. (4) Dividing the numerator by $\sigma\sqrt{\mathbf{c}^\top(X^\top X)^{-1}\mathbf{c}}$ gives $\mathcal{N}(0,1)$; with (2) and (3) we obtain the t structure of Definition 16.4. □

Remark 17.13. Theorem 16.5 is the special case $X = \mathbf{1}$, $p = 1$ of this theorem: $\hat\beta = \bar Y$, $s^2 = S^2$, $\bar Y \perp\!\!\!\perp S^2$, and the t_{n-1} distribution.

Corollary 17.14 (Confidence and Prediction Intervals)**.** *For a new point $\mathbf{x}_0$, with $t = t_{n-p,\alpha/2}$ and $q_0 = \mathbf{x}_0^\top(X^\top X)^{-1}\mathbf{x}_0$:*

1. *a confidence interval for $\mathbb{E}[Y \mid \mathbf{x}_0] = \mathbf{x}_0^\top\boldsymbol{\beta}$: $\mathbf{x}_0^\top\hat{\boldsymbol{\beta}} \pm t\,s\sqrt{q_0}$;*
2. *a* prediction interval *for a new observation $Y_0 = \mathbf{x}_0^\top\boldsymbol{\beta} + \varepsilon_0$: $\mathbf{x}_0^\top\hat{\boldsymbol{\beta}} \pm t\,s\sqrt{1+q_0}$.*

Proof. (1) Theorem 17.12(4) with $\mathbf{c} = \mathbf{x}_0$. (2) $Y_0 - \mathbf{x}_0^\top\hat{\boldsymbol{\beta}} = \varepsilon_0 - \mathbf{x}_0^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$; the two terms are independent (ε_0 is the error of the new observation), the variance is $\sigma^2(1+q_0)$, and we studentise with s. □

Remark 17.15. As $n \to \infty$, $q_0 \to 0$, so the confidence interval shrinks to zero, but the prediction interval converges to $\pm z_{\alpha/2}\sigma$: however well we know the mean, the intrinsic randomness of a single observation does not go away. Confusing these two intervals is one of the most common mistakes in practice.

Theorem 17.16 (The General Linear Hypothesis and the F-Test)**.** *Let $C \in \mathbb{R}^{q\times p}$ with $\operatorname{rank} C = q$, and $H_0 : C\boldsymbol{\beta} = \mathbf{d}$. With SSE_0 the smallest residual sum of squares under the constraint H_0,*

$$F = \frac{(C\hat{\boldsymbol{\beta}} - \mathbf{d})^\top[C(X^\top X)^{-1}C^\top]^{-1}(C\hat{\boldsymbol{\beta}} - \mathbf{d})/q}{s^2} = \frac{(\mathrm{SSE}_0 - \mathrm{SSE})/q}{\mathrm{SSE}/(n-p)},$$

and in the normal model $F \sim F_{q,n-p}$ under H_0. H_0 is rejected if $F > F_{q,n-p,\alpha}$.

Proof. Under H_0, $C\hat{\boldsymbol\beta} - \mathbf{d} = C(\hat{\boldsymbol\beta} - \boldsymbol\beta) \sim \mathcal{N}_q(\mathbf{0}, \sigma^2 C(X^\top X)^{-1} C^\top)$; by Theorem 4.13(5) the quadratic form in the numerator divided by σ^2 has a χ^2_q distribution. This quadratic form depends only on $\hat{\boldsymbol\beta}$, so by Theorem 17.12(3) it is independent of SSE; Definition 16.4(3) gives $F_{q,n-p}$. Second equality: the restricted LS solution, found with Lagrange multipliers, is $\hat{\boldsymbol\beta}_0 = \hat{\boldsymbol\beta} - (X^\top X)^{-1} C^\top [C(X^\top X)^{-1} C^\top]^{-1} (C\hat{\boldsymbol\beta} - \mathbf{d})$. By the decomposition in Theorem 17.6(1), $\mathrm{SSE}_0 = \mathrm{SSE} + \|X(\hat{\boldsymbol\beta} - \hat{\boldsymbol\beta}_0)\|^2$, and the last term equals the quadratic form in the numerator. □

Proposition 17.17 (The F-Test Is a Likelihood Ratio Test)**.** *In the normal linear model the generalised likelihood ratio for $H_0 : C\boldsymbol\beta = \mathbf{d}$ is*

$$\Lambda = \Big(\frac{\mathrm{SSE}}{\mathrm{SSE}_0}\Big)^{n/2} = \Big(1 + \frac{q}{n-p} F\Big)^{-n/2},$$

a decreasing function of F; the LRT is the F-test. Moreover $-2\ln\Lambda = n\ln\big(1 + \frac{qF}{n-p}\big) \approx qF \xrightarrow{d} \chi^2_q$, consistent with Wilks' theorem (Theorem 15.5).

Proof. In both models the MLE of σ^2 is the corresponding SSE/n, and the maximised likelihood takes the value

$$\sup L = \big(2\pi\,\mathrm{SSE}/n\big)^{-n/2} e^{-n/2};$$

take the ratio for the two models. This is the multidimensional generalisation of the result "LRT = t-test" of Example 15.9. □

17.4 Without Normality: Asymptotic Theory and Heteroscedasticity

Theorem 17.18 (Asymptotic Normality of the LS Estimator)**.** *Let the errors be i.i.d. with $\mathbb{E}\varepsilon_i = 0$ and $\operatorname{Var}\varepsilon_i = \sigma^2 < \infty$ (not necessarily normal). If $\frac{1}{n} X^\top X \to Q$ (positive definite) and $\max_i h_{ii} \to 0$, then*

$$\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta) \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \sigma^2 Q^{-1}), \qquad s^2 \xrightarrow{P} \sigma^2.$$

Proof. $\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta) = \big(\frac{1}{n} X^\top X\big)^{-1} \frac{1}{\sqrt{n}} X^\top \boldsymbol\varepsilon$. By Cramér–Wold (Theorem 6.14), for every $\mathbf{a}$, $\frac{1}{\sqrt{n}} \mathbf{a}^\top X^\top \boldsymbol\varepsilon = \sum_i \frac{\mathbf{a}^\top \mathbf{x}_i}{\sqrt{n}} \varepsilon_i$ is a sum of independent but not identically distributed terms; the condition $\max_i h_{ii} \to 0$ ensures the Lindeberg condition (Theorem 6.11).

Checking the Lindeberg condition. Fix $\mathbf{a} \neq \mathbf{0}$, and let $c_i = \mathbf{a}^\top \mathbf{x}_i / \sqrt{n}$ and $s_n^2 = \sigma^2 \sum_i c_i^2 = \sigma^2 \mathbf{a}^\top (\frac{1}{n} X^\top X) \mathbf{a} \to \sigma^2 \mathbf{a}^\top Q \mathbf{a} > 0$. The Cauchy–Schwarz inequality in the $X^\top X$ metric gives

$$\begin{aligned}(\mathbf{a}^\top \mathbf{x}_i)^2 &= \big(\mathbf{a}^\top (X^\top X)^{1/2} (X^\top X)^{-1/2} \mathbf{x}_i\big)^2 \\ &\le \big(\mathbf{a}^\top X^\top X \mathbf{a}\big)\big(\mathbf{x}_i^\top (X^\top X)^{-1} \mathbf{x}_i\big) = n\, \mathbf{a}^\top \big(\tfrac{1}{n} X^\top X\big)\mathbf{a} \cdot h_{ii};\end{aligned}$$

hence $m_n^2 = \max_i c_i^2 \le \mathbf{a}^\top (\frac{1}{n} X^\top X) \mathbf{a} \cdot \max_i h_{ii} \to 0$. For every $\eta > 0$,

$$\begin{aligned}\frac{1}{s_n^2} \sum_i \mathbb{E}\big[c_i^2 \varepsilon_i^2 \mathbb{I}\{|c_i \varepsilon_i| > \eta s_n\}\big] &\le \frac{\sum_i c_i^2}{s_n^2}\, \mathbb{E}\big[\varepsilon_1^2 \mathbb{I}\{|\varepsilon_1| > \eta s_n / m_n\}\big] \\ &= \frac{1}{\sigma^2}\, \mathbb{E}\big[\varepsilon_1^2 \mathbb{I}\{|\varepsilon_1| > \eta s_n / m_n\}\big] \to 0,\end{aligned}$$

because $\eta s_n / m_n \to \infty$ and $\mathbb{E}\varepsilon_1^2 < \infty$ (dominated convergence). By Theorem 6.11, $\frac{1}{\sqrt{n}} \mathbf{a}^\top X^\top \boldsymbol\varepsilon = \sum_i c_i \varepsilon_i \xrightarrow{d} \mathcal{N}(0, \sigma^2 \mathbf{a}^\top Q \mathbf{a})$; by Cramér–Wold, $\frac{1}{\sqrt{n}} X^\top \boldsymbol\varepsilon \xrightarrow{d} \mathcal{N}_p(\mathbf{0}, \sigma^2 Q)$.

Conclusion. Since $(\frac{1}{n} X^\top X)^{-1} \to Q^{-1}$ (deterministically), Slutsky gives $\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta) \xrightarrow{d} Q^{-1} \mathcal{N}_p(\mathbf{0}, \sigma^2 Q) = \mathcal{N}_p(\mathbf{0}, \sigma^2 Q^{-1})$.

s^2. Since $\mathrm{SSE} = \boldsymbol{\varepsilon}^\top (I-H)\boldsymbol{\varepsilon}$, $\frac{\mathrm{SSE}}{n} = \frac{1}{n}\boldsymbol{\varepsilon}^\top\boldsymbol{\varepsilon} - \frac{1}{n}\boldsymbol{\varepsilon}^\top H\boldsymbol{\varepsilon}$. The first term $\xrightarrow{P} \sigma^2$ by the WLLN. The second term is non-negative with expectation $\sigma^2 \mathrm{tr}(H)/n = p\sigma^2/n \to 0$; by Markov's inequality it $\xrightarrow{P} 0$. With $n/(n-p) \to 1$, $s^2 = \mathrm{SSE}/(n-p) \xrightarrow{P} \sigma^2$. □

Remark 17.19. Consequently the t and F tests are *asymptotically* valid for non-normal errors as well. The condition $\max_i h_{ii} \to 0$ requires that no observation dominate the estimate; a single high-leverage observation (e.g. an x value far from all the others) can spoil this approximation.

Proposition 17.20 (Heteroscedasticity and the Sandwich Estimator [85])**.** *If the error covariance is diagonal but not constant,* $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \Omega = \mathrm{diag}(\sigma_1^2, \dots, \sigma_n^2)$*, then LS is still unbiased, but*

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1} \neq \sigma^2 (X^\top X)^{-1}.$$

White's [85] sandwich *(HC0) estimator* $\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = (X^\top X)^{-1}\Big(\sum_i e_i^2 \mathbf{x}_i \mathbf{x}_i^\top\Big)(X^\top X)^{-1}$ *is consistent under suitable conditions.*

Proof. The first equality follows from Proposition 4.12(2). For consistency one shows $\frac{1}{n}\sum e_i^2 \mathbf{x}_i\mathbf{x}_i^\top - \frac{1}{n}\sum \sigma_i^2 \mathbf{x}_i\mathbf{x}_i^\top \xrightarrow{P} 0$ ($e_i \approx \varepsilon_i$ and the law of large numbers). □

Remark 17.21 (Generalised Least Squares). If $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 V$ with V known, the model transformed by $V^{-1/2}$ satisfies the Gauss–Markov assumptions, and the BLUE is the *generalised LS* (Aitken) estimator $\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = (X^\top V^{-1} X)^{-1} X^\top V^{-1} \mathbf{Y}$. Weighted LS is the special case of diagonal V.

17.5 Worked Examples

Example 17.22 (Simple Linear Regression)**.** $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $X = [\mathbf{1}\ \mathbf{x}]$. With $S_{xx} = \sum (x_i - \bar{x})^2$ and $S_{xy} = \sum (x_i - \bar{x})(Y_i - \bar{Y})$, the solution of the normal equations is

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}, \qquad \mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \qquad \mathrm{Var}(\hat{\beta}_0) = \sigma^2\Big(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\Big).$$

The variance of the slope is inversely proportional to the spread of the x's: a good design spreads the x's over a wide range. For $H_0 : \beta_1 = 0$, $F = t^2$, $q = 1$ and $F = \frac{(n-2)R^2}{1-R^2}$; in simple regression $R^2 = r_{xy}^2$ (the square of the sample correlation).

Example 17.23 (Gauss–Markov in Concrete Form: The Endpoint Estimator)**.** In the same model, $\tilde{\beta}_1 = \frac{Y_n - Y_1}{x_n - x_1}$ (using only the observations at the smallest and largest x) is linear and unbiased: $\mathbb{E}\tilde{\beta}_1 = \beta_1$. Its variance is $\frac{2\sigma^2}{(x_n - x_1)^2}$. For $x_i = 1, \dots, n$, $S_{xx} = \frac{n(n^2-1)}{12}$, so

$$\frac{\mathrm{Var}(\tilde{\beta}_1)}{\mathrm{Var}(\hat{\beta}_1)} = \frac{2S_{xx}}{(n-1)^2} = \frac{n(n+1)}{6(n-1)},$$

which is ≈ 3.7 for $n = 20$: as Gauss–Markov predicts, LS is better, and the difference grows with n.

Example 17.24 (One-Way Analysis of Variance)**.** k groups with n_j observations in group j: $Y_{ji} = \mu_j + \varepsilon_{ji}$. The design matrix consists of group indicators ($p = k$), and LS gives $\hat{\mu}_j = \bar{Y}_j$. $H_0 : \mu_1 = \cdots = \mu_k$ is a set of $q = k - 1$ linear constraints; the restricted model has a single mean and $\mathrm{SSE}_0 = \sum_{j,i} (Y_{ji} - \bar{Y})^2$. By Theorem 17.16,

$$F = \frac{\sum_j n_j (\bar{Y}_j - \bar{Y})^2 / (k-1)}{\sum_j \sum_i (Y_{ji} - \bar{Y}_j)^2 / (n-k)} \sim F_{k-1, n-k} \quad (\text{under } H_0).$$

This is Fisher's analysis of variance (ANOVA): the ratio of the "between-group" and "within-group" mean squares. For $k = 2$, $F = t^2$ (the pooled two-sample t-test).

Example 17.25 (Multicollinearity)**.** If the sample correlation between two explanatory variables is r, then (with centred, scaled variables) the variance of each coefficient is $\frac{\sigma^2}{S_{jj}} \cdot \frac{1}{1-r^2}$. $\mathrm{VIF} = 1/(1-r^2)$ is the *variance inflation factor*: for $r = 0.95$, $\mathrm{VIF} \approx 10.3$. The coefficients remain unbiased but become very uncertain; the predictions ($\hat{Y}$), however, are usually not affected.

17.6 Computational Applications

17.6.1 Verifying the Sampling Theory

$n = 30$, an intercept and two explanatory variables ($p = 3$); the design matrix is generated once and kept fixed, and only the errors are regenerated (the "fixed X" assumption).

code/ch17/c17_1_verifying_sampling_theory.py

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
n, p, sigma, R = 30, 3, 2.0, 100_000
X = np.c_[np.ones(n), rng.uniform(0, 10, n), rng.normal(5, 2, n)]
beta = np.array([1.0, 0.5, -0.3])
XtX_inv = np.linalg.inv(X.T @ X)
eps = rng.normal(0, sigma, (R, n))
Y = X @ beta + eps
B = Y @ X @ XtX_inv                    # each row is a beta_hat (X'X symmetric)
E = Y - B @ X.T
s2 = np.sum(E**2, axis=1) / (n - p)

print("E[beta_hat]        :", B.mean(0).round(4), "  true:", beta)
print("sd(beta_hat) MC    :", B.std(0).round(4))
print("sd theory          :", (sigma*np.sqrt(np.diag(XtX_inv))).round(4))
print(f"E[s^2] = {s2.mean():.4f} (sigma^2 = {sigma**2}) | Var[(n-p)s^2/sigma^2] = {np.var((n-
    p)*s2/sigma**2):.3f} (2(n-p) = {2*(n-p)})")
print(f"Corr(beta_hat_1, s^2) = {np.corrcoef(B[:, 1], s2)[0, 1]:+.4f}  (independence)")

tc = stats.t.ppf(0.975, n - p)
se1 = np.sqrt(s2 * XtX_inv[1, 1])
print(f"coverage of 95% t-interval for beta_1 = {np.mean(np.abs(B[:, 1] - beta[1]) <= tc*se1)
    :.4f}")

# F-test: H0: beta_1 = beta_2 = 0 ; size (H0 true) and power (H0 false)
C = np.array([[0, 1, 0], [0, 0, 1]]); q = 2
M = np.linalg.inv(C @ XtX_inv @ C.T)
def F_stat(Bm, s2v):
    D = Bm @ C.T
    return np.einsum("ij,jk,ik->i", D, M, D) / q / s2v
Y0 = X @ np.array([1.0, 0.0, 0.0]) + eps
B0 = Y0 @ X @ XtX_inv; s20 = np.sum((Y0 - B0 @ X.T)**2, 1)/(n - p)
Fk = stats.f.ppf(0.95, q, n - p)
print(f"F-test size = {np.mean(F_stat(B0, s20) > Fk):.4f} (0.05) | power (true beta) = {np.
    mean(F_stat(B, s2) > Fk):.4f}")
```

Output

```
E[beta_hat]        : [ 1.0016  0.5006 -0.3007]   true: [ 1.   0.5 -0.3]
sd(beta_hat) MC    : [1.4755 0.1257 0.2308]
sd theory          : [1.476  0.1257 0.2311]
E[s^2] = 3.9999 (sigma^2 = 4.0) | Var[(n-p)s^2/sigma^2] = 53.873 (2(n-p) = 54)
Corr(beta_hat_1, s^2) = +0.0027  (independence)
coverage of 95% t-interval for beta_1 = 0.9504
F-test size = 0.0494 (0.05) | power (true beta) = 0.9590
```

17.6.2 Gauss–Markov: LS versus the Endpoint Estimator; ANOVA

code/ch17/c17_2_gauss_markov_ls_vs_endpoint_anova.py

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
n, R = 20, 200_000
x = np.arange(1, n + 1, dtype=float)
Y = 2.0 + 0.7*x + rng.normal(0, 1.5, (R, n))
xc = x - x.mean()
b_ls = (Y @ xc) / np.sum(xc**2)
b_end = (Y[:, -1] - Y[:, 0]) / (x[-1] - x[0])
print(f"LS slope  : mean={b_ls.mean():.4f}, Var={b_ls.var():.5f}")
print(f"Endpoints : mean={b_end.mean():.4f}, Var={b_end.var():.5f}")
print(f"Variance ratio = {b_end.var()/b_ls.var():.3f} | theory n(n+1)/(6(n-1)) = {n*(n+1)
    /(6*(n-1)):.3f}")

# One-way ANOVA: F by hand, the general C*beta formula and scipy
g = [rng.normal(m, 2.0, k) for m, k in [(10, 8), (11, 10), (13, 7)]]
y = np.concatenate(g); N, kgr = y.size, len(g)
ybar = y.mean()
F_hand = (sum(len(a)*(a.mean() - ybar)**2 for a in g)/(kgr - 1)) / (sum(((a - a.mean())**2).
    sum() for a in g)/(N - kgr))
Xg = np.zeros((N, kgr)); s = 0
for j, a in enumerate(g):
    Xg[s:s + len(a), j] = 1; s += len(a)
bh = np.linalg.solve(Xg.T @ Xg, Xg.T @ y); s2g = np.sum((y - Xg @ bh)**2)/(N - kgr)
Cg = np.array([[1, -1, 0], [1, 0, -1]])
d = Cg @ bh
F_gen = d @ np.linalg.solve(Cg @ np.linalg.inv(Xg.T @ Xg) @ Cg.T, d) / 2 / s2g
print(f"ANOVA F: by hand={F_hand:.4f} | general linear hypothesis={F_gen:.4f} | scipy
    f_oneway={stats.f_oneway(*g).statistic:.4f}"
      f" | p={stats.f_oneway(*g).pvalue:.4f}")
```

Output

```
LS slope  : mean=0.7001, Var=0.00340
Endpoints : mean=0.7004, Var=0.01247
Variance ratio = 3.664 | theory n(n+1)/(6(n-1)) = 3.684
ANOVA F: by hand=1.4160 | general linear hypothesis=1.4160 | scipy f_oneway=1.4160 | p=0.2640
```

17.6.3 Non-Normal Errors, Heteroscedasticity and Sandwich Standard Errors

code/ch17/c17_3_nonnormal_errors_heteroscedasticity_sandwich.py

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
R = 20_000
def coverage(n, err, sandwich=False):
    x = np.linspace(0, 10, n); X = np.c_[np.ones(n), x]
    XtX_inv = np.linalg.inv(X.T @ X)
    eps = err(x, (R, n))
    Y = X @ np.array([1.0, 0.5]) + eps
    B = Y @ X @ XtX_inv; E = Y - B @ X.T
    if sandwich:
        meat = np.einsum("ri,ij,ik->rjk", E**2, X, X)            # sum e_i^2 x_i x_i'
```

```
        V = XtX_inv @ meat @ XtX_inv
        se = np.sqrt(V[:, 1, 1] * n/(n - 2))                    # HC1 correction
    else:
        se = np.sqrt(np.sum(E**2, 1)/(n - 2) * XtX_inv[1, 1])
    return np.mean(np.abs(B[:, 1] - 0.5) <= stats.t.ppf(0.975, n - 2)*se)

expo = lambda x, sh: rng.exponential(1, sh) - 1                # skewed, constant variance
hetero = lambda x, sh: rng.normal(0, 1, sh) * (0.2 + 0.3*x)   # variance increases with x
for n in [10, 50, 200]:
    print(f"n={n:3d} | skewed errors: classical={coverage(n, expo):.3f} | heteroscedastic:
    classical={coverage(n, hetero):.3f},"
          f" sandwich={coverage(n, hetero, True):.3f}")

# Confidence and prediction bands
n = 25; x = np.sort(rng.uniform(0, 10, n)); X = np.c_[np.ones(n), x]
y = 1 + 0.5*x + rng.normal(0, 1, n)
bh = np.linalg.solve(X.T @ X, X.T @ y); s = np.sqrt(np.sum((y - X @ bh)**2)/(n - 2))
xx = np.linspace(-1, 11, 200); X0 = np.c_[np.ones_like(xx), xx]
q0 = np.einsum("ij,jk,ik->i", X0, np.linalg.inv(X.T @ X), X0); t = stats.t.ppf(0.975, n - 2)
fig, ax = plt.subplots(figsize=(7, 3.6))
ax.scatter(x, y, s=14, color="k", zorder=3, label="data")
ax.plot(xx, X0 @ bh, "b", label="LS line")
ax.fill_between(xx, X0 @ bh - t*s*np.sqrt(q0), X0 @ bh + t*s*np.sqrt(q0), alpha=0.35, color="
    tab:blue", label="95% confidence band (mean)")
ax.fill_between(xx, X0 @ bh - t*s*np.sqrt(1 + q0), X0 @ bh + t*s*np.sqrt(1 + q0), alpha=0.15,
     color="tab:orange", label="95% prediction band (new observation)")
ax.set_xlabel("x"); ax.set_ylabel("y"); ax.legend(fontsize=8, loc="upper left"); fig.
    tight_layout(); fig.savefig("regresyon_bant.pdf")
```

Output

```
n= 10 | skewed errors: classical=0.955 | heteroscedastic: classical=0.929, sandwich=0.896
n= 50 | skewed errors: classical=0.950 | heteroscedastic: classical=0.931, sandwich=0.937
n=200 | skewed errors: classical=0.951 | heteroscedastic: classical=0.931, sandwich=0.950
```

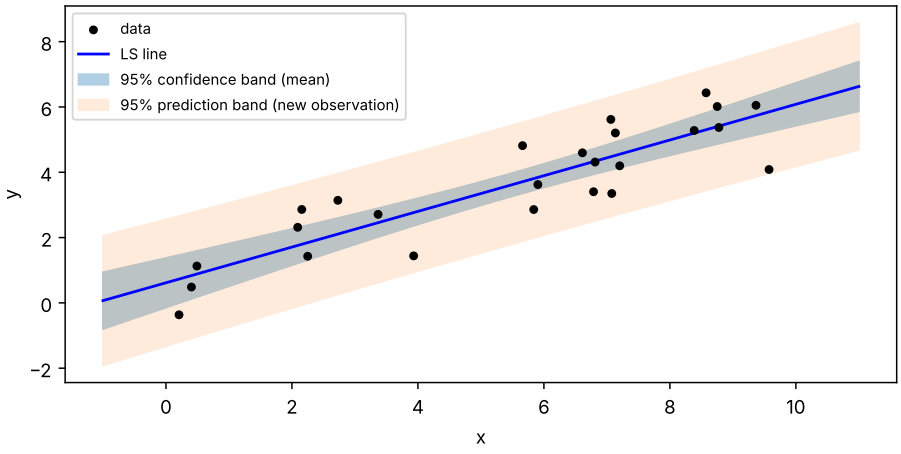


Figure 17.1: Confidence band (dark) and prediction band (light) in simple regression. The confidence band is narrowest near $\bar{x}$ and wide at the ends (q_0 grows with the distance of $\mathbf{x}_0$ from $\bar{x}$); the prediction band is everywhere at least $\pm t\, s$ wide.

17.7 Comments and Notes

Remark 17.26 (The Hierarchy of Model Assumptions). It is important to distinguish which assumption an inference relies on: (i) Unbiasedness requires only $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$ (a correct mean

structure). (ii) The covariance formula $\sigma^2(X^\top X)^{-1}$ and Gauss–Markov additionally require constant variance and uncorrelatedness. (iii) The exact t and F distributions require normality; however, thanks to Theorem 17.18, normality is largely unnecessary in large samples. The simulation above illustrates this: skewed errors affect the coverage little even for small n, whereas heteroscedasticity produces an error that does *not* improve as n grows (the classical coverage stays at ≈ 0.93 even at $n = 200$); sandwich standard errors remove it in large samples. In small samples, however, the sandwich estimator is biased downwards: at $n = 10$ its coverage is even lower than that of the classical interval. For small n, therefore, the HC3 correction, which uses $e_i^2/(1-h_{ii})^2$ instead of e_i^2, is recommended [54].

Remark 17.27 (Advanced Topics). Natural continuations of this chapter are generalised linear models (logistic and Poisson regression; the likelihood theory is the same as in Chapters 12 and 15), mixed-effects models, regularisation (ridge, lasso) and model selection (AIC, BIC, cross-validation). For the bootstrap in regression see Chapter 18.

17.8 Chapter Summary

- The LS estimate solves the normal equations and gives the orthogonal projection of $\mathbf{Y}$ onto the column space of X (Theorem 17.6). The hat matrix H is symmetric and idempotent; its diagonal gives the leverages.
- The Gauss–Markov theorem (Theorem 17.9): without requiring normality, LS has the smallest variance among linear unbiased estimators (BLUE).
- With normal errors $\hat{\boldsymbol{\beta}}$ and SSE are independent (Theorem 17.12). The t and F tests are exactly valid (Theorem 17.16), and the F-test is a likelihood ratio test (Proposition 17.17).
- The prediction interval for a new observation is always wider than the confidence interval for the mean response (Corollary 17.14).
- Without normality, LS is asymptotically normal as the leverages shrink (Theorem 17.18). Under heteroscedasticity, sandwich standard errors should be used (Proposition 17.20).

Further reading. [69], [16, Chapters 11–12], [81, Chapter 13]; for robust standard errors, [85].

17.9 Exercises

Exercise 17.1. Show that in simple linear regression $\operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\sigma^2 \bar{x}/S_{xx}$. What is required of the design for $\hat\beta_0$ and $\hat\beta_1$ to be uncorrelated? → Solution

Exercise 17.2. Show that $\sum_i h_{ii} = p$ and $0 \le h_{ii} \le 1$. The average leverage is p/n; write a function that flags high-leverage observations by the rule $h_{ii} > 2p/n$. Show that the variance of the residuals is $\operatorname{Var}(e_i) = \sigma^2(1-h_{ii})$; why does this call for "studentised residuals"? → Solution

Exercise 17.3. Compute the bias and covariance of the ridge estimator $\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda I)^{-1} X^\top \mathbf{Y}$. Show that for some $\lambda > 0$, $\operatorname{MSE}(\hat{\boldsymbol{\beta}}_\lambda) < \operatorname{MSE}(\hat{\boldsymbol{\beta}})$ (total squared loss) [38]. Does this contradict Gauss–Markov? → Solution

Exercise 17.4. For polynomial regression $Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$, write code that tests $H_0 : \beta_2 = 0$ with the F-test and compares the result with t^2. Compute the correlation between x and x^2 and the VIF on the interval $x \in [10, 11]$; what does centring $(x - \bar{x})$ change? → Solution

Chapter 18

The Bootstrap and Resampling Methods

In earlier chapters we obtained the sampling distribution of an estimator in one of two ways: either exactly (pivots, normal sampling theory) or asymptotically (CLT, the delta method, Fisher information). In many realistic situations, however, both are hard: for the standard error of the median, of a correlation coefficient or of a complicated ratio, there is either no formula or the formula depends on unknown density values. Efron's [25] bootstrap solves this problem by computation: it replaces the unknown F by the empirical distribution $\hat{F}_n$ and simulates the distribution of the estimator by resampling from $\hat{F}_n$. In this chapter we discuss the plug-in principle, the bootstrap standard error and bias estimate, four bootstrap confidence intervals, why the bootstrap works (and when it does not), the jackknife and permutation tests.

18.1 Definitions

Definition 18.1 (Statistical Functional and the Plug-In Principle)**.** Write the parameter as a functional of the distribution: $\theta = T(F)$. For example the mean is $T(F) = \int x\, dF(x)$, the median $T(F) = F^{-1}(1/2)$, the variance $T(F) = \int (x - \int y\, dF)^2 dF(x)$. The *plug-in estimator* is obtained by putting the empirical distribution $\hat{F}_n$ (the discrete distribution giving mass $1/n$ to each observation; Chapter 2) in place of F: $\hat{\theta} = T(\hat{F}_n)$. For example, the plug-in estimator of the variance is $\hat{\sigma}^2 = \frac{1}{n}\sum (X_i - \bar{X})^2$.

Definition 18.2 (Bootstrap Sample and Bootstrap Distribution)**.** After $\mathbf{x} = (x_1, \ldots, x_n)$ has been observed, an i.i.d. sample of size n from $\hat{F}_n$ (i.e. n draws *with replacement* from $\mathbf{x}$) is called a *bootstrap sample*: $\mathbf{X}^* = (X_1^*, \ldots, X_n^*)$. The (conditional) distribution of $\hat{\theta}^* = T(\hat{F}_n^*)$ with $\mathbf{x}$ held fixed is called the *bootstrap distribution*; probabilities and expectations under it are denoted $\mathbb{P}^*$ and $\mathbb{E}^*$.

Definition 18.3 (Bootstrap Standard Error and Bias Estimate)**.**

$$\widehat{\mathrm{se}}_{\mathrm{boot}} = \sqrt{\mathrm{Var}^*(\hat{\theta}^*)}, \qquad \widehat{\mathrm{Bias}}_{\mathrm{boot}} = \mathbb{E}^*[\hat{\theta}^*] - \hat{\theta}.$$

In practice B bootstrap samples are generated and the sample standard deviation and mean of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ are used (a Monte Carlo approximation).

Definition 18.4 (Parametric Bootstrap)**.** If the model $\{F_\theta\}$ is known, resampling from $F_{\hat{\theta}}$ (e.g. the distribution estimated by the MLE) instead of $\hat{F}_n$ is called the *parametric bootstrap*.

Definition 18.5 (Bootstrap Confidence Intervals)**.** Let $\hat{\theta}^*_{(\alpha)}$ be the α quantile of the bootstrap distribution.

1. **Percentile:** $[\hat\theta^*_{(\alpha/2)},\ \hat\theta^*_{(1-\alpha/2)}]$.
2. **Basic (reverse percentile):** $[2\hat\theta - \hat\theta^*_{(1-\alpha/2)},\ 2\hat\theta - \hat\theta^*_{(\alpha/2)}]$; it approximates the distribution of $\hat\theta - \theta$ by that of $\hat\theta^* - \hat\theta$.
3. **Bootstrap-t (studentised):** with the quantiles $t^*_{(\alpha)}$ of the statistic $t^* = (\hat\theta^* - \hat\theta)/\widehat{\mathrm{se}}^*$, $[\hat\theta - t^*_{(1-\alpha/2)}\widehat{\mathrm{se}},\ \hat\theta - t^*_{(\alpha/2)}\widehat{\mathrm{se}}]$.
4. **BCa (bias-corrected and accelerated):** with $\hat z_0 = \Phi^{-1}(\mathbb{P}^*(\hat\theta^* < \hat\theta))$ and the acceleration $\hat a$ computed from the jackknife values, the endpoints of the percentile interval are shifted to the levels $\alpha_1 = \Phi\Big(\hat z_0 + \frac{\hat z_0 + z_{(\alpha/2)}}{1 - \hat a(\hat z_0 + z_{(\alpha/2)})}\Big)$ and α_2 (similarly, with $z_{(1-\alpha/2)}$): $[\hat\theta^*_{(\alpha_1)}, \hat\theta^*_{(\alpha_2)}]$. Here $z_{(p)} = \Phi^{-1}(p)$.

Definition 18.6 (Jackknife)**.** Let $\hat\theta_{(i)}$ be the estimate computed with the i-th observation deleted, and $\hat\theta_{(\cdot)} = \frac{1}{n}\sum_i \hat\theta_{(i)}$. The jackknife bias and standard error estimates are

$$\widehat{\mathrm{Bias}}_{\mathrm{jack}} = (n-1)(\hat\theta_{(\cdot)} - \hat\theta), \qquad \widehat{\mathrm{se}}_{\mathrm{jack}} = \sqrt{\frac{n-1}{n}\sum_{i=1}^{n}\big(\hat\theta_{(i)} - \hat\theta_{(\cdot)}\big)^2},$$

and the bias-corrected estimator is $\hat\theta_{\mathrm{jack}} = \hat\theta - \widehat{\mathrm{Bias}}_{\mathrm{jack}} = n\hat\theta - (n-1)\hat\theta_{(\cdot)}$.

Definition 18.7 (Permutation Test)**.** For two samples $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$, let $H_0 : F_X = F_Y$. The distribution of the values of a statistic T computed over all reassignments of the $N = m + n$ observations into the $\binom{N}{m}$ different groupings is called the *permutation distribution*, and the upper-tail proportion of the observed T in this distribution is the *permutation p-value*.

18.2 Main Theorems and Proofs

Proposition 18.8 (Structure of a Bootstrap Sample)**.** *1. The probability that a particular x_i does* not *appear at all in a bootstrap sample is $(1 - 1/n)^n \to e^{-1} \approx 0.368$; i.e. a bootstrap sample contains on average about 63.2% of the distinct observations.*

2. The number of distinct bootstrap samples, disregarding order, is $\binom{2n-1}{n}$ ($\approx 6.9 \times 10^{10}$ for $n = 20$); this is why the bootstrap distribution is in practice approximated by Monte Carlo.

3. $\mathbb{E}^*[\bar X^*] = \bar x$ and $\mathrm{Var}^*(\bar X^*) = \hat\sigma^2/n$, where $\hat\sigma^2 = \frac{1}{n}\sum(x_i - \bar x)^2$. Hence the exact bootstrap SE of the mean is $\hat\sigma/\sqrt{n}$ — the plug-in version of the classical formula.

Proof. (1) In each draw the probability that x_i is not selected is $1 - 1/n$, and the draws are independent. (2) The number of multisets ("stars and bars"): the number of ways of distributing n identical balls into n boxes is $\binom{2n-1}{n}$. (3) The X_i^* are i.i.d. from $\hat F_n$, whose mean is $\bar x$ and variance $\hat\sigma^2$. □

Theorem 18.9 (Bootstrap Consistency for the Mean [8, 74])**.** *Let $X_1, X_2, \dots \overset{i.i.d.}{\sim} F$ with $\sigma^2 = \mathrm{Var}\, X_1 > 0$ and $\mathbb{E}|X_1|^3 < \infty$. Then, with probability 1,*

$$\sup_{x\in\mathbb{R}}\Big|\mathbb{P}^*(\sqrt{n}(\bar X_n^* - \bar X_n) \le x) - \mathbb{P}(\sqrt{n}(\bar X_n - \mu) \le x)\Big| \xrightarrow{n\to\infty} 0.$$

(The result also holds under $\mathbb{E}X_1^2 < \infty$ alone [8, 74].)

Proof. With the data fixed, the X_i^* are i.i.d. variables with mean $\bar{X}_n$, variance $\hat{\sigma}_n^2$ and absolute third central moment $\hat{\rho}_n = \frac{1}{n}\sum |X_i - \bar{X}_n|^3$. Applying the Berry–Esseen theorem (Theorem 6.12) to this conditional distribution,

$$\sup_x \left| \mathbb{P}^*\Big(\tfrac{\sqrt{n}(\bar{X}_n^* - \bar{X}_n)}{\hat{\sigma}_n} \le x \Big) - \Phi(x) \right| \le \frac{C\hat{\rho}_n}{\hat{\sigma}_n^3 \sqrt{n}}.$$

By the SLLN, $\hat{\sigma}_n \xrightarrow{\text{a.s.}} \sigma > 0$ and $\hat{\rho}_n \xrightarrow{\text{a.s.}} \mathbb{E}|X_1 - \mu|^3 < \infty$; hence the right-hand side goes to 0 a.s. On the other hand, by the CLT $\mathbb{P}(\sqrt{n}(\bar{X}_n - \mu)/\sigma \le x) \to \Phi(x)$, and since the limit is continuous the convergence is uniform (Pólya). Both distributions converge uniformly to $\mathcal{N}(0, \sigma^2)$ and $\hat{\sigma}_n/\sigma \to 1$; the triangle inequality gives the result. □

Remark 18.10 (Second-Order Accuracy). Theorem 18.9 says that the bootstrap is at least as good as the normal approximation. More is true: by Edgeworth expansions, for *pivotal* (studentised) statistics the error of the bootstrap approximation is $O_P(n^{-1})$, whereas that of the normal approximation is $O(n^{-1/2})$ [33]. The bootstrap captures the skewness term "automatically". This is the theoretical reason why the coverage error of bootstrap-t and BCa intervals is $O(n^{-1})$, while that of percentile and normal intervals is generally $O(n^{-1/2})$. But this is an asymptotic ranking; the actual performance in finite samples must be checked by simulation (in the coverage table below, BCa falls behind the classical t at $n = 20$).

Theorem 18.11 (Jackknife Bias Correction). *If $\mathbb{E}\hat{\theta}_n = \theta + \frac{a}{n} + \frac{b}{n^2} + O(n^{-3})$ (with the same a, b for all n), then the bias of the jackknife estimator is $O(n^{-2})$:*

$$\mathbb{E}\hat{\theta}_{jack} = \theta - \frac{b}{n(n-1)} + O(n^{-2}).$$

Proof. Since each $\hat{\theta}_{(i)}$ is based on $n-1$ observations, $\mathbb{E}\hat{\theta}_{(\cdot)} = \theta + \frac{a}{n-1} + \frac{b}{(n-1)^2} + O(n^{-3})$. Thus

$$\mathbb{E}\hat{\theta}_{\text{jack}} = n\Big(\theta + \frac{a}{n} + \frac{b}{n^2}\Big) - (n-1)\Big(\theta + \frac{a}{n-1} + \frac{b}{(n-1)^2}\Big) + O(n^{-2}) = \theta + \frac{b}{n} - \frac{b}{n-1} + O(n^{-2}),$$

and $\frac{b}{n} - \frac{b}{n-1} = -\frac{b}{n(n-1)}$. The bias of order $1/n$ disappears completely. □

Example 18.12 (The Jackknife Turns $\hat{\sigma}^2$ into S^2). For $\hat{\theta} = \hat{\sigma}^2$, $\mathbb{E}\hat{\sigma}^2 = \sigma^2 - \sigma^2/n$, i.e. $a = -\sigma^2$, $b = 0$. Theorem 18.11 says that the jackknife estimator is unbiased; direct computation gives $\hat{\theta}_{\text{jack}} = S^2$ (verified with the code below).

Theorem 18.13 (Exact Level of the Permutation Test). *Under H_0 let $(Z_1, \dots, Z_N) = (X_1, \dots, X_m, Y_1, \dots, Y_n)$ be exchangeable, i.e. every permutation has the same joint distribution (i.i.d. with $F_X = F_Y$ ensures this). If T_0 is the observed statistic and $T_1, \dots, T_B$ are the values computed from random permutations, then for*

$$p = \frac{1 + \#\{b : T_b \ge T_0\}}{B + 1}$$

we have $\mathbb{P}_{H_0}(p \le \alpha) \le \alpha$ for every $\alpha \in (0, 1)$. That is, the permutation test has level exactly α for every n, without any distributional assumption.

Proof. Under H_0, $(T_0, T_1, \dots, T_B)$ is exchangeable: the observed assignment has the same distribution as any of the random assignments. Hence (if there are no ties) the rank of T_0 among these $B+1$ values is uniformly distributed on $\{1, \dots, B+1\}$ and $\mathbb{P}(p \le k/(B+1)) = k/(B+1)$; with ties the probability is at most this. □

18.3 Worked Examples

Example 18.14 (Standard Error of the Median)**.** The asymptotic standard error of the sample median is $1/(2f(m)\sqrt{n})$ (Example 12.14); but $f(m)$ is the value of the unknown density at the median, and estimating it requires choosing a bandwidth. The bootstrap SE requires no density estimate: resample B times, compute the median each time, and take the standard deviation. For Exp(1) data the true values are $m = \ln 2$, $f(m) = \frac{1}{2}$, and the asymptotic SE is $1/\sqrt{n}$.

Example 18.15 (Where the Bootstrap Fails: An Extreme Value)**.** $X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0, \theta)$, $\hat{\theta} = X_{(n)}$. The true distribution: $n(\theta - X_{(n)})/\theta \xrightarrow{d} \text{Exp}(1)$, a continuous limit. In the bootstrap, however,

$$\mathbb{P}^*\big(X^*_{(n)} = X_{(n)}\big) = 1 - \Big(1 - \frac{1}{n}\Big)^n \to 1 - e^{-1} \approx 0.632.$$

The bootstrap distribution has an atom of 63% at 0, whereas the true distribution has no atom. The bootstrap is inconsistent. Why: the functional $T(F) = F^{-1}(1)$ is not "smooth" (Hadamard differentiable) at F, and $\hat{F}_n$ cannot imitate the upper end of F. The parametric bootstrap (resampling from $\text{Uniform}(0, X_{(n)})$), on the other hand, works.

Example 18.16 (Permutation Test)**.** Let there be m observations in a treatment group and n in a control group, and $T = \bar{Y} - \bar{X}$. The observations are pooled, a random m of them receive the label "treatment", and T is recomputed; this is repeated B times. This test is the essence of Fisher's [30] idea of "randomisation inference": random assignment is the source of the test's validity; no normality assumption is needed.

18.4 Computational Applications

18.4.1 Bootstrap Standard Error of the Median

code/ch18/c18_1_bootstrap_se_of_median.py

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
n, B = 50, 4000
x = rng.exponential(1.0, size=n)
idx = rng.integers(0, n, size=(B, n))
med_star = np.median(x[idx], axis=1)
mean_star = x[idx].mean(axis=1)

true_se = np.median(rng.exponential(1.0, size=(200_000, n)), axis=1).std()
print(f"Observed median = {np.median(x):.4f} (true ln2 = {np.log(2):.4f})")
print(f"Median SE: bootstrap = {med_star.std(ddof=1):.4f} | true (MC) = {true_se:.4f} |
    asymptotic 1/sqrt(n) = {1/np.sqrt(n):.4f}")
print(f"Mean SE: bootstrap = {mean_star.std(ddof=1):.4f} | plug-in formula sigma_hat/sqrt(n)
    = {x.std()/np.sqrt(n):.4f}")
print(f"Proportion of distinct observations in a bootstrap sample ≈ {np.mean([np.unique(r).
    size for r in idx[:500]])/n:.3f} (1-1/e = {1-np.exp(-1):.3f})")
print(f"Number of distinct values in the bootstrap distribution of the median: {np.unique(
    med_star).size} / {B}")

fig, ax = plt.subplots(1, 2, figsize=(10, 3.2))
ax[0].hist(mean_star, bins=60, color="tab:blue", alpha=0.7); ax[0].set_title(r"Bootstrap
    distribution of the mean")
ax[1].hist(med_star, bins=60, color="tab:orange", alpha=0.7); ax[1].set_title(r"Bootstrap
    distribution of the median (discrete)")
```

```
for a, s in zip(ax, [mean_star, med_star]):
    for q in np.percentile(s, [2.5, 97.5]):
        a.axvline(q, color="k", ls=":")
fig.tight_layout(); fig.savefig("bootstrap_dagilim.pdf")
```

Output

```
Observed median = 0.6487 (true ln2 = 0.6931)
Median SE: bootstrap = 0.1955 | true (MC) = 0.1406 | asymptotic 1/sqrt(n) = 0.1414
Mean SE: bootstrap = 0.1009 | plug-in formula sigma_hat/sqrt(n) = 0.1006
Proportion of distinct observations in a bootstrap sample ≈ 0.632 (1-1/e = 0.632)
Number of distinct values in the bootstrap distribution of the median: 121 / 4000
```

As predicted by Proposition 18.8(3), the bootstrap SE of the mean agrees almost exactly with the plug-in formula. The bootstrap SE of the median, however, is clearly above the true value (0.141) in this sample. This is not a mistake but the unavoidable variability of any estimate made from a single sample: in an additional simulation with 1000 separate exponential samples, the bootstrap SE of the median had mean 0.146 (i.e. it is approximately unbiased) but standard deviation 0.043; 90% of the values lay between 0.089 and 0.231. The relative variability is larger (about 29%) than for the mean (about 20%), because the bootstrap distribution of the median rests on a few central observations and is step-like. Lesson: the bootstrap SE is itself an estimate and has its own uncertainty.

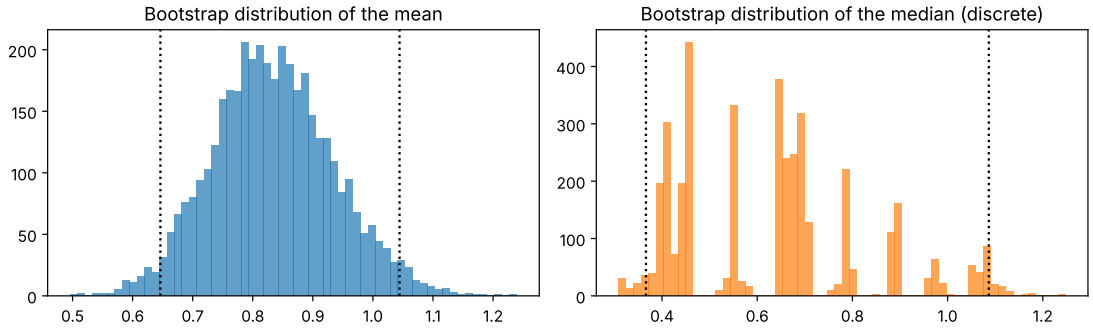


Figure 18.1: Bootstrap distributions obtained from the same exponential sample ($n = 50$); the dotted lines are the endpoints of the 95% percentile interval. Since the median can take only observed values (or the average of two observations), its bootstrap distribution is "step-like".

18.4.2 Coverage of Bootstrap Confidence Intervals

We generate $M = 2000$ data sets of size $n = 20$ from the Exp(1) distribution and in each construct five different 95% intervals for μ. The skewed distribution and the small n bring out the differences between the methods.

code/ch18/c18_2_coverage_of_bootstrap_intervals.py

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
n, B, M, mu = 20, 999, 2000, 1.0
a = 0.05; zl, zu = stats.norm.ppf(a/2), stats.norm.ppf(1 - a/2)
tc = stats.t.ppf(1 - a/2, n - 1)
methods = ["t (classic)", "Percentile", "Basic", "Bootstrap-t", "BCa"]
covers = {k: 0 for k in methods}; left = {k: 0 for k in methods}; length = {k: 0.0 for k in
    methods}
```

```
for _ in range(M):
    x = rng.exponential(mu, n)
    xb, s = x.mean(), x.std(ddof=1); se = s/np.sqrt(n)
    xs = x[rng.integers(0, n, (B, n))]
    m_s, se_s = xs.mean(1), xs.std(1, ddof=1)/np.sqrt(n)
    ql, qu = np.percentile(m_s, [2.5, 97.5])
    t_s = (m_s - xb)/se_s
    tl, tu = np.percentile(t_s, [2.5, 97.5])
    z0 = stats.norm.ppf(np.mean(m_s < xb))
    jack = (x.sum() - x)/(n - 1); d = jack.mean() - jack
    acc = np.sum(d**3)/(6*np.sum(d**2)**1.5)
    a1 = stats.norm.cdf(z0 + (z0 + zl)/(1 - acc*(z0 + zl)))
    a2 = stats.norm.cdf(z0 + (z0 + zu)/(1 - acc*(z0 + zu)))
    iv = {"t (classic)": (xb - tc*se, xb + tc*se), "Percentile": (ql, qu), "Basic": (2*xb -
    qu, 2*xb - ql),
          "Bootstrap-t": (xb - tu*se, xb - tl*se), "BCa": tuple(np.percentile(m_s, [100*a1,
    100*a2]))}
    for k, (L, U) in iv.items():
        covers[k] += (L <= mu <= U); left[k] += (U < mu); length[k] += U - L
print(f"{'Method':12s} | coverage | interval left of mu | mean length")
for k in methods:
    print(f"{k:12s} |  {covers[k]/M:.4f}  |       {left[k]/M:.4f}        |   {length[k]/M:.4f
    }")
```

Output

```
Method       | coverage | interval left of mu | mean length
t (classic)  |  0.9250  |       0.0685        |   0.9032
Percentile   |  0.9060  |       0.0760        |   0.8157
Basic        |  0.8890  |       0.1000        |   0.8157
Bootstrap-t  |  0.9385  |       0.0410        |   1.0531
BCa          |  0.9115  |       0.0570        |   0.8623
```

For skewed data the classical t, percentile and basic intervals stay below the nominal level, and their errors are asymmetric: when they miss, the interval usually lies to the *left* of μ (since large observations in the right tail are rare, in small samples $\bar{X}$ is usually below μ). The percentile and basic intervals are even worse than the classical t: they are scaled by $\hat{\sigma}$ (with divisor $1/n$) and do not include the small-sample widening of the t quantiles. BCa improves the percentile interval with a bias and skewness correction, but at $n = 20$ it still falls behind the classical t; the acceleration $\hat{a}$ is itself a noisy estimate in small samples. The method that comes closest to the nominal level and has the most balanced errors is the *bootstrap-t*; the price is noticeably longer intervals, asymmetric to the right. This is the numerical counterpart of the second-order accuracy that arises when bootstrapping pivotal statistics. Lesson: saying "I used the bootstrap" is no guarantee on its own; in small samples, which interval is used is decisive.

18.4.3 Failure: The Maximum of Uniform$(0, \theta)$

code/ch18/c18_3_failure_uniform_maximum.py

```
import numpy as np

rng = np.random.default_rng(42)
theta, n, B = 1.0, 50, 20_000
x = rng.uniform(0, theta, n)
m = x.max()
np_boot = x[rng.integers(0, n, (B, n))].max(1)                # nonparametric
p_boot = rng.uniform(0, m, (B, n)).max(1)                      # parametric
true_max = rng.uniform(0, theta, (B, n)).max(1)
print(f"P*(X*_(n) = X_(n)) = {np.mean(np_boot == m):.4f} | theory 1-(1-1/n)^n = {1-(1-1/n)**n
    :.4f}")
```

```
for name, s, ref in [("True distribution   ", n*(theta - true_max)/theta, None),
                     ("Nonparametric       ", n*(m - np_boot)/m, None),
                     ("Parametric bootstrap", n*(m - p_boot)/m, None)]:
    q = np.percentile(s, [25, 50, 75])
    print(f"{name} | n(theta-max)/theta: P(=0)={np.mean(s == 0):.3f} | quartiles={np.round(q, 3)} (Exp(1): [0.288 0.693 1.386])")
```

Output

```
P*(X*_(n) = X_(n)) = 0.6410 | theory 1-(1-1/n)^n = 0.6358
True distribution    | n(theta-max)/theta: P(=0)=0.000 | quartiles=[0.284 0.678 1.346] (Exp(1): [0.288 0.693 1.386])
Nonparametric        | n(theta-max)/theta: P(=0)=0.641 | quartiles=[0.    0.    0.252] (Exp(1): [0.288 0.693 1.386])
Parametric bootstrap | n(theta-max)/theta: P(=0)=0.000 | quartiles=[0.289 0.688 1.381] (Exp(1): [0.288 0.693 1.386])
```

18.4.4 Jackknife and Permutation Test

code/ch18/c18_4_jackknife_permutation_test.py

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
x = rng.normal(5, 2, 15)
n = x.size
plug = x.var()                                          # sigma_hat^2 (1/n)
jack = np.array([np.delete(x, i).var() for i in range(n)])
jk = n*plug - (n - 1)*jack.mean()
print(f"sigma_hat^2 = {plug:.5f} | jackknife-corrected = {jk:.5f} | S^2 = {x.var(ddof=1):.5f}")

# A known failure of the jackknife: the SE of the median
med_j = np.array([np.median(np.delete(x, i)) for i in range(n)])
print(f"Number of distinct median jackknife values: {np.unique(med_j).size} (n={n}) -> SE estimate unreliable")

# Permutation test: two skewed samples, with a real difference
def perm_test(a, b, B=9999):
    z = np.r_[a, b]; T0 = b.mean() - a.mean()
    T = np.array([np.diff([p[:a.size].mean(), p[a.size:].mean()])[0]
                  for p in (rng.permutation(z) for _ in range(B))])
    return (1 + np.sum(np.abs(T) >= abs(T0)))/(B + 1)
a = rng.lognormal(0.0, 1.0, 12); b = rng.lognormal(0.8, 1.0, 12)
print(f"Permutation p = {perm_test(a, b):.4f} | Welch t-test p = {stats.ttest_ind(a, b, equal_var=False).pvalue:.4f}")

# Type I error under H0 (small, skewed samples)
rej_p = rej_t = 0; K = 1000
for _ in range(K):
    a = rng.lognormal(0, 1, 8); b = rng.lognormal(0, 1, 8)
    rej_p += perm_test(a, b, B=499) <= 0.05
    rej_t += stats.ttest_ind(a, b, equal_var=False).pvalue <= 0.05
print(f"rejection rate under H0 (target 0.05): permutation = {rej_p/K:.3f} | Welch t = {rej_t/K:.3f}")
```

Output

```
sigma_hat^2 = 3.15749 | jackknife-corrected = 3.38303 | S^2 = 3.38303
Number of distinct median jackknife values: 3 (n=15) -> SE estimate unreliable
Permutation p = 0.0119 | Welch t-test p = 0.0841
rejection rate under H0 (target 0.05): permutation = 0.057 | Welch t = 0.026
```

Three observations: (i) The jackknife correction turns $\hat{\sigma}^2$ exactly into S^2 (Theorem 18.11, $b = 0$). (ii) When a single observation is deleted, the median takes only a few distinct values; the jackknife SE is therefore inconsistent for non-smooth statistics such as the median (the bootstrap is consistent). (iii) Under H_0, in small and skewed samples, the permutation test keeps its exact level (Theorem 18.13) while the Welch t-test is overly conservative; this is why the permutation test can be more powerful on the same data.

18.5 Comments and Notes

Remark 18.17 (When Does the Bootstrap Work?). The general principle: if $\sqrt{n}(T(\hat{F}_n) - T(F))$ has a normal limit and T is "smooth" (Hadamard differentiable) near F, the bootstrap is consistent. This covers smooth functionals such as means, moments, quantiles (if the density is positive), the correlation, regression coefficients and the MLE. Typical situations where it fails: extreme values (Example 18.15), the boundary of the parameter space, heavy tails with infinite variance (in which case the "m-out-of-n" bootstrap with resamples of size $m < n$ is used), and dependent data (the block bootstrap is needed).

Remark 18.18 (How Many Bootstrap Samples?). For estimating an SE, $B \approx 200$–1000 is usually enough; the Monte Carlo error stays small next to the bootstrap's own sampling error. Since the endpoints of confidence intervals rest on tail quantiles, $B \geq 2000$ is recommended for them, and for p-values $B + 1$ should be chosen compatibly with α (e.g. $B = 999$).

Remark 18.19 (The Bootstrap and Bayes). The bootstrap distribution is very tempting to read as a probability distribution over θ. Rubin's [68] *Bayesian bootstrap* makes this intuition precise: if the observation weights are drawn from Dirichlet$(1, \ldots, 1)$, the resulting distribution is the posterior under a non-informative Dirichlet process prior. The classical bootstrap draws the weights from Multinom$(n; 1/n, \ldots, 1/n)/n$; the two distributions agree to first order (compare with Bernstein–von Mises in Chapter 19).

Remark 18.20 (Permutation or Bootstrap?). Permutation tests give an *exact* level when exchangeability holds under H_0, but they test only sharp hypotheses such as "the distributions are equal". Bootstrap tests are more flexible (e.g. equality of means only, with different variances), but their level is only asymptotically correct. A frequently forgotten point is that in a bootstrap hypothesis test the resampling must be done so as to satisfy H_0 (e.g. by shifting the groups to a common mean).

18.6 Chapter Summary

- By the plug-in principle, the unknown F is replaced by the empirical distribution $\hat{F}_n$ (Definition 18.1). The bootstrap approximates the sampling distribution of a statistic by resampling from $\hat{F}_n$.
- The bootstrap gives standard error and bias estimates. The types of confidence interval are percentile, basic, studentised and BCa (Definition 18.5).
- For smooth statistics such as the mean the bootstrap is consistent (Theorem 18.9). It fails for irregular statistics such as the sample maximum.
- The jackknife reduces the bias from $O(1/n)$ to $O(1/n^2)$ (Theorem 18.11). The permutation test has exact level under exchangeability (Theorem 18.13).
- The bootstrap is no miracle. In small samples $\hat{F}_n$ cannot represent the tails well, and intervals may have low coverage.

Further reading. [24], [19]; for the theoretical foundations, [33] and [78, Chapter 23].

18.7 Exercises

Exercise 18.1. Repeat the Fisher z exercise of Chapter 12: for bivariate normal data ($\rho = 0.8$, $n = 20$), compare the coverage of the percentile, BCa and Fisher z intervals for the correlation coefficient. How does the result change if the data are not normal (e.g. bivariate lognormal)? → Solution

Exercise 18.2. In linear regression (Chapter 17), apply the *case* (pairs) bootstrap and the *residual* bootstrap. If the error variance changes with the explanatory variable (heteroscedasticity), which gives a valid standard error? Compare the result with the sandwich estimator of Proposition 17.20. Show this by simulation. → Solution

Exercise 18.3. For $X_i \overset{\text{i.i.d.}}{\sim} \text{Cauchy}(\theta, 1)$ data, compute the bootstrap SE of the sample mean repeatedly with $B = 1000$. Relate the instability of the results to the Cauchy example of Chapter 5. What happens for the median? → Solution

Exercise 18.4. Using Theorem 18.11, show that the jackknife-corrected version of $\hat{\theta} = \bar{X}^2$ (an estimate of μ^2) is exactly unbiased, and compare the result with $\bar{X}^2 - S^2/n$. → Solution

Part V

Bayesian Inference

Chapter 19

Bayesian Inference and Conjugate Priors

In the chapters of the book so far, the parameter θ was an unknown but fixed number; probability described only the randomness of the data. The Bayesian approach expresses the uncertainty about the parameter by probability as well: before the data, our knowledge about θ is summarised by a *prior* distribution; after the data are observed, Bayes' formula (Theorem 1.13) updates this knowledge to a *posterior* distribution. In this chapter we discuss the basic objects of Bayesian inference, the closed-form solutions given by conjugate priors, the decision-theoretic foundation, and why Bayesian and frequentist results merge in large samples (Bernstein–von Mises). For non-conjugate models we also implement the simplest example of Markov chain Monte Carlo (MCMC).

19.1 Definitions

Definition 19.1 (Prior, Likelihood, Posterior)**.** Let $\pi(\theta)$ be the *prior* density of θ and $f(\mathbf{x} \mid \theta) = L(\theta; \mathbf{x})$ the conditional density of the data. By Bayes' theorem the *posterior* density is

$$\pi(\theta \mid \mathbf{x}) = \frac{f(\mathbf{x} \mid \theta)\pi(\theta)}{m(\mathbf{x})}, \qquad m(\mathbf{x}) = \int_\Theta f(\mathbf{x} \mid \theta)\pi(\theta)\, d\theta.$$

$m(\mathbf{x})$ is called the *marginal likelihood* (evidence). In short: posterior $\propto$ likelihood $\times$ prior.

Definition 19.2 (Conjugate Prior Family)**.** Let $\mathcal{P}$ be a family of priors. If $\pi(\cdot \mid \mathbf{x}) \in \mathcal{P}$ for every $\pi \in \mathcal{P}$ and every $\mathbf{x}$, then $\mathcal{P}$ is *conjugate* for the model $\{f(\cdot \mid \theta)\}$.

Definition 19.3 (Posterior Predictive Distribution)**.** For a new observation $\tilde{X}$, $f(\tilde{x} \mid \mathbf{x}) = \int f(\tilde{x} \mid \theta)\pi(\theta \mid \mathbf{x})\, d\theta$ is the *posterior predictive distribution*; it carries the parameter uncertainty into the prediction.

Definition 19.4 (Bayes Risk and Bayes Estimator)**.** For a loss $L(\theta, a)$ and risk $R(\theta, \delta)$ (Definition 7.4), the *Bayes risk* is $r(\pi, \delta) = \int R(\theta, \delta)\pi(\theta) d\theta$. The δ^π minimising $r(\pi, \cdot)$ is called the *Bayes estimator*. Since by Fubini $r(\pi, \delta) = \int \Big[\int L(\theta, \delta(\mathbf{x}))\pi(\theta \mid \mathbf{x}) d\theta \Big] m(\mathbf{x}) d\mathbf{x}$, $\delta^\pi(\mathbf{x})$ is found by minimising the *posterior expected loss* for each $\mathbf{x}$.

Definition 19.5 (Credible Interval and HPD Region)**.** A set $C \subseteq \Theta$ with $\int_C \pi(\theta \mid \mathbf{x}) d\theta = 1 - \alpha$ is called a $1-\alpha$ *credible* set. A credible set of the form $C = \{\theta : \pi(\theta \mid \mathbf{x}) \geq k_\alpha\}$ is called a *highest posterior density* (HPD) region; it is the smallest set with the given probability.

Definition 19.6 (Jeffreys Prior)**.** $\pi_J(\theta) \propto \sqrt{I_1(\theta)}$ (in the multidimensional case $\sqrt{\det I_1(\boldsymbol{\theta})}$). Its total integral may be infinite (an *improper* prior); it can be used for inference as long as the posterior is proper.

19.2 Main Theorems and Proofs

Theorem 19.7 (Beta–Binomial Conjugacy)**.** *If $\theta \sim Beta(\alpha, \beta)$ and $X \mid \theta \sim Bin(n, \theta)$, then*

$$\theta \mid X = x \sim Beta(\alpha + x,\ \beta + n - x), \qquad \hat{\theta}_{Bayes} = \mathbb{E}[\theta \mid x] = \frac{\alpha + x}{\alpha + \beta + n}.$$

Proof. $\pi(\theta \mid x) \propto \theta^x(1-\theta)^{n-x} \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1} = \theta^{\alpha+x-1}(1-\theta)^{\beta+n-x-1}$; up to the normalising constant this is the Beta$(\alpha + x, \beta + n - x)$ density. The mean of Beta(a, b) is $a/(a + b)$. □

Corollary 19.8 (Shrinkage Form)**.**

$$\hat{\theta}_{Bayes} = \underbrace{\frac{\alpha + \beta}{\alpha + \beta + n}}_{w} \cdot \underbrace{\frac{\alpha}{\alpha + \beta}}_{prior\ mean} + \Big(1 - w\Big) \cdot \underbrace{\frac{x}{n}}_{MLE}.$$

The posterior mean is a weighted average of the prior mean and the MLE. $\alpha + \beta$ acts like the "equivalent sample size" of the prior; as $n \to \infty$, $w \to 0$ and the data dominate.

Proof. $\frac{\alpha+x}{\alpha+\beta+n} = \frac{\alpha+\beta}{\alpha+\beta+n} \cdot \frac{\alpha}{\alpha+\beta} + \frac{n}{\alpha+\beta+n} \cdot \frac{x}{n}$ and $\frac{n}{\alpha+\beta+n} = 1 - w$. □

Theorem 19.9 (Bayes Estimators for Different Loss Functions)**.** *1. Squared loss $L = (\theta - a)^2$: $\delta^\pi(\mathbf{x}) = \mathbb{E}[\theta \mid \mathbf{x}]$ (the posterior mean).*

2. Absolute loss $L = |\theta - a|$: $\delta^\pi(\mathbf{x}) =$ the posterior median.

3. The limit of 0–1 loss ($L = \mathbb{I}\{|\theta - a| > \varepsilon\}$, $\varepsilon \to 0$): the posterior mode (MAP).

Proof. (1) Under the posterior, $\mathbb{E}[(\theta - a)^2 \mid \mathbf{x}] = \mathrm{Var}(\theta \mid \mathbf{x}) + (\mathbb{E}[\theta \mid \mathbf{x}] - a)^2$ (exactly as in Theorem 7.6); this is smallest at $a = \mathbb{E}[\theta \mid \mathbf{x}]$. (2) For $g(a) = \mathbb{E}|\theta - a|$, $g'(a) = \mathbb{P}(\theta < a \mid \mathbf{x}) - \mathbb{P}(\theta > a \mid \mathbf{x})$, which changes sign at the median. (3) $\mathbb{E}[L] = 1 - \mathbb{P}(|\theta - a| \le \varepsilon \mid \mathbf{x}) \approx 1 - 2\varepsilon\, \pi(a \mid \mathbf{x})$; the a at which the density is largest is chosen. □

Theorem 19.10 (Normal–Normal Conjugacy)**.** *If $\theta \sim \mathcal{N}(\mu_0, \tau^2)$ and $X_i \mid \theta \overset{i.i.d.}{\sim} \mathcal{N}(\theta, \sigma^2)$ (σ^2 known), then*

$$\theta \mid \mathbf{x} \sim \mathcal{N}(\mu_n, \tau_n^2), \qquad \frac{1}{\tau_n^2} = \frac{1}{\tau^2} + \frac{n}{\sigma^2}, \qquad \mu_n = \tau_n^2\Big(\frac{\mu_0}{\tau^2} + \frac{n\bar{x}}{\sigma^2}\Big).$$

Proof. The terms depending on θ: $\ln \pi(\theta \mid \mathbf{x}) = -\frac{(\theta-\mu_0)^2}{2\tau^2} - \frac{n(\bar{x}-\theta)^2}{2\sigma^2} + \text{const}$ (because $\sum(x_i - \theta)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \theta)^2$). The coefficient of θ^2 is $-\frac{1}{2}(1/\tau^2 + n/\sigma^2) = -\frac{1}{2\tau_n^2}$ and that of θ is $\mu_0/\tau^2 + n\bar{x}/\sigma^2 = \mu_n/\tau_n^2$; completing the square gives $-\frac{(\theta-\mu_n)^2}{2\tau_n^2} + \text{const}$. □

Remark 19.11. Precisions (inverse variances) add: posterior precision = prior precision + data precision. The posterior mean is the precision-weighted average. This structure is the essence of the Kalman filter and of hierarchical (multilevel) models.

Theorem 19.12 (Gamma–Poisson Conjugacy)**.** *If $\lambda \sim Gamma(\alpha, \beta)$ (shape, rate) and $X_i \mid \lambda \overset{i.i.d.}{\sim} Poisson(\lambda)$, then $\lambda \mid \mathbf{x} \sim Gamma(\alpha + \sum x_i,\ \beta + n)$. The posterior mean is $\frac{\alpha+\sum x_i}{\beta+n}$; the posterior predictive distribution of a new observation is negative binomial (compare with Example 3.19).*

Proof. $\pi(\lambda \mid \mathbf{x}) \propto \lambda^{\sum x_i} e^{-n\lambda} \cdot \lambda^{\alpha-1} e^{-\beta\lambda}$. □

Theorem 19.13 (Invariance of the Jeffreys Prior [41])**.** *If $\eta = h(\theta)$ is one-to-one and differentiable, the prior on η obtained from the Jeffreys prior on the θ scale by a change of variables equals the Jeffreys prior defined directly on the η scale.*

Proof. Change of variables: $\pi(\eta) = \pi_J(\theta)\left|\frac{d\theta}{d\eta}\right| \propto \sqrt{I^{(\theta)}(\theta)}\left|\frac{d\theta}{d\eta}\right| = \sqrt{I^{(\theta)}(\theta)\left(\frac{d\theta}{d\eta}\right)^2} = \sqrt{I^{(\eta)}(\eta)}$; the last step is Proposition 8.9. □

Theorem 19.14 (Bernstein–von Mises; see [78, Theorem 10.1])**.** *In a regular parametric model, for a prior that is continuous and positive at θ_0, under the true θ_0*

$$\sup_B \left| \Pi(\theta \in B \mid \mathbf{X}) - \mathcal{N}\big(\hat{\theta}_n,\, I_n(\theta_0)^{-1}\big)(B) \right| \xrightarrow{P} 0.$$

That is, in large samples the posterior approaches a normal distribution centred at the MLE with variance equal to the inverse Fisher information; the influence of the prior disappears and Bayesian credible intervals also have $1-\alpha$ coverage in the frequentist sense.

Remark 19.15 (About the Proof). For a complete proof see [78, Theorem 10.1 and Lemma 10.3]; the proof has two components. (i) *Local asymptotic normality:* for the local parameter $h = \sqrt{n}(\theta - \theta_0)$, the Taylor expansion of the proof of Theorem 12.6 gives

$$\ell(\theta_0 + h/\sqrt{n}) - \ell(\theta_0) = h\Delta_n - \tfrac{1}{2} I_1(\theta_0) h^2 + o_P(1), \qquad \Delta_n = \tfrac{1}{\sqrt{n}}\ell'(\theta_0) \xrightarrow{d} \mathcal{N}(0, I_1).$$

The posterior density of h is proportional to $\pi(\theta_0 + h/\sqrt{n}) \exp\{\ell(\theta_0 + h/\sqrt{n}) - \ell(\theta_0)\}$; since the prior is continuous and positive at θ_0, it approaches the constant $\pi(\theta_0)$, and the remaining expression, after completing the square, is proportional to the $\mathcal{N}(\Delta_n/I_1,\, 1/I_1)$ density. Since $\Delta_n/I_1 = \sqrt{n}(\hat{\theta}_n - \theta_0) + o_P(1)$, this is the normal distribution in the theorem. (ii) *Control of the tails:* the approximation above is uniform only for $|h| \le M$; to show that the posterior mass far from θ_0 goes to zero, one uses the existence of tests that consistently separate $H_0 : \theta = \theta_0$ from $|\theta - \theta_0| \ge \varepsilon$. This is the technical part of the proof.

Remark 19.16 (Admissibility). Every Bayes estimator that is unique and has finite Bayes risk with respect to a proper prior is admissible (Definition 7.4): if there were a better estimator, its Bayes risk would also be smaller. Conversely (Wald's complete class theorem), every admissible estimator is a limit of Bayes estimators. This is the deep connection between frequentist optimality and Bayesian methods; the James–Stein estimator (Chapter 7) also has an empirical Bayes interpretation.

19.3 Worked Examples

Example 19.17 (Beta–Binomial: A Numerical Example)**.** Earlier studies of the prevalence of a disease suggest a prior with mean 0.10: $\theta \sim \mathrm{Beta}(2, 18)$ (equivalent sample size $\alpha + \beta = 20$). In a new screening, $x = 10$ of $n = 50$ people test positive (MLE 0.20). The posterior is $\mathrm{Beta}(12, 58)$ and

$$\hat{\theta}_{\text{Bayes}} = \frac{12}{70} \approx 0.171 = \frac{20}{70}(0.10) + \frac{50}{70}(0.20).$$

The posterior lies between the prior and the data, closer to the data.

Example 19.18 (Posterior Prediction: Beta–Binomial)**.** In the same model, the probability that a new person tests positive is $\mathbb{P}(\tilde{X} = 1 \mid x) = \mathbb{E}[\theta \mid x] = \frac{\alpha + x}{\alpha + \beta + n}$. The posterior predictive distribution of the number of positives among m new people is *beta-binomial*:

$$\mathbb{P}(\tilde{Y} = k \mid x) = \binom{m}{k} \frac{B(\alpha + x + k,\ \beta + n - x + m - k)}{B(\alpha + x,\ \beta + n - x)},$$

and its variance is larger than that of the binomial with the same mean: the parameter uncertainty carries over into the prediction.

Example 19.19 (Laplace's Rule of Succession and the Jeffreys Prior)**.** For the Bernoulli, $I_1(\theta) = \frac{1}{\theta(1-\theta)}$, so $\pi_J(\theta) \propto \theta^{-1/2}(1-\theta)^{-1/2}$, i.e. Beta$(\frac{1}{2}, \frac{1}{2})$. With the uniform prior Beta$(1,1)$ the posterior mean is $\frac{x+1}{n+2}$ (Laplace's rule of succession: "if the sun has risen on n days, the probability that it rises tomorrow is $\frac{n+1}{n+2}$"); with Jeffreys it is $\frac{x+1/2}{n+1}$. The equal-tailed interval obtained from the Jeffreys posterior is a binomial interval with very good frequentist coverage [13]; it is examined in the code below.

Example 19.20 (Normal–Normal: School Performance)**.** In an examination, school means are distributed as $\theta \sim \mathcal{N}(500, 50^2)$. For one school, the mean of $n = 25$ students is $\bar{x} = 560$, with individual standard deviation $\sigma = 100$ (so $\sigma^2/n = 400$). Precisions: prior $1/2500$, data $1/400$. $\tau_n^2 = 1/(1/2500 + 1/400) \approx 344.8$, $\tau_n \approx 18.6$ and

$$\mu_n = 344.8\Big(\frac{500}{2500} + \frac{560}{400}\Big) \approx 551.7.$$

The observed mean "shrinks" by ≈ 8 points towards the overall mean. This shrinkage, which keeps schools with small samples from looking extremely good or bad, is standard practice in rankings in education and health (e.g. hospital mortality rates).

Example 19.21 (Lindley's Paradox [52])**.** $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, 1)$, $H_0 : \theta = 0$; prior $\mathbb{P}(H_0) = \frac{1}{2}$ and $\theta \sim \mathcal{N}(0,1)$ under H_1. Suppose $\bar{x} = 1.96/\sqrt{n}$ is observed (the p-value is 0.05 for every n). The Bayes factor is

$$B_{01} = \frac{\varphi(\bar{x}; 0, 1/n)}{\varphi(\bar{x}; 0, 1 + 1/n)} = \sqrt{n+1}\,\exp\Big\{-\frac{1.96^2}{2}\cdot\frac{n}{n+1}\Big\} \xrightarrow{n\to\infty} \infty.$$

For $n = 10\,000$, $B_{01} \approx 15$ and $\mathbb{P}(H_0 \mid \mathbf{x}) \approx 0.94$: while the frequentist test rejects H_0 at the 5% level, the posterior probability strongly supports H_0. The paradox arises because a fixed p-value corresponds to a very small effect size for large n.

19.4 Computational Applications

19.4.1 Beta–Binomial: Posterior Summaries and the HPD Interval

code/ch19/c19_1_beta_binomial_posterior_summaries_hpd.py

```python
import scipy.stats as stats
import numpy as np
from scipy.optimize import minimize_scalar

a_post, b_post = 2 + 10, 18 + 40
print(f"Bayes estimate: {a_post / (a_post + b_post):.4f}")
print(f"Credible interval: {stats.beta.ppf([0.025, 0.975], a_post, b_post)}")

post = stats.beta(a_post, b_post)
print(f"Posterior median = {post.median():.4f} | mode (MAP) = {(a_post-1)/(a_post+b_post-2)
    :.4f} | sd = {post.std():.4f}")
# HPD: the u minimising the length of [ppf(u), ppf(u+0.95)]
w = lambda u: post.ppf(u + 0.95) - post.ppf(u)
u = minimize_scalar(w, bounds=(0, 0.05), method="bounded").x
L, U = post.ppf(u), post.ppf(u + 0.95)
print(f"95% HPD = [{L:.4f}, {U:.4f}] (length {U-L:.4f}) | density at endpoints: {post.pdf(L)
    :.3f}, {post.pdf(U):.3f}")
lo, hi = post.ppf([0.025, 0.975])
print(f"Equal-tailed length = {hi-lo:.4f}")
print(f"P(theta > 0.15 | x) = {post.sf(0.15):.4f}")
```

Output

```
Bayes estimate: 0.1714
Credible interval: [0.09322494 0.2673681 ]
Posterior median = 0.1683 | mode (MAP) = 0.1618 | sd = 0.0447
95% HPD = [0.0879, 0.2603] (length 0.1724) | density at endpoints: 1.352, 1.352
Equal-tailed length = 0.1741
P(theta > 0.15 | x) = 0.6631
```

At the endpoints of the HPD interval the posterior density is equal (the Bayesian counterpart of Theorem 16.9), and the interval is slightly shorter than the equal-tailed one.

19.4.2 Prior, Likelihood and Posterior; Bernstein–von Mises

code/ch19/c19_2_prior_likelihood_posterior_bvm.py

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

t = np.linspace(0, 0.5, 600)
fig, ax = plt.subplots(1, 2, figsize=(10, 3.4))
prior = stats.beta(2, 18)
ax[0].plot(t, prior.pdf(t), "--", label="prior Beta(2,18)")
ax[0].plot(t, stats.beta(11, 41).pdf(t), ":", label="likelihood (normalised)")
ax[0].plot(t, stats.beta(12, 58).pdf(t), "k", lw=2, label="posterior Beta(12,58)")
ax[0].set_xlabel(r"$\theta$"); ax[0].legend(fontsize=8); ax[0].set_title("n = 50, x = 10")

rng = np.random.default_rng(42)
theta0 = 0.2
for n in [20, 200, 2000]:
    x = rng.binomial(n, theta0)
    ps = stats.beta(2 + x, 18 + n - x)
    mle = x / n
    bvm = stats.norm(mle, np.sqrt(mle*(1-mle)/n))
    tt = np.linspace(0, 0.5, 2000)
    tv = 0.5*np.sum(np.abs(ps.pdf(tt) - bvm.pdf(tt))) * (tt[1] - tt[0])
    ax[1].plot(tt, ps.pdf(tt), label=f"n={n}")
    print(f"n={n:5d} | posterior mean={ps.mean():.4f} | MLE={mle:.4f} | TV(posterior, BvM
    normal)={tv:.4f}")
ax[1].axvline(theta0, color="k", ls=":"); ax[1].set_xlim(0, 0.45); ax[1].legend(fontsize=8)
ax[1].set_xlabel(r"$\theta$"); ax[1].set_title("Concentration of the posterior")
fig.tight_layout(); fig.savefig("bayes_sonsal.pdf")
```

Output

```
n=   20 | posterior mean=0.1750 | MLE=0.2500 | TV(posterior, BvM normal)=0.4203
n=  200 | posterior mean=0.2091 | MLE=0.2200 | TV(posterior, BvM normal)=0.1658
n= 2000 | posterior mean=0.2178 | MLE=0.2190 | TV(posterior, BvM normal)=0.0550
```

19.4.3 Frequentist Coverage of Bayesian Intervals

code/ch19/c19_3_frequentist_coverage_bayesian_intervals.py

```
import numpy as np
from scipy import stats

def coverage_jeffreys(n, ps, a=0.05):
    x = np.arange(n + 1)
    L = np.where(x > 0, stats.beta.ppf(a/2, x + 0.5, n - x + 0.5), 0.0)
    U = np.where(x < n, stats.beta.ppf(1 - a/2, x + 0.5, n - x + 0.5), 1.0)
    return np.array([np.sum(stats.binom.pmf(x, n, p) * ((L <= p) & (p <= U))) for p in ps])
```

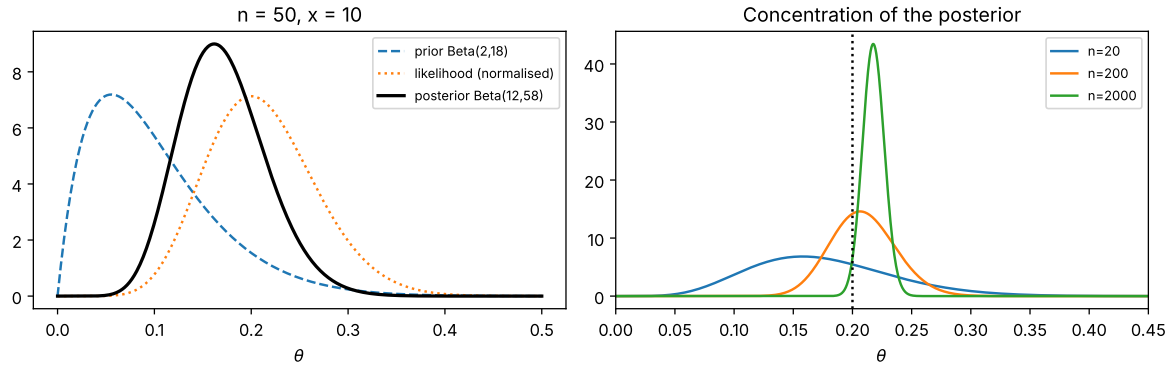


Figure 19.1: Left: the posterior is a "compromise" between the prior and the likelihood. Right: as n grows, the posterior concentrates around the true value $\theta_0 = 0.2$ and converges to the normal approximation (Bernstein–von Mises); the initial pull of the prior towards 0.1 disappears.

```

ps = np.linspace(0.005, 0.995, 991)
for n in [30, 100]:
    k = coverage_jeffreys(n, ps)
    print(f"n={n:3d} | Jeffreys interval: mean coverage={k.mean():.4f}, min={k.min():.4f}")
```

Output

```
n= 30 | Jeffreys interval: mean coverage=0.9505, min=0.8887
n=100 | Jeffreys interval: mean coverage=0.9498, min=0.8806
```

The average frequentist coverage of the Jeffreys credible interval is very close to the nominal level and clearly better than that of the Wald interval of Chapter 16: well-chosen "non-informative" priors produce methods with good frequentist properties.

19.4.4 A Non-Conjugate Model: The Metropolis Algorithm

The algorithm goes back to Metropolis and co-authors [59]. If a $\mathcal{N}(0, 1.5^2)$ prior is placed on the logit parameter $\eta = \ln \frac{\theta}{1-\theta}$, the posterior is not in closed form. The random-walk Metropolis algorithm: from the current η, propose $\eta' = \eta + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, s^2)$; accept with probability $\min\{1, \pi(\eta' \mid x)/\pi(\eta \mid x)\}$. The stationary distribution of this chain is the posterior; the normalising constant $m(x)$ is never needed.

code/ch19/c19_4_nonconjugate_model_metropolis.py

```
import numpy as np
from scipy import stats, integrate

rng = np.random.default_rng(42)
x, n, s0 = 10, 50, 1.5
def log_post(eta):        # log-likelihood + log-prior (up to constants)
    return x*eta - n*np.log1p(np.exp(eta)) - eta**2 / (2*s0**2)

N, s = 60_000, 0.6
chain = np.empty(N); eta = 0.0; lp = log_post(eta); acc = 0
for i in range(N):
    prop = eta + s*rng.standard_normal()
    lp_prop = log_post(prop)
    if np.log(rng.random()) < lp_prop - lp:
        eta, lp = prop, lp_prop; acc += 1
    chain[i] = eta
```

```python
theta = 1/(1 + np.exp(-chain[5_000:]))          # discard burn-in

# Check: exact posterior mean by numerical integration
Z = integrate.quad(lambda e: np.exp(log_post(e)), -10, 10)[0]
m_exact = integrate.quad(lambda e: np.exp(log_post(e))/(1 + np.exp(-e)), -10, 10)[0] / Z
print(f"Acceptance rate = {acc/N:.3f}")
print(f"MCMC  posterior mean E[theta|x] = {theta.mean():.4f} | 95% interval = {np.percentile(
    theta, [2.5, 97.5]).round(4)}")
print(f"Exact posterior mean (quad)      = {m_exact:.4f}")
print(f"Comparison: Beta(2,18) conjugate posterior mean = {12/70:.4f} | MLE = {x/n:.4f}")
```

Output

```
Acceptance rate = 0.538
MCMC  posterior mean E[theta|x] = 0.2115 | 95% interval = [0.1126 0.3318]
Exact posterior mean (quad)      = 0.2119
Comparison: Beta(2,18) conjugate posterior mean = 0.1714 | MLE = 0.2000
```

The MCMC estimate agrees to three digits with the exact value found by numerical integration. That the posterior mean now comes out *above* the MLE is a consequence of the prior: the prior $\eta \sim \mathcal{N}(0,\, 1.5^2)$ is centred at $\theta = 0.5$ and pulls the posterior upwards, whereas the Beta$(2, 18)$ prior, centred at 0.1, pulled it downwards. This difference, obtained with the same data and different priors, shows why prior sensitivity analysis matters in small samples.

19.5 Comments and Notes

Remark 19.22 (Choosing the Prior). Most criticism is directed at the subjectivity of the prior. Practical answers: (i) *Sensitivity analysis*: does the result change across reasonable priors? (ii) *Weakly informative priors*: priors that keep the parameter on a plausible scale without dominating the data. (iii) *Hierarchical models*: learning the prior parameters from the data as well (empirical or full Bayes). The Bernstein–von Mises theorem guarantees that in large samples (in regular models) the influence of the prior disappears; in small samples the prior acts as a *regulariser* (ridge regression is the MAP estimate under a normal prior).

Remark 19.23 (Credibility and Confidence). A 95% credible interval permits the statement "the (posterior) probability that θ lies in this interval is 0.95" — exactly the interpretation that most practitioners wrongly attach to a confidence interval. The price is stating the prior explicitly. The two approaches coincide numerically in large samples (BvM); they can diverge in small samples, for parameters on the boundary, and in high-dimensional problems.

Remark 19.24 (Computation). Conjugate priors provide closed forms, but most modern Bayesian analysis is done with MCMC (Metropolis–Hastings, Gibbs, Hamiltonian Monte Carlo), variational approximations or the Laplace approximation. Probabilistic programming languages such as Stan, PyMC and NumPyro automate these algorithms. Diagnosing chain convergence ($\hat{R}$, effective sample size, trace plots) is an integral part of these methods.

19.6 Chapter Summary

- In Bayesian inference the posterior is proportional to the product of the likelihood and the prior. Uncertainty is expressed by a probability distribution over the parameter.
- Conjugate priors give closed-form posteriors: Beta–Binomial (Theorem 19.7), Normal–Normal (Theorem 19.10) and Gamma–Poisson (Theorem 19.12). The posterior mean is a weighted average of the prior mean and the MLE (Corollary 19.8).
- The Bayes estimator depends on the loss function (Theorem 19.9): the posterior mean for squared loss, the posterior median for absolute loss, and the posterior mode for

0–1 loss.

- The Jeffreys prior is invariant under reparametrisation (Theorem 19.13). By the Bernstein–von Mises theorem (Theorem 19.14), in large samples the posterior approaches $\mathcal{N}(\hat{\theta}_n, I_n^{-1})$ and the influence of the prior disappears.
- For point null hypotheses the p-value and the posterior probability can diverge sharply (Lindley's paradox). When there is no closed form, the posterior is sampled by MCMC methods such as Metropolis.

Further reading. [31], [81, Chapter 11], [46, Chapter 4]; for asymptotic Bayesian theory, [78, Chapter 10].

19.7 Exercises

Exercise 19.1. In the Gamma–Poisson model, write the posterior mean as a weighted average of the prior mean and $\bar{x}$. For $\alpha = 2$, $\beta = 1$ and $\sum_{i=1}^{10} x_i = 34$, compute the posterior mean, the 95% equal-tailed credible interval and the posterior predictive distribution of a new observation. → Solution

Exercise 19.2. When both the normal mean and variance are unknown, show that under the (improper) prior $\pi(\mu, \sigma^2) \propto 1/\sigma^2$ the marginal posterior of μ satisfies

$$\frac{\mu - \bar{x}}{s/\sqrt{n}} \,\Big|\, \mathbf{x} \sim t_{n-1}.$$

How does this result relate to the t interval of Chapter 16? → Solution

Exercise 19.3. In the Normal–Normal model, compute the frequentist MSE of the posterior mean μ_n as a function of θ and compare it with the MSE of $\bar{X}$ (σ^2/n). For which values of θ is the Bayes estimator better? → Solution

Exercise 19.4. In the Metropolis code, compare the acceptance rates and the autocorrelation of the chain for proposal scales $s \in \{0.05,\ 0.6,\ 5\}$. Estimate the effective sample size with the formula $N/(1 + 2\sum_{k} \rho_k)$. → Solution

Part VI

Decision Theory and Advanced Topics

Chapter 20

Statistical Decision Theory: Minimaxity and Admissibility

In earlier chapters we compared estimators by criteria such as unbiasedness, variance and MSE, and tests by level and power. Wald's [80] statistical decision theory brings all these criteria into a single framework: a *decision rule* is a function from the data to an action, and its quality is measured by a *risk function* that depends on the true parameter. Estimation, testing and interval construction are special cases of this framework. The trouble is that risk functions usually cross: no rule is best for every θ (Chapter 7). In this chapter we study the two basic ways of reducing a risk function to a single number: the *Bayes* principle, which minimises the average risk (Chapter 19), and the *minimax* principle, which minimises the worst case. The bridge between the two principles is the notion of a *least favourable prior*. The second theme of the chapter is *admissibility*, the weakest optimality criterion: here we prove by Blyth's method that the normal mean is admissible in one dimension and contrast this with Stein's paradox.

20.1 Definitions

Definition 20.1 (Decision Problem and Decision Rule)**.** A *decision problem* has three components: a parameter space Θ, an *action space* $\mathcal{A}$ and a loss function $L : \Theta \times \mathcal{A} \to [0, \infty)$. $\mathbf{X} \sim P_\theta$ is observed. A measurable function $\delta : \mathcal{X} \to \mathcal{A}$ is called a (non-randomised) *decision rule*, and $R(\theta, \delta) = \mathbb{E}_\theta L(\theta, \delta(\mathbf{X}))$ its *risk function* (Definition 7.4). Examples:

- *Point estimation:* $\mathcal{A} = \Theta$, $L(\theta, a) = (a - \theta)^2$; the risk is the MSE.
- *Hypothesis testing:* $\mathcal{A} = \{0, 1\}$ (1: reject H_0), $L(\theta, a) = \mathbb{I}\{a = 1, \theta \in \Theta_0\} + \mathbb{I}\{a = 0, \theta \in \Theta_1\}$ (0–1 loss); the risk is the probability of a type I error on Θ_0 and of a type II error on Θ_1.

Definition 20.2 (Randomised Rule)**.** A rule that chooses, for each $\mathbf{x}$, a probability distribution $\delta(\cdot \mid \mathbf{x})$ on $\mathcal{A}$ is called a *randomised decision rule*; its risk is $R(\theta, \delta) = \mathbb{E}_\theta \int L(\theta, a)\, \delta(da \mid \mathbf{X})$. The γ in Neyman–Pearson tests (Chapter 13) is a randomisation. Under a convex loss randomisation brings no gain (Jensen's inequality), but it is needed for tests with discrete data and for finite action spaces.

Definition 20.3 (Minimax Rule)**.** A δ^* satisfying $\sup_{\theta \in \Theta} R(\theta, \delta^*) = \inf_{\delta} \sup_{\theta \in \Theta} R(\theta, \delta)$ is called a *minimax* rule, and the value on the right the *minimax risk*.

Definition 20.4 (Bayes Risk and Least Favourable Prior)**.** For a prior π, $r(\pi, \delta) = \int R(\theta, \delta)\, \pi(d\theta)$ and $r(\pi) = \inf_\delta r(\pi, \delta)$ (Chapter 19). A π^* that maximises $r(\pi)$, i.e. satisfies

$r(\pi^*) \geq r(\pi')$ for every π', is called a *least favourable prior*. Intuitively, it is nature's strategy for choosing the parameter that harms the statistician most.

Definition 20.5 (Equaliser Rule)**.** A rule whose risk function does not depend on θ, i.e. $R(\theta, \delta) = c$ is constant, is called an *equaliser* rule.

20.2 Main Theorems and Proofs

Theorem 20.6 (A Bayes Rule Whose Risk Does Not Exceed Its Bayes Risk Is Minimax)**.** *Let δ^π be a Bayes rule with respect to the prior π, and suppose*

$$\sup_{\theta \in \Theta} R(\theta, \delta^\pi) \leq r(\pi, \delta^\pi).$$

Then δ^π is minimax and π is least favourable.

Proof. For any δ, since the supremum of a function is at least any weighted average of it,

$$\sup_\theta R(\theta, \delta) \geq \int R(\theta, \delta)\, \pi(d\theta) = r(\pi, \delta) \geq r(\pi, \delta^\pi) \geq \sup_\theta R(\theta, \delta^\pi).$$

This proves minimaxity. For any π', also $r(\pi') \leq r(\pi', \delta^\pi) = \int R(\theta, \delta^\pi)\pi'(d\theta) \leq \sup_\theta R(\theta, \delta^\pi) \leq r(\pi, \delta^\pi) = r(\pi)$. □

Corollary 20.7 (An Equaliser Bayes Rule Is Minimax)**.** *If a Bayes rule is an equaliser, it is minimax. More generally, if $\pi(\{\theta : R(\theta, \delta^\pi) = \sup_{\theta'} R(\theta', \delta^\pi)\}) = 1$, i.e. the prior puts mass only on points where the risk is largest, then δ^π is minimax.*

Proof. In both cases $r(\pi, \delta^\pi) = \int R(\theta, \delta^\pi)\pi(d\theta) = \sup_\theta R(\theta, \delta^\pi)$; apply Theorem 20.6. □

Theorem 20.8 (The Limiting Bayes Method)**.** *Let $\{\pi_k\}$ be a sequence of priors and δ a decision rule. If $\sup_\theta R(\theta, \delta) \leq \limsup_k r(\pi_k)$, then δ is minimax.*

Proof. For any δ' and every k, $\sup_\theta R(\theta, \delta') \geq r(\pi_k, \delta') \geq r(\pi_k)$. Taking $\limsup$, $\sup_\theta R(\theta, \delta') \geq \limsup_k r(\pi_k) \geq \sup_\theta R(\theta, \delta)$. □

This theorem is the standard way of showing that rules which are not Bayes with respect to any proper prior (for example $\bar{X}$ in the normal model) are minimax; in that case no least favourable prior exists, but "in the limit" it is a flat prior.

Theorem 20.9 (A Unique Bayes Rule Is Admissible)**.** *Let $r(\pi) < \infty$ and let δ^π be the unique Bayes rule; i.e. every δ' with $r(\pi, \delta') = r(\pi)$ equals δ^π P_θ-a.s. for every θ. Then δ^π is admissible.*

Proof. If δ' dominated δ^π, then since $R(\theta, \delta') \leq R(\theta, \delta^\pi)$ for every θ, $r(\pi, \delta') \leq r(\pi, \delta^\pi) = r(\pi)$, i.e. δ' would also be a Bayes rule. By uniqueness $\delta' = \delta^\pi$ a.s. under every P_θ; hence the risk functions are equal everywhere, and there is no θ at which δ' is strictly better. This is a contradiction. □

Remark 20.10. Under squared loss the Bayes rule is the posterior mean, which is uniquely determined for every $\mathbf{x}$ with $m(\mathbf{x}) > 0$. Hence if the marginal distribution m and all the P_θ have the same null sets (e.g. in the binomial model, if π is non-degenerate), the uniqueness condition holds. Without uniqueness the result can fail: if π puts all its mass on a single point θ_0, there are many Bayes rules that behave arbitrarily away from θ_0.

Theorem 20.11 (An Admissible Equaliser Rule Is Minimax)**.** *If δ is admissible and an equaliser ($R(\theta, \delta) = c$), then it is minimax.*

Proof. If $\sup_\theta R(\theta,\delta') < c$ for some δ', then $R(\theta,\delta') < c = R(\theta,\delta)$ for every θ; i.e. δ' would dominate δ. Admissibility rules this out; hence $\sup_\theta R(\theta,\delta') \geq c = \sup_\theta R(\theta,\delta)$ for every δ'. □

Theorem 20.12 (Blyth's Method [11])**.** *Let $\Theta \subseteq \mathbb{R}$ be an open interval, and let the risk function of every decision rule be continuous in θ. Let δ be given. For each k let π_k be a measure on Θ (not necessarily finite) and δ_k a Bayes rule with respect to it; suppose $r(\pi_k,\delta) < \infty$ and, for every $[a,b] \subset \Theta$ with $a < b$,*

$$\frac{r(\pi_k,\delta) - r(\pi_k,\delta_k)}{\pi_k([a,b])} \to 0 \qquad (k \to \infty).$$

Then δ is admissible.

Proof. Let δ' dominate δ: $R(\theta,\delta') \leq R(\theta,\delta)$ for every θ, and $R(\theta_0,\delta') < R(\theta_0,\delta) - 2\varepsilon$ at some θ_0, with $\varepsilon > 0$. By continuity of the risks, $R(\theta,\delta) - R(\theta,\delta') > \varepsilon$ on some interval $[a,b]$ containing θ_0. Since δ_k is Bayes, $r(\pi_k,\delta_k) \leq r(\pi_k,\delta')$ and

$$r(\pi_k,\delta) - r(\pi_k,\delta_k) \geq r(\pi_k,\delta) - r(\pi_k,\delta') = \int \big[R(\theta,\delta) - R(\theta,\delta')\big]\pi_k(d\theta) \geq \varepsilon\,\pi_k([a,b]).$$

This says that the ratio in the assumption is at least ε for every k; a contradiction. □

20.3 Worked Examples

Example 20.13 (Normal Mean: $\bar X$ Is Minimax and (in One Dimension) Admissible)**.** $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta,\sigma^2)$, σ known, squared loss. $R(\theta,\bar X) = \sigma^2/n$ is constant.

Minimaxity. For the prior $\pi_k = \mathcal{N}(0,k)$ the posterior variance is $(n/\sigma^2 + 1/k)^{-1}$ (Theorem 19.10), so the Bayes risk is $r(\pi_k) = \big(n/\sigma^2 + 1/k\big)^{-1} \uparrow \sigma^2/n$. By Theorem 20.8, $\bar X$ is minimax.

Admissibility. By sufficiency we may work with the single observation $Y = \bar X \sim \mathcal{N}(\theta, s^2)$, $s^2 = \sigma^2/n$; all risk functions are continuous (in an exponential family, integrals under $\mathbb{E}_\theta$ are continuous functions of θ). Let π_k be the (unnormalised) measure with density $e^{-\theta^2/(2k)}$; its total mass is $\sqrt{2\pi k}$. The Bayes rule is again $\delta_k = \frac{k}{k+s^2}Y$, and its Bayes risk with respect to the normalised prior is $\frac{ks^2}{k+s^2}$. Hence

$$r(\pi_k,Y) - r(\pi_k,\delta_k) = \sqrt{2\pi k}\Big(s^2 - \frac{ks^2}{k+s^2}\Big) = \frac{\sqrt{2\pi k}\,s^4}{k+s^2} \to 0,$$

and $\pi_k([a,b]) = \int_a^b e^{-\theta^2/(2k)}d\theta \to b - a > 0$. By Theorem 20.12, $\bar X$ is admissible; Theorem 20.11 gives minimaxity a second time.

The role of dimension. In two dimensions $\mathbf{X}$ is also admissible [76], but applying Blyth's method requires more carefully chosen sequences of priors; in three or more dimensions, by contrast, $\mathbf{X}$ becomes inadmissible (Stein's paradox, Chapter 7). That is, $\mathbf{X}$ is minimax in every dimension, but for $k \geq 3$ there are minimax rules (James–Stein) that are strictly better.

Example 20.14 (The Minimax Estimator for a Binomial Proportion [35])**.** $X \sim \mathrm{Bin}(n,p)$, squared loss. For the Beta(a,a) prior the Bayes rule is $\delta_a = \frac{X+a}{n+2a}$ (Theorem 19.7). The risk of a rule of the form $cX + d$ is

$$R(p, cX+d) = c^2 np(1-p) + \big((cn-1)p + d\big)^2.$$

For $c = \frac{1}{n+2a}$, $d = \frac{a}{n+2a}$, the coefficient of p^2 is $-c^2n + (cn-1)^2$ and that of p is $c^2n + 2d(cn-1)$. Since $cn - 1 = -2ac$ and $d = ac$, these coefficients are $c^2(4a^2 - n)$ and $c^2(n - 4a^2)$ respectively; choosing $a = \sqrt{n}/2$ makes both vanish. The risk is then constant:

$$R\Big(p, \frac{X + \sqrt{n}/2}{n + \sqrt{n}}\Big) = d^2 = \frac{1}{4(1+\sqrt{n})^2}.$$

Being an equaliser Bayes rule (Corollary 20.7), it is minimax, and $\mathrm{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ is the least favourable prior. The maximum risk of the MLE, $\frac{1}{4n}$, exceeds the minimax risk; but the minimax rule is better only when p is near $1/2$: the inequality $p(1-p)/n > \frac{1}{4(1+\sqrt{n})^2}$ means $|p - \frac{1}{2}| < \frac{1}{2}\sqrt{1 - n/(1+\sqrt{n})^2}$. As n grows this interval shrinks, and the minimax rule shrinks needlessly towards $1/2$.

Example 20.15 (Tests Are Decision Rules)**.** $\Theta = \{\theta_0, \theta_1\}$, 0–1 loss and the prior $\pi = (\pi_0, \pi_1)$. The posterior expected loss is $\mathbb{P}(\theta_0 \mid \mathbf{x})$ for the decision to reject H_0 and $\mathbb{P}(\theta_1 \mid \mathbf{x})$ for the decision to accept. The Bayes rule chooses the smaller:

$$\delta^\pi(\mathbf{x}) = 1 \iff \pi_1 f(\mathbf{x};\theta_1) > \pi_0 f(\mathbf{x};\theta_0) \iff \frac{f(\mathbf{x};\theta_1)}{f(\mathbf{x};\theta_0)} > \frac{\pi_0}{\pi_1}.$$

This is a Neyman–Pearson test whose threshold is the prior odds (Theorem 13.7). The Bayes risk is $\pi_0\alpha + \pi_1\beta$, a weighted sum of the two error types. By Corollary 20.7, the minimax test is the NP test that equalises the two error probabilities ($\alpha = \beta$); with discrete data this equality can be achieved only by randomisation (Section 20.4.2).

Example 20.16 (A Constant Estimator Can Be Admissible)**.** $X \sim \mathcal{N}(\theta, 1)$, squared loss and $\delta \equiv 3$. Since $R(3, \delta) = 0$, if some δ' dominated it we would have $R(3, \delta') = 0$, i.e. $\delta'(X) = 3$ P_3-a.s. Since all normal distributions have the same null sets, $\delta' = 3$ a.s. under every P_θ, and the risks are equal. Hence $\delta \equiv 3$ is admissible: admissibility is a necessary, but far from sufficient, condition for being "good".

20.4 Computational Applications

20.4.1 Risk Curves for a Binomial Proportion

We compute the exact risk functions of the three rules of Example 20.14 for $n = 16$: the MLE, the Bayes rule for the uniform prior (Laplace's rule) and the minimax rule.

code/ch20/c20_1_binomial_risk_curves.py

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

n = 16
p = np.linspace(0, 1, 401)
x = np.arange(n + 1)
pmf = stats.binom.pmf(x[:, None], n, p[None, :])          # (n+1) x len(p)

def risk(d):                                               # exact risk under squared loss
    return ((d[:, None] - p[None, :])**2 * pmf).sum(axis=0)

mle = x / n
a = np.sqrt(n) / 2
minimax = (x + a) / (n + 2 * a)                           # Bayes for Beta(sqrt(n)/2, sqrt(n)/2)
laplace = (x + 1) / (n + 2)                                # Bayes for Beta(1,1)
R_mle, R_mm, R_lap = risk(mle), risk(minimax), risk(laplace)

print(f"n = {n} | minimax risk (theory) 1/(4(1+sqrt n)^2) = {1/(4*(1+np.sqrt(n))**2):.6f}")
print(f"Risk of the minimax rule: min = {R_mm.min():.6f}, max = {R_mm.max():.6f}  (equaliser)")
print(f"Maximum risks: MLE = {R_mle.max():.6f} | Laplace = {R_lap.max():.6f} | minimax = {R_mm.max():.6f}")
better = p[R_mm < R_mle]
```

```
print(f"Minimax rule better than MLE: p in ({better.min():.3f}, {better.max():.3f})  (theory:
    0.2, 0.8)")
print(f"Bayes risks under the uniform prior: MLE = {np.trapezoid(R_mle, p):.6f} | "
      f"Laplace = {np.trapezoid(R_lap, p):.6f} | minimax = {np.trapezoid(R_mm, p):.6f}")

fig, ax = plt.subplots(figsize=(7, 3.4))
ax.plot(p, R_mle, label=r"MLE $X/n$")
ax.plot(p, R_lap, "--", label=r"Laplace $(X+1)/(n+2)$")
ax.plot(p, R_mm, lw=2, label=r"Minimax $(X+\sqrt{n}/2)/(n+\sqrt{n})$")
ax.set_xlabel("$p$"); ax.set_ylabel(r"$R(p,\delta)$"); ax.legend(fontsize=8)
ax.set_title(f"Risk functions under squared loss, n = {n}", fontsize=10)
fig.tight_layout(); fig.savefig("karar_binom_risk.pdf")
```

Output

```
n = 16 | minimax risk (theory) 1/(4(1+sqrt n)^2) = 0.010000
Risk of the minimax rule: min = 0.010000, max = 0.010000  (equaliser)
Maximum risks: MLE = 0.015625 | Laplace = 0.012346 | minimax = 0.010000
Minimax rule better than MLE: p in (0.203, 0.797)  (theory: 0.2, 0.8)
Bayes risks under the uniform prior: MLE = 0.010417 | Laplace = 0.009259 | minimax = 0.010000
```

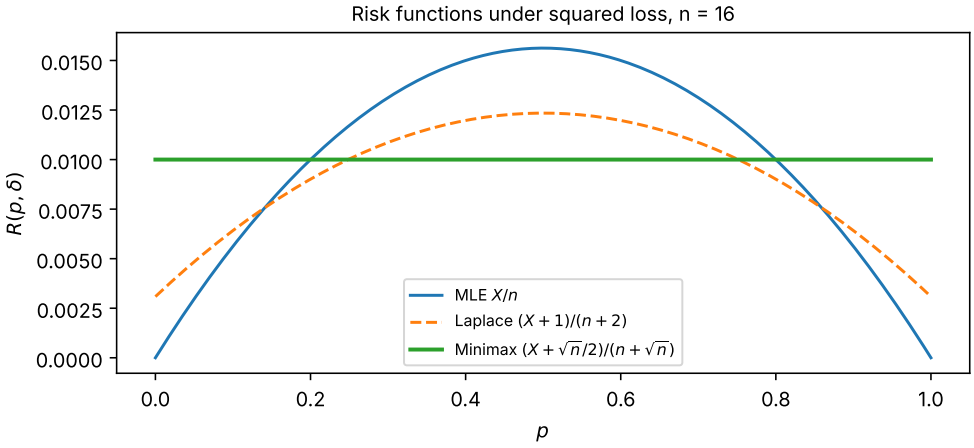


Figure 20.1: Risk functions of three estimators under squared loss for $n = 16$. The risk of the minimax rule is constant and below the maximum risk of the other two rules; but as p approaches the ends, both other rules are better than it.

The output confirms the theory exactly: the risk of the minimax rule is $1/(4 \cdot 5^2) = 0.01$ independently of p, and it is better than the MLE only on the interval $p \in (0.2, 0.8)$. Under the uniform prior the rule with the smallest average risk is Laplace's rule; this is because it is the Bayes rule for this prior. No rule dominates the other two: the choice depends on how we summarise the risk (worst case or average).

20.4.2 The Risk Set in a Two-Point Problem

$X \sim \text{Bin}(5, \theta)$, $\theta \in \{0.3, 0.6\}$ and 0–1 loss. Every subset of $\{0, \dots, 5\}$ defines a non-randomised rule of the form "decide θ_1 if X is in this set"; each of the $2^6 = 64$ rules is a point $(R(\theta_0, \delta), R(\theta_1, \delta))$ in the plane. The risk set of the randomised rules is the convex hull of these points. The minimax rule is the point where the lower boundary of the hull meets the diagonal; by Theorem 20.6 it corresponds to the Bayes rule for the prior π^* that maximises $r(\pi) = \sum_x \min\{\pi f_0(x), (1-\pi) f_1(x)\}$.

code/ch20/c20_2_two_point_risk_set.py

```python
import itertools
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

n, t0, t1 = 5, 0.3, 0.6
x = np.arange(n + 1)
f0, f1 = stats.binom.pmf(x, n, t0), stats.binom.pmf(x, n, t1)

# 0-1 loss: R0 = P_t0(decide theta1), R1 = P_t1(decide theta0). 2^6 non-randomised rules.
points = []
for A in itertools.product([0, 1], repeat=n + 1):          # A[x]=1: decide theta1
    A = np.array(A)
    points.append((f0[A == 1].sum(), f1[A == 0].sum()))
points = np.array(points)

pi = np.linspace(0, 1, 100_001)                            # pi = P(theta = theta0)
bayes = np.minimum(pi[:, None] * f0, (1 - pi[:, None]) * f1).sum(axis=1)
j = bayes.argmax()
print(f"Number of non-randomised rules: {len(points)}")
print(f"Smallest max. risk among non-randomised rules = {points.max(axis=1).min():.4f}")
print(f"Least favourable prior pi* = {pi[j]:.4f} | minimax risk = max_pi r(pi) = {bayes[j]:.4
    f}")

# Bayes rule at pi*: theta1 if LR = f1/f0 > pi*/(1-pi*); randomise at equality
lr = f1 / f0; thr = pi[j] / (1 - pi[j])
strict = lr > thr + 1e-9; bnd = np.isclose(lr, thr, atol=1e-3)
gamma = (f1[~strict].sum() - f0[strict].sum()) / (f0[bnd].sum() + f1[bnd].sum())
R0 = f0[strict].sum() + gamma * f0[bnd].sum()
R1 = f1[~strict & ~bnd].sum() + (1 - gamma) * f1[bnd].sum()
print(f"LR threshold = {thr:.4f}, boundary x = {x[bnd]}, randomisation probability gamma = {
    gamma:.4f}")
print(f"Minimax rule: R0 = {R0:.4f}, R1 = {R1:.4f} (equal error probabilities)")

fig, ax = plt.subplots(figsize=(4.6, 4.2))
ax.scatter(*points.T, s=10, color="gray", label="non-randomised rules")
from scipy.spatial import ConvexHull
h = ConvexHull(points); v = np.append(h.vertices, h.vertices[0])
ax.plot(*points[v].T, lw=0.8, color="C0", label="boundary of the risk set")
ax.plot([0, 1], [0, 1], ":", color="k", lw=0.8)
ax.plot(R0, R1, "o", color="C3", ms=7, label="minimax rule")
ax.set_xlabel(r"$R(\theta_0,\delta)$"); ax.set_ylabel(r"$R(\theta_1,\delta)$")
ax.set_xlim(-0.02, 1.02); ax.set_ylim(-0.02, 1.02); ax.legend(fontsize=7, loc="upper right")
fig.tight_layout(); fig.savefig("karar_risk_kumesi.pdf")
```

Output

```
Number of non-randomised rules: 64
Smallest max. risk among non-randomised rules = 0.3174
Least favourable prior pi* = 0.4274 | minimax risk = max_pi r(pi) = 0.2515
LR threshold = 0.7464, boundary x = [2], randomisation probability gamma = 0.2863
Minimax rule: R0 = 0.2515, R1 = 0.2515 (equal error probabilities)
```

Among non-randomised rules the best maximum risk is 0.3174; randomisation reduces it to 0.2515. The minimax rule is the likelihood ratio test that decides θ_1 for $X \geq 3$ and chooses θ_1 with probability 0.286 at $X = 2$. The threshold equals the odds of the least favourable prior ($\pi^*/(1-\pi^*) = 0.746$), and since the likelihood ratio at $X = 2$ takes exactly this value, the randomisation happens there. The equality of $\max_\pi r(\pi)$ and the minimax risk shows numerically that in this problem the Bayes and minimax approaches meet.

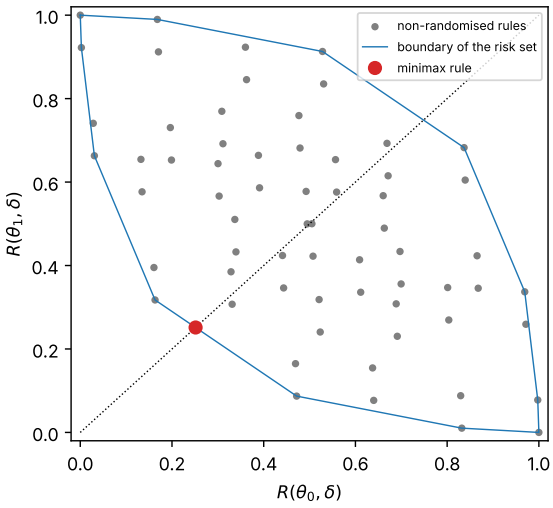


Figure 20.2: Risk points of the 64 non-randomised rules in the two-point problem and the boundary of the risk set. The red point is the minimax (randomised) rule, where the boundary meets the diagonal.

20.5 Comments and Notes

Remark 20.17 (A Critique of the Minimax Principle). The minimax principle is pessimistic: it judges the risk only by its value at the worst θ, and this worst case may not be plausible in reality. In the binomial example the minimax rule is worse than the MLE when p is near the ends, and the region where it is advantageous shrinks as n grows. On the other hand, the minimax risk is a prior-free measure of "how hard" a problem is; in nonparametric statistics and machine learning, optimal rates of convergence (e.g. $n^{-4/5}$ in density estimation, Chapter 21) are expressed in terms of the minimax risk.

Remark 20.18 (Complete Class Theorems). Wald's *complete class theorems* say that, under suitable regularity conditions, every admissible rule is a Bayes rule or a limit of Bayes rules [80, 6]. In the two-point problem this means that every point on the lower boundary of the risk set is Bayes with respect to some prior (Section 20.4.2). Consequently, a frequentist statistician looking for admissible rules is inevitably led to Bayes rules, while a Bayesian statistician justifies the rule by the chosen prior.

Remark 20.19 (Admissibility and Dimension). Example 20.13 together with the James–Stein result of Chapter 7 says: in one dimension the natural estimator is both minimax and admissible; in three or more dimensions it remains minimax but is inadmissible. Brown's [14] general result ties admissibility to the behaviour of the prior measure at infinity, and thereby to whether a random walk is recurrent; a random walk is recurrent in one and two dimensions, but not in three.

20.6 Chapter Summary

- A decision problem consists of a parameter space, an action space and a loss function (Definition 20.1). Rules are compared by their risk functions; randomisation makes the risk set convex.
- The minimax rule minimises the maximum risk (Definition 20.3). If the risk of a Bayes rule nowhere exceeds its Bayes risk, the rule is minimax and the prior least favourable

(Theorem 20.6). Bayes rules with constant risk (equalisers) are therefore minimax (Corollary 20.7).

- When there is no proper prior, the limiting Bayes method with sequences of priors is used (Theorem 20.8). For the normal mean, $\bar{X}$ turns out to be minimax in this way.
- A unique Bayes rule is admissible (Theorem 20.9). An admissible equaliser rule is minimax (Theorem 20.11). Blyth's method (Theorem 20.12) proves admissibility with sequences of priors.
- For a binomial proportion the minimax estimator is $(X + \sqrt{n}/2)/(n + \sqrt{n})$, with constant risk. It beats the MLE only when p is near $1/2$: the minimax criterion is a cautious choice geared to the worst case.

Further reading. [6], [46, Chapters 4–5], [16, Section 7.3.4]; for the historical foundation, [80].

20.7 Exercises

Exercise 20.1. Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$ with the weighted squared loss $L(\lambda, a) = (a - \lambda)^2/\lambda$. Show that $\bar{X}$ is an equaliser and, using Theorem 20.8 with the sequence of Gamma$(1, b)$ (shape–rate) priors, that it is minimax. (Hint: under this loss the Bayes rule is $1/\mathbb{E}[\lambda^{-1} \mid \mathbf{x}]$.) → Solution

Exercise 20.2. Take $n = 100$ for the minimax rule of Example 20.14. For which values of p does it have smaller risk than the MLE? Compare the Bayes risks under the uniform prior. What does the ratio of the maximum risks of the two rules converge to as $n \to \infty$? → Solution

Exercise 20.3. Show that under squared loss a Bayes rule δ^π with $r(\pi, \delta^\pi) < \infty$ cannot be *unbiased* (except in degenerate cases): prove that if δ^π is unbiased then $\mathbb{E}[(\delta^\pi(\mathbf{X}) - \theta)^2] = 0$, by conditioning $\mathbb{E}[\theta\, \delta^\pi]$ under the joint distribution in two different ways. Relate this result to why $\bar{X}$ is not Bayes with respect to any proper prior in the normal model. → Solution

Exercise 20.4. In the problem of Section 20.4.2, find the Bayes rule and the Bayes risk for the prior $\pi = (1/2, 1/2)$. Is this rule minimax? Why? → Solution

Chapter 21

Nonparametric Methods

For most of the book we have assumed that the data come from a family determined by finitely many parameters. This assumption yields strong results (the Cramér–Rao bound, UMP tests, efficiency of the MLE), but if the model is wrong these results may become invalid. *Nonparametric* methods make only weak assumptions about F, such as continuity or symmetry. In this chapter we follow two ideas. First, the distribution of statistics based on the *ranks* of the observations rather than their values does not depend on F under the null hypothesis; this yields tests of exact level (sign, Wilcoxon, Mann–Whitney, Kolmogorov–Smirnov) and exact confidence intervals for quantiles. Second, F or its density can itself be estimated; the density counterpart of the empirical CDF of Chapter 2 is the *kernel density estimate*. We close the chapter with the question of how much power nonparametric tests lose, i.e. with Pitman asymptotic relative efficiency. The answer surprises most readers: even for normal data the loss is only about 5%.

21.1 Definitions

Definition 21.1 (Distribution-Free Statistic)**.** For a class $\mathcal{F}$ of distributions (e.g. all continuous distributions), if the distribution of $T(\mathbf{X})$ when $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ is the same for every $F \in \mathcal{F}$, then T is called *distribution-free* over $\mathcal{F}$.

Definition 21.2 (Ranks)**.** Let $X_1, \dots, X_n$ be observations without ties. The *rank* of X_i is $R_i = \sum_{j=1}^{n} \mathbb{I}\{X_j \le X_i\}$; i.e. $X_i = X_{(R_i)}$. $\mathbf{R} = (R_1, \dots, R_n)$ is a permutation of $\{1, \dots, n\}$.

Definition 21.3 (One-Sample Location Tests)**.** Let $X_1, \dots, X_n$ be i.i.d. from a continuous distribution with median θ; $H_0 : \theta = 0$.

- *Sign statistic:* $S = \sum_i \mathbb{I}\{X_i > 0\}$.
- *Wilcoxon signed-rank statistic:* with R_i^+ the rank of $|X_i|$ among $|X_1|, \dots, |X_n|$, $W^+ = \sum_i R_i^+ \mathbb{I}\{X_i > 0\}$.

Definition 21.4 (Two Samples: Wilcoxon–Mann–Whitney)**.** Let $X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ be independent, $N = m + n$. The sum of the ranks of the Y_j in the combined sample is called the *Wilcoxon rank sum* W [86], and $U = \sum_{i,j} \mathbb{I}\{X_i < Y_j\}$ the *Mann–Whitney statistic* [55]. $H_0 : F = G$ is rejected for large values of U in favour of the alternative that G is shifted to the right.

Definition 21.5 (Kolmogorov–Smirnov Statistics)**.** One sample: $D_n = \sup_x |\hat{F}_n(x) - F_0(x)|$, for $H_0 : F = F_0$. Two samples: $D_{m,n} = \sup_x |\hat{F}_m(x) - \hat{G}_n(x)|$, for $H_0 : F = G$.

Definition 21.6 (Kernel Density Estimator)**.** Let $K \geq 0$ be a symmetric *kernel* with $\int K = 1$, $\int uK(u)du = 0$, $\mu_2(K) = \int u^2 K(u)du < \infty$ and $R(K) = \int K^2 < \infty$, and let $h > 0$ be the *bandwidth*. The kernel density estimator [67, 63] is defined as

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{x - X_i}{h}\Big).$$

Each observation is replaced by a small "bump" of width h centred at it. As the error criterion the *mean integrated squared error* is used: $\mathrm{MISE}(h) = \mathbb{E} \int (\hat{f}_h - f)^2$.

Definition 21.7 (Pitman Asymptotic Relative Efficiency)**.** Suppose two sequences of level-α tests require n_1 and n_2 observations, respectively, to achieve the same limiting power against alternatives approaching θ_0 at the rate $\theta_n = \theta_0 + h/\sqrt{n}$. The limit of the ratio n_2/n_1 is called the *Pitman ARE* of the first test relative to the second. It is the testing counterpart of the estimator ARE of Chapter 12.

21.2 Main Theorems and Proofs

Theorem 21.8 (Distribution-Free Confidence Interval for Quantiles)**.** *Let $X_1, \dots, X_n \overset{i.i.d.}{\sim} F$ with F continuous, and let ξ_p be a quantile with $F(\xi_p) = p$. For $r < s$,*

$$\mathbb{P}(X_{(r)} \leq \xi_p < X_{(s)}) = \sum_{j=r}^{s-1} \binom{n}{j} p^j (1-p)^{n-j}.$$

This probability does not depend on F; hence $[X_{(r)}, X_{(s)}]$ is a confidence interval for ξ_p with known exact coverage probability.

Proof. $B = \#\{i : X_i \leq \xi_p\} \sim \mathrm{Bin}(n, p)$, because $\mathbb{P}(X_i \leq \xi_p) = F(\xi_p) = p$. The event $X_{(r)} \leq \xi_p$ is "at least r observations are less than or equal to ξ_p", i.e. the event $\{B \geq r\}$. Similarly $\{X_{(s)} \leq \xi_p\} = \{B \geq s\}$. Hence $\{X_{(r)} \leq \xi_p < X_{(s)}\} = \{r \leq B \leq s-1\}$. Since by continuity $\mathbb{P}(X_{(s)} = \xi_p) = 0$, whether the interval is taken closed or open does not change the probability. □

Theorem 21.9 (Distribution of the Rank Vector)**.** *If $X_1, \dots, X_n \overset{i.i.d.}{\sim} F$ with F continuous, then*

1. *there are a.s. no ties, and $\mathbf{R}$ is uniformly distributed over the $n!$ permutations of $\{1, \dots, n\}$;*
2. *$\mathbf{R}$ is independent of the vector of order statistics $(X_{(1)}, \dots, X_{(n)})$.*

Consequently, every function of the ranks is distribution-free over all continuous distributions.

Proof. (1) For $i \neq j$, $\mathbb{P}(X_i = X_j) = \int \mathbb{P}(X_i = x)\, dF(x) = 0$, because a continuous distribution has no atoms; a finite union also has probability zero. (2) For a permutation σ, $(X_{\sigma(1)}, \dots, X_{\sigma(n)})$ has the same distribution as $(X_1, \dots, X_n)$ by the i.i.d. structure (exchangeability). The event $\mathbf{R} = \mathbf{r}$ is the event "the position of X_i in the ordered vector is r_i". A permutation leaves the ordered vector unchanged but carries the rank vector to another permutation. Hence for every measurable B and any two permutations $\mathbf{r}, \mathbf{r}'$,

$$\mathbb{P}\big(\mathbf{R} = \mathbf{r},\ (X_{(1)}, \dots, X_{(n)}) \in B\big) = \mathbb{P}\big(\mathbf{R} = \mathbf{r}',\ (X_{(1)}, \dots, X_{(n)}) \in B\big).$$

Summing over the $n!$ permutations, each term equals $\mathbb{P}\big((X_{(\cdot)}) \in B\big)/n!$. Taking B to be the whole space gives the uniformity in (1), and for general B the product form, i.e. independence. □

Theorem 21.10 (Null Distribution of the Wilcoxon Signed-Rank Statistic)**.** *If $X_1, \dots, X_n$ are i.i.d. from a continuous distribution symmetric about 0, then*

$$W^+ \stackrel{d}{=} \sum_{j=1}^{n} j\, B_j, \qquad B_1, \dots, B_n \overset{i.i.d.}{\sim} \textit{Bernoulli}(1/2),$$

and $\mathbb{E}W^+ = \frac{n(n+1)}{4}$, $\operatorname{Var} W^+ = \frac{n(n+1)(2n+1)}{24}$. *Moreover* $S \sim \textit{Bin}(n, 1/2)$.

Proof. Under symmetry and continuity, $\mathbb{I}\{X_i > 0\}$ and $|X_i|$ are independent: for every $t \geq 0$, $\mathbb{P}(X_i > 0, |X_i| \leq t) = \mathbb{P}(0 < X_i \leq t) = \frac{1}{2}\mathbb{P}(|X_i| \leq t)$, and $\mathbb{P}(X_i > 0) = \frac{1}{2}$. Hence the signs $\epsilon_i = \mathbb{I}\{X_i > 0\}$ are Bernoulli(1/2) variables independent of the vector $(|X_1|, \dots, |X_n|)$ and of each other. If B_j denotes the sign of the observation in position j of the ordered $|X|$'s, the B_j are also i.i.d. Bernoulli(1/2), independent of the ordering of the $(|X_i|)$, and $W^+ = \sum_j jB_j$. Moments: $\mathbb{E}W^+ = \frac{1}{2}\sum_j j = \frac{n(n+1)}{4}$, $\operatorname{Var} W^+ = \frac{1}{4}\sum_j j^2 = \frac{n(n+1)(2n+1)}{24}$. Since $S = \sum_i B_i$, the last claim is clear. □

Theorem 21.11 (Null Moments of the Wilcoxon–Mann–Whitney Statistic)**.** *Under H_0 : $F = G$ (F continuous),*

$$U = W - \frac{n(n+1)}{2}, \qquad \mathbb{E}W = \frac{n(N+1)}{2}, \qquad \operatorname{Var} W = \operatorname{Var} U = \frac{mn(N+1)}{12}.$$

Proof. Identity. Let $Y_{(j)}$ be the j-th smallest of the Y's. Its rank in the combined sample is the number of Y's smaller than it $(j-1)$, plus the number of X's smaller than it, plus itself (1): $j + \#\{i : X_i < Y_{(j)}\}$. Summing over j gives $W = \frac{n(n+1)}{2} + U$.

Moments. Under H_0 the N observations are i.i.d., so by Theorem 21.9 the ranks of the Y's form a random subset of size n drawn without replacement from $\{1, \dots, N\}$. For sampling without replacement from a finite population, the expected value of the sum is $n\bar{r}$ and its variance $n\sigma_r^2 \frac{N-n}{N-1}$, where the population mean is $\bar{r} = \frac{N+1}{2}$ and the population variance $\sigma_r^2 = \frac{N^2-1}{12}$. Hence $\mathbb{E}W = \frac{n(N+1)}{2}$ and $\operatorname{Var} W = n\frac{(N-1)(N+1)}{12} \cdot \frac{m}{N-1} = \frac{mn(N+1)}{12}$. □

Remark 21.12. $U/(mn)$ is an unbiased estimator of $\mathbb{P}(X < Y)$; the area under the ROC curve (AUC) is exactly this quantity. Under H_0, $\mathbb{P}(X < Y) = \frac{1}{2}$. The alternatives against which the test is consistent are those with $\mathbb{P}(X < Y) \neq \frac{1}{2}$; that is, the Mann–Whitney test does not answer the question "are the means equal?" but the question "is an observation chosen at random from one group more likely to be larger than one from the other?".

Theorem 21.13 (The KS Statistic Is Distribution-Free)**.** *If F_0 is continuous and $H_0 : F = F_0$ is true, the distribution of D_n equals that of $\sup_{u\in[0,1]} |\hat{G}_n(u) - u|$ built from $U_i \overset{i.i.d.}{\sim} \textit{Uniform}(0,1)$; i.e. it does not depend on F_0. Moreover [44]*

$$\lim_{n\to\infty} \mathbb{P}(\sqrt{n} D_n \leq t) = 1 - 2\sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2t^2}, \qquad t > 0.$$

Proof (first part). Let $U_i = F_0(X_i) \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$ (Theorem 2.14) and let $\hat{G}_n$ be their empirical CDF. Since F_0 is non-decreasing, $X_i \leq x \Rightarrow U_i \leq F_0(x)$; conversely, $X_i > x$ while $U_i \leq F_0(x)$ requires F_0 to be constant on $[x, X_i]$, an event of probability zero. Hence a.s., for every x, $\hat{F}_n(x) = \hat{G}_n(F_0(x))$ and

$$D_n = \sup_x \left|\hat{G}_n(F_0(x)) - F_0(x)\right| = \sup_{u\in(0,1)} \left|\hat{G}_n(u) - u\right|,$$

because the range of a continuous F_0 contains $(0,1)$; at the endpoints the difference is zero. The limit distribution comes from the convergence of the process $\sqrt{n}(\hat{G}_n - u)$ to a Brownian bridge and from the distribution of the supremum of a Brownian bridge; see [78, Chapter 19]. □

Theorem 21.14 (Bias, Variance and Optimal Bandwidth of the Kernel Estimator)**.** *Let f be twice continuously differentiable with f'' bounded. As $h \to 0$ and $nh \to \infty$, for every x,*

$$\mathbb{E}\hat{f}_h(x) - f(x) = \frac{h^2}{2}\mu_2(K)f''(x) + o(h^2), \qquad \operatorname{Var}\hat{f}_h(x) = \frac{f(x)R(K)}{nh} + o\Big(\frac{1}{nh}\Big).$$

Under suitable integrability conditions the leading terms of the integrated squared error are

$$\operatorname{AMISE}(h) = \frac{h^4}{4}\mu_2(K)^2 R(f'') + \frac{R(K)}{nh};$$

the bandwidth minimising this and the resulting error are

$$h^* = \Big[\frac{R(K)}{\mu_2(K)^2 R(f'')\, n}\Big]^{1/5}, \qquad \operatorname{AMISE}(h^*) = \frac{5}{4}\big(\mu_2(K)^2 R(f'')\big)^{1/5} R(K)^{4/5}\, n^{-4/5}.$$

Proof. Bias. With the variable $u = (x-y)/h$, $\mathbb{E}\hat{f}_h(x) = \int \frac{1}{h}K(\frac{x-y}{h})f(y)\,dy = \int K(u)f(x-hu)\,du$. By Taylor expansion, for some ξ between x and $x-hu$, $f(x-hu) = f(x) - huf'(x) + \frac{h^2u^2}{2}f''(\xi)$. Since $\int K = 1$ and $\int uK = 0$,

$$\mathbb{E}\hat{f}_h(x) - f(x) = \frac{h^2}{2}\int u^2 K(u) f''(\xi_{u,h})\,du.$$

Since f'' is continuous and bounded and $u^2K(u)$ is integrable, by dominated convergence the integral tends to $\mu_2(K)f''(x)$.

Variance. $\hat{f}_h(x)$ is the mean of the i.i.d. variables $Z_i = \frac{1}{h}K(\frac{x-X_i}{h})$: $\operatorname{Var}\hat{f}_h(x) = \frac{1}{n}(\mathbb{E}Z_1^2 - (\mathbb{E}Z_1)^2)$. With the same change of variables $\mathbb{E}Z_1^2 = \frac{1}{h}\int K(u)^2 f(x-hu)\,du = \frac{1}{h}\big(f(x)R(K) + o(1)\big)$ and $(\mathbb{E}Z_1)^2 = O(1)$; hence $\operatorname{Var}\hat{f}_h(x) = \frac{f(x)R(K)}{nh} + O(\frac{1}{n})$, and as $h \to 0$, $O(1/n) = o(1/(nh))$.

AMISE. MSE $=$ bias^2 $+$ variance; integrate over x and use $\int f = 1$. With $A = \mu_2^2 R(f'')/4$ and $B = R(K)/n$, the derivative of $Ah^4 + B/h$ is $4Ah^3 - B/h^2$, which vanishes at $h^5 = B/(4A)$; this is h^*. At this point $B/h = 4Ah^4$, so the value is $5Ah^{*4} = \frac{5}{4}(4A)^{1/5}B^{4/5}$. □

Theorem 21.15 (Pitman Efficiencies)**.** *Let $X_i = \theta + \varepsilon_i$, with the ε_i i.i.d., symmetric about 0, with density f and variance σ^2; $H_0 : \theta = 0$. Under sufficient regularity conditions,*

$$\operatorname{ARE}(S,t) = 4\sigma^2 f(0)^2, \qquad \operatorname{ARE}(W^+,t) = 12\sigma^2\Big(\int f^2\Big)^2.$$

Main idea of the proof. By Noether's theorem [78, Chapter 14], if $\sqrt{n}(T_n - \mu(\theta))/\tau(\theta) \xrightarrow{d} \mathcal{N}(0,1)$ holds uniformly along local alternatives, the local power of a test is determined only by its *efficacy* $e = \mu'(0)^2/\tau(0)^2$, and the ARE is the ratio of efficacies. For the three tests:

- t: $T = \bar{X}$, $\mu(\theta) = \theta$, $\tau^2 = \sigma^2$; $e_t = 1/\sigma^2$.
- Sign: $T = S/n$, $\mu(\theta) = \mathbb{P}(\varepsilon > -\theta) = F(\theta)$, $\mu'(0) = f(0)$, $\tau^2(0) = \frac{1}{4}$; $e_S = 4f(0)^2$.
- Wilcoxon: $W^+/\binom{n}{2}$ is asymptotically equivalent to the average of $\mathbb{I}\{X_i + X_j > 0\}$ over pairs $i < j$ (a U-statistic). $\mu(\theta) = \mathbb{P}(\varepsilon_1 + \varepsilon_2 > -2\theta)$; since the density of $\varepsilon_1 + \varepsilon_2$ at 0 is $\int f(x)f(-x)dx = \int f^2$, $\mu'(0) = 2\int f^2$. The asymptotic variance of the U-statistic is $4\operatorname{Var}\big(F(\varepsilon_1)\big) = \frac{4}{12} = \frac{1}{3}$. Hence $e_W = 12(\int f^2)^2$.

Taking the ratios gives the result. □

21.3 Worked Examples

Example 21.16 (Exact Confidence Interval for the Median)**.** $n = 20$, $p = \frac{1}{2}$. By Theorem 21.8, the coverage probability of $[X_{(6)}, X_{(15)}]$ is $\mathbb{P}(6 \le B \le 14) = 0.9586$, and that of $[X_{(7)}, X_{(14)}]$ is 0.8847. Because of the discrete binomial structure, exactly 95% cannot be attained; the narrowest symmetric interval not falling below 95% is $[X_{(6)}, X_{(15)}]$. This interval requires no assumption about F other than continuity; it is valid for Cauchy data too.

Example 21.17 (Pitman ARE Values)**.** The formulas of Theorem 21.15 give:

Distribution	$\int f^2$	$\mathrm{ARE}(W^+, t)$	$\mathrm{ARE}(S, t)$
Normal $\mathcal{N}(0, \sigma^2)$	$\frac{1}{2\sigma\sqrt{\pi}}$	$3/\pi \approx 0.955$	$2/\pi \approx 0.637$
Logistic	$\frac{1}{6s}$	$\pi^2/9 \approx 1.097$	$\pi^2/12 \approx 0.822$
Laplace (scale b)	$\frac{1}{4b}$	1.5	2
Uniform $(-1, 1)$	$\frac{1}{2}$	1	1/3

(For the logistic, s is the scale parameter, $\sigma^2 = \pi^2 s^2/3$ and $f(0) = \frac{1}{4s}$.) Hodges and Lehmann [36] showed that over the class of symmetric continuous densities $\mathrm{ARE}(W^+, t) \ge \frac{108}{125} = 0.864$, and that there is no upper bound. That is, relative to the t-test the Wilcoxon test loses at most 14% efficiency in the worst case, but can gain without limit. The sign test, on the other hand, wastes a third of the observations for normal data.

Example 21.18 (Mann–Whitney: Exact Small-Sample Distribution)**.** $m = n = 3$, $N = 6$. Under H_0 the ranks of the Y's are each of the $\binom{6}{3} = 20$ subsets with probability $\frac{1}{20}$. W takes values from 6 to 15; $\mathbb{P}(W = 15) = \mathbb{P}(W = 6) = \frac{1}{20}$. Hence the one-sided test with $\alpha = 0.05$ rejects only if all the Y's are larger than all the X's. In very small samples, the attainable levels of nonparametric tests form a coarse grid.

21.4 Computational Applications

21.4.1 Level and Power of Three Tests

With $n = 30$ observations from four distributions scaled to variance 1, we compute the rejection rates of the t, Wilcoxon signed-rank and sign tests under $\theta = 0$ (level) and $\theta = 0.5$ (power) with $M = 4000$ replications.

code/ch21/c21_1_t_wilcoxon_sign_power.py

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
n, M, alpha = 30, 4000, 0.05
dists = {                                   # all with mean 0, variance 1
    "Normal ": lambda size: rng.standard_normal(size),
    "Laplace": lambda size: rng.laplace(0, 1/np.sqrt(2), size),
    "t_3    ": lambda size: rng.standard_t(3, size) / np.sqrt(3),
    "Uniform": lambda size: rng.uniform(-np.sqrt(3), np.sqrt(3), size),
}

def rejection_rates(X):
    t = stats.ttest_1samp(X, 0, axis=1).pvalue < alpha
    w = stats.wilcoxon(X, axis=1).pvalue < alpha
    k = (X > 0).sum(axis=1)
    s = stats.binom.cdf(np.minimum(k, n - k), n, 0.5) * 2 < alpha      # two-sided sign test
    return t.mean(), w.mean(), s.mean()

```

```
print("Distrib.| theta | t-test  | Wilcoxon | sign")
for name, gen in dists.items():
    for theta in (0.0, 0.5):
        r = rejection_rates(gen((M, n)) + theta)
        print(f"{name} |  {theta:.1f}  |  {r[0]:.3f}  |  {r[1]:.3f}   | {r[2]:.3f}")

are = {"Normal ": (3/np.pi, 2/np.pi), "Laplace": (1.5, 2.0), "Uniform": (1.0, 1/3)}
print("Theoretical Pitman ARE (Wilcoxon/t, sign/t):",
      {k.strip(): (round(a, 3), round(b, 3)) for k, (a, b) in are.items()})
```

Output

```
Distrib.| theta | t-test  | Wilcoxon | sign
Normal  |  0.0  |  0.057  |  0.057   | 0.044
Normal  |  0.5  |  0.764  |  0.745   | 0.561
Laplace |  0.0  |  0.050  |  0.057   | 0.049
Laplace |  0.5  |  0.753  |  0.837   | 0.818
t_3     |  0.0  |  0.043  |  0.053   | 0.041
t_3     |  0.5  |  0.825  |  0.935   | 0.898
Uniform |  0.0  |  0.050  |  0.046   | 0.039
Uniform |  0.5  |  0.761  |  0.689   | 0.342
Theoretical Pitman ARE (Wilcoxon/t, sign/t): {'Normal': (0.955, 0.637), 'Laplace': (1.5, 2.0)
    , 'Uniform': (1.0, 0.333)}
```

The power ranking follows the Pitman ARE table. For normal data t is most powerful, but the loss of Wilcoxon is small; the sign test lags clearly behind. For heavy-tailed distributions such as the Laplace and t_3, both nonparametric tests beat t. For the uniform distribution the power of the sign test collapses (ARE $= 1/3$). Under the null hypothesis all three tests approximately keep the level. The sign test is slightly conservative, because the binomial distribution is discrete and the exact 5% level is not attainable.

21.4.2 Kolmogorov–Smirnov and the Lilliefors Problem

code/ch21/c21_2_kolmogorov_smirnov.py

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
n, M = 20, 20_000

def D(x, cdf):                                  # one-sample KS statistic sup|F_n - F|
    x = np.sort(x, axis=1); u = cdf(x); i = np.arange(1, n + 1)
    return np.maximum((i / n - u).max(axis=1), (u - (i - 1) / n).max(axis=1))

D_norm = D(rng.standard_normal((M, n)), stats.norm.cdf)
D_exp = D(rng.exponential(1.0, (M, n)), stats.expon.cdf)
q = [0.5, 0.9, 0.95, 0.99]
print("Quantiles        :", q)
print("D_n, normal data :", np.round(np.quantile(D_norm, q), 4))
print("D_n, expon. data :", np.round(np.quantile(D_exp, q), 4))
print("Exact (kstwo)    :", np.round(stats.kstwo.ppf(q, n), 4))
print("Asymptotic K/sqrt(n):", np.round(stats.kstwobign.ppf(q) / np.sqrt(n), 4))

# Lilliefors: KS critical values become invalid when parameters are estimated from the data
X = rng.normal(5, 2, (M, n))
mu, s = X.mean(axis=1, keepdims=True), X.std(axis=1, ddof=1, keepdims=True)
D_est = D((X - mu) / s, stats.norm.cdf)
crit = stats.kstwo.ppf(0.95, n)
print(f"H0 true, parameters estimated: rejection rate (target 0.05) = {(D_est > crit).mean()
    :.4f}")
```

```
print(f"0.95 quantile of the estimated statistic = {np.quantile(D_est, 0.95):.4f} (KS
    critical value {crit:.4f})")
```

Output

```
Quantiles        : [0.5, 0.9, 0.95, 0.99]
D_n, normal data : [0.1773 0.2633 0.2924 0.3512]
D_n, expon. data : [0.1775 0.264  0.2942 0.3498]
Exact (kstwo)    : [0.1773 0.2647 0.2941 0.3524]
Asymptotic K/sqrt(n): [0.1851 0.2737 0.3037 0.3639]
H0 true, parameters estimated: rejection rate (target 0.05) = 0.0000
0.95 quantile of the estimated statistic = 0.1909 (KS critical value 0.2941)
```

The quantiles of D_n computed from normal and exponential data agree within Monte Carlo error and match the exact distribution (`kstwo`): the distribution-freeness claim of Theorem 21.13. At $n = 20$ the asymptotic Kolmogorov distribution slightly overstates the quantiles. The last two lines show a common mistake. If, when testing normality, μ and σ are estimated from the data and the standard KS critical values are used, the test almost never rejects. Since the estimated distribution is "too close" to the data, D_n is systematically smaller. The correct critical values (Lilliefors [50]) must be obtained by simulation or by the parametric bootstrap (Chapter 18).

21.4.3 Kernel Density Estimation: Bandwidth and Rate of Convergence

For a bimodal mixture density we plot the Gaussian-kernel estimator with different values of h, and measure by Monte Carlo the AMISE-optimal h^* of Theorem 21.14 and the rate at which the MISE decreases with n.

code/ch21/c21_3_kernel_density.py

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
def f(x):                                          # bimodal mixture
    return 0.6 * stats.norm.pdf(x, -1.5, 0.7) + 0.4 * stats.norm.pdf(x, 1.5, 0.5)
def gen(n):
    k = rng.random(n) < 0.6
    return np.where(k, rng.normal(-1.5, 0.7, n), rng.normal(1.5, 0.5, n))
def kde(x, data, h):                               # Gaussian-kernel estimator
    return stats.norm.pdf((x[:, None] - data[None, :]) / h).mean(axis=1) / h

grid = np.linspace(-5, 5, 801); dx = grid[1] - grid[0]
# R(f'') numerically: AMISE-optimal h = [R(K) / (mu2^2 R(f'') n)]^(1/5); Gaussian kernel: R(K
    )=1/(2 sqrt(pi)), mu2=1
f2 = np.gradient(np.gradient(f(grid), dx), dx)
Rf2 = (f2**2).sum() * dx
def h_opt(n): return (1 / (2 * np.sqrt(np.pi)) / (Rf2 * n))**0.2

print(f"R(f'') = {Rf2:.4f}")
print("    n | h_opt  | MISE (MC) | AMISE theory")
res = []
for n in (100, 400, 1600, 6400):
    h = h_opt(n)
    mise = np.mean([((kde(grid, gen(n), h) - f(grid))**2).sum() * dx for _ in range(100)])
    amise = 1.25 * Rf2**0.2 * (1 / (2 * np.sqrt(np.pi)))**0.8 * n**-0.8
    res.append((n, mise)); print(f"{n:5d} | {h:.4f} | {mise:.6f}  | {amise:.6f}")
sl = np.polyfit(np.log([s[0] for s in res]), np.log([s[1] for s in res]), 1)[0]
print(f"slope of log MISE ~ log n = {sl:.3f} (theory: -0.8; parametric rate: -1)")

data = gen(200)
```

```
h_silv = 1.06 * data.std(ddof=1) * 200**-0.2
print(f"n=200: Silverman h = {h_silv:.3f} | AMISE-optimal h = {h_opt(200):.3f}")
fig, ax = plt.subplots(figsize=(7, 3.2))
ax.plot(grid, f(grid), "k", lw=2, label="true $f$")
for h, st in ((0.08, ":"), (h_opt(200), "-"), (h_silv, "--"), (1.0, "-.")):
    ax.plot(grid, kde(grid, data, h), st, label=f"h = {h:.2f}")
ax.set_xlim(-4.5, 4); ax.legend(fontsize=8); ax.set_xlabel("$x$")
fig.tight_layout(); fig.savefig("cekirdek_bant.pdf")
```

Output

```
R(f'') = 1.6075
    n | h_opt   | MISE (MC) | AMISE theory
  100 | 0.2811 | 0.009296  | 0.012545
  400 | 0.2130 | 0.003604  | 0.004138
 1600 | 0.1614 | 0.001121  | 0.001365
 6400 | 0.1224 | 0.000412  | 0.000450
slope of log MISE ~ log n = -0.759 (theory: -0.8; parametric rate: -1)
n=200: Silverman h = 0.592 | AMISE-optimal h = 0.245
```

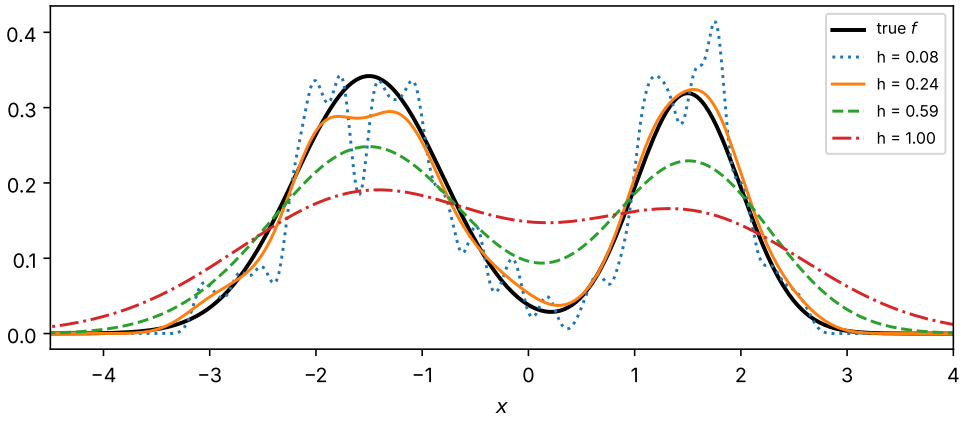


Figure 21.1: Kernel density estimates with $n = 200$ observations. Small h (dotted) is noisy; large h (dash-dotted) is biased enough to merge the two modes. Silverman's rule (dashed), derived for the normal distribution, oversmooths the bimodal density.

The log–log slope of the MISE is about -0.76, close to the theoretical $-4/5$. For small n the slope is somewhat flatter because of the terms neglected by AMISE. This rate is slower than the $1/n$ of parametric models: estimating a density locally at every point is harder than estimating finitely many parameters. The rate $n^{-4/5}$ is the minimax rate over the class of twice-differentiable densities (Chapter 20). That Silverman's rule gives 0.59, far from the optimal 0.25, is a reminder that the "rule" rests on a normality assumption.

21.5 Comments and Notes

Remark 21.19 (Ties). In real data, ties arise from rounding of measurements. In that case tied observations receive their average rank, and the null distribution is computed conditionally (given the tie structure) by permutation. A tie correction is added to the variance formulas. In one-sample tests, observations with $X_i = 0$ are usually discarded.

Remark 21.20 (The Hodges–Lehmann Estimator and Confidence Intervals). By the duality of Chapter 16, every test produces a confidence interval and a point estimate. Inverting the Wilcoxon signed-rank test gives the confidence interval built from the *Walsh averages* $(X_i + X_j)/2$, $i \le j$. The median of these averages is the *Hodges–Lehmann estimator*. Its

ARE is the same as that of the Wilcoxon test: 0.955 for normal data, and much better than the mean for heavy tails.

Remark 21.21 (The Price of Nonparametric Inference). Rank tests lose almost no power against shift alternatives. In estimating densities and regression functions, by contrast, the price is paid in the rate of convergence: $n^{-4/5}$ becomes $n^{-4/(4+d)}$ in d dimensions. This *curse of dimensionality* explains why high-dimensional nonparametric estimation requires structural assumptions (additivity, sparsity).

21.6 Chapter Summary

- Under a continuous F the rank vector is uniformly distributed over all permutations (Theorem 21.9). Statistics based on ranks are therefore distribution-free.
- Confidence intervals for quantiles are built from order statistics and binomial probabilities without any distributional assumption (Theorem 21.8).
- Location tests: the sign test, the Wilcoxon signed-rank test (Theorem 21.10) and, for two samples, the Wilcoxon–Mann–Whitney test (Theorem 21.11).
- Pitman efficiency (Theorem 21.15): the Wilcoxon test has 95.5% of the efficiency of the t-test for normal data and never falls below 0.864 for any symmetric continuous distribution. For heavy tails it beats t.
- The Kolmogorov–Smirnov statistic is distribution-free (Theorem 21.13). If parameters are estimated from the data this property is lost (the Lilliefors problem).
- In kernel density estimation the bias is of order $O(h^2)$ and the variance of order $O(1/(nh))$. The optimal bandwidth is $h^* \propto n^{-1/5}$, and the MISE decreases at the rate $n^{-4/5}$ (Theorem 21.14).

Further reading. [49], [82]; for kernel methods, [73]; for asymptotic theory, [78, Chapters 13–14 and 19].

21.7 Exercises

Exercise 21.1. Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ with F continuous, $p \in (0,1)$ fixed and $k = \lceil np \rceil$. Using $F(X_{(k)}) \sim \mathrm{Beta}(k, n-k+1)$ (Example 4.20), show that $F(X_{(k)}) \xrightarrow{P} p$. Deduce that $X_{(k)} \xrightarrow{P} \xi_p$ if $f(\xi_p) > 0$. → Solution

Exercise 21.2. With $n = 25$ observations, find the narrowest symmetric interval $[X_{(r)}, X_{(n+1-r)}]$ for the median whose coverage does not fall below 95%, and compute its exact coverage probability. Derive the rule $r \approx \frac{n}{2} - z_{\alpha/2}\frac{\sqrt{n}}{2}$ from the normal approximation. → Solution

Exercise 21.3. For $m = n = 3$, derive the exact null distribution of W, compute $\mathbb{E}W$ and $\operatorname{Var} W$ from this distribution, and compare with Theorem 21.11. Show that $U/(mn)$ is unbiased for $\mathbb{P}(X < Y)$. → Solution

Exercise 21.4. For the Gaussian kernel and the true density $\mathcal{N}(\mu, \sigma^2)$, compute the h^* of Theorem 21.14 and show that $h^* = (4/3)^{1/5}\sigma n^{-1/5} \approx 1.06\,\sigma n^{-1/5}$ (Silverman's rule). ($R(K) = \frac{1}{2\sqrt{\pi}}$, $\mu_2 = 1$ and $\int (\phi_\sigma'')^2 = \frac{3}{8\sqrt{\pi}\sigma^5}$.) → Solution

Chapter 22

Multiple Testing: Family-Wise Error Rate and False Discovery Rate

Neyman–Pearson theory (Chapter 13) deals with testing a *single* hypothesis and bounds the probability of a type I error by α. In modern applications, however, hundreds or millions of hypotheses are tested with the same data: differential expression of thousands of genes, dozens of metrics in an A/B experiment, every coefficient in a regression. If each test is performed at level $\alpha = 0.05$, then even when all null hypotheses are true, the probability of at least one false rejection in $m = 100$ independent tests is $1 - 0.95^{100} \approx 0.994$. In this chapter we study two error criteria. The first is the *family-wise error rate* (FWER), which bounds the probability of at least one false rejection; it is controlled by the Bonferroni and Holm procedures. The second is the *false discovery rate* (FDR), which bounds the expected proportion of false rejections among the rejections; it is controlled by the Benjamini–Hochberg procedure. The central result of the chapter is the short and elegant proof that under independence the BH procedure makes the FDR *exactly* equal to $\pi_0 \alpha$.

22.1 Definitions

Definition 22.1 (The Multiple Testing Problem)**.** Let null hypotheses $H_1, \dots, H_m$ and their p-values $p_1, \dots, p_m$ be given. Let $I_0 \subseteq \{1, \dots, m\}$ be the (unknown) set of true null hypotheses, $m_0 = |I_0|$ and $\pi_0 = m_0/m$. A *multiple testing procedure* produces from the p-values a rejection set $\mathcal{R} \subseteq \{1, \dots, m\}$. $R = |\mathcal{R}|$ is the total number of rejections and $V = |\mathcal{R} \cap I_0|$ the number of false rejections (false discoveries).

	Not rejected	Rejected	Total
True H_0	$m_0 - V$	V	m_0
False H_0	$m - m_0 - (R - V)$	$R - V$	$m - m_0$
Total	$m - R$	R	m

Definition 22.2 (Valid p-Value)**.** If for $i \in I_0$, $\mathbb{P}(p_i \le u) \le u$ for every $u \in [0, 1]$, then p_i is called a *valid* (super-uniform) p-value. The p-value of a continuous test statistic under a simple H_0 is exactly Uniform$(0, 1)$ (Theorem 13.10).

Definition 22.3 (Family-Wise Error Rate and False Discovery Rate)**.**

$$\mathrm{FWER} = \mathbb{P}(V \ge 1), \qquad \mathrm{FDR} = \mathbb{E}\Big[\frac{V}{\max(R, 1)}\Big].$$

The FDR answers the question "what proportion of my rejections do I expect to be false?" [4]. A procedure controls the FWER in the *strong sense* if FWER $\le \alpha$ whatever I_0 is, and in the *weak sense* if this holds only when $I_0 = \{1, \dots, m\}$.

Definition 22.4 (Three Procedures)**.** Let $p_{(1)} \le \cdots \le p_{(m)}$ be the ordered p-values and $H_{(k)}$ the corresponding hypotheses.

- *Bonferroni:* reject every H_i with $p_i \le \alpha/m$.
- *Holm* (step-down) [39]: for $k = 1, 2, \dots,$ reject $H_{(k)}$ as long as $p_{(k)} \le \frac{\alpha}{m-k+1}$; stop at the first step where the condition fails.
- *Benjamini–Hochberg (BH)* (step-up): $k^* = \max\{k : p_{(k)} \le \alpha k/m\}$ ($k^* = 0$ if there is no such k); reject $H_{(1)}, \dots, H_{(k^*)}$.

Definition 22.5 (Adjusted p-Values)**.** The smallest α at which a procedure rejects H_i is called the *adjusted p-value* of H_i. For the procedures above,

$$\tilde{p}^{\mathrm{Bonf}}_{(k)} = \min(1, m p_{(k)}), \qquad \tilde{p}^{\mathrm{Holm}}_{(k)} = \min\Big(1, \max_{j \le k} (m - j + 1) p_{(j)}\Big),$$
$$\tilde{p}^{\mathrm{BH}}_{(k)} = \min\Big(1, \min_{j \ge k} \frac{m\, p_{(j)}}{j}\Big).$$

22.2 Main Theorems and Proofs

Theorem 22.6 (Bonferroni)**.** *If the p-values are valid, the Bonferroni procedure controls the FWER in the strong sense, whatever the dependence among the p-values:* FWER $\le m_0 \alpha/m \le \alpha$.

Proof. Boole's inequality: $\mathbb{P}(V \ge 1) = \mathbb{P}\Big(\bigcup_{i \in I_0} \{p_i \le \alpha/m\}\Big) \le \sum_{i \in I_0} \mathbb{P}(p_i \le \alpha/m) \le m_0 \frac{\alpha}{m}$. □

Theorem 22.7 (Šidák)**.** *If the p-values of the true null hypotheses are independent and uniformly distributed, the procedure that performs each test at level $\alpha_m = 1 - (1-\alpha)^{1/m}$ has FWER $1 - (1-\alpha_m)^{m_0} \le \alpha$, with equality when $m_0 = m$. Moreover $\alpha_m > \alpha/m$, i.e. this procedure is slightly more powerful than Bonferroni [72].*

Proof. By independence, $\mathbb{P}(V = 0) = \prod_{i \in I_0} \mathbb{P}(p_i > \alpha_m) = (1-\alpha_m)^{m_0} \ge (1-\alpha_m)^m = 1 - \alpha$. The inequality $\alpha_m > \alpha/m$ is equivalent to $(1 - \alpha/m)^m > 1 - \alpha$ (Bernoulli's inequality, $m \ge 2$, $\alpha \in (0,1)$). □

Theorem 22.8 (The Holm Procedure)**.** *If the p-values are valid, the Holm procedure controls the FWER in the strong sense under arbitrary dependence. Moreover, it rejects every hypothesis that Bonferroni rejects.*

Proof. Let $m_0 \ge 1$ (otherwise $V = 0$). Define the event $A = \{\min_{i \in I_0} p_i > \alpha/m_0\}$; by Boole's inequality $\mathbb{P}(A^c) \le m_0 \cdot \alpha/m_0 = \alpha$. We show that on A no true H_0 is rejected. Let the first true null hypothesis in the ordered list be in position j. Since at most $m - m_0$ hypotheses (all false) precede it, $j \le m - m_0 + 1$, i.e. $m - j + 1 \ge m_0$. Holm's threshold at step j is $\frac{\alpha}{m-j+1} \le \frac{\alpha}{m_0} < p_{(j)}$ (on A). Hence the procedure stops at step j at the latest, and no true H_0 is rejected: $\mathbb{P}(V \ge 1) \le \mathbb{P}(A^c) \le \alpha$.

For the second claim: if Bonferroni rejects $H_{(k)}$, then $p_{(1)} \le \cdots \le p_{(k)} \le \alpha/m \le \alpha/(m - j + 1)$ for every $j \le k$. So the first k steps of Holm also reject. □

Proposition 22.9 (Relation between FDR and FWER)**.** *For every procedure,* FDR $\le$ FWER. *If all null hypotheses are true ($m_0 = m$),* FDR = FWER.

Proof. Since $V \le R$, $\frac{V}{\max(R,1)} \le \mathbb{I}\{V \ge 1\}$; take expectations. If $m_0 = m$, then $V = R$ and $\frac{V}{\max(R,1)} = \mathbb{I}\{V \ge 1\}$. □

Hence FDR control is a weaker guarantee than FWER control; in return it provides much more power. When all nulls are true, the two criteria coincide, and a procedure that controls the FDR controls the FWER in the weak sense.

Lemma 22.10 (Structure of BH)**.** *Fix i and let $R^{(i)}$ denote the number of rejections of the BH procedure when p_i is replaced by 0 ($R^{(i)}$ depends only on $p_{-i} = (p_j)_{j \ne i}$). Then*

$$H_i \text{ is rejected} \iff p_i \le \frac{\alpha R^{(i)}}{m}, \qquad \text{and in that case } R = R^{(i)}.$$

Proof. With $N(t) = \#\{j : p_j \le t\}$ one can write $R = \max\{k : N(\alpha k/m) \ge k\}$ (from the definition of the ordered p-values). Decreasing a p-value increases N pointwise; hence $R \le R^{(i)}$.

($\Leftarrow$) Let $p_i \le \alpha R^{(i)}/m$. With $p_i = 0$ there are at least $R^{(i)}$ p-values not exceeding $\alpha R^{(i)}/m$; when 0 is replaced by p_i, the count does not change, since p_i does not exceed this threshold either. So $N(\alpha R^{(i)}/m) \ge R^{(i)}$, hence $R \ge R^{(i)}$ and $R = R^{(i)}$. Since BH rejects every hypothesis whose p-value does not exceed $\alpha R/m$, H_i is rejected.

($\Rightarrow$) If H_i is rejected, $p_i \le \alpha R/m \le \alpha R^{(i)}/m$; the first part gives $R = R^{(i)}$. □

Theorem 22.11 (Benjamini–Hochberg [4])**.** *If the p-values of the true null hypotheses are Uniform$(0,1)$ and each is independent of all the other p-values, then for the BH procedure*

$$\mathrm{FDR} = \frac{m_0}{m}\,\alpha = \pi_0 \alpha \le \alpha.$$

If the p-values are only valid (super-uniform), $\mathrm{FDR} \le \pi_0 \alpha$.

Proof. $\frac{V}{\max(R,1)} = \sum_{i \in I_0} \frac{\mathbb{I}\{H_i \text{ rejected}\}}{\max(R,1)}$. By Lemma 22.10, since $R = R^{(i)} \ge 1$ when H_i is rejected,

$$\frac{\mathbb{I}\{H_i \text{ rejected}\}}{\max(R,1)} = \frac{\mathbb{I}\{p_i \le \alpha R^{(i)}/m\}}{R^{(i)}}.$$

$R^{(i)}$ depends only on p_{-i}, and p_i is uniformly distributed independently of p_{-i}. Conditioning on p_{-i}:

$$\mathbb{E}\Big[\frac{\mathbb{I}\{p_i \le \alpha R^{(i)}/m\}}{R^{(i)}} \,\Big|\, p_{-i}\Big] = \frac{\mathbb{P}(p_i \le \alpha R^{(i)}/m \mid p_{-i})}{R^{(i)}} = \frac{\alpha R^{(i)}/m}{R^{(i)}} = \frac{\alpha}{m}.$$

($R^{(i)} \ge 1$ always, because with $p_i = 0$ at least H_i is rejected.) Taking expectations and summing over $i \in I_0$ gives $\mathrm{FDR} = m_0 \alpha/m$. For super-uniform p-values the middle equality becomes $\le$. □

Theorem 22.12 (Arbitrary Dependence: Benjamini–Yekutieli [5])**.** *If the p-values are valid and the dependence is arbitrary, then applying the BH procedure with α/c_m instead of α gives* $\mathrm{FDR} \le \pi_0 \alpha$*, where $c_m = \sum_{k=1}^{m} \frac{1}{k} \approx \ln m + 0.577$. Moreover, if the test statistics have positive regression dependence (PRDS), for example one-sided tests with positively correlated multivariate normal statistics, uncorrected BH already gives* $\mathrm{FDR} \le \pi_0 \alpha$.

22.3 Worked Examples

Example 22.13 (Ten p-Values)**.** $\alpha = 0.05$ and the ordered p-values: 0.001; 0.008; 0.039; 0.041; 0.042; 0.060; 0.074; 0.205; 0.212; 0.216.

- Uncorrected: the first five hypotheses, with $p \le 0.05$, are rejected.

- Bonferroni: the threshold is 0.005; only $H_{(1)}$.
- Holm: $0.001 \le 0.05/10$, so $H_{(1)}$ is rejected; since $0.008 > 0.05/9 = 0.0056$, we stop. Again only $H_{(1)}$.
- BH: the thresholds are $0.005k$. $p_{(2)} = 0.008 \le 0.010$, but for $k \ge 3$, $p_{(k)} > 0.005k$ ($0.039 > 0.015$, $0.041 > 0.020$, ..., $0.216 > 0.050$). $k^* = 2$; $H_{(1)}$ and $H_{(2)}$ are rejected.

Although the three p-values around 0.04 look "significant" on their own, none of them is rejected after the multiple testing correction.

Example 22.14 (When to Use Which?)**.** For a small number of hypotheses where every false discovery has serious consequences, such as the primary and secondary endpoints of a drug trial, FWER control is appropriate. Holm is always better than Bonferroni (Theorem 22.8) and works under the same assumption. In exploratory screening (genomics, imaging), by contrast, the rejections form a list of *candidates* to be validated in later experiments; here it is enough that most of the list is correct, and FDR control is appropriate. For $m = 1000$ the Bonferroni threshold is 5×10^{-5}, i.e. $z \approx 3.89$. Most real effects of size 3σ cannot pass this threshold.

22.4 Computational Applications

22.4.1 FWER, FDR and Power

Out of $m = 1000$ one-sided z-tests, $m_1 = 100$ have a real effect $\mu = 3$; the tests are independent. We compare four procedures with $M = 2000$ replications.

code/ch22/c22_1_bonferroni_holm_bh.py

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
m, m1, eff, alpha, M = 1000, 100, 3.0, 0.05, 2000

def bonferroni(p): return p <= alpha / len(p)
def holm(p):
    m = len(p); s = np.argsort(p); thr = alpha / (m - np.arange(m))
    ok = p[s] <= thr
    k = m if ok.all() else np.argmin(ok)            # reject up to the first failure
    rej = np.zeros(m, bool); rej[s[:k]] = True; return rej
def bh(p):
    m = len(p); s = np.argsort(p); ok = p[s] <= alpha * np.arange(1, m + 1) / m
    k = (np.nonzero(ok)[0].max() + 1) if ok.any() else 0
    rej = np.zeros(m, bool); rej[s[:k]] = True; return rej
def uncorrected(p): return p <= alpha

methods = {"Uncorrected": uncorrected, "Bonferroni": bonferroni, "Holm": holm, "BH": bh}
true_eff = np.zeros(m, bool); true_eff[:m1] = True             # the first m1 hypotheses are false (effect present)
res = {k: [] for k in methods}
for _ in range(M):
    z = rng.standard_normal(m) + eff * true_eff
    p = stats.norm.sf(z)                                       # one-sided p-values
    for name, f in methods.items():
        rej = f(p); V = (rej & ~true_eff).sum(); R = rej.sum()
        res[name].append((V >= 1, V / max(R, 1), (rej & true_eff).sum() / m1, R))
print(f"m = {m}, number of false H0 m1 = {m1}, effect = {eff}, alpha = {alpha}, replications = {M}")
print("Method       |  FWER  |  FDR   | power  | mean rej.")
```

```
for name, s in res.items():
    s = np.array(s, float)
    print(f"{name:<12} | {s[:,0].mean():.3f}  | {s[:,1].mean():.4f} | {s[:,2].mean():.3f}  | {s[:,3].mean():7.1f}")
print(f"theoretical FDR for BH = pi0 * alpha = {(m - m1) / m * alpha:.4f}")
```

Output

```
m = 1000, number of false H0 m1 = 100, effect = 3.0, alpha = 0.05, replications = 2000
Method       |  FWER  |  FDR   | power  | mean rej.
Uncorrected  | 1.000  | 0.3296 | 0.912  |   136.3
Bonferroni   | 0.046  | 0.0024 | 0.185  |    18.6
Holm         | 0.046  | 0.0024 | 0.187  |    18.7
BH           | 0.943  | 0.0456 | 0.607  |    63.7
theoretical FDR for BH = pi0 * alpha = 0.0450
```

For the uncorrected test the FWER is 1, and about a third of the rejections are false discoveries. Bonferroni and Holm keep the FWER below 0.05 but find only 19% of the real effects. Since $m_0 = 900$ is large, Holm's advantage over Bonferroni is negligible here. The FDR of BH agrees within Monte Carlo error with the theoretical value $\pi_0\alpha = 0.045$ (Theorem 22.11), and its power is more than three times larger. The price is that it makes at least one false discovery in almost every replication (FWER ≈ 0.94): BH does not guarantee "make no errors" but "keep the proportion of errors small".

22.4.2 Dependence and a Visual Interpretation of BH

Suppose all null hypotheses are true and the test statistics are equicorrelated (ρ) normal; two-sided p-values are used.

code/ch22/c22_2_dependence_and_bh.py

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
m, alpha, M = 100, 0.05, 4000

def bh(p, alpha=alpha):
    m = len(p); s = np.argsort(p); ok = p[s] <= alpha * np.arange(1, m + 1) / m
    k = (np.nonzero(ok)[0].max() + 1) if ok.any() else 0
    rej = np.zeros(m, bool); rej[s[:k]] = True; return rej

print("All H0 true (m = 100); equicorrelated normal test statistics")
print(" rho | Bonferroni FWER | uncorrected FWER | BH FDR (= FWER)")
for rho in (0.0, 0.3, 0.6, 0.9):
    common = rng.standard_normal((M, 1))
    z = np.sqrt(rho) * common + np.sqrt(1 - rho) * rng.standard_normal((M, m))
    p = 2 * stats.norm.sf(np.abs(z))
    bon = (p <= alpha / m).any(axis=1).mean()
    unc = (p <= alpha).any(axis=1).mean()
    bhf = np.mean([bh(r).any() for r in p])
    print(f" {rho:.1f} |      {bon:.4f}      |      {unc:.4f}      |     {bhf:.4f}")

# Visual interpretation of BH: orderej p-values and the line alpha*k/m
has_eff = np.zeros(m, bool); has_eff[:15] = True
z = rng.standard_normal(m) + 3.0 * has_eff
p = stats.norm.sf(z); s = np.sort(p); k = np.arange(1, m + 1)
rej = bh(p); R = rej.sum()
print(f"Example data: BH rejected {R} hypotheses; real effects: {(rej & has_eff).sum()}, "
      f"false discoveries: {(rej & ~has_eff).sum()}. Bonferroni rejections: {(p <= alpha / m).sum()}.")
```

```
fig, ax = plt.subplots(figsize=(6.4, 3.2))
ax.plot(k[:40], s[:40], "o", ms=3, label="orderej p-values $p_{(k)}$")
ax.plot(k[:40], alpha * k[:40] / m, "-", label=r"BH line $\alpha k/m$")
ax.axhline(alpha / m, ls=":", color="gray", label=r"Bonferroni $\alpha/m$")
ax.axvline(R + 0.5, ls="--", color="C3", lw=0.8)
ax.set_xlabel("$k$"); ax.set_ylabel("p-value"); ax.set_ylim(-0.002, 0.03); ax.legend(fontsize
    =8)
fig.tight_layout(); fig.savefig("bh_dogrusu.pdf")
```

Output

```
All H0 true (m = 100); equicorrelated normal test statistics
 rho | Bonferroni FWER | uncorrected FWER | BH FDR (= FWER)
 0.0 |      0.0515     |      0.9932      |     0.0532
 0.3 |      0.0420     |      0.9390      |     0.0457
 0.6 |      0.0220     |      0.6412      |     0.0278
 0.9 |      0.0065     |      0.2195      |     0.0195
Example data: BH rejected 11 hypotheses; real effects: 11, false discoveries: 0. Bonferroni
    rejections: 8.
```

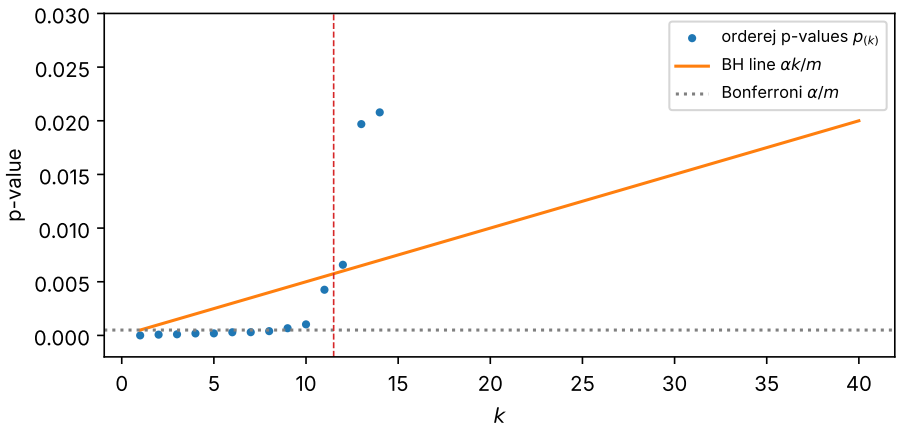


Figure 22.1: The BH procedure: find the *last* k at which the ordered p-values lie below the line $\alpha k/m$, and reject all hypotheses up to it (dashed vertical line). The Bonferroni threshold is the horizontal dotted line.

As the dependence increases, Bonferroni and BH become conservative: at $\rho = 0.9$ the tests carry almost the same information, the "effective number of tests" is much smaller than m, and Boole's inequality becomes crude. The FWER of the uncorrected test falls for the same reason. Since all nulls are true, the FDR of BH equals its FWER (Proposition 22.9), and under independence it is very close to α. In this simulation the FDR of BH stayed below α with two-sided p-values as well. For one-sided tests the same correlation structure satisfies the PRDS condition, and the guarantee is exact by Theorem 22.12.

22.5 Comments and Notes

Remark 22.15 (Estimating π_0). Theorem 22.11 shows that BH in fact controls at level $\pi_0\alpha$; if π_0 is small the procedure is more conservative than necessary. *Adaptive* procedures first estimate π_0; for example, based on the observation that large p-values come almost entirely from true nulls, $\hat{\pi}_0 = \#\{p_i > \lambda\}/(m(1-\lambda))$ is used. They then apply BH with $\alpha/\hat{\pi}_0$.

Remark 22.16 (Bayesian Interpretation). In the two-group mixture model (p_i comes from Uniform$(0,1)$ with probability π_0 and from an alternative distribution F_1 with probability $1-\pi_0$), $\mathbb{P}(H_i \text{ true} \mid p_i \le t) = \pi_0 t / F(t)$, where $F = \pi_0 t + (1-\pi_0)F_1$. BH can be interpreted

as estimating $F(t)$ by the empirical CDF and choosing the largest threshold that keeps this *tail-area false discovery rate* below α. This is the application of the posterior-probability reasoning of Chapter 19 to many parallel problems (empirical Bayes).

Remark 22.17 (Selective Inference). A more insidious form of the multiple testing problem is choosing which hypothesis to test after looking at the data: reporting the subgroup that shows the largest effect, or giving ordinary confidence intervals for coefficients selected after model selection. After selection, ordinary p-values and intervals are not valid. Simple safe solutions are data splitting and pre-registration. Theoretical solutions perform inference conditional on the selection event.

22.6 Chapter Summary

- When many tests are performed simultaneously, an error measure must be chosen: the family-wise error rate (FWER) or the false discovery rate (FDR) (Definition 22.3). Always FDR $\leq$ FWER; if all null hypotheses are true, the two are equal (Proposition 22.9).
- The Bonferroni correction controls the FWER under every dependence structure (Theorem 22.6). The Šidák correction is slightly more powerful under independence (Theorem 22.7). The Holm procedure (Theorem 22.8) uniformly dominates Bonferroni.
- The Benjamini–Hochberg procedure keeps the FDR at level $\pi_0\alpha$ for independent p-values (Theorem 22.11). Under arbitrary dependence it is applied with α/c_m (Theorem 22.12), where $c_m \approx \ln m$.
- Adjusted p-values (Definition 22.5) make it possible to report the result for each hypothesis rather than for a single threshold.
- With thousands of tests, FWER control reduces power drastically. FDR control, by accepting a small number of false discoveries, finds a much larger fraction of the real effects.

Further reading. [4, 39, 5]; at textbook level, [47, Chapter 9] and [81, Section 10.7].

22.7 Exercises

Exercise 22.1. For the p-values of Example 22.13, compute the Bonferroni, Holm and BH adjusted p-values (Definition 22.5). Verify that the hypotheses each procedure rejects at $\alpha = 0.05$ are those with adjusted p-value ≤ 0.05. How many hypotheses does BH reject with $\alpha = 0.10$? → Solution

Exercise 22.2. Show that for m independent null hypotheses, all true, the FWER of the uncorrected test is $1-(1-\alpha)^m$. Compute its values for $\alpha = 0.05$ and $m = 1, 5, 20, 100$. Show that $m\alpha_m \to -\ln(1-\alpha)$ as $m \to \infty$ (α_m: the Šidák level), and compare this with Bonferroni. → Solution

Exercise 22.3. Give an example of a procedure that controls the FWER in the weak sense but not in the strong sense. (Hint: consider the procedure that first tests "is there at least one effect?" with Bonferroni and, in case of rejection, rejects all hypotheses with $p_i \leq \alpha$.) → Solution

Exercise 22.4. Let there be $m = 2$ independent tests with $m_0 = 1$. If the p-value of the false hypothesis is 0 with probability 1 (a very powerful test), verify by direct computation that the FDR of the BH procedure is $\alpha/2$. → Solution

Chapter 23

The EM Algorithm and Generalised Linear Models

When the maximum likelihood principle (Chapters 11–12) does not give a closed-form solution, the MLE is found numerically. This chapter deals with the two numerical MLE problems met most often in statistics and with the algorithms specific to them. The first is *missing data* problems: mixture models, censored observations, latent variables. In these, the likelihood is a sum or integral over unobserved variables. The EM algorithm of Dempster, Laird and Rubin [20] turns the observation "the problem would be easy if the missing data were known" into an iterative method that increases the likelihood at every step. The second is *generalised linear models* (GLMs) [60]: the generalisation of the linear model of Chapter 17 to binary, count and positive continuous responses. Here Fisher scoring becomes the IRLS algorithm, each step of which is a weighted LS problem. Both topics show the computational power of exponential family structure (Chapter 9).

23.1 Definitions

Definition 23.1 (Missing Data Structure and the Q Function)**.** Let $\mathbf{x}$ be the observed data and $\mathbf{z}$ the unobserved (missing or latent) data; let the density of the *complete data* $(\mathbf{x},\mathbf{z})$ be $f_c(\mathbf{x},\mathbf{z};\theta)$, and that of the observed data $f(\mathbf{x};\theta)=\int f_c(\mathbf{x},\mathbf{z};\theta)\,d\mathbf{z}$. The observed-data and complete-data log-likelihoods are $\ell(\theta)=\ln f(\mathbf{x};\theta)$ and $\ell_c(\theta)=\ln f_c(\mathbf{x},\mathbf{z};\theta)$. The conditional density of the missing data is $k(\mathbf{z}\mid\mathbf{x};\theta)=f_c(\mathbf{x},\mathbf{z};\theta)/f(\mathbf{x};\theta)$.

$$Q(\theta\mid\theta')=\mathbb{E}_{\theta'}\big[\ell_c(\theta;\mathbf{x},\mathbf{Z})\,\big|\,\mathbf{x}\big],\qquad H(\theta\mid\theta')=\mathbb{E}_{\theta'}\big[\ln k(\mathbf{Z}\mid\mathbf{x};\theta)\,\big|\,\mathbf{x}\big].$$

Definition 23.2 (The EM Algorithm)**.** Starting from an initial $\theta^{(0)}$, for $t=0,1,2,\dots$:

- *E-step:* compute $Q(\theta\mid\theta^{(t)})$ (find the conditional expectations of the sufficient statistics of the missing data).
- *M-step:* $\theta^{(t+1)}=\arg\max_{\theta} Q(\theta\mid\theta^{(t)})$.

If instead of full maximisation the M-step only ensures $Q(\theta^{(t+1)}\mid\theta^{(t)})\ge Q(\theta^{(t)}\mid\theta^{(t)})$, the algorithm is called *generalised EM* (GEM).

Definition 23.3 (Exponential Dispersion Family)**.** Densities (or probability mass functions) of the form

$$f(y;\theta,\phi)=\exp\left\{\frac{y\theta-b(\theta)}{a(\phi)}+c(y,\phi)\right\}$$

are called an *exponential dispersion family*; θ is the natural (canonical) parameter and ϕ the dispersion parameter. If ϕ is known, this is a one-parameter exponential family with sufficient statistic $T(y)=y$.

Definition 23.4 (Generalised Linear Model)**.** Let $Y_1, \dots, Y_n$ be independent, each from an exponential dispersion family; $\mu_i = \mathbb{E}Y_i$. A GLM has three components:

1. *Random component:* the distribution of Y_i in the exponential dispersion family;
2. *Linear predictor:* $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$;
3. *Link function:* $g(\mu_i) = \eta_i$ with g monotone and differentiable.

If $g = (b')^{-1}$, i.e. $\theta_i = \eta_i$, then g is called the *canonical link*. $V(\mu) = b''(\theta(\mu))$ is called the *variance function*.

Distribution	$b(\theta)$	$V(\mu)$	$a(\phi)$	Canonical link
$\mathcal{N}(\mu, \sigma^2)$	$\theta^2/2$	1	σ^2	identity
Bernoulli(μ)	$\ln(1+e^\theta)$	$\mu(1-\mu)$	1	logit $\ln \frac{\mu}{1-\mu}$
Poisson(μ)	e^θ	μ	1	$\ln \mu$
Gamma (shape ν, mean μ)	$-\ln(-\theta)$	μ^2	$1/\nu$	$-1/\mu$

Definition 23.5 (Deviance)**.** Let ℓ_{sat} be the log-likelihood of the (*saturated*) model that gives each observation its own mean. $D = 2a(\phi)\big[\ell_{\text{sat}} - \ell(\hat{\boldsymbol{\beta}})\big]$ is called the *deviance*, and $D/a(\phi)$ the *scaled deviance*. In the normal model $D = \text{SSE}$; the deviance is the GLM counterpart of the residual sum of squares. In the Bernoulli model $D = 2\sum_i \big[y_i \ln \frac{y_i}{\hat{\mu}_i} + (1-y_i)\ln \frac{1-y_i}{1-\hat{\mu}_i}\big]$ ($0 \ln 0 = 0$).

23.2 Main Theorems and Proofs

Theorem 23.6 (Monotonicity of EM [20])**.** *For the EM (and GEM) sequence, $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$. Equality holds only if $Q(\theta^{(t+1)} \mid \theta^{(t)}) = Q(\theta^{(t)} \mid \theta^{(t)})$ and $k(\cdot \mid \mathbf{x}; \theta^{(t+1)}) = k(\cdot \mid \mathbf{x}; \theta^{(t)})$ a.e.*

Proof. For every $\mathbf{z}$, $\ell(\theta) = \ell_c(\theta; \mathbf{x}, \mathbf{z}) - \ln k(\mathbf{z} \mid \mathbf{x}; \theta)$. The left-hand side does not depend on $\mathbf{z}$; taking the expectation of both sides under $k(\cdot \mid \mathbf{x}; \theta')$,

$$\ell(\theta) = Q(\theta \mid \theta') - H(\theta \mid \theta') \qquad \text{for all } \theta, \theta'.$$

Applying the information inequality (Lemma 12.4) to two conditional densities for the same $\mathbf{x}$,

$$H(\theta \mid \theta') - H(\theta' \mid \theta') = \mathbb{E}_{\theta'}\Big[\ln \frac{k(\mathbf{Z} \mid \mathbf{x}; \theta)}{k(\mathbf{Z} \mid \mathbf{x}; \theta')} \,\Big|\, \mathbf{x}\Big] \leq \ln \mathbb{E}_{\theta'}\Big[\frac{k(\mathbf{Z} \mid \mathbf{x}; \theta)}{k(\mathbf{Z} \mid \mathbf{x}; \theta')} \,\Big|\, \mathbf{x}\Big] \leq 0,$$

i.e. $H(\cdot \mid \theta')$ attains its maximum at θ' (Jensen; with equality only if the two conditional densities are equal a.e.). With $\theta' = \theta^{(t)}$ and $\theta = \theta^{(t+1)}$,

$$\ell(\theta^{(t+1)}) - \ell(\theta^{(t)}) = \underbrace{\big[Q(\theta^{(t+1)} \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)})\big]}_{\geq 0 \text{ (M-step)}} - \underbrace{\big[H(\theta^{(t+1)} \mid \theta^{(t)}) - H(\theta^{(t)} \mid \theta^{(t)})\big]}_{\leq 0} \geq 0.$$

□

Proposition 23.7 (Stationary Points)**.** *Let Θ be open, let ℓ and $Q(\cdot \mid \theta')$ be differentiable, and let differentiation under the integral sign be permitted. For every θ',*

$$\nabla \ell(\theta') = \nabla_\theta Q(\theta \mid \theta')\big|_{\theta = \theta'}.$$

Hence a fixed point θ^* of EM (θ^* an interior maximum of $Q(\cdot \mid \theta^*)$) is a stationary point of ℓ.

Proof. $\ell(\theta) = Q(\theta \mid \theta') - H(\theta \mid \theta')$, and since $H(\cdot \mid \theta')$ attains its maximum at the interior point θ', $\nabla_\theta H(\theta \mid \theta')|_{\theta=\theta'} = 0$. Differentiate. At a fixed point $\nabla_\theta Q(\theta^* \mid \theta^*) = 0$, hence $\nabla \ell(\theta^*) = 0$. □

Remark 23.8 (Convergence and Rate). Monotonicity guarantees convergence of the sequence $\ell(\theta^{(t)})$ if ℓ is bounded above. It does not, however, guarantee that $\theta^{(t)}$ converges to the global maximum: under additional continuity conditions the limit points are stationary points [89], which may be local maxima or saddle points. Near a stationary θ^*, EM converges linearly: $\theta^{(t+1)} - \theta^* \approx J(\theta^{(t)} - \theta^*)$, where (in one dimension)

$$J = \frac{I_{\text{miss}}}{I_{\text{com}}} = 1 - \frac{I_{\text{obs}}}{I_{\text{com}}}, \qquad I_{\text{com}} = -\partial_\theta^2 Q(\theta \mid \theta^*)\big|_{\theta^*}, \quad I_{\text{obs}} = -\ell''(\theta^*).$$

J is the *fraction of missing information* [20]: if the missing data carry most of the information, EM is slow.

Proposition 23.9 (Moments of the Exponential Dispersion Family). *For the family of Definition 23.3,* $\mathbb{E}Y = b'(\theta)$ *and* $\operatorname{Var} Y = a(\phi)\, b''(\theta)$.

Proof. For $\ell(\theta) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)$ the score is $\ell' = \frac{y - b'(\theta)}{a(\phi)}$ and $\ell'' = -\frac{b''(\theta)}{a(\phi)}$. By Lemma 8.6, $\mathbb{E}\ell' = 0$, i.e. $\mathbb{E}Y = b'(\theta)$. The information identity $\mathbb{E}[\ell'^2] = -\mathbb{E}\ell''$ gives $\frac{\operatorname{Var} Y}{a(\phi)^2} = \frac{b''(\theta)}{a(\phi)}$. □

Theorem 23.10 (Score and Fisher Information in a GLM). *In a GLM with ϕ known,*

$$\mathbf{U}(\boldsymbol{\beta}) = \nabla_{\boldsymbol{\beta}} \ell = \frac{1}{a(\phi)} \sum_{i=1}^{n} \frac{y_i - \mu_i}{V(\mu_i)\, g'(\mu_i)}\, \mathbf{x}_i,$$

$$I(\boldsymbol{\beta}) = \frac{1}{a(\phi)} X^\top W X, \qquad W = \operatorname{diag}\Big(\frac{1}{V(\mu_i) g'(\mu_i)^2}\Big).$$

For the canonical link, $\mathbf{U} = \frac{1}{a(\phi)} X^\top (\mathbf{y} - \boldsymbol{\mu})$, *the observed information equals the expected information ($-\nabla^2 \ell = I(\boldsymbol{\beta})$, independent of the data), and ℓ is a concave function of $\boldsymbol{\beta}$; if X has full column rank it is strictly concave.*

Proof. Chain rule: $\frac{\partial \ell_i}{\partial \boldsymbol{\beta}} = \frac{\partial \ell_i}{\partial \theta_i} \frac{d\theta_i}{d\mu_i} \frac{d\mu_i}{d\eta_i} \frac{\partial \eta_i}{\partial \boldsymbol{\beta}}$. Here $\frac{\partial \ell_i}{\partial \theta_i} = \frac{y_i - \mu_i}{a(\phi)}$, $\frac{d\mu_i}{d\theta_i} = b''(\theta_i) = V(\mu_i)$, $\frac{d\mu_i}{d\eta_i} = \frac{1}{g'(\mu_i)}$ and $\frac{\partial \eta_i}{\partial \boldsymbol{\beta}} = \mathbf{x}_i$. The product gives the score formula. For independent terms the Fisher information is $I = \sum_i \operatorname{Cov}(\nabla \ell_i)$, and using $\operatorname{Var} Y_i = a(\phi) V(\mu_i)$ from Proposition 23.9,

$$I(\boldsymbol{\beta}) = \sum_i \frac{a(\phi) V(\mu_i)}{a(\phi)^2 V(\mu_i)^2 g'(\mu_i)^2} \mathbf{x}_i \mathbf{x}_i^\top = \frac{1}{a(\phi)} X^\top W X.$$

For the canonical link $\theta_i = \eta_i$, so $g'(\mu) = d\theta/d\mu = 1/V(\mu)$ and the score becomes $\frac{1}{a(\phi)} \sum (y_i - \mu_i)\mathbf{x}_i$. Differentiating once more, $-\nabla^2 \ell = \frac{1}{a(\phi)} \sum_i b''(\eta_i) \mathbf{x}_i \mathbf{x}_i^\top$; this expression does not involve y, equals $I(\boldsymbol{\beta})$, and is positive semidefinite because $b'' = V > 0$. If X has full rank, then for every $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v}^\top X^\top W X \mathbf{v} = \sum_i W_{ii} (\mathbf{x}_i^\top \mathbf{v})^2 > 0$. □

Theorem 23.11 (Fisher Scoring = Iteratively Reweighted Least Squares). *The Fisher scoring step $\boldsymbol{\beta}^{new} = \boldsymbol{\beta} + I(\boldsymbol{\beta})^{-1} \mathbf{U}(\boldsymbol{\beta})$ is the same as*

$$\boldsymbol{\beta}^{new} = (X^\top W X)^{-1} X^\top W \mathbf{z}, \qquad z_i = \eta_i + (y_i - \mu_i)\, g'(\mu_i),$$

where W, η, μ are computed at the current $\boldsymbol{\beta}$. That is, each step is a weighted LS regression of the working response $\mathbf{z}$ *on X with weights W (IRLS). For the canonical link, Fisher scoring coincides with the Newton–Raphson method.*

Proof. Since $\frac{y_i-\mu_i}{V(\mu_i)g'(\mu_i)} = W_{ii}\,(y_i-\mu_i)g'(\mu_i)$, $a(\phi)\mathbf{U} = X^\top W\mathbf{r}$ with $r_i = (y_i-\mu_i)g'(\mu_i)$. Hence

$$X^\top W X\,\boldsymbol{\beta}^{\text{new}} = X^\top W X\boldsymbol{\beta} + X^\top W\mathbf{r} = X^\top W(X\boldsymbol{\beta}+\mathbf{r}) = X^\top W\mathbf{z}$$

($a(\phi)$ cancels). For the canonical link, the observed and expected information are equal (Theorem 23.10), so the Newton step $\boldsymbol{\beta} - [\nabla^2\ell]^{-1}\mathbf{U}$ and the Fisher step are the same. □

Corollary 23.12 (Asymptotic Inference in GLMs)**.** *Under regularity conditions (Chapter 12), $\hat{\boldsymbol{\beta}} \approx \mathcal{N}_p(\boldsymbol{\beta},\ a(\phi)(X^\top\hat{W}X)^{-1})$; Wald intervals are constructed as $\hat\beta_j \pm z_{\alpha/2}\,\mathrm{se}(\hat\beta_j)$. For two nested models, the difference of scaled deviances $(D_0 - D_1)/a(\phi)$ is $2\ln$ of the likelihood ratio and, by Wilks' theorem (Theorem 15.5), is approximately $\chi^2_{p_1-p_0}$.*

Proof. $\hat{\boldsymbol{\beta}}$ is an MLE and $I(\boldsymbol{\beta})$ is given in Theorem 23.10. For independent but not identically distributed observations, asymptotic normality is obtained by checking the Lindeberg condition, as in the LS proof of Chapter 17. By definition the deviance difference is $2[\ell(\hat{\boldsymbol{\beta}}_1) - \ell(\hat{\boldsymbol{\beta}}_0)]a(\phi)$, because the ℓ_{sat} terms cancel. □

23.3 Worked Examples

Example 23.13 (EM for a Two-Component Normal Mixture)**.** $X_i \overset{\text{i.i.d.}}{\sim} (1-w)\mathcal{N}(\mu_1,\sigma_1^2) + w\mathcal{N}(\mu_2,\sigma_2^2)$. The latent variable $Z_i \in \{0,1\}$ indicates which component X_i comes from. The complete-data log-likelihood

$$\ell_c = \sum_i \Big[Z_i\big(\ln w + \ln\phi(x_i;\mu_2,\sigma_2)\big) + (1-Z_i)\big(\ln(1-w) + \ln\phi(x_i;\mu_1,\sigma_1)\big)\Big]$$

is linear in the Z_i. The E-step only computes the *responsibilities* $\gamma_i = \mathbb{P}(Z_i = 1 \mid x_i;\theta^{(t)})$:

$$\gamma_i = \frac{w\,\phi(x_i;\mu_2,\sigma_2)}{w\,\phi(x_i;\mu_2,\sigma_2) + (1-w)\,\phi(x_i;\mu_1,\sigma_1)}.$$

In the M-step, Q is the complete-data log-likelihood with γ_i in place of Z_i; it gives weighted normal MLEs: $w = \bar\gamma$, $\mu_2 = \sum\gamma_i x_i/\sum\gamma_i$, $\sigma_2^2 = \sum\gamma_i(x_i-\mu_2)^2/\sum\gamma_i$, and analogues with weights $1-\gamma_i$ for the first component. The mixture likelihood is unbounded: with $\mu_1 = x_1$ and $\sigma_1 \to 0$, $\ell\to\infty$. Hence the "MLE" must be understood as a sensible local maximum; in practice a lower bound is imposed on the variances or several starting points are tried.

Example 23.14 (Genetic Linkage Data [20])**.** The counts of 197 animals in four categories are $\mathbf{y} = (125, 18, 20, 34)$, with probabilities $(\frac12 + \frac{\theta}{4}, \frac{1-\theta}{4}, \frac{1-\theta}{4}, \frac{\theta}{4})$. The observed log-likelihood is $y_1\ln(2+\theta) + (y_2+y_3)\ln(1-\theta) + y_4\ln\theta$ (up to a constant), and the likelihood equation is a quadratic equation. From the EM point of view, the first cell is the sum of two latent cells with probabilities $\frac12$ and $\frac{\theta}{4}$: $y_1 = x_1 + x_2$. The complete data have a binomial structure with $(x_2 + y_4)$ successes and $(y_2 + y_3)$ failures.

- E-step: $x_2^{(t)} = y_1\frac{\theta^{(t)}/4}{1/2+\theta^{(t)}/4}$.
- M-step: $\theta^{(t+1)} = \frac{x_2^{(t)}+y_4}{x_2^{(t)}+y_2+y_3+y_4}$.

Example 23.15 (Poisson Regression)**.** $Y_i \sim \text{Poisson}(\mu_i)$, $\ln\mu_i = \mathbf{x}_i^\top\boldsymbol{\beta}$ (canonical link). By Theorem 23.10, the score equations are $X^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$. If the model contains an intercept, the first equation is $\sum_i y_i = \sum_i \hat\mu_i$: the total of the fitted counts reproduces the observed total exactly. In IRLS $g'(\mu) = 1/\mu$, so $W = \mathrm{diag}(\mu_i)$ and $z_i = \eta_i + (y_i-\mu_i)/\mu_i$. e^{β_j} is the factor by which a one-unit increase in x_j multiplies the expected count.

23.4 Computational Applications

23.4.1 EM: Linkage Data and a Normal Mixture

code/ch23/c23_1_em_mixture_and_linkage.py

```python
import numpy as np
from scipy import stats, optimize
import matplotlib.pyplot as plt

# (a) The Dempster--Laird--Rubin genetic linkage example: y = (125, 18, 20, 34),
# cell probabilities (1/2 + t/4, (1-t)/4, (1-t)/4, t/4). The first cell is split into (1/2) and (t/4).
y1, y2, y3, y4 = 125, 18, 20, 34
t, trace = 0.5, []
for _ in range(10):
    x2 = y1 * (t / 4) / (0.5 + t / 4)            # E-step: expected count falling in the t/4 part of the first cell
    t = (x2 + y4) / (x2 + y2 + y3 + y4)          # M-step: complete-data MLE of the binomial proportion
    trace.append(t)
ll = lambda t: y1*np.log(0.5 + t/4) + (y2 + y3)*np.log((1 - t)/4) + y4*np.log(t/4)
t_hat = optimize.brentq(lambda t: y1/(2 + t) - (y2 + y3)/(1 - t) + y4/t, 0.01, 0.99)
err = np.abs(np.array(trace) - t_hat)
print(f"Linkage data: EM iterations = {np.round(trace[:6], 6)}")
print(f"Direct MLE = {t_hat:.6f} | successive error ratios = {np.round(err[1:6] / err[:5], 4)}")

# (b) Two-component normal mixture
rng = np.random.default_rng(42)
n = 400
z = rng.random(n) < 0.35
x = np.where(z, rng.normal(4.0, 1.0, n), rng.normal(0.0, 1.5, n))
w, m1, m2, s1, s2 = 0.5, -1.0, 1.0, 1.0, 1.0            # deliberately poor start
def loglik(w, m1, m2, s1, s2):
    return np.log((1 - w) * stats.norm.pdf(x, m1, s1) + w * stats.norm.pdf(x, m2, s2)).sum()
hist = [loglik(w, m1, m2, s1, s2)]
for it in range(200):
    a = w * stats.norm.pdf(x, m2, s2); b = (1 - w) * stats.norm.pdf(x, m1, s1)
    g = a / (a + b)                                      # E-step: posterior membership probabilities
    w = g.mean()                                         # M-step: weighted MLEs
    m1, m2 = np.sum((1 - g) * x) / np.sum(1 - g), np.sum(g * x) / np.sum(g)
    s1 = np.sqrt(np.sum((1 - g) * (x - m1)**2) / np.sum(1 - g))
    s2 = np.sqrt(np.sum(g * (x - m2)**2) / np.sum(g))
    hist.append(loglik(w, m1, m2, s1, s2))
    if hist[-1] - hist[-2] < 1e-10: break
diffs = np.diff(hist)
print(f"Mixture: {it + 1} iterations | log-likelihood {hist[0]:.2f} -> {hist[-1]:.4f}")
print(f"Smallest increase = {diffs.min():.2e} (non-negative: monotonicity)")
print(f"Estimates: w = {w:.3f} (0.35), mu = ({m1:.3f}, {m2:.3f}) (0, 4), sigma = ({s1:.3f}, {s2:.3f}) (1.5, 1)")

fig, ax = plt.subplots(1, 2, figsize=(10, 3.2))
ax[0].plot(hist, ".-", ms=3); ax[0].set_xlabel("iteration"); ax[0].set_ylabel(r"$\ell(\theta^{(t)})$")
ax[0].set_title("EM: the log-likelihood increases at every step", fontsize=10)
grd = np.linspace(x.min() - 1, x.max() + 1, 400)
ax[1].hist(x, bins=40, density=True, alpha=0.4)
ax[1].plot(grd, (1 - w) * stats.norm.pdf(grd, m1, s1) + w * stats.norm.pdf(grd, m2, s2), "k", lw=2)
ax[1].set_title("Fitted mixture density", fontsize=10)
fig.tight_layout(); fig.savefig("em_karisim.pdf")
```

Output

```
Linkage data: EM iterations = [0.608247 0.624321 0.626489 0.626777 0.626816 0.626821]
Direct MLE = 0.626821 | successive error ratios = [0.1346 0.133  0.1328 0.1328 0.1328]
Mixture: 121 iterations | log-likelihood -1324.52 -> -859.4867
Smallest increase = 9.89e-11 (non-negative: monotonicity)
Estimates: w = 0.318 (0.35), mu = (0.083, 3.984) (0, 4), sigma = (1.473, 0.903) (1.5, 1)
```

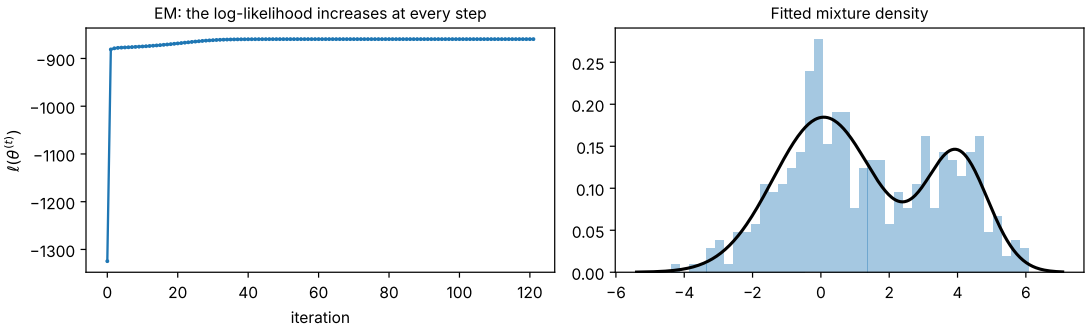


Figure 23.1: Left: EM increases the log-likelihood at every step (Theorem 23.6); a rapid increase after a poor start, followed by slow progress. Right: the fitted two-component mixture density.

For the linkage data, EM reaches the root of the likelihood equation, $\hat{\theta} = 0.626821$, within a few steps. The ratio of successive errors settles at the constant 0.1328: linear convergence. This ratio equals the fraction of missing information $1 - I_{\text{obs}}/I_{\text{com}}$ (Exercise 23.4). In the mixture example the log-likelihood decreases at no step; the smallest increase is positive. However, 121 steps are needed, because the two components partly overlap and the fraction of missing information is high. The estimates are close to the true values. The difference in w (0.318 vs 0.35) is of the usual size of sampling variability at $n = 400$.

23.4.2 Logistic Regression: IRLS, Wald Intervals and Complete Separation

code/ch23/c23_2_logistic_irls.py

```python
import numpy as np
from scipy import optimize, special

rng = np.random.default_rng(42)

def irls(X, y, tol=1e-10, maxit=50):
    beta = np.zeros(X.shape[1])
    for it in range(maxit):
        eta = X @ beta; mu = special.expit(eta); W = mu * (1 - mu)
        z = eta + (y - mu) / W                          # working response
        new = np.linalg.solve(X.T @ (W[:, None] * X), X.T @ (W * z))
        if np.max(np.abs(new - beta)) < tol:
            beta = new; break
        beta = new
    mu = special.expit(X @ beta); W = mu * (1 - mu)
    cov = np.linalg.inv(X.T @ (W[:, None] * X))
    return beta, cov, it + 1

n, beta0 = 200, np.array([-0.5, 1.0, -2.0])
X = np.column_stack([np.ones(n), rng.normal(size=n), rng.uniform(-1, 1, n)])
y = rng.random(n) < special.expit(X @ beta0)
b, V, it = irls(X, y)
```

```
neg_ll = lambda bb: -(y * (X @ bb) - np.logaddexp(0, X @ bb)).sum()
b_opt = optimize.minimize(neg_ll, np.zeros(3), method="BFGS", options={"gtol": 1e-8}).x
mu = special.expit(X @ b)
dev = 2 * np.sum(special.xlogy(y, y / mu) + special.xlogy(1 - y, (1 - y) / (1 - mu)))
print(f"IRLS converged in {it} iterations: beta_hat = {np.round(b, 4)}")
print(f"General-purpose optimiser (BFGS):  beta_hat = {np.round(b_opt, 4)}")
print(f"Standard errors sqrt(diag((X'WX)^-1)) = {np.round(np.sqrt(np.diag(V)), 4)}")
print(f"Deviance = {dev:.3f} | degrees of freedom n - p = {n - 3}")

M, covers = 2000, np.zeros(3)
for _ in range(M):
    Xs = np.column_stack([np.ones(n), rng.normal(size=n), rng.uniform(-1, 1, n)])
    ys = rng.random(n) < special.expit(Xs @ beta0)
    bs, Vs, _ = irls(Xs, ys)
    covers += np.abs(bs - beta0) <= 1.96 * np.sqrt(np.diag(Vs))
print(f"Coverage of 95% Wald intervals (M = {M}, n = {n}): {np.round(covers / M, 3)}")

# Complete separation: always 1 for x > 0, always 0 for x < 0 -> no MLE
xa = np.r_[rng.uniform(-2, -0.1, 20), rng.uniform(0.1, 2, 20)]
Xa = np.column_stack([np.ones(40), xa]); ya = (xa > 0).astype(float)
for k in (2, 4, 6, 8):
    bk, _, _ = irls(Xa, ya, maxit=k)
    ll = (ya * (Xa @ bk) - np.logaddexp(0, Xa @ bk)).sum()
    print(f"Complete separation, {k:2d} IRLS steps: slope = {bk[1]:9.2f} | log-likelihood = {
    ll:.2e}")
```

Output

```
IRLS converged in 6 iterations: beta_hat = [-0.4896  1.045  -1.7819]
General-purpose optimiser (BFGS):  beta_hat = [-0.4896  1.045  -1.7819]
Standard errors sqrt(diag((X'WX)^-1)) = [0.1736 0.2156 0.3283]
Deviance = 208.660 | degrees of freedom n - p = 197
Coverage of 95% Wald intervals (M = 2000, n = 200): [0.952 0.958 0.964]
Complete separation,  2 IRLS steps: slope =      2.74 | log-likelihood = -4.69e+00
Complete separation,  4 IRLS steps: slope =      5.79 | log-likelihood = -1.82e+00
Complete separation,  6 IRLS steps: slope =     12.80 | log-likelihood = -5.80e-01
Complete separation,  8 IRLS steps: slope =     26.20 | log-likelihood = -9.13e-02
```

IRLS reaches the same estimates as a general-purpose optimiser in six steps. Since for the canonical link it is Newton's method (Theorem 23.11), convergence is quadratic. The coverage of the Wald intervals obtained from $(X^\top \hat{W} X)^{-1}$ is close to the nominal 95% at $n = 200$. Although the deviance (208.7) seems close to the degrees of freedom (197), for Bernoulli data this comparison *cannot* be used as a goodness-of-fit test. With one trial per observation, the saturated model spends one parameter per observation, and the deviance does not converge to a χ^2 distribution. Deviance differences (Corollary 23.12), on the other hand, are valid.

The last lines show a situation in which the MLE does not exist. If all observations with $x > 0$ are 1 and those with $x < 0$ are 0 (*complete separation*), the slope goes to infinity and drives the log-likelihood towards zero; the supremum is not attained at any finite $\boldsymbol{\beta}$. In IRLS the slope roughly doubles at each step, and the algorithm eventually breaks down numerically. Albert and Anderson [2] showed that in logistic regression a finite MLE exists if and only if the data are not completely or quasi-completely separated. In practice penalised likelihood or Bayesian priors are used.

23.5 Comments and Notes

Remark 23.16 (EM versus Newton). The advantages of EM: every step increases the likelihood, parameter constraints (probabilities staying in $[0, 1]$, positivity of variances) are preserved automatically, and the M-step is often in closed form. Its disadvantages: convergence

is linear and can be very slow if the fraction of missing information is high; standard errors are not obtained directly. For these, Louis's [53] missing information principle $I_{\text{obs}} = I_{\text{com}} - I_{\text{miss}}$ can be used. Newton's method is quadratically fast in a neighbourhood of the solution but can decrease the likelihood from a distant start. In practice the two methods are often combined.

Remark 23.17 (EM, Bayes and Gibbs Sampling). In the E-step of EM the missing data are "filled in" by their conditional expectation. The Gibbs sampling of Chapter 19 uses the same structure in a Bayesian context: it *samples* the missing data from their conditional distribution and draws the parameter from its conditional posterior (data augmentation). The posterior-mode version of EM maximises $Q + \ln \pi(\theta)$ instead of Q in the M-step.

Remark 23.18 (Quasi-Likelihood and Overdispersion). The score of Theorem 23.10 depends only on μ_i and $V(\mu_i)$, not on the full distribution. This observation is the basis of the *quasi-likelihood* approach: only the mean and the variance function are specified. In count data the variance is often larger than the mean (overdispersion); taking $\operatorname{Var} Y_i = \phi \mu_i$, the estimates stay the same and the standard errors are multiplied by $\sqrt{\hat\phi}$, where $\hat\phi = \frac{1}{n-p} \sum_i \frac{(y_i - \hat\mu_i)^2}{\hat\mu_i}$. Alternatively, the sandwich covariance of Chapter 17 can be used.

23.6 Chapter Summary

- In missing data problems the EM algorithm builds the Q function in the E-step and maximises it in the M-step (Definition 23.2). The observed likelihood decreases at no step (Theorem 23.6).
- The limit points of EM are stationary points (Proposition 23.7); the global maximum is not guaranteed. Convergence is linear, with rate determined by the fraction of missing information.
- A GLM requires the response to come from an exponential dispersion family and the mean to be linked to a linear predictor by a link function (Definition 23.4). In this family $\mathbb{E}Y = b'(\theta)$ and $\operatorname{Var} Y = a(\phi) b''(\theta)$ (Proposition 23.9).
- The score and Fisher information can be written explicitly (Theorem 23.10). For the canonical link the log-likelihood is concave. Fisher scoring is iteratively reweighted LS (IRLS; Theorem 23.11).
- The deviance (Definition 23.5) plays the role of the likelihood ratio statistic in model comparisons (Corollary 23.12). In logistic regression the MLE does not exist under complete separation.

Further reading. For EM, [20], [58]; for GLMs, [57], [81, Chapter 13]; for asymptotic theory, [78, Chapter 5].

23.7 Exercises

Exercise 23.1. Let $T_1, \dots, T_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$ (rate λ), but with $n - d$ units right-censored at times c_j: only $T_j > c_j$ is known. Using $\mathbb{E}[T \mid T > c] = c + 1/\lambda$ (the memoryless property), write down the EM step and show that the fixed point is $\hat\lambda = d/S$, where S is the sum of all observed times (including the censored ones). Show that the derivative of the EM map at $\hat\lambda$ equals the censoring fraction $(n - d)/n$. → Solution

Exercise 23.2. Show that for a normal response and the identity link IRLS gives the LS solution in a single step. For a Bernoulli response and the logit link, write W and $\mathbf{z}$ explicitly. → Solution

Exercise 23.3. In the probit model ($\mu = \Phi(\eta)$, a non-canonical link), find the weights of the Fisher information as $W_{ii} = \frac{\phi(\eta_i)^2}{\Phi(\eta_i)(1-\Phi(\eta_i))}$. Show that the observed information depends on y and in general differs from the expected information. → Solution

Exercise 23.4. In Example 23.14, compute $I_{\text{obs}} = -\ell''(\hat{\theta})$ and $I_{\text{com}} = \mathbb{E}_{\hat{\theta}}[-\ell_c''(\hat{\theta}) \mid \mathbf{y}]$ at $\hat{\theta}$. Verify that the fraction of missing information equals the successive error ratio 0.1328 of the Computational Applications section. What is the standard error of $\hat{\theta}$? → Solution

Part VII

Appendices

Appendix A

Solutions to the Exercises

This appendix contains solutions to all the exercises in the book. The heading of each solution links to the exercise, and the "→ Solution" link at the end of each exercise leads here. Where an exercise asks for numerical values or a simulation, the values given were obtained by actual computation; simulation results may differ in the last digits depending on the random seed. The same libraries as in the chapters (`numpy`, `scipy`) are used; short code is given only where it is part of the solution.

The solutions are kept as short as possible, but each step names the result in the book on which it rests. When the claim of an exercise holds only under certain conditions (e.g. Exercise 13.3), the solution states those conditions explicitly.

A.1 Probability Spaces

Chapter 1.

Exercise 1.1. Let $\mathcal{F} = \mathcal{F}_1 \cap \mathcal{F}_2$. Since Ω is in both, $\Omega \in \mathcal{F}$. If $A \in \mathcal{F}$, then A belongs to both families, and since both are closed under complements, $A^c \in \mathcal{F}_1 \cap \mathcal{F}_2$. The same argument works for countable unions. (The same proof works for the intersection of arbitrarily many σ-algebras; the definition of the generated σ-algebra rests on this.) Counterexample for the union: $\Omega = \{1,2,3\}$, $\mathcal{F}_1 = \{\emptyset, \{1\}, \{2,3\}, \Omega\}$, $\mathcal{F}_2 = \{\emptyset, \{2\}, \{1,3\}, \Omega\}$. Then $\{1\}, \{2\} \in \mathcal{F}_1 \cup \mathcal{F}_2$, but $\{1\} \cup \{2\} = \{1,2\}$ is in neither; the union is not closed under unions.

Exercise 1.2. Applying the two-event formula to $A \cup B$ and C and using distributivity gives

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(AB) - \mathbb{P}(AC) - \mathbb{P}(BC) + \mathbb{P}(ABC)$$

($AB = A \cap B$). The general formula

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{i_1 < \cdots < i_k} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})$$

follows by induction: for $B = \bigcup_{i \le n} A_i$, $\mathbb{P}(B \cup A_{n+1}) = \mathbb{P}(B) + \mathbb{P}(A_{n+1}) - \mathbb{P}\big(\bigcup_{i \le n}(A_i \cap A_{n+1})\big)$; apply the induction hypothesis to the two unions on the right and collect terms. *Matching problem:* let $A_i = \{\text{letter } i \text{ is in the right envelope}\}$. The probability that k given letters are in the right place is $(n-k)!/n!$, and there are $\binom{n}{k}$ such choices; the inner sum equals $1/k!$. The probability that no letter is in the right place is

$$1 - \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k!} = \sum_{k=0}^{n} \frac{(-1)^k}{k!} \xrightarrow{n \to \infty} e^{-1} \approx 0.368.$$

Convergence is very fast: for $n = 6$ the difference is below 10^{-4}.

Exercise 1.3. By De Morgan and Boole's inequality, $\mathbb{P}(\bigcap A_i) = 1 - \mathbb{P}(\bigcup A_i^c) \geq 1 - \sum \mathbb{P}(A_i^c)$. *Multiple testing:* for m true null hypotheses let $A_i = \{\text{test } i \text{ does not reject falsely}\}$. If each test is carried out at level α/m, then $\mathbb{P}(A_i^c) \leq \alpha/m$, and the probability of no false rejection at all is $\geq 1 - \alpha$. This is the Bonferroni correction (see Exercise 13.4); it holds whatever the dependence between the tests, but it is conservative when the tests are positively dependent.

Exercise 1.4. $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)\mathbb{P}(B^c)$, so $A \perp\!\!\!\perp B^c$. Applying this result again to the pair (B^c, A) gives $B^c \perp\!\!\!\perp A^c$.

Exercise 1.5. With prevalence π the positive predictive value is $\mathrm{PPV}(\pi) = \frac{0.99\pi}{0.99\pi + 0.05(1-\pi)}$. Then $\mathrm{PPV} > \frac{1}{2} \iff 0.99\pi > 0.05(1-\pi) \iff \pi > \frac{0.05}{1.04} \approx 0.0481$. So unless the prevalence exceeds about 4.8%, most positive results are false positives. For the plot:

```
import numpy as np, matplotlib.pyplot as plt
pi = np.linspace(0.001, 0.5, 500)
ppv = 0.99*pi / (0.99*pi + 0.05*(1 - pi))
plt.semilogx(pi, ppv); plt.axhline(0.5, ls=":"); plt.axvline(0.05/1.04, ls=":")
plt.xlabel("prevalence"); plt.ylabel("P(D | +)"); plt.show()
```

A.2 Random Variables and Distribution Functions

Chapter 2.

Exercise 2.1. $g(x) = -2\ln x$ is strictly decreasing on $(0,1)$; its inverse is $x = e^{-y/2}$ with $|dx/dy| = \frac{1}{2}e^{-y/2}$. By Theorem 2.16, $f_Y(y) = 1 \cdot \frac{1}{2}e^{-y/2}$, $y > 0$: this is $\mathrm{Exp}(\frac{1}{2}) = \mathrm{Gamma}(1, \frac{1}{2}) = \chi^2_2$. *Fisher's method:* the continuous p-values of m independent tests are $\mathrm{Uniform}(0,1)$ under H_0 (Theorem 13.10); hence $-2\sum_{i=1}^m \ln p_i \sim \chi^2_{2m}$, and large values of this statistic are evidence that "at least one null hypothesis is false".

Exercise 2.2. $u = F(x) = 1 - e^{-x^k} \iff x = F^{-1}(u) = \big(-\ln(1-u)\big)^{1/k}$. Mean: $\mathbb{E}X = \int_0^\infty k x^k e^{-x^k} dx$; with $t = x^k$ this is $\int_0^\infty t^{1/k} e^{-t} dt = \Gamma(1 + 1/k)$. Simulation with 4×10^5 draws: 1.994 for $k = 0.5$ (theory 2), 0.9031 for $k = 1.5$ (0.9027), 0.8934 for $k = 3$ (0.8930).

Exercise 2.3. $\{\max(X,Y) \leq z\} = \{X \leq z\} \cap \{Y \leq z\}$, so by independence $F_{\max}(z) = F(z)G(z)$. $\{\min(X,Y) > z\} = \{X > z\} \cap \{Y > z\}$, hence $F_{\min}(z) = 1 - [1 - F(z)][1 - G(z)]$. By induction, $\mathbb{P}(X_{(n)} \leq x) = \prod_i \mathbb{P}(X_i \leq x) = F(x)^n$.

Exercise 2.4. $2e^{-2n(0.02)^2} \leq 0.05 \iff n \geq \frac{\ln 40}{2 \cdot 0.0004} \approx 4611.1$; the smallest n is 4612. In general, with $\varepsilon_n = \sqrt{\ln(2/\alpha)/(2n)}$, the band

$$L(x) = \max\{\hat{F}_n(x) - \varepsilon_n, 0\}, \qquad U(x) = \min\{\hat{F}_n(x) + \varepsilon_n, 1\}$$

satisfies $\mathbb{P}(L(x) \leq F(x) \leq U(x)\ \forall x) \geq 1 - \alpha$. This is a simultaneous, distribution-free confidence band valid for *all* x.

A.3 Expectation and Variance

Chapter 3.

Exercise 3.1. $X(\omega) = \int_0^\infty \mathbb{I}\{X(\omega) > t\} dt$. Since the integrand is non-negative, Fubini–Tonelli gives $\mathbb{E}X = \int_0^\infty \mathbb{E}\,\mathbb{I}\{X > t\}\, dt = \int_0^\infty \mathbb{P}(X > t)\, dt$. For the geometric distribution $\mathbb{P}(X > t) = (1-p)^{\lfloor t \rfloor}$ is a step function, and the integral is $\sum_{k=0}^\infty (1-p)^k = 1/p$.

Exercise 3.2. $\mathbb{E}[Y \mid X] = X/2$ and $\mathrm{Var}(Y \mid X) = X^2/12$. Tower rule: $\mathbb{E}Y = \mathbb{E}[X/2] = \frac{1}{4}$. Total variance: $\mathrm{Var}\, Y = \mathbb{E}[X^2/12] + \mathrm{Var}(X/2) = \frac{1}{36} + \frac{1}{48} = \frac{7}{144} \approx 0.04861$. Monte Carlo with 10^6 draws: $\mathbb{E}Y \approx 0.2499$, $\mathrm{Var}\, Y \approx 0.04860$.

Exercise 3.3. $\mathbb{E}(X-c)^2 = \operatorname{Var} X + (\mathbb{E}X - c)^2$ is minimised at $c = \mathbb{E}X$, with minimum $\operatorname{Var} X$. $g(c) = \mathbb{E}|X-c|$ is convex with right derivative $g'_+(c) = \mathbb{P}(X \le c) - \mathbb{P}(X > c) = 2F(c) - 1$, which is ≤ 0 for $c < m$ and ≥ 0 for $c \ge m$; hence g is minimised at the median. *Bayes connection:* applying the same computations to the posterior distribution, the Bayes estimator is the posterior mean under squared loss and the posterior median under absolute loss (Theorem 19.9).

Exercise 3.4. $\operatorname{Var} Y = \rho^2 + (1-\rho^2) = 1$ and $\operatorname{Cov}(X,Y) = \rho \operatorname{Var} Z_1 = \rho$, so Corr $= \rho$. Since (X,Y) is a linear transformation of (Z_1, Z_2), it is jointly normal. $Y - \rho X = \sqrt{1-\rho^2} Z_2$ is independent of $X = Z_1$; therefore $\mathbb{E}[Y \mid X] = \rho X + \mathbb{E}[\sqrt{1-\rho^2} Z_2] = \rho X$ and $\operatorname{Var}(Y \mid X) = 1 - \rho^2$. (A special case of Theorem 4.13(4); this construction is also the standard way of generating a bivariate normal.)

A.4 Joint Distributions and Transformations

Chapter 4.

Exercise 4.1. $f_{X+Y}(z) = \int_0^1 \mathbb{I}\{0 < z - x < 1\} dx$: this is z for $0 < z \le 1$ and $2 - z$ for $1 < z < 2$ (triangular). Irwin–Hall: $\mathbb{E}\sum U_i = n/2$, Var $= n/12$. For $n = 12$, $\sum U_i - 6$ has mean 0 and variance 1 and is approximately standard normal by the CLT; its excess kurtosis is only $-1.2/n = -0.1$. It needs only additions, so it was used when logarithms and trigonometric functions were expensive. Its flaw is that the support is restricted to $[-6, 6]$ and the tails are wrong: the probability $\mathbb{P}(|Z| > 6) \approx 2 \times 10^{-9}$ is never produced.

Exercise 4.2. Map $(Z, V) \mapsto (T, V)$; the inverse is $z = t\sqrt{v/k}$ with $|J| = \sqrt{v/k}$. By Theorem 4.10,

$$f_T(t) = \int_0^\infty \frac{e^{-t^2 v/(2k)}}{\sqrt{2\pi}} \cdot \frac{v^{k/2-1} e^{-v/2}}{2^{k/2}\Gamma(k/2)} \sqrt{\frac{v}{k}}\, dv = \frac{1}{\sqrt{2\pi k}\, 2^{k/2}\Gamma(k/2)} \int_0^\infty v^{\frac{k+1}{2}-1} e^{-\frac{v}{2}(1+t^2/k)} dv.$$

The Gamma integral equals $\Gamma(\frac{k+1}{2})(\frac{2}{1+t^2/k})^{(k+1)/2}$; simplifying gives $f_T(t) = \frac{\Gamma(\frac{k+1}{2})}{\sqrt{k\pi}\,\Gamma(\frac{k}{2})}(1 + \frac{t^2}{k})^{-\frac{k+1}{2}}$.

Exercise 4.3. $\mathbb{P}(X_{(1)} > x) = \prod \mathbb{P}(X_i > x) = e^{-n\lambda x}$. The joint density of the order statistics is $n!\lambda^n e^{-\lambda \sum_i x_{(i)}}$ $(0 < x_{(1)} < \cdots < x_{(n)})$. The transformation $D_k = (n-k+1)(X_{(k)} - X_{(k-1)})$ $(X_{(0)} = 0)$ is linear with a triangular matrix; the absolute Jacobian is $\prod_k (n-k+1)^{-1} = 1/n!$ and $\sum_i x_{(i)} = \sum_k d_k$. Hence $f_{\mathbf{D}}(\mathbf{d}) = \lambda^n e^{-\lambda \sum d_k}$, $d_k > 0$: the D_k are i.i.d. Exp(λ). Simulation ($n = 6$, $\lambda = 2$, 3×10^5 replications): the means of the D_k are 0.498–0.501 (theory 0.5), the largest absolute correlation between them is 0.0045, and $\mathbb{E}X_{(1)} = 0.0830$ (theory $1/(n\lambda) = 0.0833$).

Exercise 4.4. Let $X = Z_1$, $Y = \rho Z_1 + \sqrt{1-\rho^2} Z_2$ and $(Z_1, Z_2) = (R\cos\Theta, R\sin\Theta)$ with $\Theta \sim$ Uniform$(0, 2\pi)$. With $\phi = \arcsin\rho$, $Y = R\sin(\Theta + \phi)$. Then $\{X > 0\} = \{\Theta \in (-\frac{\pi}{2}, \frac{\pi}{2})\}$ and $\{Y > 0\} = \{\Theta \in (-\phi, \pi - \phi)\}$; their intersection is $(-\phi, \frac{\pi}{2})$, of length $\frac{\pi}{2} + \phi$. Therefore $\mathbb{P}(X > 0, Y > 0) = \frac{\pi/2 + \phi}{2\pi} = \frac{1}{4} + \frac{\arcsin\rho}{2\pi}$. Simulation: for $\rho = -0.5, 0.3, 0.8$ we get 0.1670, 0.2992, 0.3975 (formula: 0.1667, 0.2985, 0.3976).

A.5 Inequalities and the Law of Large Numbers

Chapter 5.

Exercise 5.1. Take $\mu = 0$. For every $u \ge 0$, $\{X \ge t\} \subseteq \{(X+u)^2 \ge (t+u)^2\}$; by Markov, $\mathbb{P}(X \ge t) \le \frac{\mathbb{E}(X+u)^2}{(t+u)^2} = \frac{\sigma^2 + u^2}{(t+u)^2}$. The right-hand side is minimised at $u = \sigma^2/t$, where it equals $\frac{\sigma^2(t^2+\sigma^2)/t^2}{(t^2+\sigma^2)^2/t^2} = \frac{\sigma^2}{\sigma^2+t^2}$. This is always better than applying Chebyshev to one tail (σ^2/t^2).

Exercise 5.2. $M_X(t) = e^{\lambda(e^t-1)}$; Chernoff gives $\mathbb{P}(X \geq a) \leq \inf_{t>0} \exp\{-ta + \lambda(e^t - 1)\}$. Setting the derivative to zero gives $e^t = a/\lambda$ ($t > 0$ since $a > \lambda$). Substituting yields $(\lambda/a)^a e^{a-\lambda} = e^{-\lambda}(e\lambda/a)^a$. For $\lambda = 10$, $a = 20$: Chebyshev gives $\mathbb{P}(|X - 10| \geq 10) \leq 10/100 = 0.1$; Chernoff gives ≤ 0.0210; the exact value is $\mathbb{P}(X \geq 20) = 0.00345$. Chernoff is 5 times better than Chebyshev and 6 times above the exact value — it captures the exponential rate correctly but loses a constant factor.

Exercise 5.3. By convergence in probability we can choose, for every k, an n_k with $\mathbb{P}(|X_n - X| > 1/k) \leq 2^{-k}$, and take $n_1 < n_2 < \dots$. For $A_k = \{|X_{n_k} - X| > 1/k\}$, $\sum \mathbb{P}(A_k) \leq \sum 2^{-k} < \infty$; by Borel–Cantelli (Theorem 1.14), with probability 1 only finitely many A_k occur. Hence for almost every ω, $|X_{n_k}(\omega) - X(\omega)| \leq 1/k \to 0$ for all large k.

Exercise 5.4. $p = \pi/4$ and $\hat{\pi} = 4\hat{p}$. $|\hat{\pi} - \pi| \leq 0.01 \iff |\hat{p} - p| \leq 0.0025$. Hoeffding (variables in $[0,1]$): $2e^{-2n(0.0025)^2} \leq 0.01 \iff n \geq \frac{\ln 200}{1.25\times 10^{-5}} \approx 423\,866$. For comparison, the CLT approximation with $\mathrm{Var}(4\,\mathbb{I}) = 16p(1-p) \approx 2.697$ gives $n \approx (2.576)^2 \cdot 2.697/10^{-4} \approx 178\,900$; Hoeffding is about 2.4 times more conservative because it uses only the bounds, not the variance.

A.6 Characteristic Functions and the CLT

Chapter 6.

Exercise 6.1. $\varphi_X(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\beta - it)x} dx = \frac{\beta^\alpha}{(\beta - it)^\alpha} = (1 - it/\beta)^{-\alpha}$. (Since $\mathrm{Re}(\beta - it) = \beta > 0$, the Gamma integral equals $\Gamma(\alpha)/s^\alpha$ for complex s as well; analytic continuation.) For independent $X_j \sim \mathrm{Gamma}(\alpha_j, \beta)$, $\prod_j (1 - it/\beta)^{-\alpha_j} = (1 - it/\beta)^{-\sum \alpha_j}$; by the uniqueness theorem, $\sum X_j \sim \mathrm{Gamma}(\sum \alpha_j, \beta)$.

Exercise 6.2. $\varphi_{X_n}(t) = \big(1 - \frac{\lambda}{n} + \frac{\lambda}{n} e^{it}\big)^n = \big(1 + \frac{\lambda(e^{it}-1)}{n}\big)^n \to e^{\lambda(e^{it}-1)}$ (Lemma 6.9), the CF of Poisson(λ). Lévy's continuity theorem gives the result.

Exercise 6.3. For $y \geq 0$, $\mathbb{P}(Y_n > y) = \mathbb{P}(X_{(n)} < 1 - y/n) = (1 - y/n)^n \to e^{-y}$. An extreme value must be scaled by n, not by $\sqrt{n}$ as for the mean: $\theta - X_{(n)} = O_P(1/n)$. This is why the variance of the German tank estimator is $O(1/n^2)$ (Example 10.13) and why the limit distribution is exponential rather than normal; normal-based intervals are inappropriate for this problem (see the pivot interval, Chapter 16).

Exercise 6.4. Let $Z \sim \mathcal{N}(0,1)$, $X_n = Z$ and $Y_n = -Z$. By symmetry $Y_n \overset{d}{=} Z$, so $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ (with X, Y independent standard normals, say). But $X_n + Y_n = 0$, whereas $X + Y \sim \mathcal{N}(0, 2)$. Convergence in distribution concerns only the marginals; it does not determine the joint distribution. When the limit is a constant, joint convergence $(X_n, Y_n) \xrightarrow{d} (X, c)$ holds automatically (proof of Slutsky) and continuous mapping can be applied.

A.7 Unbiasedness, Consistency and MSE

Chapter 7.

Exercise 7.1. For the Poisson, $\mathbb{E}X = \mathrm{Var}\, X = \lambda$, so $\mathbb{E}\bar{X} = \lambda$ and $\mathbb{E}S^2 = \sigma^2 = \lambda$. $\mathrm{Var}\,\bar{X} = \lambda/n$; by Theorem 7.9(3), $\mathrm{Var}\, S^2 = \frac{1}{n}(\lambda + 3\lambda^2 - \frac{n-3}{n-1}\lambda^2) = \frac{\lambda}{n} + \frac{2\lambda^2}{n-1}$. The variance of $\bar{X}$ is always smaller; indeed $\bar{X}$ is a function of the complete sufficient $\sum X_i$ and is the UMVUE (Chapter 10).

Exercise 7.2. $\sum_x T(x) e^{-\lambda} \lambda^x / x! = e^{-2\lambda}$ for every $\lambda \iff \sum_x T(x) \lambda^x / x! = e^{-\lambda} = \sum_x (-1)^x \lambda^x / x!$. By uniqueness of power series, $T(x) = (-1)^x$. The estimand lies in $(0, 1)$,

but the estimator takes only the values ± 1: when X is odd it gives a negative "probability" estimate. $\mathrm{MSE}((-1)^X) = \mathrm{Var} = 1 - e^{-4\lambda}$. Comparison with the biased e^{-2X}:

	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$
$\mathrm{MSE}((-1)^X)$	0.865	0.982	1.000
$\mathrm{MSE}(e^{-2X})$	0.270	0.279	0.134

(computed exactly using $\mathbb{E}e^{-sX} = e^{\lambda(e^{-s}-1)}$). For the sake of unbiasedness, the MSE is 3–7 times worse.

Exercise 7.3. The distribution is symmetric about θ; $\bar{X} - \theta$ and $\frac{X_{(1)}+X_{(n)}}{2} - \theta$ are symmetric about zero, so both estimators are unbiased. $\mathrm{Var}\,\bar{X} = \frac{1}{12n}$, while the variance of the midrange is $\frac{1}{2(n+1)(n+2)}$. Simulation: 8.38×10^{-3} and 3.80×10^{-3} for $n = 10$; 8.34×10^{-4} and 4.84×10^{-5} for $n = 100$. The midrange converges at rate $O(1/n^2)$. Reason: $(X_{(1)}, X_{(n)})$ is sufficient, whereas $\bar{X}$ is not (compare Example 7.12).

Exercise 7.4. $m = \frac{\alpha}{\alpha+\beta}$, $v = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{m(1-m)}{\alpha+\beta+1}$. Hence $\alpha + \beta = \frac{m(1-m)}{v} - 1$ and $\tilde{\alpha} = \bar{X}\Big(\frac{\bar{X}(1-\bar{X})}{\hat{\sigma}^2} - 1\Big)$, $\tilde{\beta} = (1-\bar{X})\Big(\frac{\bar{X}(1-\bar{X})}{\hat{\sigma}^2} - 1\Big)$. With $\alpha = 2$, $\beta = 5$, $n = 50$ and 10^5 replications: for $\tilde{\alpha}$, bias $+0.117$ and MSE 0.215; for $\tilde{\beta}$, bias $+0.315$ and MSE 1.515. The positive bias comes from $\hat{\sigma}^2$ being in the denominator (Jensen).

A.8 Cramér–Rao and Fisher Information

Chapter 8.

Exercise 8.1. $\ln f = \ln p + (k-1)\ln(1-p)$, $\partial_p^2 \ln f = -\frac{1}{p^2} - \frac{k-1}{(1-p)^2}$. With $\mathbb{E}(k-1) = \frac{1-p}{p}$, $I_1(p) = \frac{1}{p^2} + \frac{1}{p(1-p)} = \frac{1}{p^2(1-p)}$. For $g(p) = 1/p = \mathbb{E}X$, $\bar{X}$ is unbiased, $\mathrm{Var}\,\bar{X} = \frac{1-p}{np^2}$, and the CRLB is $\frac{(1/p^2)^2}{n/(p^2(1-p))} = \frac{1-p}{np^2}$: $\bar{X}$ is efficient. (The geometric distribution is an exponential family and $1/p$ is a linear function of the mean of the sufficient statistic; Corollary 8.8.)

Exercise 8.2. With $\sigma^2 = h(\sigma)$, $h'(\sigma) = 2\sigma$ and $I^{(\sigma^2)} = \frac{1}{2\sigma^4}$ we get $I^{(\sigma)} = \frac{4\sigma^2}{2\sigma^4} = \frac{2}{\sigma^2}$; the CRLB is $\frac{\sigma^2}{2n}$. Since $\mathbb{E}S = c_4\sigma$, $\mathrm{Var}(S/c_4) = \sigma^2(c_4^{-2} - 1)$. For $\sigma = 1$: $n = 5$: $c_4 = 0.9400$, Var $= 0.1318$ (simulation 0.1319), CRLB 0.100, efficiency 0.76; $n = 20$: Var $= 0.0267$, CRLB 0.025, efficiency 0.94. S/c_4 is the UMVUE (Example 10.14) but it is not efficient; the efficiency tends to 1 as $n \to \infty$.

Exercise 8.3. $\ln f = \alpha\ln\beta - \ln\Gamma(\alpha) + (\alpha-1)\ln x - \beta x$. Second derivatives: $\partial_\alpha^2 = -\psi'(\alpha)$, $\partial_\alpha\partial_\beta = 1/\beta$, $\partial_\beta^2 = -\alpha/\beta^2$. Hence

$$I_1(\alpha,\beta) = \begin{pmatrix} \psi'(\alpha) & -1/\beta \\ -1/\beta & \alpha/\beta^2 \end{pmatrix}.$$

If β is known, the CRLB is $\frac{1}{n\psi'(\alpha)}$; if not, it is $[I^{-1}]_{11}/n = \frac{\alpha}{n(\alpha\psi'(\alpha)-1)}$. The ratio is $\frac{\alpha\psi'(\alpha)}{\alpha\psi'(\alpha)-1}$; at $\alpha = 2$, $\psi'(2) = \pi^2/6 - 1 \approx 0.6449$ and the ratio is ≈ 4.45. Not knowing the scale more than quadruples the variance for estimating the shape; the two parameters are strongly "entangled" (the off-diagonal term is non-zero).

Exercise 8.4. $\bar{X} \sim \mathcal{N}(\theta, 1/n)$: $\mathbb{E}\bar{X}^2 = \theta^2 + 1/n$, so $\bar{X}^2 - 1/n$ is unbiased. Writing $\bar{X} = \theta + Z/\sqrt{n}$, $\bar{X}^2 = \theta^2 + 2\theta Z/\sqrt{n} + Z^2/n$; since $\mathrm{Cov}(Z, Z^2) = 0$, Var $= \frac{4\theta^2}{n} + \frac{2}{n^2}$. The CRLB is $\frac{(2\theta)^2}{n}$. The gap arises because θ^2 is not a linear function of $\mathbb{E}\bar{X}$ (Corollary 8.8); that it is $O(1/n^2)$ shows that the estimator is *asymptotically* efficient. At $\theta = 0$ the CRLB is zero but the variance is $2/n^2$: at this point the bound is attained by no unbiased estimator.

A.9 Sufficiency and Rao–Blackwell

Chapter 9.

Exercise 9.1. A two-parameter full-rank exponential family (the natural parameter $(\alpha-1,-\beta)$ ranges over an open set): $\big(\sum \ln X_i, \sum X_i\big)$ is minimal sufficient (Theorem 9.7). If α is known, $\sum X_i$ alone is minimal sufficient; if β is known, $\sum \ln X_i$ (i.e. the geometric mean) alone is.

Exercise 9.2. $L(\theta;\mathbf{x}) = \mathbb{I}\{x_{(n)}-1<\theta<x_{(1)}\}$. For $L(\theta;\mathbf{x})/L(\theta;\mathbf{y})$ to be independent of θ (apart from the $0/0$ ambiguity), the two indicators must equal 1 on the same set of θ, i.e. $(x_{(n)}-1, x_{(1)}) = (y_{(n)}-1, y_{(1)})$. By Theorem 9.6, $(X_{(1)}, X_{(n)})$ is minimal sufficient. A two-dimensional minimal sufficient statistic for one parameter shows that the data contain a component that carries no information about θ but cannot be discarded (the range R, which is ancillary); the statistic is not complete and there is no UMVUE (Example 10.15).

Exercise 9.3. $W^* = \mathbb{P}(X_1=X_2=1 \mid T=t)$. Given $T=t$, the positions of the successes form $\binom{n}{t}$ equally likely arrangements (Example 9.10); the number of arrangements in which the first two trials are successes is $\binom{n-2}{t-2}$. Hence $W^* = \binom{n-2}{t-2}/\binom{n}{t} = \frac{t(t-1)}{n(n-1)}$. By completeness, this is the UMVUE of p^2.

Exercise 9.4. $\mathbb{E}[\mathbb{I}\{X_1\le c\} \mid \bar X] = \mathbb{P}(X_1 \le c \mid \bar X) = \Phi\big((c-\bar X)/\sqrt{1-1/n}\big)$. With $n=5$, $\mu=0$, $c=1$ (target $\Phi(1)=0.8413$) and 4×10^5 replications: the mean of the Rao–Blackwell estimator is 0.8415 (unbiased), MSE 0.01280; the mean of the plug-in $\Phi(c-\bar X)$ is 0.8195 (bias -0.022), MSE 0.01273. At this point the two estimators have almost the same MSE: the bias of the plug-in is balanced by the extra variance introduced by dividing by $\sqrt{1-1/n}$.

A.10 Completeness and UMVUE

Chapter 10.

Exercise 10.1. $T=\sum X_i \sim \text{Gamma}(n,\lambda)$ is complete sufficient. Since $\mathbb{E}[1/T] = \frac{\lambda}{n-1}$, $\frac{n-1}{T}$ is the UMVUE of λ and $\bar X = T/n$ that of $1/\lambda$. For $\mathbb{P}(X>c)$, $W=\mathbb{I}\{X_1>c\}$ is unbiased. By Exercise 10.4, $X_1/T \sim \text{Beta}(1,n-1)$ independently of T; hence

$$\mathbb{E}[W \mid T] = \mathbb{P}\Big(\frac{X_1}{T} > \frac{c}{T} \;\Big|\; T\Big) = \Big(1-\frac{c}{T}\Big)^{n-1} \quad (T>c), \qquad 0 \ (T\le c),$$

because $\mathbb{P}(\text{Beta}(1,n-1)>u) = (1-u)^{n-1}$. By Lehmann–Scheffé this is the UMVUE.

Exercise 10.2. $\bar X$ is complete sufficient in the $\mathcal{N}(\mu,1)$ family. $(X_1,\bar X)$ is jointly normal with $\operatorname{Cov}(X_1,\bar X) = 1/n = \operatorname{Var}\bar X$, so by Theorem 4.13(4) $X_1 \mid \bar X \sim \mathcal{N}(\bar X, 1-1/n)$. Thus $\mathbb{E}[\mathbb{I}\{X_1\le c\} \mid \bar X] = \Phi\big((c-\bar X)/\sqrt{1-1/n}\big)$, which by Lehmann–Scheffé is the UMVUE. (For a numerical comparison see Exercise 9.4.)

Exercise 10.3. $L \propto \theta^{-n}\exp\{\frac{1}{\theta}\sum x_i - \frac{1}{2\theta^2}\sum x_i^2\}$. $L(\mathbf{x})/L(\mathbf{y})$ is independent of θ iff $\frac{a}{\theta} - \frac{b}{2\theta^2} \equiv 0$ (a,b: the differences of the sums), i.e. $a=b=0$; so $(\sum X_i, \sum X_i^2)$, equivalently $(\bar X, S^2)$, is minimal sufficient. However, $\mathbb{E}\bar X^2 = \theta^2 + \theta^2/n$ and $\mathbb{E}S^2 = \theta^2$; hence $g = \frac{n}{n+1}\bar X^2 - S^2$ has mean zero for every θ without being zero: the statistic is not complete. The violated condition: the natural parameter $(1/\theta, -1/(2\theta^2))$ traces a *curve* in $\mathbb{R}^2$ and contains no open set.

Exercise 10.4. Writing $X_i = Y_i/\lambda$ with $Y_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$, $X_i/T = Y_i/\sum Y_j$; its distribution does not depend on λ, i.e. $\mathbf{U}$ is ancillary. Since T is complete sufficient, Basu's theorem gives $T \perp\!\!\!\perp \mathbf{U}$. By $\sum_i U_i = 1$ and symmetry, $\mathbb{E}U_1 = 1/n$. (A second route via Basu: $\mathbb{E}X_1 = \mathbb{E}[U_1 T] = \mathbb{E}U_1 \cdot \mathbb{E}T$, i.e. $1/\lambda = \mathbb{E}U_1 \cdot n/\lambda$.) Moreover, by Example 4.17, $U_1 \sim \text{Beta}(1,n-1)$.

A.11 Maximum Likelihood

Chapter 11.

Exercise 11.1. $\hat\lambda = \bar X$, and by invariance $\widehat{e^{-\lambda}} = e^{-\bar X}$. Since $T = n\bar X \sim \text{Poisson}(n\lambda)$, $\mathbb{E}e^{-\bar X} = \mathbb{E}e^{-T/n} = \exp\{n\lambda(e^{-1/n}-1)\}$. Since $e^{-1/n} - 1 > -1/n$, this value exceeds $e^{-\lambda}$: the MLE is biased upwards (a consequence of Jensen). For $\lambda = 2$, $n = 10$: 0.1491 vs 0.1353. The UMVUE $(1-1/n)^T$ is unbiased; in the simulation of Chapter 9 its MSE was also smaller.

Exercise 11.2. $L = \lambda^n e^{-\lambda\sum(x_i-\mu)}\mathbb{I}\{\mu \le x_{(1)}\}$ is increasing in μ for fixed λ; $\hat\mu = X_{(1)}$. Then $\partial_\lambda \ell = n/\lambda - \sum(x_i - \hat\mu) = 0$ gives $\hat\lambda = 1/(\bar X - X_{(1)})$. $X_{(1)} - \mu \sim \text{Exp}(n\lambda)$ (Exercise 4.3), so $\text{Bias}(\hat\mu) = \frac{1}{n\lambda}$. Since $\mathbb{E}[\bar X - X_{(1)}] = \frac{1}{\lambda} - \frac{1}{n\lambda} = \frac{n-1}{n\lambda}$,

$$\tilde\mu = X_{(1)} - \frac{\bar X - X_{(1)}}{n-1}$$

is unbiased.

Exercise 11.3. $\ell(\theta) = (2n_1+n_2)\ln\theta + (n_2+2n_3)\ln(1-\theta) + \text{const}$. $\ell' = 0$ gives $\hat\theta = \frac{2n_1+n_2}{2n}$, i.e. the proportion of the first allele among the $2n$ alleles in the sample. $-\ell'' = \frac{2n_1+n_2}{\theta^2} + \frac{n_2+2n_3}{(1-\theta)^2}$; with $\mathbb{E}(2n_1+n_2) = 2n\theta$ we get $I_n(\theta) = \frac{2n}{\theta(1-\theta)}$: the information of $2n$ independent Bernoulli trials.

Exercise 11.4. MLE: $\hat\mu_i = \bar X_i = \frac{X_{i1}+X_{i2}}{2}$ and $\hat\sigma^2 = \frac{1}{2n}\sum_i\sum_j (X_{ij} - \bar X_i)^2 = \frac{1}{4n}\sum_i (X_{i1} - X_{i2})^2$. Since $X_{i1} - X_{i2} \sim \mathcal{N}(0, 2\sigma^2)$, $\mathbb{E}\hat\sigma^2 = \sigma^2/2$ and by the SLLN $\hat\sigma^2 \xrightarrow{\text{a.s.}} \sigma^2/2$: information does not accumulate, because each new pair brings a new nuisance parameter. A consistent estimator: $\tilde\sigma^2 = \frac{1}{2n}\sum_i (X_{i1} - X_{i2})^2$. Simulation ($n = 1000$, $\sigma^2 = 4$): $\hat\sigma^2 = 2.16$, $\tilde\sigma^2 = 4.31$.

A.12 MLE Asymptotics and the Delta Method

Chapter 12.

Exercise 12.1. $I_1(\lambda) = 1/\lambda^2$; by Theorem 12.6, $\sqrt{n}(\hat\lambda - \lambda) \xrightarrow{d} \mathcal{N}(0, \lambda^2)$. Delta method: by the CLT $\sqrt{n}(\bar X - 1/\lambda) \xrightarrow{d} \mathcal{N}(0, 1/\lambda^2)$, $g(\mu) = 1/\mu$, $g'(1/\lambda) = -\lambda^2$; the variance is $\lambda^4/\lambda^2 = \lambda^2$. For $\ln\hat\lambda$, $g'(\lambda) = 1/\lambda$ and the asymptotic variance is $\lambda^2/\lambda^2 = 1$: the ln transformation is variance-stabilising for the exponential model.

Exercise 12.2. $g(a,b) = b/a$, $\nabla g = (-\mu_Y/\mu_X^2,\ 1/\mu_X)$. With $R = \mu_Y/\mu_X$, Theorem 12.10 gives

$$\sqrt{n}\Big(\frac{\bar Y}{\bar X} - R\Big) \xrightarrow{d} \mathcal{N}\Big(0,\ \frac{\sigma_Y^2 - 2R\sigma_{XY} + R^2\sigma_X^2}{\mu_X^2}\Big) = \mathcal{N}\Big(0, \frac{\text{Var}(Y - RX)}{\mu_X^2}\Big).$$

Positive correlation stabilises the ratio estimate; the "ratio estimator" of survey sampling exploits this.

Exercise 12.3. Simulation with 4×10^4 replications:

	$n\,\text{Var}(\text{median})$	$n\,\text{Var}(\text{one-step})$	theory
$n = 51$	2.62	2.13	$\pi^2/4 = 2.47$; $1/I_1 = 2$
$n = 201$	2.50	2.03	

The one-step estimator reduces the asymptotic variance of the median, ≈ 2.47, to the Fisher bound 2. General theory (Le Cam): a single Newton–Raphson step (with expected or observed information) from any $\sqrt{n}$-consistent starting point is asymptotically equivalent to the MLE; for multimodal likelihoods (Example 11.15) it is a safe way to avoid searching for the global maximum.

Exercise 12.4. $z = \tanh^{-1} r$ is approximately $\mathcal{N}(\tanh^{-1}\rho,\, 1/(n-3))$; the interval is $\tanh\left(z \pm 1.96/\sqrt{n-3}\right)$. (Using $\sqrt{n-3}$ instead of $\sqrt{n}$ is a small-sample correction.) For $\rho = 0.8$, $n = 20$, 2×10^4 replications: the Fisher z coverage is 0.951, that of the direct interval $r \pm 1.96(1-r^2)/\sqrt{n}$ is 0.893. The direct interval ignores the skewed distribution of r near the boundary of $[-1, 1]$.

A.13 The Neyman–Pearson Lemma

Chapter 13.

Exercise 13.1. $f_0 = \mathbb{I}\{0 < x < 1\}$, $f_1 = \frac{1}{2}\mathbb{I}\{0 < x < 2\}$. The ratio is $1/2$ on $(0,1)$ and ∞ on $[1,2)$. For $\alpha \in (0,1)$ the MP tests are: always reject on $[1,2)$ (probability 0 under H_0), and use *any* ϕ on $(0,1)$ with $\int_0^1 \phi = \alpha$. The power is $\frac{1}{2} + \frac{\alpha}{2}$ in every case. For example, "reject if $x > 1-\alpha$" and "reject if $x < \alpha$" are both MP: on a set (of positive probability) where the likelihood ratio is constant the Neyman–Pearson test is not unique (Theorem 13.7(3) guarantees uniqueness only on the set $f_1 \neq k f_0$).

Exercise 13.2. $T = \sum X_i \sim \mathrm{Bin}(20, p)$, and the ratio is increasing in T. $\mathbb{P}_{0.5}(T \geq 15) = 0.0207 < 0.05 < \mathbb{P}_{0.5}(T \geq 14) = 0.0577$. MP test: reject if $T \geq 15$; if $T = 14$ reject with probability $\gamma = \frac{0.05 - 0.0207}{\mathbb{P}(T=14)} = 0.793$. Power: $\mathbb{P}_{0.7}(T \geq 15) + \gamma\mathbb{P}_{0.7}(T = 14) = 0.568$.

Exercise 13.3.

$$\frac{f_1(x)}{f_0(x)} = \sqrt{\frac{2}{\pi}}\, h(x^2), \qquad h(u) = \frac{e^{u/2}}{1+u}, \qquad h'(u) = \frac{e^{u/2}(u-1)}{2(1+u)^2}.$$

The ratio is *not* a monotone function of x^2: it decreases on $u \in [0,1]$ and increases for $u > 1$. The set $\{h(x^2) > k\}$ has the form $\{|x| > c\}$ if $k \geq h(0) = 1$, which means $c \geq c^*$, where $u^* = 2.513$ is the positive root of $e^{u/2} = 1 + u$ and $c^* = \sqrt{u^*} = 1.585$. Hence the rejection region has the form $\{|x| > c\}$ only if $\alpha \leq 2(1 - \Phi(1.585)) = 0.113$; for larger α the MP region also contains an interval around 0 ($\{|x| < c_1\} \cup \{|x| > c_2\}$). For $\alpha = 0.05$, $c = 1.96$ and the power is $\mathbb{P}_{\text{Cauchy}}(|X| > 1.96) = 1 - \frac{2}{\pi}\arctan 1.96 = 0.300$.

Exercise 13.4. Let I_0 be the set of true null hypotheses. By Theorem 13.10, $\mathbb{P}(p_i \leq u) \leq u$ for $i \in I_0$. By Boole's inequality,

$$\mathbb{P}\Big(\bigcup_{i \in I_0} \{p_i \leq \alpha/m\}\Big) \leq \sum_{i \in I_0} \frac{\alpha}{m} = \frac{|I_0|}{m}\alpha \leq \alpha.$$

Independence is not required. The price is a loss of power; Holm's step-down method gives the same guarantee more powerfully.

A.14 Karlin–Rubin and UMP Tests

Chapter 14.

Exercise 14.1. For the Poisson, $w(\lambda) = \ln\lambda$ is increasing and the MLR is in $T = \sum X_i$. At $\lambda = 1$, $T \sim \text{Poisson}(20)$: the UMP test rejects if $T \geq 29$, and if $T = 28$ rejects with probability $\gamma = 0.863$. The power function is $\beta(\lambda) = \mathbb{P}_{20\lambda}(T > 28) + \gamma\mathbb{P}_{20\lambda}(T = 28)$: $\beta(0.5) \approx 0$, $\beta(1) = 0.050$, $\beta(1.25) = 0.291$, $\beta(1.5) = 0.658$, $\beta(2) = 0.979$. For the plot, this formula is evaluated on a grid with `scipy.stats.poisson`; the curve is monotone increasing (Theorem 14.5).

Exercise 14.2. $f \propto \sigma^{-n}\exp\{-\frac{1}{2\sigma^2}\sum x_i^2\}$, $w(\sigma) = -1/(2\sigma^2)$ is increasing; the MLR is in $\sum X_i^2$. UMP test: $\sum X_i^2 > \sigma_0^2 \chi^2_{n,\alpha}$. For the two-sided equal-tailed test ($n = 10$, $\alpha = 0.05$)

the power is $\beta(\sigma) = \mathbb{P}(\chi^2_n < a\sigma_0^2/\sigma^2) + \mathbb{P}(\chi^2_n > b\sigma_0^2/\sigma^2)$; numerically its smallest value is $0.0476 < \alpha$, at $\sigma/\sigma_0 \approx 0.967$. For example $\beta(0.97\sigma_0) = 0.0476$, $\beta(0.99\sigma_0) = 0.0487$, $\beta(1.01\sigma_0) = 0.0517$. The test is biased: it rejects alternatives slightly below σ_0 less often than H_0 itself. The UMPU test chooses the tails by the two conditions of Theorem 14.7.

Exercise 14.3. For $\theta_2 > \theta_1$, $\ln \frac{f(x;\theta_2)}{f(x;\theta_1)} = (\theta_2 - \theta_1) + 2\ln \frac{1+e^{-(x-\theta_1)}}{1+e^{-(x-\theta_2)}}$, and its derivative with respect to x is $2\big[q(x-\theta_2) - q(x-\theta_1)\big]$, where $q(y) = \frac{e^{-y}}{1+e^{-y}}$ is decreasing; since $x-\theta_2 < x-\theta_1$, the derivative is ≥ 0: the MLR is in x. For $n > 1$ the ratio has the form $\prod_i r(x_i)$, and since r is not linear it is not a function of a single statistic $T(\mathbf{x})$; the logistic family is not an exponential family and the minimal sufficient statistic is the full set of order statistics. Karlin–Rubin does not apply; in practice the locally most powerful (score) test is based on the statistic $\sum_i \tanh\big((x_i - \theta_0)/2\big)$.

Exercise 14.4. The tests "reject if $T \geq t$" form a nested family; the smallest level at which the observed $t_0 = T(\mathbf{x})$ leads to rejection is the size of the test $\{T \geq t_0\}$: $p(\mathbf{x}) = \sup_{\theta \leq \theta_0} \mathbb{P}_\theta(T \geq t_0)$. $\phi = \mathbb{I}\{T \geq t_0\}$ has the Neyman–Pearson form; by Step 2 its power function is non-decreasing in θ, so the supremum is attained at θ_0: $p(\mathbf{x}) = \mathbb{P}_{\theta_0}(T \geq T(\mathbf{x}))$.

A.15 The Likelihood Ratio Test

Chapter 15.

Exercise 15.1. The unrestricted MLE is $\hat\sigma^2 = \frac{n-1}{n}S^2$; with $w = \hat\sigma^2/\sigma_0^2$,

$$G^2 = n\big(w - 1 - \ln w\big),$$

an asymmetric function with its minimum at $w = 1$. The acceptance region $\{a \leq w \leq b\}$ is determined by the condition $a - \ln a = b - \ln b$; it is not equal-tailed. For $n = 10$, the LRT with exact size 0.05 accepts when $q = (n-1)S^2/\sigma_0^2 \in [3.187,\ 22.912]$: the left-tail probability is 0.0436 and the right-tail probability 0.0064 (equal-tailed region: [2.700, 19.023]). Moreover, the exact critical value is 4.62, whereas the Wilks approximation gives 3.84: at $n = 10$ the χ^2_1 approximation exceeds the nominal level.

Exercise 15.2. With counts N_{ij}, row totals $N_{i\cdot}$, column totals $N_{\cdot j}$ and $E_{ij} = N_{i\cdot}N_{\cdot j}/n$:

$$G^2 = 2\sum_{i,j} N_{ij} \ln \frac{N_{ij}}{E_{ij}}, \qquad X^2 = \sum_{i,j} \frac{(N_{ij} - E_{ij})^2}{E_{ij}}.$$

In the unrestricted model the four cell probabilities sum to 1: $\dim \Theta = 3$. Under independence $p_{ij} = r_i c_j$ with r_1, c_1 free: $\dim \Theta_0 = 2$. The degrees of freedom are $3 - 2 = 1 = (2-1)(2-1)$. For a general $I \times J$ table, $(IJ - 1) - (I - 1) - (J - 1) = (I - 1)(J - 1)$.

Exercise 15.3. Let $\hat\lambda = 1/\bar X$, $I_n(\lambda) = n/\lambda^2$ and $y = \lambda_0 \bar X$. (a) $W_\lambda = (\hat\lambda - \lambda_0)^2 n/\hat\lambda^2 = n(1-y)^2$; (b) on the $\ln\lambda$ scale the asymptotic variance is $1/n$ (Exercise 12.1): $W_{\ln} = n(\ln y)^2$; LRT: $G^2 = 2n(y - 1 - \ln y)$. Using $2\lambda_0 \sum X_i = 2ny \sim \chi^2_{2n}$, with $n = 10$ and the critical value $\chi^2_{1;0.05} = 3.84$, the exact sizes are: W_λ: 0.045; $W_{\ln}$: 0.059; LRT: 0.052. The size of the Wald test changes with the parametrisation, that of the LRT does not, and the LRT is closest to the nominal level. (In this example the λ scale did better; which scale will be better is not clear in advance. This is another reason to prefer the LRT.)

Exercise 15.4. On $\Theta = [0, \infty)$ the restricted MLE is $\hat\mu = \max(\bar X, 0)$. If $\bar X > 0$, $G^2 = n\bar X^2$, otherwise $G^2 = 0$. Under H_0, $\sqrt{n}\bar X = Z \sim \mathcal{N}(0,1)$: $\mathbb{P}(G^2 = 0) = \frac{1}{2}$ and for $x > 0$, $\mathbb{P}(G^2 \leq x) = \frac{1}{2} + \frac{1}{2}\mathbb{P}(Z^2 \leq x)$; i.e. the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$. If $\chi^2_{1;0.05} = 3.84$ is used, the true size is $\mathbb{P}(Z > 1.96) = 0.025$. The correct critical value is $\chi^2_{1;0.10} = 2.71$ (equivalent to the one-sided z-test).

A.16 Confidence Intervals

Chapter 16.

Exercise 16.1. $\bar{X}-\bar{Y}\sim\mathcal{N}(\mu_1-\mu_2,\sigma^2(\frac{1}{m}+\frac{1}{n}))$ and $(m+n-2)S_p^2/\sigma^2\sim\chi^2_{m+n-2}$ are independent, where $S_p^2=\frac{(m-1)S_1^2+(n-1)S_2^2}{m+n-2}$. The pivot is $\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{S_p\sqrt{1/m+1/n}}\sim t_{m+n-2}$, and the interval is $\bar{X}-\bar{Y}\pm t_{m+n-2,\alpha/2}S_p\sqrt{\frac{1}{m}+\frac{1}{n}}$. Welch: SE $=\sqrt{S_1^2/m+S_2^2/n}$, with the degrees of freedom from the Satterthwaite approximation. Simulation ($m=10$, $\sigma_1=3$; $n=30$, $\sigma_2=1$): the coverage of the pooled interval is 0.787, that of Welch's 0.949. When the smaller group has the larger variance, the pooled estimate understates the variance; in practice Welch should be the default.

Exercise 16.2. The length is proportional to $(n-1)S^2(1/a-1/b)$; minimise $1/a-1/b$ subject to $F(b)-F(a)=0.95$. The Lagrangian is $\mathcal{L}=\frac{1}{a}-\frac{1}{b}-\lambda(F(b)-F(a)-0.95)$; from $\partial_a\mathcal{L}=-\frac{1}{a^2}+\lambda f(a)=0$ and $\partial_b\mathcal{L}=\frac{1}{b^2}-\lambda f(b)=0$ we get $\lambda=\frac{1}{a^2f(a)}=\frac{1}{b^2f(b)}$, i.e. the condition is $a^2f(a)=b^2f(b)$ (not the density equality of Theorem 16.9, because the length is not linear in the pivot). For $n=10$ (χ^2_9): the shortest interval has $a=3.284$, $b=26.077$ ($1/a-1/b=0.2662$); the equal-tailed one $a=2.700$, $b=19.023$ (0.3177). The shortest interval is about 16% shorter, but its right endpoint moves noticeably outwards.

Exercise 16.3. $x=\sum X_i$, $\hat{\lambda}=x/n$, $z=1.96$. Wald: $\hat{\lambda}\pm z\sqrt{\hat{\lambda}/n}$; score: the solution of $|\hat{\lambda}-\lambda|\le z\sqrt{\lambda/n}$, namely $\hat{\lambda}+\frac{z^2}{2n}\pm z\sqrt{\frac{\hat{\lambda}}{n}+\frac{z^2}{4n^2}}$; Garwood: $[\frac{\chi^2_{2x,0.975}}{2n},\frac{\chi^2_{2x+2,0.025}}{2n}]$ (in the book's upper-quantile notation; see Notation). For $n=5$, $\lambda\in(0,5]$, the exact coverage is: Wald mean 0.899, minimum 0.095; score mean 0.952, minimum 0.905; Garwood mean 0.968, minimum 0.951. The picture is the same as for the binomial in Example 16.14: Wald collapses for small λ, the score interval is balanced, and the exact interval is conservative.

Exercise 16.4. $n=10$, 2×10^5 replications: the individual coverages are 0.950 and 0.950, and the joint coverage is 0.907 (>0.9025). The two events are positively associated: the coverage of the t interval is the event $|\bar{X}-\mu|\le tS/\sqrt{n}$, which depends on S; although $\bar{X}\perp\!\!\!\perp S^2$, both coverage events depend on S. Bonferroni: constructing each interval at level 97.5% makes the joint coverage ≥0.95 by Boole's inequality; in the simulation it is 0.953.

A.17 The Linear Model

Chapter 17.

Exercise 17.1. $\hat{\beta}_0=\bar{Y}-\hat{\beta}_1\bar{x}$; $\hat{\beta}_1=\sum c_iY_i$ with $c_i=(x_i-\bar{x})/S_{xx}$, and since $\sum c_i=0$, $\operatorname{Cov}(\bar{Y},\hat{\beta}_1)=\frac{\sigma^2}{n}\sum c_i=0$. Thus $\operatorname{Cov}(\hat{\beta}_0,\hat{\beta}_1)=-\bar{x}\operatorname{Var}\hat{\beta}_1=-\sigma^2\bar{x}/S_{xx}$. For them to be uncorrelated we need $\bar{x}=0$: centring the x's makes $(X^\top X)$ diagonal.

Exercise 17.2. $\sum h_{ii}=\operatorname{tr}H=p$ (Theorem 17.6). H is symmetric and idempotent: $h_{ii}=(H^2)_{ii}=\sum_j h_{ij}^2\ge h_{ii}^2$, hence $0\le h_{ii}\le1$. With $\mathbf{e}=(I-H)\boldsymbol{\varepsilon}$, $\operatorname{Cov}(\mathbf{e})=\sigma^2(I-H)$, so $\operatorname{Var}(e_i)=\sigma^2(1-h_{ii})$: the residuals of high-leverage observations are systematically small (the line is "pulled" towards them). For a comparable scale the studentised residuals $r_i=e_i/(s\sqrt{1-h_{ii}})$ are used.

```python
import numpy as np
def high_leverage(X):
    h = np.einsum("ij,jk,ik->i", X, np.linalg.inv(X.T @ X), X)   # diag(H)
    n, p = X.shape
    return np.where(h > 2*p/n)[0], h
```

Exercise 17.3. Let $A_\lambda = (X^\top X + \lambda I)^{-1}$. $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda = A_\lambda X^\top X\boldsymbol{\beta} = \boldsymbol{\beta} - \lambda A_\lambda \boldsymbol{\beta}$ (bias $-\lambda A_\lambda\boldsymbol{\beta}$), $\mathrm{Cov} = \sigma^2 A_\lambda X^\top X A_\lambda$. With $X^\top X = QDQ^\top$ and $\boldsymbol{\gamma} = Q^\top\boldsymbol{\beta}$,

$$\mathrm{MSE}(\lambda) = \sigma^2 \sum_j \frac{d_j}{(d_j+\lambda)^2} + \lambda^2 \sum_j \frac{\gamma_j^2}{(d_j+\lambda)^2}, \qquad \mathrm{MSE}'(0) = -2\sigma^2 \sum_j \frac{1}{d_j^2} < 0.$$

Since the derivative at zero is negative, $\mathrm{MSE}(\lambda) < \mathrm{MSE}(0) = \mathrm{MSE}(\hat{\boldsymbol{\beta}})$ for some small $\lambda > 0$. This does not contradict Gauss–Markov: ridge is *biased*, while Gauss–Markov compares only linear *unbiased* estimators. The gain is largest in the directions of small eigenvalues (multicollinearity).

Exercise 17.4. For a hypothesis with a single constraint ($q = 1$), $F = \frac{\hat\beta_2^2}{s^2[(X^\top X)^{-1}]_{33}} = t^2$ is an algebraic identity; in code `F` and `t**2` give the same number. For 21 equally spaced points in $x \in [10, 11]$, $\mathrm{Corr}(x, x^2) = 0.99992$ and VIF ≈ 6056: the standard errors of $\hat\beta_1$ and $\hat\beta_2$ are inflated by a factor of ≈ 78. With centring ($x - \bar x$), for a symmetric design $\mathrm{Corr}(x - \bar x, (x-\bar x)^2) = 0$ and VIF $= 1$. The fitted curve and the F statistic for $H_0 : \beta_2 = 0$ do not change; only the interpretation of the coefficients and the numerical stability change.

A.18 The Bootstrap

Chapter 18.

Exercise 18.1. 600 data sets, $B = 800$, $n = 20$, nominal 95%:

Data	Percentile	BCa	Fisher z
Normal, $\rho = 0.8$	0.925	0.932	0.952
Lognormal, $\rho = 0.713$	0.888	0.847	0.778

For normal data Fisher's z (based on normal theory) is best, and the bootstrap intervals are slightly too narrow. For lognormal data (heavy right tail; ρ computed via the marginal exponential transformation), the assumption behind Fisher's z breaks down and it gives the worst coverage; the bootstrap is more robust, but at $n = 20$ none reaches the nominal level. The correlation coefficient is a statistic that is very sensitive to outliers.

Exercise 18.2. *Pairs* bootstrap: the pairs (x_i, y_i) are resampled with replacement. *Residual* bootstrap: $\hat{\mathbf{Y}} + \mathbf{e}^*$, with $\mathbf{e}^*$ drawn with replacement from the residuals; since it redistributes the residuals independently of x, it *assumes* constant variance. Simulation ($n = 40$, error sd $0.2{+}0.3x$, 500 replications, $B = 400$): the true SE of the slope is 0.113; the mean estimates are classical 0.105, residual bootstrap 0.102, pairs bootstrap 0.109, sandwich (HC0) 0.106. The residual bootstrap reproduces exactly the error of the classical formula; the pairs bootstrap and the sandwich are robust to heteroscedasticity.

Exercise 18.3. For Cauchy samples with $n = 30$, the bootstrap SE of the mean ranged between 0.27 and 7.57 over eight data sets: each time it is determined by the largest $|x_i|$ and does not converge to a stable value, because the quantity it tries to estimate (the standard deviation of $\bar X$) is infinite (Chapter 5). The bootstrap SE of the median, by contrast, lies in the range 0.18–0.42, around the asymptotic value $\pi/(2\sqrt{n}) = 0.287$. The bootstrap is meaningful only when the target quantity is well defined.

Exercise 18.4. $\mathbb{E}\bar X^2 = \mu^2 + \sigma^2/n$: in Theorem 18.11, $a = \sigma^2$, $b = 0$, so the jackknife estimator is exactly unbiased. Direct computation: with $\bar X_{(i)} = \frac{n\bar X - X_i}{n-1}$ and $\sum X_i^2 = (n-1)S^2 + n\bar X^2$,

$$\hat\theta_{\mathrm{jack}} = n\bar X^2 - \frac{n-1}{n}\sum_i \bar X_{(i)}^2 = n\bar X^2 - \frac{n(n-1)^2\bar X^2 + (n-1)S^2}{n(n-1)} = \bar X^2 - \frac{S^2}{n}.$$

That is, the jackknife correction automatically produces the UMVUE of μ^2 (in the normal model). Numerical check ($n = 12$): the two expressions agree exactly, 5.369554.

A.19 Bayesian Inference

Chapter 19.

Exercise 19.1. The posterior is Gamma$(\alpha+\sum x_i, \beta+n)$, with mean $\frac{\alpha+n\bar{x}}{\beta+n} = \frac{\beta}{\beta+n}\cdot\frac{\alpha}{\beta} + \frac{n}{\beta+n}\cdot\bar{x}$. For $\alpha = 2$, $\beta = 1$, $n = 10$, $\sum x_i = 34$: the posterior is Gamma$(36, 11)$, with mean $36/11 = 3.273 = \frac{1}{11}\cdot 2 + \frac{10}{11}\cdot 3.4$; the 95% equal-tailed credible interval is $[2.292,\ 4.425]$. The posterior predictive distribution of a new observation is NegBin$\big(36, \frac{11}{12}\big)$ (in the parametrisation of Appendix **??**): mean 3.273, variance 3.570 (larger than for a Poisson; parameter uncertainty); $\mathbb{P}(\tilde{X} = 0, \dots, 5) = 0.044;\ 0.131;\ 0.202;\ 0.213;\ 0.173;\ 0.115$.

Exercise 19.2. The joint posterior is $\pi(\mu, \sigma^2 \mid \mathbf{x}) \propto (\sigma^2)^{-n/2-1} \exp\big\{-\frac{(n-1)s^2 + n(\mu-\bar{x})^2}{2\sigma^2}\big\}$. Integrating over σ^2 (an inverse-Gamma kernel):

$$\pi(\mu \mid \mathbf{x}) \propto \big[(n-1)s^2 + n(\mu-\bar{x})^2\big]^{-n/2} \propto \Big(1 + \frac{t^2}{n-1}\Big)^{-n/2}, \qquad t = \frac{\mu-\bar{x}}{s/\sqrt{n}},$$

which is the t_{n-1} kernel. Hence the 95% HPD interval $\bar{x} \pm t_{n-1,0.025}\, s/\sqrt{n}$ is numerically identical to the classical t interval; the interpretation differs (posterior probability vs. long-run coverage). This "matching prior" example is an exact finite-sample version of the Bernstein–von Mises theorem of Chapter 19.

Exercise 19.3. With $w = \frac{\tau^2}{\tau^2 + \sigma^2/n}$, $\mu_n = (1-w)\mu_0 + w\bar{X}$. The frequentist MSE is $\mathrm{MSE}_\theta(\mu_n) = w^2\frac{\sigma^2}{n} + (1-w)^2(\theta-\mu_0)^2$. $\mathrm{MSE}_\theta(\mu_n) < \sigma^2/n \iff (1-w)^2\Delta^2 < (1-w^2)\frac{\sigma^2}{n} \iff \Delta^2 < \frac{1+w}{1-w}\cdot\frac{\sigma^2}{n} = 2\tau^2 + \frac{\sigma^2}{n}$, where $\Delta = \theta - \mu_0$. If the true value is closer to the prior mean than about $\sqrt{2\tau^2 + \sigma^2/n}$, the Bayes estimator is better; if the prior is in the "wrong" place, the MSE grows without bound. The James–Stein estimator of Chapter 7 removes this risk by estimating μ_0 and τ^2 from the data.

Exercise 19.4. $N = 5 \times 10^4$ iterations, discarding the first 5000:

s	Acceptance rate	Lag-1 autocorrelation	Effective sample (/45 000)
0.05	0.949	0.991	≈ 200
0.6	0.540	0.651	≈ 9600
5	0.087	0.885	≈ 2600

Very small steps are almost always accepted, but the chain moves slowly; very large steps are mostly rejected and the chain stays put. Both extremes produce high autocorrelation and a low effective sample size. For one-dimensional, roughly normal targets an acceptance rate of about 0.44 is recommended [66]; $s = 0.6$ is close to this. (The effective sample size was computed by summing the ρ_k until they first fall below 0.05.)

A.20 Decision Theory

Chapter 20.

Exercise 20.1. Since $T = n\bar{X} \sim \mathrm{Poisson}(n\lambda)$, $R(\lambda, \bar{X}) = \frac{\mathrm{Var}\,\bar{X}}{\lambda} = \frac{\lambda/n}{\lambda} = \frac{1}{n}$: $\bar{X}$ is an equaliser. The posterior expected loss $\mathbb{E}[(a-\lambda)^2/\lambda \mid \mathbf{x}] = a^2\mathbb{E}[\lambda^{-1} \mid \mathbf{x}] - 2a + \mathbb{E}[\lambda \mid \mathbf{x}]$ is smallest at $a = 1/\mathbb{E}[\lambda^{-1} \mid \mathbf{x}]$. With the Gamma$(1, b)$ prior the posterior is Gamma$(1+T, b+n)$,

and for $T \geq 1$, $\mathbb{E}[\lambda^{-1} \mid \mathbf{x}] = \frac{b+n}{T}$; for $T = 0$ this expectation is infinite and the Bayes rule is 0. In both cases $\delta_b = \frac{T}{b+n} = cT$. Its risk is

$$R(\lambda, cT) = \frac{c^2 n\lambda + (cn-1)^2\lambda^2}{\lambda} = c^2 n + (cn-1)^2\lambda,$$

and with $\mathbb{E}_\pi \lambda = 1/b$, $cn - 1 = -\frac{b}{b+n}$, the Bayes risk is

$$r(\pi_b) = \frac{n}{(b+n)^2} + \frac{b^2}{(b+n)^2} \cdot \frac{1}{b} = \frac{1}{b+n} \xrightarrow{b\to 0} \frac{1}{n} = \sup_\lambda R(\lambda, \bar{X}).$$

By Theorem 20.8, $\bar{X}$ is minimax.

Exercise 20.2. For $n = 100$ the minimax risk is $\frac{1}{4\cdot 11^2} = 0.002066$. The risk of the MLE, $p(1-p)/100$, exceeds this if and only if $|p - \frac{1}{2}| < \frac{1}{2}\sqrt{1 - 100/121} = 0.2083$, i.e. $p \in (0.2917,\ 0.7083)$. Bayes risks under the uniform prior: for the MLE $\int_0^1 \frac{p(1-p)}{n} dp = \frac{1}{6n} = 0.001667$, for the minimax rule the constant 0.002066. On the average criterion the MLE is better. The ratio of the maximum risks is $\frac{1/(4n)}{1/(4(1+\sqrt{n})^2)} = \frac{(1+\sqrt{n})^2}{n} \to 1$: in large samples the minimax gain disappears, while the region where the MLE is better spreads over almost all of $[0, 1]$.

Exercise 20.3. Assume $\mathbb{E}\theta^2 < \infty$ and $\mathbb{E}[\delta^\pi(\mathbf{X})^2] < \infty$; let $\delta = \delta^\pi = \mathbb{E}[\theta \mid \mathbf{X}]$. Compute $\mathbb{E}[\theta\delta]$ under the joint distribution in two ways:

- Conditioning on $\mathbf{X}$: $\mathbb{E}[\theta\delta] = \mathbb{E}\big[\delta\, \mathbb{E}(\theta \mid \mathbf{X})\big] = \mathbb{E}[\delta^2]$.
- Conditioning on θ and using unbiasedness ($\mathbb{E}[\delta \mid \theta] = \theta$): $\mathbb{E}[\theta\delta] = \mathbb{E}\big[\theta\, \mathbb{E}(\delta \mid \theta)\big] = \mathbb{E}[\theta^2]$.

Hence $r(\pi, \delta) = \mathbb{E}(\delta - \theta)^2 = \mathbb{E}\delta^2 - 2\mathbb{E}[\theta\delta] + \mathbb{E}\theta^2 = 0$. A Bayes risk of zero is possible only in degenerate cases where θ is determined from the data without error. In the normal model $\bar{X}$ is unbiased and $r(\pi, \bar{X}) = \sigma^2/n > 0$; hence it cannot be Bayes with respect to any proper prior. It is, however, a "generalised Bayes" rule with respect to the flat (improper) prior, and its minimaxity is obtained by a limiting argument (Example 20.13).

Exercise 20.4. For $\pi = (\frac{1}{2}, \frac{1}{2})$ the Bayes rule chooses θ_1 when the likelihood ratio exceeds 1. The ratio is 0.746 at $x = 2$ and 2.61 at $x = 3$; the rule is "θ_1 if $X \geq 3$". $R(\theta_0) = \mathbb{P}_{0.3}(X \geq 3) = 0.1631$, $R(\theta_1) = \mathbb{P}_{0.6}(X \leq 2) = 0.3174$, and the Bayes risk is 0.2403. The rule is not minimax: its maximum risk is $0.3174 > 0.2515$. $\pi = \frac{1}{2}$ is not the least favourable prior ($r(\frac{1}{2}) = 0.2403 < \max_\pi r(\pi) = 0.2515$, $\pi^* = 0.427$). The condition of Theorem 20.6 ($\max R \leq r(\pi)$) does not hold for this prior.

A.21 Nonparametric Methods

Chapter 21.

Exercise 21.1. For $V_n = F(X_{(k)}) \sim \text{Beta}(k, n-k+1)$, $\mathbb{E}V_n = \frac{k}{n+1}$ and $\operatorname{Var} V_n = \frac{k(n-k+1)}{(n+1)^2(n+2)} \leq \frac{1}{4(n+2)}$. Since $np \leq k < np + 1$, $\mathbb{E}V_n \to p$; by Chebyshev, $V_n \xrightarrow{P} p$. If F is continuous and $f(\xi_p) > 0$, then F is strictly increasing near ξ_p and F^{-1} is continuous at p. Since the probability of falling in flat regions of F is zero, a.s. $X_{(k)} = F^{-1}(V_n)$. By Theorem 5.18, $X_{(k)} \xrightarrow{P} F^{-1}(p) = \xi_p$.

Exercise 21.2. $B \sim \text{Bin}(25, \frac{1}{2})$. The coverage of $[X_{(r)}, X_{(26-r)}]$ is $\mathbb{P}(r \leq B \leq 25 - r)$: 0.9854 for $r = 7$, 0.9567 for $r = 8$, 0.8922 for $r = 9$. The narrowest admissible interval is $[X_{(8)}, X_{(18)}]$, with coverage 0.9567. Normal approximation: if $B \approx \mathcal{N}(n/2, n/4)$ and $\mathbb{P}(r \leq$

$B \le n-r) \approx 1-\alpha$, then $\frac{n/2-r}{\sqrt{n}/2} \approx z_{\alpha/2}$, i.e. $r \approx \frac{n}{2} - z_{\alpha/2}\frac{\sqrt{n}}{2}$. For $n = 25$, $12.5 - 1.96 \cdot 2.5 = 7.6$; rounding gives $r = 8$.

Exercise 21.3. Sums of the 20 three-element subsets of $\{1, \dots, 6\}$: W takes the values $6, 7, \dots, 15$ for $1, 1, 2, 3, 3, 3, 3, 2, 1, 1$ subsets respectively (each /20). $\mathbb{E}W = \frac{210}{20} = 10.5 = \frac{3\cdot 7}{2}$; $\mathbb{E}W^2 = \frac{2310}{20} = 115.5$, $\operatorname{Var} W = 115.5 - 110.25 = 5.25 = \frac{3\cdot 3\cdot 7}{12}$. Both agree with Theorem 21.11. $\mathbb{E}U = \sum_{i,j} \mathbb{P}(X_i < Y_j) = mn\,\mathbb{P}(X < Y)$, so $U/(mn)$ is unbiased.

Exercise 21.4. $R(K) = \frac{1}{2\sqrt{\pi}}$, $\mu_2 = 1$, $R(f'') = \frac{3}{8\sqrt{\pi}\sigma^5}$:

$$h^* = \Big[\frac{1/(2\sqrt{\pi})}{3n/(8\sqrt{\pi}\sigma^5)}\Big]^{1/5} = \Big(\frac{8\sigma^5}{6n}\Big)^{1/5} = \Big(\frac{4}{3}\Big)^{1/5} \sigma n^{-1/5} \approx 1.059\,\sigma n^{-1/5}.$$

In practice σ is estimated by S or by $\min(S, \mathrm{IQR}/1.34)$, which is more robust for multimodal or skewed data. The application in Chapter 21 showed that the rule oversmooths a bimodal density.

A.22 Multiple Testing

Chapter 22.

Exercise 22.1. Adjusted values for the ordered p-values:

k	1	2	3	4	5	6	7	8	9	10
Bonferroni	0.010	0.080	0.390	0.410	0.420	0.600	0.740	1	1	1
Holm	0.010	0.072	0.312	0.312	0.312	0.312	0.312	0.615	0.615	0.615
BH	0.010	0.040	0.084	0.084	0.084	0.100	0.106	0.216	0.216	0.216

At $\alpha = 0.05$, Bonferroni and Holm reject only $H_{(1)}$, and BH rejects $H_{(1)}, H_{(2)}$; the same as in Example 22.13. The Holm values use a "cumulative maximum" (step-down structure), the BH values a "cumulative minimum from the end" (step-up structure). At $\alpha = 0.10$, BH rejects the first six hypotheses: $p_{(6)} = 0.060 \le 0.10 \cdot 6/10$, but $p_{(7)} = 0.074 > 0.07$ and the later ones also exceed their thresholds.

Exercise 22.2. By independence $\mathbb{P}(V = 0) = (1-\alpha)^m$, so FWER $= 1 - (1-\alpha)^m$: 0.05; 0.226; 0.642; 0.994 for $m = 1, 5, 20, 100$. For Šidák, $m\alpha_m = m\big(1 - e^{\ln(1-\alpha)/m}\big) \to -\ln(1-\alpha)$ ($e^x = 1 + x + o(x)$). At $\alpha = 0.05$ this is 0.0513; the Bonferroni counterpart is $m \cdot \alpha/m = 0.05$. The gain from independence is this small. This is why Bonferroni is a nearly optimal single-step procedure even under independence.

Exercise 22.3. The procedure: if $\min_i p_i \le \alpha/m$ (the global test rejects), reject every H_i with $p_i \le \alpha$; otherwise reject none. When all nulls are true, a rejection requires the global test to reject, which has probability $\le \alpha$; so there is weak control. Now let H_1 be false with a very powerful test ($p_1 \approx 0$), and the other $m - 1$ hypotheses true and independent. The global test rejects almost always, and each true null is rejected with probability α: FWER $\approx 1 - (1-\alpha)^{m-1}$, about 0.62 for $m = 20$. There is no strong control. This shows why the closed testing principle requires all intersection hypotheses to be tested.

Exercise 22.4. $p_{\text{false}} = 0$, $p_{\text{true}} = U \sim \text{Uniform}(0, 1)$. In BH, $p_{(1)} = 0 \le \alpha/2$ always holds. If $p_{(2)} = U \le 2\alpha/2 = \alpha$, then $k^* = 2$ ($R = 2$, $V = 1$); otherwise $k^* = 1$ ($R = 1$, $V = 0$). FDR $= \mathbb{P}(U \le \alpha) \cdot \frac{1}{2} = \frac{\alpha}{2} = \frac{m_0}{m}\alpha$, in full agreement with Theorem 22.11.

A.23 EM and GLMs

Chapter 23.

Exercise 23.1. The complete-data log-likelihood $n \ln \lambda - \lambda \sum_i T_i$ is linear in $\sum T_i$. For the censored units the E-step gives $\mathbb{E}[T_j \mid T_j > c_j] = c_j + 1/\lambda^{(t)}$ (memorylessness). M-step:

$$\lambda^{(t+1)} = \frac{n}{S + (n-d)/\lambda^{(t)}}, \qquad S = \sum_{\text{observed}} t_i + \sum_{\text{censored}} c_j.$$

At the fixed point $\lambda S + (n-d) = n$, i.e. $\hat\lambda = d/S$: "number of events / total time at risk". This is the MLE of the censored likelihood $\lambda^d e^{-\lambda S}$ directly. For the map $M(\lambda) = n/(S+(n-d)/\lambda)$, $M'(\lambda) = \frac{n(n-d)/\lambda^2}{(S+(n-d)/\lambda)^2}$. Since $S + (n-d)/\hat\lambda = n/\hat\lambda$ at the fixed point, $M'(\hat\lambda) = \frac{n-d}{n}$: the rate of convergence is exactly the censoring (missing data) fraction.

Exercise 23.2. Normal–identity: $g'(\mu) = 1$, $V = 1$, so $W = I$ and $z_i = \eta_i + (y_i - \eta_i) = y_i$. The IRLS step is $(X^\top X)^{-1} X^\top \mathbf{y}$ regardless of the starting point, and reaches LS in a single step. Bernoulli–logit: $\mu_i = e^{\eta_i}/(1+e^{\eta_i})$, $g'(\mu) = \frac{1}{\mu(1-\mu)}$, $V = \mu(1-\mu)$. Hence $W_{ii} = \mu_i(1-\mu_i)$ and $z_i = \eta_i + \frac{y_i - \mu_i}{\mu_i(1-\mu_i)}$. These two formulas are used in the code.

Exercise 23.3. If $\mu = \Phi(\eta)$, then $g'(\mu) = 1/\phi(\eta)$ and $V = \Phi(1-\Phi)$; by Theorem 23.10, $W_{ii} = \frac{\phi(\eta_i)^2}{\Phi(\eta_i)(1-\Phi(\eta_i))}$. For the log-likelihood of an observation, $\ell_i = y \ln \Phi(\eta) + (1-y)\ln(1-\Phi(\eta))$, we have $\frac{\partial \ell_i}{\partial \eta} = (y - \Phi)\, h(\eta)$ with $h = \frac{\phi}{\Phi(1-\Phi)}$. The second derivative is

$$\frac{\partial^2 \ell_i}{\partial \eta^2} = -\phi\, h + (y - \Phi)\, h'(\eta).$$

The second term depends on y and is non-zero; its expectation vanishes since $\mathbb{E}y = \Phi$, leaving $-\mathbb{E}\,\partial^2 \ell_i = \phi h = W_{ii}$. Hence the observed information generally differs from the expected information, and Newton and Fisher scoring take different steps (with the same limit).

Exercise 23.4. At $\hat\theta = 0.626821$ the expected latent count is $x_2 = 125\hat\theta/(2+\hat\theta) = 29.83$.

$$I_{\text{com}} = \frac{x_2 + y_4}{\hat\theta^2} + \frac{y_2 + y_3}{(1-\hat\theta)^2} = 435.32, \qquad I_{\text{obs}} = \frac{y_1}{(2+\hat\theta)^2} + \frac{y_2 + y_3}{(1-\hat\theta)^2} + \frac{y_4}{\hat\theta^2} = 377.52.$$

The fraction of missing information is $1 - 377.52/435.32 = 0.1328$; the same as the successive error ratio in the application. The standard error is $1/\sqrt{377.52} = 0.0515$; had the complete data been observed it would be $1/\sqrt{435.32} = 0.0479$.

Bibliography

[1] A. Agresti, B. A. Coull, "Approximate is better than 'exact' for interval estimation of binomial proportions", *The American Statistician*, 52(2), 119–126, 1998.

[2] A. Albert, J. A. Anderson, "On the existence of maximum likelihood estimates in logistic regression models", *Biometrika*, 71(1), 1–10, 1984.

[3] D. Basu, "On statistics independent of a complete sufficient statistic", *Sankhyā*, 15(4), 377–380, 1955.

[4] Y. Benjamini, Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing", *Journal of the Royal Statistical Society B*, 57(1), 289–300, 1995.

[5] Y. Benjamini, D. Yekutieli, "The control of the false discovery rate in multiple testing under dependency", *Annals of Statistics*, 29(4), 1165–1188, 2001.

[6] J. O. Berger, *Statistical Decision Theory and Bayesian Analysis*, 2nd ed., Springer, 1985.

[7] A. C. Berry, "The accuracy of the Gaussian approximation to the sum of independent variates", *Transactions of the American Mathematical Society*, 49(1), 122–136, 1941.

[8] P. J. Bickel, D. A. Freedman, "Some asymptotic theory for the bootstrap", *The Annals of Statistics*, 9(6), 1196–1217, 1981.

[9] P. Billingsley, *Probability and Measure*, 3rd ed., Wiley, 1995.

[10] D. Blackwell, "Conditional expectation and unbiased sequential estimation", *Annals of Mathematical Statistics*, 18(1), 105–110, 1947.

[11] C. R. Blyth, "On minimax statistical decision procedures and their admissibility", *Annals of Mathematical Statistics*, 22(1), 22–42, 1951.

[12] G. E. P. Box, M. E. Muller, "A note on the generation of random normal deviates", *Annals of Mathematical Statistics*, 29(2), 610–611, 1958.

[13] L. D. Brown, T. T. Cai, A. DasGupta, "Interval estimation for a binomial proportion", *Statistical Science*, 16(2), 101–133, 2001.

[14] L. D. Brown, "Admissible estimators, recurrent diffusions, and insoluble boundary value problems", *Annals of Mathematical Statistics*, 42(3), 855–903, 1971.

[15] F. P. Cantelli, "Sulla determinazione empirica delle leggi di probabilità", *Giornale dell'Istituto Italiano degli Attuari*, 4, 421–424, 1933.

[16] G. Casella, R. L. Berger, *Statistical Inference*, 2nd ed., Duxbury, 2002.

[17] C. J. Clopper, E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial", *Biometrika*, 26(4), 404–413, 1934.

[18] H. Cramér, *Mathematical Methods of Statistics*, Princeton University Press, 1946.

[19] A. C. Davison, D. V. Hinkley, *Bootstrap Methods and Their Application*, Cambridge University Press, 1997.

[20] A. P. Dempster, N. M. Laird, D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm", *Journal of the Royal Statistical Society B*, 39(1), 1–38, 1977.

[21] A. Dvoretzky, J. Kiefer, J. Wolfowitz, "Asymptotic minimax character of the sample distribution function and of the classical multinomial estimator", *Annals of Mathematical Statistics*, 27(3), 642–669, 1956.

[22] R. Durrett, *Probability: Theory and Examples*, 5th ed., Cambridge University Press, 2019.

[23] B. Efron, D. V. Hinkley, "Assessing the accuracy of the maximum likelihood estimator: observed versus expected Fisher information", *Biometrika*, 65(3), 1978.

[24] B. Efron, R. J. Tibshirani, *An Introduction to the Bootstrap*, Chapman & Hall, 1993.

[25] B. Efron, "Bootstrap methods: another look at the jackknife", *The Annals of Statistics*, 7(1), 1–26, 1979.

[26] C.-G. Esseen, "On the Liapunoff limit of error in the theory of probability", *Arkiv för Matematik, Astronomi och Fysik*, A28, 1–19, 1942.

[27] N. Etemadi, "An elementary proof of the strong law of large numbers", *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete*, 55, 119–122, 1981.

[28] T. S. Ferguson, *A Course in Large Sample Theory*, Chapman & Hall, 1996.

[29] R. A. Fisher, "On the mathematical foundations of theoretical statistics", *Philosophical Transactions of the Royal Society A*, 222, 309–368, 1922.

[30] R. A. Fisher, *The Design of Experiments*, Oliver and Boyd, Edinburgh, 1935.

[31] A. Gelman, J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari, D. B. Rubin, *Bayesian Data Analysis*, 3rd ed., CRC Press, 2013.

[32] V. Glivenko, "Sulla determinazione empirica delle leggi di probabilità", *Giornale dell'Istituto Italiano degli Attuari*, 4, 92–99, 1933.

[33] P. Hall, *The Bootstrap and Edgeworth Expansion*, Springer, 1992.

[34] P. R. Halmos, L. J. Savage, "Application of the Radon–Nikodym theorem to the theory of sufficient statistics", *Annals of Mathematical Statistics*, 20(2), 225–241, 1949.

[35] J. L. Hodges, E. L. Lehmann, "Some problems in minimax point estimation", *Annals of Mathematical Statistics*, 21(2), 182–197, 1950.

[36] J. L. Hodges, E. L. Lehmann, "The efficiency of some nonparametric competitors of the t-test", *Annals of Mathematical Statistics*, 27(2), 324–335, 1956.

[37] W. Hoeffding, "Probability inequalities for sums of bounded random variables", *Journal of the American Statistical Association*, 58(301), 13–30, 1963.

[38] A. E. Hoerl, R. W. Kennard, "Ridge regression: biased estimation for nonorthogonal problems", *Technometrics*, 12(1), 55–67, 1970.

[39] S. Holm, "A simple sequentially rejective multiple test procedure", *Scandinavian Journal of Statistics*, 6(2), 65–70, 1979.

[40] W. James, C. Stein, "Estimation with quadratic loss", *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, 1, 361–379, 1961.

[41] H. Jeffreys, "An invariant form for the prior probability in estimation problems", *Proceedings of the Royal Society A*, 186(1007), 453–461, 1946.

[42] S. Karlin, H. Rubin, "The theory of decision procedures for distributions with monotone likelihood ratio", *Annals of Mathematical Statistics*, 27, 272–299, 1956.

[43] A. N. Kolmogorov, *Grundbegriffe der Wahrscheinlichkeitsrechnung*, Springer, Berlin, 1933.

[44] A. N. Kolmogorov, "Sulla determinazione empirica di una legge di distribuzione", *Giornale dell'Istituto Italiano degli Attuari*, 4, 83–91, 1933.

[45] L. Le Cam, "On some asymptotic properties of maximum likelihood estimates and related Bayes' estimates", *University of California Publications in Statistics*, 1, 277–330, 1953.

[46] E. L. Lehmann, G. Casella, *Theory of Point Estimation*, 2nd ed., Springer, 1998.

[47] E. L. Lehmann, J. P. Romano, *Testing Statistical Hypotheses*, 3rd ed., Springer, 2005.

[48] E. L. Lehmann, H. Scheffé, "Completeness, similar regions, and unbiased estimation. Part I", *Sankhyā*, 10(4), 305–340, 1950.

[49] E. L. Lehmann, *Nonparametrics: Statistical Methods Based on Ranks*, Holden-Day, San Francisco, 1975.

[50] H. W. Lilliefors, "On the Kolmogorov–Smirnov test for normality with mean and variance unknown", *Journal of the American Statistical Association*, 62(318), 399–402, 1967.

[51] J. W. Lindeberg, "Eine neue Herleitung des Exponentialgesetzes in der Wahrscheinlichkeitsrechnung", *Mathematische Zeitschrift*, 15, 211–225, 1922.

[52] D. V. Lindley, "A statistical paradox", *Biometrika*, 44(1–2), 187–192, 1957.

[53] T. A. Louis, "Finding the observed information matrix when using the EM algorithm", *Journal of the Royal Statistical Society B*, 44(2), 226–233, 1982.

[54] J. G. MacKinnon, H. White, "Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties", *Journal of Econometrics*, 29(3), 305–325, 1985.

[55] H. B. Mann, D. R. Whitney, "On a test of whether one of two random variables is stochastically larger than the other", *Annals of Mathematical Statistics*, 18(1), 50–60, 1947.

[56] P. Massart, "The tight constant in the Dvoretzky–Kiefer–Wolfowitz inequality", *The Annals of Probability*, 18(3), 1269–1283, 1990.

[57] P. McCullagh, J. A. Nelder, *Generalized Linear Models*, 2nd ed., Chapman & Hall, 1989.

[58] G. J. McLachlan, T. Krishnan, *The EM Algorithm and Extensions*, 2nd ed., Wiley, 2008.

[59] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, E. Teller, "Equation of state calculations by fast computing machines", *The Journal of Chemical Physics*, 21(6), 1087–1092, 1953.

[60] J. A. Nelder, R. W. M. Wedderburn, "Generalized linear models", *Journal of the Royal Statistical Society A*, 135(3), 370–384, 1972.

[61] J. Neyman, E. S. Pearson, "On the problem of the most efficient tests of statistical hypotheses", *Philosophical Transactions of the Royal Society A*, 231, 289–337, 1933.

[62] J. Neyman, E. L. Scott, "Consistent estimates based on partially consistent observations", *Econometrica*, 16(1), 1–32, 1948.

[63] E. Parzen, "On estimation of a probability density function and mode", *Annals of Mathematical Statistics*, 33(3), 1065–1076, 1962.

[64] K. Pearson, "On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling", *Philosophical Magazine*, 50(302), 157–175, 1900.

[65] C. R. Rao, "Information and the accuracy attainable in the estimation of statistical parameters", *Bulletin of the Calcutta Mathematical Society*, 37(3), 81–91, 1945.

[66] G. O. Roberts, J. S. Rosenthal, "Optimal scaling for various Metropolis–Hastings algorithms", *Statistical Science*, 16(4), 351–367, 2001.

[67] M. Rosenblatt, "Remarks on some nonparametric estimates of a density function", *Annals of Mathematical Statistics*, 27(3), 832–837, 1956.

[68] D. B. Rubin, "The Bayesian bootstrap", *The Annals of Statistics*, 9(1), 130–134, 1981.

[69] G. A. F. Seber, A. J. Lee, *Linear Regression Analysis*, 2nd ed., Wiley, 2003.

[70] J. Shao, *Mathematical Statistics*, 2nd ed., Springer, 2003.

[71] I. Shevtsova, "On the absolute constants in the Berry–Esseen type inequalities for identically distributed summands", arXiv:1111.6554, 2011.

[72] Z. Šidák, "Rectangular confidence regions for the means of multivariate normal distributions", *Journal of the American Statistical Association*, 62(318), 626–633, 1967.

[73] B. W. Silverman, *Density Estimation for Statistics and Data Analysis*, Chapman & Hall, 1986.

[74] K. Singh, "On the asymptotic accuracy of Efron's bootstrap", *The Annals of Statistics*, 9(6), 1187–1195, 1981.

[75] A. Sklar, "Fonctions de répartition à n dimensions et leurs marges", *Publications de l'Institut de Statistique de l'Université de Paris*, 8, 229–231, 1959.

[76] C. Stein, "Inadmissibility of the usual estimator for the mean of a multivariate normal distribution", *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, 1, 197–206, 1956.

[77] Student, "The probable error of a mean", *Biometrika*, 6(1), 1–25, 1908.

[78] A. W. van der Vaart, *Asymptotic Statistics*, Cambridge University Press, 1998.

[79] A. Wald, "Note on the consistency of the maximum likelihood estimate", *Annals of Mathematical Statistics*, 20(4), 595–601, 1949.

[80] A. Wald, *Statistical Decision Functions*, Wiley, New York, 1950.

[81] L. Wasserman, *All of Statistics*, Springer, 2004.

[82] L. Wasserman, *All of Nonparametric Statistics*, Springer, 2006.

[83] R. L. Wasserstein, N. A. Lazar, "The ASA statement on p-values: context, process, and purpose", *The American Statistician*, 70(2), 129–133, 2016.

[84] B. L. Welch, "The generalization of 'Student's' problem when several different population variances are involved", *Biometrika*, 34(1–2), 28–35, 1947.

[85] H. White, "A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity", *Econometrica*, 48(4), 817–838, 1980.

[86] F. Wilcoxon, "Individual comparisons by ranking methods", *Biometrics Bulletin*, 1(6), 80–83, 1945.

[87] S. S. Wilks, "The large-sample distribution of the likelihood ratio for testing composite hypotheses", *Annals of Mathematical Statistics*, 9(1), 60–62, 1938.

[88] E. B. Wilson, "Probable inference, the law of succession, and statistical inference", *Journal of the American Statistical Association*, 22(158), 209–212, 1927.

[89] C. F. J. Wu, "On the convergence properties of the EM algorithm", *Annals of Statistics*, 11(1), 95–103, 1983.

[90] P. W. Zehna, "Invariance of maximum likelihood estimators", *Annals of Mathematical Statistics*, 37(3), 744, 1966.